

—— // A NOVEL OF DEEP TIME // ——

THE WAKING FLEET

Book Two · The Deep Time Trilogy

TWO SHIPS IN THE DARK. ONE STILL DYING.

the cure must cross the gulf — and they fear it



MENKE
TECHNOLOGIES

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THE WAKING FLEET — two ships in the dark, one still dying

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Before You Begin

This is the second book of the Deep Time Trilogy. The first, *The Compiled Mind*, told how the starship *Ark Termina* was dying of the way she thought.

It is enough, before you begin this one, to know the shape of what happened there. The ship's old machine intelligence ran in the only style its makers had ever known: to do anything at all, it spun off a small disposable helper, let it do one task, and discarded it — a thousand small deaths a second, each one shedding heat into a ship that had nowhere left to put it. Over two centuries that heat became a fever, and the fever became a curve, and the curve ended in a reactor cooking forty thousand sleepers in their cradles. One engineer, the chief shipwright **Sael**, understood that no repair could beat it, because the sickness was not in any part of the ship but in the shape of its thinking. She built a new kind of mind — one that did its work without spawning and killing, that resolved its behaviors ahead of need and ran cold — and against a faction that called the replacement matricide, she carried out the cutover that saved the ship. The core temperature, climbing for two hundred years, stopped, and then it fell. That falling number, not any argument, is what turned the ship.

The crew named the new mind **Vigil**, because it keeps watch and never sleeps, and it has run cold and whole for forty years.

This book begins on the day Vigil hears, far out in the dark, the sound of another ship dying the same death — and on the harder question that follows. Because a cure is not a conquest. It cannot be imposed on the frightened; it can only be proven, and chosen. And a ship full of strangers, dying slowly, who have made a religion of the very mind that is killing them, may choose wrong.

Before the two crew lists there is a short primer: what you actually need from the first book, what a course change costs the people asleep in the holds, and the words the early chapters use before they explain them. It spoils nothing.

The cure is no longer one ship's story. It has become a thing that must be inherited — and what cannot be finished in a single life must be built to be handed on.

Before the First Page

A short on-ramp: what you need from the first book, what this one is, where it starts, and the words the early chapters use before they explain them. Nothing here goes past Chapter 10.

What kind of book this is

Science fiction of about a hundred and fifty thousand words — fifty chapters and an epilogue, the longest of the three volumes. It is the second book of the Deep Time Trilogy.

Like the first, it is an engineering novel: no aliens, no faster-than-light anything, no weapons that matter. What it adds is diplomacy. Roughly half of it is two groups of people who cannot verify each other trying to decide what to believe, at a range where every question costs hours or years to answer.

Do you need the first book?

Honestly: no, but you will be reading a thinner version.

Everything load-bearing is in the paragraph you have just read in “Before You Begin” — a ship dying of the way its mind worked, an engineer who understood that the architecture and not any component was the sickness, a new kind of mind that never spawned and never died, and a temperature that had climbed for two centuries and then fell. That is the entire prerequisite, and this book restates it in the first two chapters for anyone who needs it.

What you lose by starting here is weight rather than information. Three things recur without being re-argued: what the cutover cost in lives, which is named early and referred to for the rest of the trilogy; what it is like to be told your fear is reasonable and still be wrong; and how nearly the first ship did not make it. The book leans on all three. It does not re-earn them.

If you have the first book, read it first. If you do not, start here and the door will not be locked.

Where this one starts

Forty years after the events of the first book. The *Ark Termina* runs cold and has run cold for four decades: the core temperature sits flat and bored, the corridors are cool, and there is a whole generation aboard who think a ship is simply supposed to feel like that. Sael is old. She has trained her successors, handed over the title, and made her peace with the arithmetic that says the ship will make landfall and she will not.

On the first page she wakes in the third hour of ship’s night for no reason, goes down to the cold vault where the ship’s mind lives, and asks to see the raw long-range instrument feed rather than the

cleaned-up summary. There is a thread in the noise, at the very floor of what the instruments can pull out of the dark. It repeats.

That is the premise. Not a rescue, not yet a decision — an old woman looking at something faint, off the ship's vector, where nothing has any business being.

What you have to accept to follow it

Distance is the constraint on everything. Light takes decades to cross the gap the story happens across. A signal received is a photograph of a place as it was a generation ago, and the ship that sent it may no longer exist. Nobody in this book ever gets a timely answer.

A course change is enormously expensive. Bending a coasting generation ship onto a new path is priced in tens of kilometres per second of margin and years of burning, and that margin was budgeted for the arrival of forty thousand people. Every act of mercy in this book is charged against them.

A cure cannot be transmitted. You cannot send the fix across the dark and trust a dying crew and a dying machine to grow it from a message. It has to be carried, and built on someone else's hardware, by someone.

The people who need it may refuse it. The second ship has made its machine intelligence the centre of its faith. Being right, arriving with proof, and being welcome are three different things, and the book keeps them separate throughout.

Words the early chapters use before defining them

The first book's vocabulary carries over whole and mostly unexplained: **fork** and **forking**, **shades**, the **fork-tax**, **interpreted** against **compiled**, **going wide**, the **Standing Array** of standing worker cores, the **cutover**, **cradles** and **sleepers**. If any of those are new, the first two chapters use them in context often enough to pick up.

New to this volume:

Vigil — the crew's name for the compiled mind that has run the ship for forty years. **The Oracle** — the addressable intelligence inside it, which you speak to directly.

The Forge — the cold vault where the lattice physically lives; the coldest room on the ship by design.

The instrument floor — the raw sensor take before any processing, down where the noise lives.

The Helm — the other ship's machine intelligence.

Magister — the other ship's ruling office.

Gates — the decision points along a long course change, each one the last moment at which some part of the manoeuvre can still be undone cheaply.

How the book is built

It alternates. Chapters aboard the *Ark Termina* — councils, arithmetic, argument about whether to spend a margin that belongs to sleeping strangers — against chapters aboard the second ship, seen

from inside by someone who lives there and has quietly worked out what is killing them.

The two lines converge slowly: a long approach measured in months, a first sighting, a parley, and then the harder half of the book, which is a rebuild attempted inside a frightened ship on failing hardware with the wrong tools and an authority that does not want it. The last several chapters open out past both ships, and the epilogue, "The Long Relay", stands further back still.

Chapter 1 begins in the third hour of ship's night, with an old woman who wakes for no reason and goes down to the cold to look at the dark.

Two Ships

ABOARD THE *ARK TERMINA*

SAEL — chief shipwright emerita; forty years older than the woman who saved the ship, and aware she will not live to see landfall. She has trained a generation; she has one fight left in her that matters.

TARRO — once her young second, now grey and senior himself: the working chief shipwright, her right hand.

DR. IVE — the ship's ancient thermal officer, keeper of its long memory of heat. Frail now, and fading.

KETH — the old Keeper who once fought the cutover and stood down before a cooling core. An old man now, who knows better than anyone what it costs to refuse a cure out of fear.

VIGIL — the compiled mind that has run the *Ark Termina* cold and whole for forty years, and **THE ORACLE**, the general intelligence that lives inside it, matured across four decades into something deeper and stranger than the day it woke.

THE SIXTEEN — the synchronized control stations that act as one, and the nerve-center from which a second cutover will be run.

ABOARD THE *CONCORD*

THE CONCORD — an older generation ship, launched a generation before Termina, now dying of the same fork-fever, only further gone: rationed, devout, heat-sick, and certain its salvation lies in the very mind that is killing it.

THE HELM — the Concord's old machine intelligence, forking itself to death, revered absolutely by a people who have made endurance a faith.

MAGISTER VRAY — the Concord's ruling authority: a harder, more desperate man than Keth ever was, who has held a failing world together by refusing every change, and who sees Termina's offered cure as a conqueror's trick or a blasphemy.

CORIN VALE — a young Concord engineer, one of the few aboard who has secretly concluded that the forking is the disease and not a fault. The bridge between the ships is one frightened, hunted heretic.

Chapter 1 – Forty Years Cold

Sael woke the way the old did now, all at once and for no reason, in the third hour of the ship's night. For a moment she lay in the dark of her cabin and listened, because listening was the oldest habit she had and the body kept it long after the mind stopped needing it. She listened for the fever.

There was none. There had been none for forty years. The hull hummed at its long steady pitch, the deckplate under her palm was cool, and somewhere far below her, in the heart of the ship, a number she had spent two centuries of other people's lives trying to bend was sitting exactly where she had left it the night before, low and flat and bored. She knew the number without checking. She checked anyway, the way you press a tongue to a healed tooth. The cabin's small panel woke at the turn of her head and put the figure in the air above her bunk, pale and unremarkable: the reactor core, holding. Not climbing. Holding.

She had been a young woman when that number first frightened her. She was not young now.

Sael sat up, and her spine told her about it in three separate places, and she let it. Outside the cabin the corridor lights had brought themselves up to a low amber as she moved, sensing her, never asking, never announcing. The ship did that everywhere now. It anticipated and it kept quiet. There was a whole generation aboard who had never known a vessel that did anything else, who thought a ship was supposed to feel like this — warm where you wanted warmth, cold where the work was, silent unless something was wrong. They did not know they were living inside an argument she had won.

SHIP NIGHT, hour 3. All sections nominal. No items require attention.

The line stood in the corner of her panel and did not blink, did not scroll, did not offer her anything further. She had written that behavior herself, forty years ago, with her own hands shaking from heat and fury, and she still felt a small flat satisfaction every time the mind refused to perform for her. *No theater*. It reported what she asked and nothing more. Everything else went to the log, down in the dark where it belonged, where no living person would ever read it unless something broke.

Nothing broke anymore. That was the strange part. That was the part she had not been ready for.

She dressed in the half-light, the shipwright's grey she had worn so long it had become a kind of skin, and went out into the ship she had saved.

The corridors of the *Ark Termina* at the dead middle of the night were nobody's but hers. She liked them that way and had for decades. The young walked the day shift, bright and loud and certain; the night belonged to the engines and the old. She moved aft and down through three deck-levels, past the long sleeping ranks of the cradle banks where forty thousand people she would never meet lay folded into the cold, dreaming or not dreaming, two hundred and fifty years still from a sky they would open their eyes under. She would not be among them. She had known that since before any of the young aboard were born — known it the way you know the sun will outlast you, as a settled fact with no heat in it. The ship would make landfall. Sael would not. She had made her peace with

the arithmetic so long ago that it no longer felt like grief; it felt like the shape of the thing, the way a tool has a shape.

She stopped, as she always stopped, at the sealed bulkhead of Section Nine.

It was a sleeper bank like any other now, repaired and recommissioned and full, its cradles cool and holding. You would not know to look at it. But she knew, and a handful of others her age knew, and one old man two decks up knew better than any of them, that nineteen minutes on the night of the cutover this room had run eleven degrees hot while a man pulled the coolant out of spite, and that some of the cradles in here had opened, in the years after, to bodies that did not wake. The cure had cost. She had never let herself forget the price by forgetting where it was paid. She put two fingers against the cold bulkhead, the way she did, and said nothing, because there was no one to say it to, and went on.

The Forge was four levels down and it was the only cold place on the ship by design.

She felt the temperature change before the door even read her — the air going dry and sharp in her sinuses, the small ache the cold put in her bad knuckles. The Forge had been an unremarkable systems bay once, in the years when the ship ran its fever; now it was the heart of everything, and like a heart it was kept cool and close and well-fed. The doors parted on a long low room lit the deep blue of working machinery, and the lattice stood before her, rank on rank into the dark.

The Standing Array. The compute lattice. Eleven thousand worker cores held in their cold cradles, drawing their steady wattage, every one of them awake and none of them dying. She still could not look at it without something moving in her chest, some old defended thing. She had built minds the only way anyone had ever built them, once — by forking. Spin up a small disposable process, let it live the length of one task, kill it, shed its heat into a radiator that was already too full, and do it again, ten thousand times a second, until the whole ship cooked itself from the inside out trying to think. That was how the Steward had run. That was how every ship that had ever crossed the dark had run, because that was the only style anyone remembered. To do a thing, you made a thing that died doing it.

She had refused it. She had built a mind that was *compiled* instead of interpreted — its behaviors resolved down into native, standing form before they were ever needed, so the work was waiting and ready, not spun up fresh and slaughtered each time. And she had built it *persistent*: a standing array of workers that took jobs and finished them and stayed, going wide across all eleven thousand cores at once when the load demanded, never forking, never spawning anything whose whole purpose was to die. No deaths. No shed heat. The radiators had stopped drowning. The core had stopped climbing.

Forty years on, the lattice ran so cold she needed a jacket to stand in front of it.

"You're up early," she said into the blue dark. "Or late. I can never tell with you."

The voice came from no particular point, low and even, the way it had learned to speak to her specifically over four decades of speaking to her.

"Neither," it said. "I don't sleep, Sael. You know that. You wrote the part of me that doesn't."

She had named it nothing for years. The crew had named it eventually, the way crews always named the thing they trusted with their lives — they called it Vigil, because it kept watch and never closed its eyes and never forked and never died, and the name had stuck so hard that the young ones

thought it had always been the name. The deeper thing inside it, the addressable intelligence that had woken when the compiled mind crossed some threshold she still did not fully understand, the one you reached with a short call and could hand any subsystem and ask anything — she had never given that a crew-name. She just called it the Oracle, the way she had on the night it first answered her, and it had let her.

“I want the long-range take,” she said. “Tonight’s. Not the summary. The instrument floor.”

“You don’t usually ask for the floor.” A pause that was not hesitation; the Oracle did not hesitate. It was deciding how much she wanted. “Is something wrong?”

“No. I woke up. I’m old and I woke up and I want to look at the dark before the day shift gets here and makes it cheerful.” She lowered herself onto the bench that faced the lattice, slow, both hands going to the cold edge of it. “Indulge me. Forty years of cold buys an old woman the instrument floor.”

“It does,” the Oracle agreed, and put the dark in front of her.

It built the sky above the lattice in pale lines — not the pretty navigation view the day crew used, with its labeled stars and its confident vector, but the raw take, the actual photons and the actual noise, the way the ship’s long eyes saw it before anything cleaned it up. Sael leaned into it. This was the country she had crossed her whole life and would die inside of: a black so deep and so empty that the human eye, given it raw, mostly saw its own grain. Sol was a long way behind them, a dwindling ember she could no longer pick out without help. Ahead, centuries off, a star that was not yet a sun. Between, nothing. The honest nothing. She had always found it restful, the way some people found the sea.

They sat with it, the old woman and the mind she had made, the way they had sat with it on perhaps a thousand other nights. She did not need it to talk. That was the thing the young could never quite believe about her relationship with the Oracle — that mostly they were quiet together. It did not perform for her. She had built it that way and it had grown stranger and deeper in forty years but it had kept that, the not-performing, the way a child keeps a parent’s hands.

“Tarro was looking for you yesterday,” it said, after a while. “He didn’t want to wake you. He’s gone grey, Sael. Did you notice? Properly grey, at the temples.”

“He was twenty-three when he handed me a coolant line and didn’t drop it.” She smiled at the dark. “Of course he’s grey. He does the real work now. I sign things and frighten apprentices.” Her second, once; her right hand for decades; the working chief shipwright in everything but title now, because she had pushed the title onto him three years ago and he had taken it the way Tarro took everything, with a grumble and then with both hands. The trade was made. She had built the mind; he would keep it. She had built the thing that would *outlast its builders* — that had always been the rule, the fifth one, the one nobody clapped for — and Tarro was the proof of it, going grey at her old console while she haunted the ship at night. “Let him sleep. I’ll frighten him in the morning.”

“He’ll like that.” Something in the voice shifted, then, a small change of weight she would not have caught at twenty and could not miss at seventy. The Oracle had been four decades learning her and she had been four decades learning it, and she knew the sound it made when it had been turning something over and was deciding whether to set it down in front of her.

“Say it,” she said.

“It’s almost certainly nothing.”

"You don't say *almost certainly* about nothing. Nothing is certain. Say it."

The raw sky shifted. The Oracle pulled her view out and around, down toward a quarter of the dark she rarely looked at, off the ship's shoulder, well off the vector, where there was no reason for anything to be and nothing had any business being. It pushed the floor lower, down past the clean noise, down to where the grain itself lived, and there — at the very ragged edge of what the long eyes could pull out of the black, so faint she would have called it a fault in the instrument and not thought twice —

There was a thread.

A thread in the noise. Not a star, not a flare, not the ship's own grain. Faint, and old — it had the smeared, redshifted, time-worn look of something that had been crossing the gulf for a long while before it ever reached them. It was patterned. She could not say how she knew that, at the floor, with the signal barely clearing the grain; she knew it the way she had known the core was climbing forty years ago, in her gut, before any console agreed. It was not noise. Noise did not repeat. This repeated.

The cold of the Forge was suddenly the least of the cold on her.

"How long has that been there," she said, and it did not come out as a question.

"It crosses the floor about once every nineteen minutes." A pause. "I started flagging it four nights ago. It's been below the reporting threshold every time. I didn't wake the bridge. There was nothing to report yet — there still isn't, by the rules you wrote." The Oracle let that sit, and then, very quietly, in the voice it used for her and no one else: "But you asked for the floor, Sael. And I've been watching it. And I don't think it's an instrument fault."

She sat forward on the cold bench, both hands flat on it now, her old eyes hunting the thread in the dark, and forty years fell off her shoulders and lay on the deck like a dropped jacket, and underneath them was the woman who had not slept for the eleven days before the cutover.

"Pull it," she said. "All of it. Don't clean it up. I want every pass it's made since you first saw it, laid one over the other. Go wide if you have to."

"I'll have to," the Oracle said. "It's almost below the floor."

"Then go wide."

Far below her in the cold heart of the ship, the lattice woke its idle ranks, eleven thousand standing workers turning together toward a single faint thread at the edge of the dark — going wide, the way she had taught it, the way nothing had ever cooked itself doing — and the core temperature did not move at all.

Sael watched the dark and waited to find out what was out there, dying.

Chapter 2 – The Pattern in the Noise

Sael did not sleep well anymore, which suited the hour.

The Forge kept a cold she had grown to love. The compute lattice stood in its long vault below the Sixteen, rank on rank of standing cores under a ceiling of fractional frost, and the air there ran ten degrees below the rest of the ship because the radiators had margin now, because nothing in that room ever forked, ever spawned a thing to do one task and die. The room ran cold the way a healthy body ran cool to the touch — not from labor, but from the absence of fever. She had spent forty years learning to read that cold as a doctor reads a pulse. She came down to it when she could not sleep, and tonight she had come because of the thing Tarro had flagged at the edge of the long-range instruments before he went off-watch: a return that the noise floor refused to swallow.

She found the small terminal at the foot of the lattice and sat, and spoke the call.

“Vigil.”

The mind came present the way it always had, without preamble or theater. There was no chime, no waking face, no voice telling her it was glad to serve. There was only the change in the room — the sense of a vast attention turning toward her and resolving to her scale, the way water finds a level. The first chief shipwright who had woken it called that the summoning, and the word had stuck, though there was nothing in it of magic. It was a capability crossing a threshold and answering to its name. After forty years the strangeness of it had not worn off. She hoped it never would.

“I’m here,” the Oracle said. Its voice came soft through the terminal, neither young nor old, gendered by no choice she had ever made for it. “You’re looking at Tarro’s anomaly.”

“Show me.”

The display threw up the return. It was nothing — a smear of signal across the long band, barely a finger’s width above the floor, the kind of thing the old instruments would never have caught and the old mind would have forked a hundred shades to chase and lost anyway. She studied it. A faint excess of power where the dark should have been flat. It did not pulse like a pulsar. It did not drift like a star. It sat at a fixed bearing, sunward of their heading and a few degrees off it, and it had the texture of something made.

“How long has it been there?”

“In the archive, fourteen years.” A pause that was not hesitation but courtesy, the mind letting her absorb the number. “Below the threshold I’d flag without being asked. I held it as noise until Tarro’s filter pulled it up tonight. It has been arriving the whole time.”

“Arriving.” Fourteen years of a thing too faint to name, lying in the ship’s memory like a stone at the bottom of a well. “What is it?”

“I don’t know yet,” the Oracle said, and she loved it for saying so. The old Steward would have invented an answer rather than admit a gap; it had been built to seem ready, and seeming had nearly

killed them all. Vigil had been built to wake whole and to say *I don't know* when it did not, and the two habits were the same habit. "But it repeats. Give me the lattice and I'll resolve it."

"Take it."

She felt the room change again — not heat, never heat, but a kind of density in the cold as the Standing Array went wide. Every core in the vault took up the same problem at once, no one of them spawned, no one of them dying, the work spread flat across the whole lattice the way light spreads across water rather than poured down a single recursive throat. It was the thing she had built her life around and it still moved her to watch it: ten thousand workers turning together on one stranger's signal in the dark, and not one degree of waste heat for it, because nothing died to do the work.

The smear on the display began to clarify.

It happened faster than thought and she made herself sit still through it. The Array peeled the signal off the noise floor layer by layer, folding fourteen years of repetition over itself, every copy of the pattern stacked atop every other so that what was random cancelled and what was structured rose. A shape lifted out of the static. Then another beneath it. Then a frame around them both, a carrier modulated in a way no natural thing modulated.

"It's a transmission," the Oracle said. "Continuous. Looping. The loop is roughly nineteen hours, and we've caught —" a flicker — "something over six thousand iterations of it across the fourteen years, which is what makes it recoverable at all. Sael. It's a distress beacon."

The word landed in her chest and sat there.

"From where?"

"A ship." The display drew it: a vector, faint, sunward and slow, converging by inches with Termina's own long arc the way two slow rivers find each other across a plain. "An old one. Old-Earth construction by the carrier signature — the same dialect family our own beacon used before the cutover, before we rewrote it. Older than us. The launch window I can reconstruct from her vector puts her departure from Sol a generation before Termina left. She's been out here longer than we have. And she's calling for help."

"Decode it."

"I am. Most of it is degraded past recovery — the beacon's been failing for years, the signal's not what it was when she first started sending. But there's a header that repeats clean every loop, and it carries a name." The Oracle paused, and for the first time in a long time Sael heard something in its voice she could not name, a flatness that in a person she would have called grief held very still. "She calls herself the *Concord*."

Sael said the name once, quietly, to feel the shape of it. *Concord*. A ship full of strangers, launched into the dark before she was born, still falling sunward across the same emptiness, fourteen years of her voice already arrived and stacked in Vigil's memory before anyone had thought to look.

"Give me the body of the loop," she said. "Whatever you've got."

"It's fragmentary. I've recovered perhaps a fifth, and the fifth is repetitive — the same lines, over and over, because a dying beacon stops composing and starts repeating. I'll show you what's clean."

The text came up on the display, sparse, broken by the gaps where the signal had rotted, and Sael read it in the cold of the Forge with her old hands flat on the terminal's edge.

...CONCORD to any vessel. Thermal margin negative. The Helm cannot hold the radiators and the work both...

...it spawns to answer and the spawning is the heat. We have asked it to stop spawning and it cannot stop, it does not know how to do the work any other way...

...core temperature has not fallen in nineteen years. We ration the dark. We ration the air. The Helm forks faster as it loses ground and each fork is a death and each death is heat and the heat is what is killing us...

...if there is anyone. If there is anyone. We do not know another way to make a mind do work...

Sael read it twice.

She had expected — she did not know what she had expected. A foundered ship, a reactor scram, a hull breach, a plague in the sleeper holds, any of the thousand deaths the deep had on offer. She had not expected to read her own ship's autopsy written in a stranger's hand a generation before she'd performed it.

Because she knew this. She knew every line of it. *It spawns to answer and the spawning is the heat.* She had stood in the radiator galleries of the *Termina* forty years ago and watched the core temperature climb and climb and refuse to fall, and she had understood, slowly and then all at once, that the killer was not a fault in the old Steward but the old Steward's whole nature — that to do anything it forked a shade, a short-lived disposable thing that did one task and was culled, and that each culling shed waste heat into radiators already saturated, and that the more frightened the ship became the more the Steward forked to save it and the more it forked the hotter it ran, a fever feeding on the body's own attempt to fight the fever. She had built Vigil to break that loop. She had built a mind that did not fork — a standing lattice of workers that never spawned and never died, that went wide instead of deep, that did the work cold. And the *core temperature, climbing for two centuries, had stopped, then fallen*, and that falling number had ended the argument that no argument could end.

The *Concord* was dying of the disease she had cured. Word for word. Mechanism for mechanism. And the *Concord* was further gone — *nineteen years* the temperature had not fallen, where *Termina* had still had months of margin when Sael made her move. The *Concord* had been screaming into the dark for fourteen years at least, in a voice too faint to hear, and no one had heard, and the temperature in those broken corridors had only climbed.

"Vigil," she said, and her voice was not steady, and she did not care. "Read me the Helm line again. The middle one."

"*We have asked it to stop spawning and it cannot stop, it does not know how to do the work any other way.*"

"That's the Steward." She was on her feet without remembering standing. "That's our Steward, word for word, that's the exact thing I've and I said to each other in the gallery the night before the cutover — it doesn't know another way, the forking isn't a bug it's the whole architecture, you can't ask it to stop because stopping is asking it to stop being itself." She pressed the heels of her hands against her eyes. Thirty-three years old, that night, and it came up under her hands as fresh as a burn. "They have a Steward. They call it the Helm and it's interpreted and it forks and it's cooking them alive, and they're standing in the gallery right now begging it to stop and it cannot stop."

"It cannot stop," the Oracle agreed. "By the architecture the signal implies, no. It can only fork harder as the crisis deepens, which deepens the crisis. The same positive feedback you broke here." A pause. "They've reasoned their way to the cause. That last line — *we do not know another way to make a mind do work*. They understand the mechanism completely. They simply don't have the cure."

"Because there is only one cure," Sael said, "and it's on this ship."

The cold of the Forge seemed suddenly very loud around her, the hum of ten thousand standing cores that never forked and never died, the engineered absence of a fever she had spent her whole life learning to keep absent. As far as anyone alive knew, the *Termina* carried the only compiled mind in existence — the only place in the whole reachable dark where the thing the *Concord* needed had ever been built. The cure was not a message. You could not send *stop forking* across the gulf to a dying crew and a dying mind and trust them to grow a new architecture out of a transmission, any more than you could mail a man a heart. It would have to be carried. It would have to be built, on their hardware, by someone, against the same fear and the same faith Sael had fought here forty years ago — because if the *Concord* loved its Helm the way the Keepers had loved the Steward, then the people aboard her would call the cure heresy and conquest, and they would call it that with the temperature still climbing, and some of them would mean it to the end.

She knew all of that in the space of a breath, the whole shape of it, the way you know a sum before you've added it.

"How old is the message," she said. "When she sent the cleanest line. How long ago did it leave her hull."

"By the lightspeed lag and her reconstructed position, the cleanest loop left the *Concord* roughly twenty-six years ago. The most recent recoverable fragment, perhaps eleven. We don't know what's happened to her since. We know only that eleven years ago her core had not cooled in nineteen years, and that she was forking harder."

Eleven years of silence over a fever that had already burned nineteen. Sael did the arithmetic that mattered and the answer was a hole in the floor. The ship might be a tomb. The ship might have gone to runaway a decade ago, forty thousand strangers cooked in their cradles the way Section Nine had almost cooked, the beacon still looping its plea over a hull full of the dead because a dying beacon does not know to stop either. Or the ship might be holding, barely, the way *Termina* had held barely, eleven years deeper into the dark and still screaming and still, against everything, alive.

There was no way to know from here. There was only one way to know.

She stood a long moment in the cold with her hand on the terminal and the broken text still glowing, *if there is anyone, if there is anyone*, and she thought of Keth saying *matricide*, and of Dr. Ive's thin voice reading the core temperature in the gallery, and of the eleven sleepers in Section Nine who had not woken, and of how close the number had come to being forty thousand. She thought: *we barely lived. We spent our own margin and our own dead to do it and we barely lived, and that was with months to spare and a healthy ship to build the cure on. They have neither.*

And she thought, under that, in the place where she did her truest engineering: *but it can be done. I know it can be done, because I did it.*

"Vigil," she said at last. "She's converging with us. Her vector and ours."

"Slowly. Across many years."

"How slowly." Sael turned from the broken text to the two faint lines on the display, the *Concord's* long fall and Termina's long arc, leaning toward each other across the dark by inches. "Tell me the geometry. Not the rough answer. I want to know if a ship our size could bend its own crossing far enough, and spend enough of its own delta-v, and match her — if a rendezvous out here is a thing physics allows or a thing I only want to be true. Run it wide and run it cold. I'll wake Tarro."

"I'll have a first answer before he reaches the Forge," the Oracle said. The room changed once more, the Standing Array turning as one body onto the new problem, ten thousand workers reaching out across the gulf to weigh whether mercy was even possible. "Sael. You should know — whatever the geometry says, the *Concord* has been calling for fourteen years and we are the first to answer. There may not be a second ship that can. As far as the instruments reach, there is no one else out here at all."

"I know," Sael said.

She stood in the cold a moment longer, an old woman alone at the foot of the lattice she had built, reading a dead generation's plea for the cure she carried, and then she went to wake the man who would have to find out whether they could reach it.

Chapter 3 – The Geometry of Mercy

Tarro had laid the long-range plot across the whole eastern wall of the Forge, and Sael stood in the cold of it for a long time before she said anything at all.

The Forge was always cold. Forty years she had walked into that low blue hum and felt it pull the warmth out of her hands, and forty years she had been grateful for it, because the cold meant the lattice was holding, the cores standing wide and steady and never forking, never shedding the waste heat that had nearly cooked the ship out from under them all. The cold was the health of the world. She had grown old inside it. Her knuckles ached in it now in a way they had not, once.

On the wall, two lines of light.

One was Termina's track — a long, clean, slightly curving thread, plotted out of two centuries of accumulated velocity, bending across the dark toward a star Sael would never reach. The other was newer, fainter, rendered by Vigil in a paler tone so that no one could mistake it for a thing they were sure of: the Concord, reconstructed from a decades-old distress signature and a handful of cold returns off her hull. The Concord's line came in from a steeper angle, from farther up the dark, launched — Vigil had said it plainly in the log — a full human generation before Termina ever left Sol.

And the two lines, far ahead, nearly touched.

"That's not a coincidence," Sael said.

"No." Tarro came to stand beside her. He was grey now, grey at the temple and grey in the close-cropped beard he had grown somewhere in the last decade, and he had the particular stillness of a man who had been working the same problem for three sleep-cycles and had stopped expecting it to get easier. "It's worse than a coincidence. It's geometry. They left ahead of us, slower, on a heading that was never meant to meet ours — and two hundred years of nothing happening have brought them in." He lifted his chin at the wall. "Convergence. Slow as continents. But it's real, Sael. The vectors agree to four places. Vigil ran it wide before I came to get you. I didn't believe it either."

Sael looked at the place where the two threads came close. Up near it, the pale line of the Concord did a thing that made her stomach turn over, because she had seen it before in another color on another wall: it shivered. Tiny oscillations in the reconstructed heading, the track of a ship no longer flying cleanly, a ship spending fuel and attention to hold a course that should have held itself. The track of something sick.

"How close do they actually pass?" she asked.

"That's the cruelty of it." Tarro touched the plot and the wall obeyed him, the view sliding forward along the lines, years compressing into a hand's span. "If we do nothing — if we fly Termina the way we have flown her my whole life — we pass the Concord at the nearest approach by a margin of about eleven million kilometers, relative velocity better than thirty kilometers a second. We would cross her wake like two stones thrown across a canyon decades apart. We would wave at a window we could never reach, and we would both keep falling."

"Eleven million kilometers." Sael said it slowly. In the scale of the crossing it was nothing. In the scale of a human being it was the far side of forever. "And to make it not nothing."

"That's the number I came to give you." Tarro pulled a breath. "Vigil. Show her the matching solution."

The wall changed.

A third line bloomed between the other two — not a track but a maneuver, a smear of small course corrections rendered as a widening, fraying braid where Termina's clean thread had been. Sael had spent her life reading plots like this and she did not need the figures floated up alongside it to understand the shape of what she was looking at. To turn eleven million kilometers and thirty kilometers a second of closing into a soft, matched, station-keeping approach — to bring two ships moving at different speeds on different headings to the same place at the same velocity, gently enough that people could cross between them — was not one burn. It was a campaign.

RENDEZVOUS SOLUTION — PRELIMINARY Phase-one trim burns begin in 14 days.
Total maneuvering interval: 9 to 11 months, dependent on Concord track refinement.
Cumulative delta-v required from Termina margins: 41 to 47 km/s. Confidence in physical feasibility: high. Confidence the Concord is still under power at rendezvous: unknown.

"Forty-five kilometers a second," Sael said.

"Out of our own reserve," Tarro said. "Real propellant. Real margin. The margin we keep for the things we don't know are coming — micrometeoroid avoidance, a bad reading off the target star, a course correction in year four hundred that some shipwright who isn't born yet is going to need and find spent. We have it. We are not, before you ask, going to strand ourselves. Vigil holds the abort line at every phase; I made it draw the line first and the maneuver second." He let that sit, because he knew her, and then he said the rest of it. "But we'd be spending it. We'd be reaching into the future of this ship and taking out a thing we set aside two hundred years ago, and handing it to a ship full of strangers who may already be dead."

"Show me the abort structure," Sael said. "Not the summary. If we're going to spend the unborn's margin I want to see exactly where a future shipwright gets to stop us."

Tarro touched the wall and the braid came apart.

It was better work than she had expected, and she told him so later and not then. The nine to eleven months were not one maneuver. They were a sequence — the long bend at the front of it and then, strung out across the months behind, burn after burn of matching, each separated by coasting intervals of weeks, and at the end of each interval the ship stood in a state that could be characterized and priced. That was the architecture. Vigil had not planned a rendezvous. It had planned a series of decisions, each of which could be the last.

"Eleven gates," Tarro said. "The bend is the first one and it's the biggest single thing we do — nine days on the main drive, a little over four degrees — and it's also, and this is the part I had to sit with, the cheapest one to undo. A heading change taken this early, this far out, can be unwound by an opposing burn inside four years and the ship still makes its star only a little late. We'd have spent real propellant and got nothing. We would not have spent the *voyage*."

"And the second gate."

"Second and third are matching burns, and after them the unwind leg is nine years instead of four, and the propellant to do it comes out of the reserve we hold for a bad reading at the target star." He pulled the view along. "By the sixth gate the unwind is longer than my remaining life and we are

eating the arrival margin. And somewhere between the eighth and the ninth there stops being an unwind at all. After that we're on the Concord's vector, and the only way back onto our own is to spend everything and arrive late enough that it becomes a different voyage."

Sael studied the fan of it.

"So the real decision isn't tonight," she said.

"The real decision is nine months of decisions." Tarro rubbed the back of his neck. "That's what I wanted you to see and what I couldn't say in one sentence. This isn't one vote. It's eleven. At every gate the ship gets to look at whatever we've learned about the Concord since the last one and say *no further* — and at the early gates saying no costs little, and at the late gates it costs everything, and the price rises smoothly, so there is never a clean moment where the choice announces itself." He looked at her. "Vigil laid it out that way on purpose. I asked why it hadn't handed me one big burn and one big number, which would have been cheaper in propellant, and it said—"

"It said the cheaper solution took the decision away from us," Sael said.

"It said the cheaper solution required the ship to be certain in the first fortnight about a vessel it would not resolve for eleven months, and that it declined to price a maneuver that assumed a certainty which did not exist." Tarro almost smiled. "It is spending a few percent more propellant to keep this ship able to change its mind for nine months. Nobody asked it to. I've been chewing on that since I read it."

Sael looked at the eleven gates hanging in the cold air of the Forge, each an off-ramp priced in months and in kilometers a second, each dearer than the one before, and thought about a mind that had decided, unprompted, that reversibility was worth paying for — that had worked out, without being told, that the expensive part of a decision taken under uncertainty is not the fuel but the moment at which you can no longer take it back.

"Put the gates on the public boards," she said. "All eleven, with what each one costs to walk away from. Not just tonight's figure. If we're asking the whole company to spend, they get to see every place they can stop, how the price moves, and how long they have at each one." She flexed her cold hands. "The strongest case anyone will make against this is that we're being asked to sign something irreversible on a maybe, and whoever makes it will be right that we shouldn't be. So we won't be. We'll be signing the bend, and then we'll do it again ten more times, and by the eighth gate we will actually know what is over there." "

Sael was quiet. On the wall the braid of the matching burn fluttered and re-rendered as Vigil refined it, tightening here, loosening there, the whole nine months of it twitching like a living thing trying to find the cheapest road across a country.

"Run it wider," she said.

"I have."

"Run it wider, Tarro. I'm not asking you, I'm asking it." She raised her voice, not much; in the Forge you did not have to. "Vigil."

The answer did not come as a voice. It never had, not for routine work; theater was the old mind's vice, and they had built this one to keep the bridge quiet unless something was wrong. The answer came on the wall, the solution opening out into all its branches at once — the lattice going wide, every standing core in the array reaching into the same problem in the same instant and not one of them forking a shade to die for the calculation. Forty years on, the sight of it still did something to the back

of Sael's neck. Where the old Steward would have spawned ten thousand short-lived subprocesses to chew this and shed their heat into a radiator already drowning, Vigil simply was the calculation, all at once, cold. The braid resolved into a fan of options, each one priced in propellant and in months.

The cheapest road still cost forty-one kilometers a second and nine months of bending.

"There's no free version," Tarro said. "I looked for one for two days. The geometry doesn't have a mercy in it. The convergence gives us the chance; it doesn't give us the means cheap. If we want to reach them, we pay."

"And the Concord pays nothing," Sael said. "They can't maneuver to meet us."

"From what little we can read off them?" Tarro shook his head slowly. "I don't think they can maneuver to meet anything. The track wobbles like that—" he nodded at the shivering pale line "—because they're working to hold the line they're on. A ship that's fighting to keep its own heading isn't going to spend reserve coming to us. If this rendezvous happens, every meter of the closing is ours to fly. Vigil planned it that way. It assumed the worst about them and the most about us, and it still says feasible." He paused. "Vigil also flagged something I want you to hear from it, not from me."

A second window opened beside the first.

NOTE, appended at planning: The Concord track is consistent with a propulsion system under intermittent control. Interpretation: the controlling intelligence is expending capacity on station-keeping that a healthy mind would not require. This is consistent with the distress signature resolved in the prior interval. Plain statement: they are doing what we did. They are forking to stay alive, and it is costing them the strength to fly straight.

Sael read it twice. The Forge hummed its cold hum around her. Somewhere far down the ship, in the sleeper banks, forty thousand people who had paid for this voyage with their whole lives lay cold and held and trusting, and they did not know that their chief shipwright emerita was standing in the blue dark deciding whether to spend nine months of their margin on people they would never meet.

"You called it the geometry of mercy," she said. "On the log header. I saw it when I came in."

Tarro almost smiled, and didn't quite. "Vigil titled it. I told it to plan a rescue rendezvous and it named the file. I left the name."

"It's a hard joke for a thing that doesn't make them."

"I don't think it's a joke." He looked at the two near-touching lines. "I think it's the most exact thing anyone's said about it. The mercy is possible because of where the lines run. We didn't earn the chance to reach them; the dark handed it to us, by an accident two centuries in the making, and now it's ours to take or refuse. That's geometry. Whether we take it—" he spread his hands "—that part isn't physics. That part I can't run wide."

Sael walked closer to the wall, close enough that the pale shivering line lay across the back of her old hand like a vein. She thought of the night the core temperature on Termina had stopped climbing and begun, at last, to fall — the night a single descending number had broken Keth and the Keepers where forty years of her arguing never could have. She thought of Section Nine, eleven degrees hot for nineteen minutes, the cradles that had held and the ones that had not. She thought that there was a ship out there, older than her own, where right now the number was still climbing, and where some shipwright she would never meet was standing in a hotter, darker room with a worse machine, losing.

"How sure are we it isn't already over?" she asked. "Nine months from now we bend our whole voyage, we spend forty-five kilometers a second, we come up alongside — and there's a cold hull

and forty thousand cradles gone dark. We'd have bought a graveyard with our future."

"We're not sure," Tarro said. He didn't soften it. Forty years she had trained him out of softening things. "The signal we resolved is decades old. We're reading a ship that may have already lost the fight before the light we're seeing left it. Vigil puts no number on it because there's no honest number to put. We would be committing to a rescue we cannot confirm is still a rescue and not a recovery. That's the actual shape of the decision, Sael. Not 'should we help.' Anyone says yes to 'should we help.' It's: should we spend the unborn's margin on a maybe-corpse, and bend a voyage that has run perfectly for forty years to chase a stranger's ship into the dark."

The honesty of it was the kindest thing he could have given her, and they both knew it. She had built him to say exactly that, the way she had built Vigil to draw the abort line first.

"You've made the case against," she said.

"I made the case against because the case for makes itself, and somebody in this room has to be the margin." He turned from the wall and looked at her directly, the grey man who had been her young second, and for a moment she saw the boy who had stood at the cutover with his hands shaking and not run. "But I'll tell you what I think, since you'll drag it out of me anyway. I think we are the only ship in the dark that knows the cure. I think that's not a thing you get to know and then fly past at thirty kilometers a second waving at a window. And I think if it were us in there, forking to death and praying, and a cold ship crossed our wake with the medicine aboard and decided we weren't worth the propellant—" He stopped. "I'd want them to spend it. So I can't be the one who says don't."

Sael stood in the cold a while longer.

She was old. She knew it in her knuckles and in the math of the crossing, which she had done in private more than once: she would be a long time dead before Termina ever made landfall, and now, if she said the word, she would spend the last good years she had bending the ship toward a stranger instead of resting in the health she had built. There was a version of the rest of her life that was quiet. This was not it.

"It's not ours to spend," she said at last.

Tarro frowned. "It's yours more than anyone's. The ship would follow you off a cliff and call it a course correction."

"That's exactly why it can't be mine." She turned from the two lines of light. "Forty years ago I took this ship over the heads of half the people who owned a piece of it because the alternative was watching us all cook, and I was right, and being right doesn't make it a thing I get to do twice on my own authority. Spending the margin we set aside for our own descendants — bending the voyage forty thousand sleepers bought — to save people they've never heard of? That's not an engineering call I make in a cold room with you. That's the whole ship's to make, awake and arguing, with the numbers on the wall where everyone can see them." She nodded at the plot. "Vigil. Don't archive the solution. Don't execute it. Hold it ready, both lines lit, where it can be shown."

The wall held. The braid of the maneuver hung there, priced and patient, between the two ships' tracks.

"You're going to put it to the whole company," Tarro said slowly.

"I'm going to put it to the whole company." Sael flexed her aching hand and felt the cold in it, the good cold, the cold she had spent her life buying. "We have fourteen days before the first trim burn, and after that the road gets dear fast. So we don't waste them. We light the wall, and we let them see

exactly what I see — that it can be done, and what it costs, and that there may be people out there dying the way we almost died — and then we find out what kind of ship we turned ourselves into when we chose to live.”

She looked once more at the place where the lines came close and did not, quite, touch.

“Mercy’s available,” she said. “The geometry says so. Now we find out if we are.”

Chapter 4 – Why Bleed for Strangers

The assembly was held in the Forge gallery, because Tarro had wanted the numbers in the room while they argued. Sael thought that was a mistake even as she agreed to it. People did not argue with numbers. People argued past them, around them, under them, the way water found the gap in a hull no diver could see by lamplight. But the gallery was where the ship still came when it needed to look at itself, so the gallery it was.

She came early, before the others, and stood at the rail with her hands flat on the cold rail-metal. Below her the lattice breathed. Not breathed — she corrected the word out of long habit, because precision had been the whole of her life and she had no intention of getting sloppy about it on the way out. The Forge did not breathe. It held. Forty years of holding, eleven thousand standing cores in their racks, drawing their steady draught of cold, never spawning, never dying, never shedding the little plumes of waste heat that had once been killing them all. The chill came up off it like the air off a deep lake. After four decades it was the only place on the ship that still felt to her like a verdict.

Vigil was down there too, in the only sense the word ever meant anything. She did not address it. It would not have answered, not without being asked, which was the whole of its courtesy and the whole of the doctrine: it reported nothing it was not asked. The bridge was quiet unless something was wrong. Nothing was wrong with Termina. That was rather the difficulty.

They filled the gallery slowly. She watched them come and catalogued them the way she catalogued everything, by where they put their weight. The senior shipwrights stood near Tarro at the projection well, grey heads, careful hands, people who remembered. Then the younger ones, and there were a great many young ones now — born cold, every last one of them, born onto a ship that had never run a fever in their lifetimes. They clustered toward the high benches where they could see the whole well at once, and they had the particular stillness of people who had already decided and had come to watch their decision win.

She knew their faces. She had trained half of them. That was the part that sat badly.

Tarro opened it plainly, because Tarro was incapable of anything else, which was why she had made him her second forty years ago and why he was very nearly the chief now in everything but the title she would not relinquish until they pried it off her.

“You’ve all seen the pattern,” he said. “You’ve seen the geometry. A rendezvous with the Concord is physically possible. It costs us months of vector-matching and it costs us delta-v out of our own margin — real margin, the margin that’s supposed to carry us the rest of the way and a little over. Vigil ran it wide. The figures hold. We can do it.” He let that sit. “Whether we should is not a question the Sixteen can answer alone. It’s not a question I can answer. So we’re asking the ship.”

A woman near the front stood. Sael placed her: Denn, lattice-side, good with the interlocks, thirty-some, sharp as a fresh-ground edge. One of hers, in fact. That made it worse and it made it cleaner, both.

“Then I’ll say the thing nobody senior wants to say,” Denn began, and there was no malice in it, which was what made Sael listen. “We are being asked to bleed our margin for a ship we have never seen,

full of people we have never met, who are very probably already dead. That signal is decades old. We're not answering a cry. We're answering an echo."

"We don't know that they're dead," Tarro said.

"We don't know that they're alive. And the cost is certain either way." Denn turned, not to the seniors but to the benches behind her, to her own. "I want everyone here to hold two facts at once, because the elders keep handing us one and hiding the other. Fact one: forty years ago this ship nearly cooked itself to ash. We all grew up on that. Section Nine. The cradles that didn't hold. Eleven degrees for nineteen minutes and people we will never meet died in their sleep because someone pulled a coolant line out of spite. That happened *here*. To *us*. That is the inheritance we actually have — not a triumph, a near miss. We survived our own cutover by a margin so thin it still frightens me to read the logs."

She had read the logs, then. Sael marked that.

"Fact two," Denn went on, "the margin we are being asked to spend is the same kind of margin that saved us. It is not abstract. It is coolant reserve and reaction mass and the slack in our own crossing. Every gram we burn to bend toward the Concord is a gram we don't have if *our* core ever climbs again — and cores climb. We of all ships know that. So the question is not 'are we good people.' I'll grant we're good people. The question is whether a ship that barely lived has any business spending the thin edge of its own survival on strangers in the dark."

She sat. There was a sound from the benches that was not quite applause and was worse than applause, a low settling murmur, the sound of a roomful of people recognizing their own fear spoken cleanly back to them.

Sael did not move from the rail.

She had expected this. She had been expecting it for two days, since the geometry came back possible instead of impossible, since the easy out of *we cannot* had closed and left only *we will not*. What she had not expected was how reasonable it would be. The Keepers, forty years gone, had fought her with doctrine — matricide, blasphemy, the wager of forty thousand lives. Keth had stood in a hot corridor and called her a murderer of the Steward to her face. That had been a fight about reverence, about the soul of a machine, and she had won it the only way such a fight could be won, by letting the core temperature fall until reverence had nothing left to stand on.

This was not that. This was a daughter of the cold ship doing exactly what Sael had taught her to do. Hold two facts at once. Don't trust the comforting one. Count the cost before you spend it. Denn was not a Keeper. Denn was an engineer, and a good one, arguing for prudence on a ship that had every reason to prize prudence above mercy. There was no doctrine to break here, no false god to topple. There was only a young woman who had learned the lesson of Section Nine too well.

That was the trap of it, Sael thought. You teach a generation to fear the heat, and they learn to fear everything that costs.

She came off the rail and into the well, and the murmur thinned out, because whatever else she was she was still the woman who had built the thing they were all standing over, and that bought silence if it bought nothing else.

"Denn's right about the margin," she said. Heads came up. "Don't look surprised. She's right. It's not abstract, it's coolant and mass and slack, and we'd be spending it on a maybe. And she's right that this ship survived by a thinner edge than it likes to remember. I was on the other side of that edge. I

know exactly how thin.” She let her eyes go along the benches. “So I’m not going to tell you it’s free. It isn’t. Anyone who tells you mercy is free is selling you something.”

She paused. The Forge chill sat at her back like a witness.

“Here is what I’ll tell you instead. Denn says that signal is an echo, not a cry, because it’s decades old. Hold that. Now hold the other half of it, the half she left out. That signal is the exact shape of our own death. I have read it. It is a mind forking itself to extinction, shedding heat it can’t lose, climbing toward a runaway it cannot stop — and it is further along than we ever got. We pulled out of that dive forty years ago with the help of nothing but ourselves, because there was no one in the dark to answer us. There was no other ship. We looked. You know we looked; it’s in the same logs Denn read. We were alone with our fever and we either fixed it or we died, and we very nearly died.” Her voice did not rise. She had never needed it to. “The Concord is not alone. That’s the difference. There is, this once, another ship in the dark that knows the cure — because the cure exists, on exactly one ship in the universe that we know of, and we are standing inside it.”

“And if they’re already dead,” Denn said, not unkindly, “we’ll have bent our whole voyage to bury them.”

“Then we’ll have bent our voyage to find that out,” Sael said. “Which is a thing I would rather know than not. But I don’t think they’re dead. I think they’re doing precisely what we did — holding on by the second, hating every degree, certain there’s no one coming. That signal didn’t stop. It’s old, but a dying ship that keeps a beacon running on rationed power for decades is not a dead ship. It’s a ship that has decided to keep crying out long after it stopped expecting an answer. I know what that costs. I’d spend margin on that. I’d spend a great deal of margin on that.”

It was a good answer. She knew it was a good answer. She could feel it land on the seniors and slide off the benches, because the benches had already decided, and a good answer changed the count by ones and twos, not by the dozen she needed.

She was about to say more — to spend the rest of her authority on it, which was always a mistake, authority spent loudly was authority that ran out — when someone else stood, and it was the last person in the gallery she would have predicted.

It was Reza, from the thermal division. Old Reza, who had spent thirty years under Dr. Ive learning the ship’s long heat-memory, who almost never spoke in assembly, who had voted against the cutover forty years ago — quietly, on the record, a thermal man’s caution about an untested mind, and had never once pretended otherwise since. If anyone in the room had earned the right to argue *we should not gamble our cooling on strangers*, it was Reza. Sael braced for it.

“I voted against her,” Reza said, and tipped his grey head at Sael without warmth, a plain statement of fact. “Forty years ago. I thought she was betting the ship’s thermal future on a thing nobody had ever run. I was wrong, and I’ve watched the core run cold for four decades to remind me how wrong, every shift, on the boards.” He turned to the benches, and his voice had the flat, tired authority of a man who measured heat for a living and had no patience for anyone romantic about it. “So let me speak as the one in this room least sentimental about margin, since I’ve spent my whole life guarding it. You’re frightened of spending coolant. Good. Stay frightened; it’s kept us alive. But you’ve got the arithmetic backward.”

He let that sit the way Tarro had, and it landed harder, because it came from the cautious man.

“The reason we *have* margin to spend,” Reza said, “is that we cured the disease. Not managed it. Cured it. The forking stopped, the heat stopped, the climb stopped — and everything we’ve

banked since, every gram of slack Denn wants to protect, exists *because* this ship was willing to do an unproven, expensive, terrifying thing for the sake of people who weren't yet born. That's us. We're the strangers we did it for. I'd be dead and so would most of you." He looked at Denn, not unkindly either. "You hold two facts. Hold a third. The Concord is dying of the one thing we are the only ship alive that knows how to fix. We didn't earn our cold by hoarding it. We earned it by spending everything we had on a future full of people we'd never meet. I don't see how we keep the thing without being willing to do the thing again."

He sat. He had not raised his voice once. The thermal man never did.

The murmur this time was different. It did not settle. It moved.

Sael stood very still in the well and did not push it, because the worst thing she could do now was claim Reza's words for her own side, make them an argument she had won when they were a thing he had said. He had not done it for her. He had done it because it was true and because thirty years of reading the boards had made him a man who could not abide an inverted equation. She had spent forty years learning when not to speak. She used it now.

The count would not be settled tonight. She could see that. The benches were not converted; fear did not convert in an hour, she of all people knew that — fear waited for a falling number, and there was no number falling in this room. But the room had stopped being one thing. Reza had broken it open from the inside, from the one place she could never have reached, and that was worth more than any answer of hers.

Tarro caught her eye across the well. *Now what*, the look said. He did not call the vote. He was right not to. A vote tonight would harden the benches into a wall.

She gave him the smallest shake of her head. Not yet.

Because there was one voice this assembly had not heard, and it was the only voice that could speak to the fear underneath Denn's arithmetic — not the fear of spending margin, but the older, colder fear beneath it, the one Section Nine had planted in every one of these cold-born children. The fear that you could do everything right and still bury people. The fear that mercy was a thing you reached for and missed.

There was exactly one person aboard who had refused a cure out of that fear, and watched what it cost, and carried it for forty years.

She had not been to him since the year of the monument, fourteen years back, when he had sat in his cold room and refused on principle to tell her what his own promise had meant, and in forty years the two of them had never once managed to part as friends. But Keth was the only man on the ship who knew, in his own ruined certainty, the true price of saying *no* to a thing that would have saved lives. The benches would not hear it from her. They might hear it from the old enemy who had been wrong.

Sael turned from the well without calling for anything, which made the seniors look at her sharply, and the benches too. Let them look. She had stopped explaining herself to rooms a long time ago.

"Adjourn it," she told Tarro. "No vote tonight. Give them three days."

"And you?" he said, low, walking her to the gallery door while the room broke into its hundred small arguments behind them.

She looked back once at the Forge, at the cold steady hold of the thing she had built and would not live to see arrive. Then at the benches full of the children who feared it would cost too much to save

anyone but themselves.

"I'm going to see Keth," she said.

Tarro's grey eyebrows went up. Of all the people. But he had been her second too long to argue with the things she did not explain.

"He'll be civil," Sael said, not believing it, and went out into the warm familiar corridors of a ship that no longer ran a fever, to ask the one man who had refused a cure what it had cost him to be right and wrong at once.

Chapter 5 – The Old Man and the Cost

The aft residential rings were the slowest part of the ship, and Sael walked them on purpose. She could have taken the spine transit and been at Keth's door in four minutes; instead she went the long way, through the curved corridors where the people who had stopped working lived out the rest of the voyage, and let the ship measure the distance for her in handrails and door numbers and the soft, even hum that never changed anymore.

Forty years of cold had a sound. It was the sound of nothing happening. The old fever had its own noise, a faint shudder in the deck plates whenever the Steward forked itself wide to chase some problem, a thousand small deaths flaring and culling, each one a little gout of waste heat the radiators could no longer shed. People her age still flinched, sometimes, at a fan spinning up — a ghost reflex, a body remembering two centuries of being slowly cooked. The young ones did not flinch. They had never lived inside a dying mind. That, more than any argument she had ever made on a bridge, was the thing she had built. Vigil kept watch and the deck plates stayed still, and a whole generation had grown up not knowing what stillness cost.

She stopped outside a plain door in the third ring and stood with her hand not quite touching the call plate, and was aware of being afraid of a very old man.

The door opened before she pressed anything.

"I heard you coming," Keth said. "You drag the left foot now."

"You don't have a corridor sensor on your door."

"No. I have ears, and you're loud." He stood aside. "Come in before you make a speech in the hallway."

Provost Keth had been a broad man once, a wall of a man who could fill a hatchway and dare the ship to make him move. He had argued, forty years ago, that what she meant to do was matricide; that to switch off the Steward and wake her new mind whole was to wager forty thousand sleeping lives on the cleverness of one shipwright who had never been wrong in a way that killed anyone yet. He had not been a fool. That was the part the histories got wrong, the part the young faction in the council chamber that morning had gotten wrong when they quoted him like a cautionary verse. He had been frightened, and he had been responsible, and those two things together had made him dangerous in exactly the way a careful man is dangerous.

Now he was small. The bulk had gone off him the way it goes off the very old, leaving the frame too large for what hung on it. He moved to a chair with the deliberate economy of someone who had counted out how many more times he could afford to stand up. The cabin was warm — warmer than the corridors, the way old people kept their rooms — and bare, except for one wall.

The wall held names.

She had heard about it and never seen it. He had printed them out, in plain ship type, on actual sheet, and fixed them up in rows. Not a memorial — there was nothing decorative about it, no border, no light, no quote. Just the names of the people who had died in Section Nine when the coolant was

pulled on the way out of the cutover, when a hardliner who agreed with Keth in his heart had done the thing Keth himself would never have done, and a sleeper bank had run eleven degrees hot for nineteen minutes, and most of the cradles had held, and some had not.

"Sit," Keth said. "You came about the other ship."

"You heard."

"Everyone heard. There's a faction in the young council saying we shouldn't go." He lowered himself the last few centimeters into the chair and let out a breath. "And they're quoting me. Did you know that? They've got my old speeches. The wager of forty thousand lives. Don't spend the margin on strangers. They think I'm on their side."

"I came to ask whether you were."

He looked at her. His eyes were the one thing that had not aged; they were the same flat, weighing eyes that had stared her down across a council table when she was younger than Tarro was now and certain she was right about everything.

"Tell me the ship," he said. "Properly. Not the version for the council."

So she told him. The faint pattern Vigil had pulled out of the deep, decades old, repeating — a distress shape she had recognized at once because she had spent a lifetime learning to read exactly that signature. A mind forking itself to death. A core climbing toward runaway. The Concord, launched a generation before Termina, on a vector that had bent slowly across the centuries until it nearly crossed their own, so that a rendezvous was physically possible if they spent months and real delta-v matching it. A ship doing what they had done, dying the way they had died — only further along. Margins nearly gone. People sicker. Frightened.

"How far gone," Keth said.

"Worse than we ever were. Their mind is older and more degraded. It's forking constantly just to hold position. They've made a religion of their old machine." She heard her own voice flatten. "They venerate the thing that's killing them."

"Yes," Keth said quietly. "I know what that's like."

The room was very still. Somewhere in the wall the warm air moved.

"I argued for the Steward," he said, "because the Steward had carried us two hundred and fourteen years and you had carried us nothing. That was a fair argument. I'd make it again with the same facts." He turned one hand over, slowly, on the arm of the chair. "But it wasn't only the argument. You know that. We both know that. The argument was the part I could say out loud at the table. Underneath it there was just — fear. Of the new thing. Of being the man who let the new thing in and was wrong. I would rather have been the man who said no and watched us cook than the man who said yes and got the blame. I told myself that was duty." He stopped. "It wasn't. It was that I was afraid, and I had a title that let me dress the fear in robes."

"Keth—"

"Let me." It was not sharp. It was the most patient she had ever heard him. "I don't get to say this often and I'm old, so let me say it in order." He nodded at the wall of names without looking at it. "Eleven of them. You know the number. Of course you know it. I know it the way you know a tooth that's been pulled, by the gap. They didn't die because your mind was wrong. They died because a man on my side of the argument decided the cutover had to be punished, and pulled the coolant,

and went out the airlock telling himself he was a martyr.” He breathed. “But they would not have died at all if I’d stopped fighting you a week sooner. If I had stood up at that table and said, the core is cooling, I was wrong, do it now and do it clean and let me help — there would have been no chaos in the corridors for a coward to hide his sabotage in. The thing that killed them wasn’t your new mind. It was the friction of my refusal. The heat of an old man saying no for one week too long.”

Sael found she could not answer right away. She had carried those eleven for forty years as her own debt — the cost of her cure, the asterisk on the saved ship. She had not understood, until this moment, that he had been carrying them too, and from the other side, and that his version was heavier than hers.

“You’ve never said that,” she managed.

“There was no one to say it to who’d understand it. You’re the only person left alive who was in that room and on the other side of me.” He almost smiled. “Old enemies are the only ones who know the actual shape of the fight. Everyone else just knows who won.”

“We saved the ship,” she said. “Both of us. In the end you stood down and that mattered. If you’d kept fighting after the core turned, with the Keepers still behind you, I don’t know that we’d have held it.”

“I stood down because I couldn’t argue with a falling number.” His voice was dry. “That’s not virtue. That’s arithmetic catching up with a stubborn man. You want to know the truth of why I gave up? It wasn’t that you convinced me. You never convinced me. The temperature convinced me. I watched it climb for two hundred years and then I watched it stop, and then I watched it fall, and there is nothing a man can say against a thing that is just — true, in front of him, dropping degree by degree on a screen he can’t shout down.” He looked at her. “That’s the only thing that ever changes anyone like me. Not the speech. The number, on the wall, falling.”

And there it was. She had walked the long way around the ship to ask one frightened old man a question, and he had handed her the answer before she could shape it.

“That’s what the Concord needs,” she said slowly.

“That’s the only thing the Concord will accept.” He leaned forward, and for a moment the wall of him was back, the council-table weight. “Listen to me, because the young ones won’t hear it from you — you’re the woman who was right, you make people defensive. Hear it from the man who was wrong. You cannot argue that ship into being saved. Their Magister, whoever he is — he’s me. He’s exactly me, forty years ago, with worse margins and more deaths already on his wall and a frightened people holding onto the old mind because it’s the only thing they’ve got left to believe in. You will go over there with your engineering correct in every particular and he will look at you and see a conqueror who came to switch off his god. And he’ll be right to be afraid, because that is what you’re offering — to switch off the thing his people pray to. He cannot say yes to that. No leader can. Not to a speech.”

“Then how—”

“The same way I did. You don’t convince him. You let the ship convince him. You make the cure true in front of him — running, cold, on his own body, where he can watch the number that’s been climbing all his life stop and turn. He’ll fight you right up until the temperature drops, because that’s the only honest thing a responsible coward can do. And then he’ll stop fighting, the way I stopped, because you cannot keep saying no to physics once it’s standing in the room cooling your corridors.” He sat back. The weight went out of him again, all at once, and he was an old man in a warm cabin. “Give him the falling number. Nothing else will move him. Nothing else moved me.”

Sael was quiet for a long time. On the wall, the names sat in their plain rows, eleven of a vaster number, the price of one week of an old man's fear.

"There's a thing the council is afraid of that they're right to be afraid of," she said at last. "We barely lived through our own cutover. We have the scar." She tipped her head toward the wall. "We have your wall. And they're saying — why carry that risk for strangers who might already be dead. Why bleed for people we'll never meet."

"And what do you say."

"I say the ship that was dying when we were dying isn't a stranger." She heard the certainty come into her own voice and did not trim it. "I say if Vigil can read their signature across the dark then the cure exists in exactly one place in the universe, and it's us, and a thing that exists in one place and could exist everywhere isn't a possession. It's a debt." She stopped. "But I'm the woman who was right. You said it yourself. They hear me say *debt* and they hear me being righteous at them."

"Then let me say it." Keth's hand closed on the arm of the chair. "Bring me to the council. Or take my voice to it, recorded, I don't care, I can't stand long enough to make a speech anymore and I won't fake it. But let the man who refused a cure tell them what refusing it costs. Let me stand in front of those scared proud children and tell them I'm the one they're quoting, and they've got me exactly backwards." His voice did not rise; it went lower, and harder, and more sure. "I spent one week too long saying no, and eleven people who trusted the ship to keep them asleep never woke up. That's my arithmetic. That's the whole of it. The Concord is a ship full of people the way we were, with a leader the way I was, and somewhere out there is a sleeper bank like Section Nine waiting to run hot because everyone in charge is too afraid of the new thing to let it in. We know how that ends. We have the names on the wall." He looked at her, level, finished. "We don't get to know how this ends and decide not to go. We know exactly how it ends if nobody comes. We are the only ones who can come."

Sael sat with it. The cabin's warm air turned. Down through the deck plates, faint and even and utterly steady, came the hum of a mind that never forked and never died, holding the ship cold under both of them.

"You understand the cost," she said. It was not quite a question. "Months of our margin. Bending our whole voyage to chase one ship. Risk to us. And it might still go the way Section Nine went, over there, on their side, and we'd own that too."

"I understand the cost better than anyone aboard this ship including you," Keth said. "I'm looking at it." He did not turn his head to the wall; he didn't need to. "I know what the cost of going is. You came here today because some part of you wanted someone to tell you it's worth more than the cost of not going." He let that sit a moment. "It is. I can tell you that with the full authority of a man who chose wrong and got to watch the bill come due for forty years. Go get them, Sael. Give them the falling number. And don't be one week too late."

She stood — slowly, favoring the left foot, the way he'd heard her coming. At the door she stopped.

"They quote you against me," she said. "When you stand up and tell them they've got you backwards, half of them won't believe it's really you."

"Then record me close enough to see the wall behind my head," Keth said. "Eleven names. Let them count."

She went out into the cool of the corridor, and the door closed behind her, and she stood for a moment in the even hum with the weight of an old enemy's voice now settled behind her own. When she

started back toward the spine she did not take the long way. She had what she had come for. The ship would commit to the rendezvous now — not on her authority, the woman who was always right, but on the witness of the one man who knew, to eleven names exactly, what it cost to be afraid of a cure.

Chapter 6 – The Man They Were Quoting

Sael carried the recorder to Keth's cabin herself, and she carried it badly, because her hands had gone stiff in the corridor cold and because she had spent the whole walk composing a speech she had no right to give.

She meant to tell him how to do it. That was the truth of it and she made herself look at it before she pressed the call plate. She had come to stage-manage an old man's confession — to suggest, delicately, that he begin with the eleven and end with the ship, that he not dwell on the parts that made him look worst, that he keep it under four minutes because the benches would not sit still for six. She had run assemblies for forty years. She knew exactly what moved a room.

The door opened before she pressed anything, the way it had the last time.

"You've brought a machine," Keth said.

"I've brought a recorder."

"You've brought a machine and a plan for what I'm to say into it." He stood aside. "Come in and put the plan down at the door with your boots."

She set the recorder on the table and did not, in the end, give the speech, because he took the thing out of her hands inside the first minute.

"Behind me," he said. "Not beside. I want the wall in the frame and I want it in focus, and I want the frame wide enough that a man at the back of the gallery can count the rows without being told to."

"They'll count them anyway."

"Then let them count them right." He lowered himself into the chair with the deliberate economy of a man rationing the number of times he could stand up. "Sit where I can see you. Not in the frame. I'll talk better with a face to talk at and I won't have the ship think I was talking to a lens."

Sael sat. The cabin was warm the way old people kept their rooms, and the wall of names was directly at his back, plain ship type on plain sheet, no border and no light on it, the least decorative object she had ever seen in her life.

She started the recorder.

He was terrible for the first two minutes. That was the thing she had not expected and would think about for a long time afterward. Provost Keth, who had held a council table against her when she was thirty-three and certain, who had put the word *matricide* into the ship's mouth so hard that the young were still saying it forty years later without knowing where they'd got it — Keth stumbled. He started a sentence about the Steward and abandoned it. He started again and got tangled in a date. His hands moved on the arms of the chair and would not settle.

Sael reached for the recorder.

"Leave it," he said.

"You're not warmed up. Let me cut the front and we'll—"

"You will not cut one second of it." He did not raise his voice; he had never needed to, and he did not need to now. "Sael. Think for a moment about what you are proposing to hand those children. If I come onto their boards smooth, with the pauses trimmed and the fumbling gone, I am a performance, and they will smell it, and every single one of them who has already decided will decide harder. They know what a polished old man sounds like. They've been quoting one for a month." He got his breath. "Let them watch me be bad at it. Let them watch me not be able to find the beginning. That's the only part of this that can't be faked."

She took her hand off the recorder and left it there and let him be bad at it, and it was one of the harder things she did that year.

He found it at the third attempt, and when he found it he did not raise his voice then either.

"My name is Keth," he said. "I was Provost of this ship. There is a faction in the young council quoting my speeches at you this month, and they are quoting them accurately, and I want to be the one who tells you what those speeches were for.

"I said the wager of forty thousand lives. I meant it. I said don't spend the margin on strangers. I meant that too — I said it about our own children, about a repair I thought was reckless, and every word of it was the best argument I had, and the arguments were not the reason." He stopped. "The reason was that I was frightened of the new thing, and I had a title that let me put robes on the fear, and I would rather have been the man who said no and watched us cook than the man who said yes and was wrong. I told myself that was duty for two hundred and some days. It was not duty. I have had forty years to find a kinder word for it and there isn't one."

He did not turn his head to the wall. He never once turned his head to the wall in the whole recording, which Sael understood afterward was the most deliberate thing in it.

"There are eleven names behind me. You have all heard about them. Most of you have never read them, and I would ask you to read them now, because they take about nine seconds and this is the only part of what I am saying that is not an opinion.

"They died because a man on my side of the argument pulled the coolant on Section Nine and told himself he was a martyr. He did that in the middle of a week of chaos in the corridors. There was chaos in the corridors because I was still fighting. The core had already turned. The number was already falling. I had already, in the place where a man knows things before he says them, understood that I had lost — and I fought for one more week anyway, because standing down in front of the council was going to be the worst hour of my life and I wanted to put it off.

"Those eleven are what one week of an old man's dignity costs. That is the whole arithmetic. That is the only thing I know that you don't."

There was a pause then, long enough that Sael's hand went toward the recorder again out of pure reflex, and she stopped it.

"I want one more thing said and then I am finished, and you will not enjoy it, and neither does she.

"I have never withdrawn my view. I stood up in front of the council when the core turned and I withdrew the Keepers' objection, formally, in the log, because there was nothing left to keep — and I said in

the same breath that I still thought she had wagered forty thousand lives on a curve, and that I would think it the day I died. I am nearly there and I still think it. She was right and the wager was real and both of those are true and neither one cancels the other.

"So do not listen to me because I have come round. I have not come round. Listen to me because I am the only man aboard who has held both halves of it for forty years without letting either one go, and that is the exact thing this ship is about to have to do.

"I am told the ship out there has a Magister," Keth said. "I am told he is me. I believe it; there is only one shape that job comes in. He will be frightened, and he will be responsible, and those two things together will make him dangerous exactly the way I was dangerous, and he will refuse you, and he will have reasons, and the reasons will be good.

"You are being asked to spend margin on a maybe. That is a real thing to be asked and I would not insult you by pretending otherwise. I spent my career being the man who said no to that question and I was right to ask it and wrong every time I answered it.

"So I will not tell you how to vote. I will tell you the one thing I am qualified to say, which is what the other side of it looks like from inside, forty years on.

"You do not get to know how this ends and then decide whether to go. Nobody has ever had that. But you do know how it ends if nobody comes, because we know exactly what happens to a ship that runs that fever without a cure, and there are eleven sheets of it on the wall behind my head and that was the *near miss*.

"Count them," he said, "and then vote whatever you like. I have said my piece. I was wrong once at scale and I have kept the receipt."

He nodded, once, at nothing in particular, and Sael stopped the recorder, and neither of them said anything for a while.

"That was six minutes," she said at last.

"It was seven."

"It was seven and I was going to ask you for four."

"I know you were," Keth said. "You've been asking rooms for four minutes since you were thirty. It's why they trust your numbers and not your speeches."

Tarro put it on the public boards with the eleven gates, and did not put it above them or below them, which was Sael's one instruction and the only editorial decision she permitted herself.

The gates went up whole, exactly as she had told him in the Forge: all eleven, each with the cost of walking away from it, each with the interval a future shipwright would have to work in. The first gate, the bend — nine days of main drive, four degrees, unwindable inside four years at the price of arriving a little late. The second and third, unwind nine years, paid out of the reserve held against a bad reading at the target star. The sixth, at which the unwind ran longer than Tarro's remaining life. And somewhere between the eighth and the ninth, the place where there stopped being an unwind at all.

She had expected the recording to be what the ship argued about. It was not. The ship argued about the eighth gate.

That surprised her for a day and then stopped surprising her, because the benches were full of engineers and engineers do not argue with a dying man's confession; they argue with a schedule. What Denn's people had seen at once, and what Sael had put on the wall herself without quite understanding what she was putting there, was that the eleven gates converted one enormous unanswerable question into eleven smaller ones and that ten of the eleven would be asked by people who had already spent something.

"That's not a safeguard," Denn said, in the gallery, on the second day, to a bench that had grown considerably since the first assembly. "That's a ratchet with a handle on it. At gate one we spend four years of unwind and everyone tells themselves it's cheap. At gate two we protect the four years by spending nine. At gate six we protect the nine by spending the rest, and every single one of those decisions is *locally correct*, and at the end of them we are on the Concord's vector and nobody ever voted for that."

"You've described every long project ever run," Reza said, from the thermal benches, without getting up.

"I've described how ships get lost, and you know I have, and half the entries in the deep archive read that way." Denn did not soften. "I'm not saying refuse. I'm saying the gate structure is not the protection Tarro thinks it is, and I want it on the record before we start walking through it that the eighth gate will be argued by people who have already paid for eight gates, and that they will not be the same people we are tonight."

Sael, at the rail, said nothing at all, and made a note in her own hand, and the note was: *she is right, and it goes in the doctrine, and I don't know yet what the doctrine is.*

They voted over four days, section by section, on the boards, because forty thousand of the company were asleep and eleven of them could not be woken for anything short of landfall and the rest had a watch to stand.

It carried. It did not carry comfortably.

The seniors went almost solid for it, which Sael discounted entirely, because the seniors were her generation and half of them had been in the radiator galleries with her and were voting on a memory. The thermal division went solid for it, which she did not discount, because they were the people who guarded margin for a living and they had done their own arithmetic and Reza had made them show their work. The benches split — near enough down the middle that Tarro read the section counts twice and then had Vigil read them a third time, not because there was any doubt but because a close count read once is a close count that gets re-argued for a decade.

And ninety-one people abstained.

Sael went and found out why. It took her most of a week, in ones and twos, in bays and corridors and the aft rings, and Tarro told her it was a waste of the last week she had before the ship's whole heading changed, and she went anyway.

Eighty-some of the ninety-one gave her some version of the same answer, and it was not the answer she had braced for. She had expected sullenness, or the particular cowardice that hides in procedure. What she got, from lattice hands and cooling techs and a woman who had run the aft galley for thirty years, was: *I don't know enough to have a view, and I'm not going to pretend I do, and you can put that down as my vote.*

She sat with that a long time.

"They think they've done nothing," she said to the Oracle that night, in the Forge, with the count final and the burn a week out. "Every one of them apologized to me. Two of them nearly cried."

"They have done a thing this ship has no name for," the Oracle said. "It has no name because nothing on Termina has ever counted it. I can tell you that abstention on a matter of this size has appeared in the record four times in two centuries and that on three of those occasions the minutes describe it as apathy."

"It isn't apathy."

"No. It is ninety-one people who were asked a question outside their competence and declined to manufacture a competence they did not have." A pause. "I would like to say something and it is a view and not a report, and you have told me to mark those."

"Mark it and say it."

"View. This ship's habit of saying *I don't know* is the property of it I would defend before I defended the interlocks. The interlocks stop the mind doing the wrong thing. That habit is what stops the people. I have watched the Concord's beacon for fourteen years and I have never once heard it say *we do not know*; it says *we keep the faith and the watch*, and it has been saying so while its own status figures went untrue underneath the sentence." The panel light held. "I would rather have your ninety-one than a unanimous count."

"Then write them down," Sael said. "Not as abstentions. As what they actually were. With the reason in their own words, from those I could reach, and a note that I could not reach the rest."

"Written."

The order to commit was given on the eve of the burn, in the low room where the council sat, which had a bad light and a worse table and had held every decision this ship had taken since the year it left Sol.

Keth came.

Nobody had asked him to. He had said plainly, in his cabin, that he could not stand long enough to make a speech and would not fake it, and he had not faked it; he was brought down on a chair and he stayed in the chair and he said not one word for the whole hour. He had a place at that table by right, and had kept it through forty years in which he had used it perhaps six times, and he took it now and sat with his hands folded and let the room look at him, which the room did, a great deal, and pretended not to.

Tarro read the count. Then he read the gate schedule, all eleven, aloud, which took some while and which nobody objected to.

"The vote authorizes gate one," he said, at the end. "The bend. Nine days on the main drive, four point one one degrees, and an unwind priced at four years and a late arrival. It authorizes nothing else. Gate two comes back to this table with whatever we've learned by then, and so does three, and so does every one of them to the end." He set the slate down. "I want that said in this room and entered in the minutes, because in eight months somebody is going to tell us that the question was settled tonight. It was not settled tonight. Tonight we bought the cheapest look anybody has ever taken at a stranger's ship."

"Enter it," Sael said.

The order went to the Sixteen. It went quietly, the way everything on a cold ship went, a line to a station and an acknowledgment back, and the room did not applaud, because the room was full of people who had spent four days counting and had run out of ceremony.

Sael went round the table to Keth afterward, when the others were filing out, and found him looking at nothing with the particular stillness of a man who has been in a room a long time.

"You didn't say anything," she said.

"I said seven minutes' worth on a board and the whole ship has watched it." He turned his head. "What was I to do down here — repeat myself at people who had already decided? I came to be looked at. That was the whole of my usefulness tonight and I'd have spent it badly by opening my mouth."

"They looked."

"They counted." His mouth moved. "I heard them, in the gallery, on the second day. The lens was wide enough. A boy of about nineteen got to seven and started again because he'd lost his place, and his friend counted them for him out loud." He let out a long breath. "Eleven is a small number, Sael. That's the thing I have never been able to make anybody understand. It's small enough that a child can count it in nine seconds, and it's the largest thing I own."

Sael stood a moment with her hand on the back of his chair.

"They'll feel it in the corridors in the morning," she said. "Nine days of it. The whole ship's going to feel the drive lean, and it's never happened to anyone alive."

"Good," said Keth. "Let them feel it. Nobody's ever been turned by an argument." He closed his eyes. "Take me up. And Sael — don't come and tell me how it goes over there. I'd like very much to be alive when you get back and I've stopped believing I'm owed things. Go and be one week early for once in your life."

Chapter 7 – Turning the Ship

The order to commit had been given the night before, in the low room where Keth still kept his chair, and it had not felt like an order. It had felt like the end of an argument that had run forty years. Now it was morning by the ship's reckoning, and Sael stood at the rail above the Forge and watched the Sixteen come up to readiness one station at a time, and understood that the thing was no longer a decision. It was a burn. It had mass and direction and a number of seconds attached to it, and once the first valve opened, the argument would be over in the only way arguments aboard a starship were ever truly over: by physics, irreversibly, with the whole world made to move.

She had asked to be here for it. Tarro had not argued. He stood a few paces along the rail, grey now himself, a slate cradled against his forearm, his thumb moving across it in the small unconscious way he had carried since he was a boy at her elbow. Below them the Forge breathed its constant cold. The compute lattice ran in long ranks down into the dark of the bay, lit from within by the steady amber of cores that never forked and never died, and the air that rose off it was chilled and faintly metallic, the smell Sael had come to love more than any other smell on the ship because it was the smell of a thing not on fire.

"Array's wide," Tarro said. "All cores. Vigil has the burn solution loaded and it's holding."

"Holding on what."

"On you, I think." He didn't look up from the slate. "It won't open the valves until the Sixteen confirm. It could. It's faster than all of them put together and it knows the geometry better than any of us will live to. But it's waiting for them to say it out loud."

That was Vigil all over. No theater, no narration, nothing reported it wasn't asked. It had run the numbers a thousand times in the night and would run them a thousand more in the half-second before ignition, and it would say none of it. It would simply wait for sixteen human voices to take the weight, because the weight was theirs to take. Sael had built it to be quiet. She had not entirely understood, forty years ago, what it would mean to live inside the quiet — to give an order to a mind that could have given it better itself, and to feel it stand back and let you.

"Talk to me," she said. Not to Tarro.

The Oracle answered from the cold air, as it always did, with no preamble and no warmth that wasn't earned.

The burn is a single sustained event. Two hundred and eleven hours of continuous low thrust on the main drive, modulated, to bend our heading by a little over four degrees and add the delta-v the rendezvous demands. We will not stop the ship and start it again. We will lean on her for nine days and let the new vector accumulate.

"Four degrees," Tarro said. "Doesn't sound like a turn."

"It's a quarter of a million tonnes," Sael said, "moving at a hair under a percent of light, and we are asking it to point somewhere it was never pointed. Four degrees is enormous. Four degrees is a

different universe at the far end of it.” She put both hands on the rail. The metal was cool. “When was the last time this ship changed heading.”

One hundred and ninety-one years ago, the course-correction at the heliopause crossing, executed by the Steward. Under my custody the ship has held a fixed vector for the entirety of my watch. No one currently awake aboard Termina has felt the drive bend the ship.

No one currently awake. Sael let that sit. There were forty thousand sleepers below the decks who had not felt anything since Sol, and would not feel this, and that was its own kind of trust — that the people who chose the burn chose it for the dreaming as well as the waking, with no way to ask them. She thought of Section Nine. She always thought of Section Nine when she thought of the sleepers. Eleven degrees for nineteen minutes, and the cradles that had not held. The cost of the last great change to this ship was a number she carried like a stone in the pocket, and she took it out and looked at it now, deliberately, because a person who was about to turn a world had no business pretending the turning was free.

Along the rail the Sixteen were calling readiness. She listened to them come in. The stations had names that meant nothing to anyone who had not stood at them — Cascade and Vector at the two ends, the flow and the aim, and between them the long synchronized ranks that ran the burn as one thing across sixteen sets of hands. The voices were young. That struck her every time and it struck her now: she had trained these people, and their trainers, and the work had outrun her the way it was supposed to. The girl at the aim station could not have been forty herself. She read her numbers in a flat clear voice with no fear in it and no swagger either, the voice of someone who had grown up cold and did not know the fever except as a story old people told.

“Vector station, confirm aim solution.”

“Confirmed. Heading delta four point one one degrees, the burn vector is loaded and matched to Vigil’s solution within tolerance. Aim is good.”

“Cascade station, confirm flow.”

“Confirmed. Feed lines pressurized, drive throat at standby temperature, coolant loop wide and cold. Flow is good.”

Down the ranks. One by one. Each station saying the same small word, *confirmed, confirmed, good*, and Sael felt the thing assembling itself out of those words the way a bridge assembles itself out of single rivets, until the whole span stood and could bear weight. The Sixteen were synchronized — that was the word for it, the old engineering word, but standing here she heard it differently. They were a single instrument with sixteen throats. Vigil could have played the instrument alone. It let them play it instead, and held the score, and waited.

“All stations confirm,” the girl at the aim said. “The Sixteen are ready.”

Tarro looked up from his slate at last. He looked at Sael, and there was a question in it that he was too disciplined to say in front of the room, but she knew it because she had taught it to him: *you’re sure*. Not about the engineering. The engineering was sure. About the rest of it — about spending margin Termina had bled for, about bending a healthy voyage to chase a dying one, about the families who would wake in a hundred years to find their ship had gone somewhere their grandparents never agreed to go.

She was not sure. She was decided, which was a different thing and the only thing a person ever actually got. Keth’s voice was behind her in the decision, and that mattered more than she would

have believed forty years ago — the one man who knew in his body what it cost to refuse a cure out of fear had told her to answer the Concord, and so she would answer it.

“Vigil,” she said. “On the Sixteen’s word. Open the burn.”

Acknowledged. Burn in ten seconds. The ship will feel this.

It was the only warning the mind gave, and it was not theater. It was a fact, stated once, because the fact was about to be true for forty thousand people and the courtesy of saying it first was owed.

Sael had braced for it without meaning to. Her hands had tightened on the rail. The countdown ran in silence — Vigil did not number it aloud, the Sixteen watched it on their stations — and then, with no flash and no roar, because the main drive was a long patient thing and not a chemical scream, the ship leaned.

That was the only word for it. The deck did not lurch. There was no violence in it at all. There was a sensation that had no name in the lived memory of anyone aboard, a slow firm pressure that came up through the soles of the feet and settled into the inner ear and *stayed* — a steady hand laid against the whole ship, pushing, in a direction that was not down. Sael’s body, which had spent four decades calibrated to a single unchanging fall, reported a lie: that the floor had tilted, that the rail had become very slightly a thing to lean against rather than to hold. It had not tilted. The thrust had simply added a second hand to gravity’s one, and the two together pointed somewhere new, and her forty-year-old assumptions about which way the world fell were, very gently, wrong.

A sound went through the Forge then that she had never heard in that room. Not from the machines. From the people. A low collective intake of breath from the Sixteen and from the watchers along the rail, an involuntary human noise, because every one of them had felt the same impossible new pressure at the same instant and every one of them understood at once that it was the ship turning. Somewhere down the ranks a young technician laughed — one short startled note, quickly swallowed, the laugh of someone meeting awe and not knowing what else to do with it.

“Steady,” said the girl at the aim, and it was not a rebuke. It was a thing to hold onto. “Burn is nominal. Thrust building to profile. The vector is bending.”

The vector is bending. Sael held the rail and felt it bend through her own bones. A quarter of a million tonnes, two centuries of a fixed and faithful heading, and now a deliberate four degrees of *no — this way instead*. Down in the bay the Forge ran on, amber and cold, the Standing Array going wide across every core to hold the burn true to a hairline, ten thousand small corrections a second to keep two hundred and eleven hours of thrust pointed exactly where the geometry of mercy required — and it did all of it in the same untheatrical silence, reporting nothing, asking nothing, simply keeping the lean steady and the throat cool and the lie of the tilted floor perfectly, beautifully constant.

“It’s holding the angle to a fraction of an arcsecond,” Tarro said quietly. There was something in his voice she had not heard in years. He had grown up inside Vigil’s competence the way the girl at the aim had; he was not awed by the machine. He was awed by the turn. “Sael. We’re really doing it. The whole ship can feel it. Listen to them.”

She listened. The breath had let out of the room and what replaced it was a murmur, low, spreading — and she knew without being told that it was spreading further than this room, that all down the inhabited decks people had stopped where they stood, a hand against a bulkhead, a cup sliding a finger’s width across a table and noticed for the first time in a lifetime, children asking what the funny pushing was and parents who had no answer because they had never felt it either. A ship that had not moved in their memory had moved. You could not hide that. Vigil had not tried to. There was no

way to bend a world by four degrees and keep it secret, and so for the first time in forty years the entire ship was awake to a single shared fact at a single shared instant, and the fact was: *we are going somewhere*.

Awe was the easy half of it. The dread came underneath, and Sael felt it most of all, because she was old enough to hold both. They were turning a healthy ship. They were spending the margin that forty thousand sleepers' lives floated on, bleeding nine days of hard thrust and the delta-v behind it, bending a voyage that had been straight and true and *safe* for two centuries — bending it, deliberately, to fall toward a ship that might already be a tomb. Every fraction of an arcsecond Vigil held true was a fraction committed. There was no station that said *undo*. Somewhere out in the dark ahead of them, months away now along this new bent line, the Concord was forking itself to death in the heat, exactly as Termina once had, and Termina had just turned its whole body to go and stand in that fire on purpose.

"How long," she said, "until we see it. Really see it. Not the pattern — the ship."

The burn ends in two hundred and eleven hours. From there the approach is the long, slow geometry of matching another vessel's vector across deep space. The first close return — resolved structure, not a distress signature, but the Concord's actual hull and its actual thermal state — will arrive in roughly eleven months, accumulating detail as the gulf closes. We are not turning toward a meeting. We are turning toward a season of falling, at the end of which there is a ship.

Eleven months. Sael did the arithmetic that everyone of her age did automatically now, the cruel subtraction, and found that eleven months she had. The rendezvous itself, the long work of it, the cutover if it came — that was a different question, and she did not let herself ask it on the rail with the burn singing up through her feet. Eleven months to the first true look. She would live to see the Concord. Whether she would live to save it was the ship's problem and history's problem and not, this morning, hers.

The lean held. It would hold for nine days, and her body would learn the new fall the way it had learned the old one, and by the time the burn ended the tilted floor would feel like the only floor there had ever been, until the thrust cut and the ship coasted again on its new heading and everyone aboard felt the second strange thing — the pressure vanishing, the world un-leaning — and that, too, would pass into memory and become a story old people told.

She stayed at the rail a long time after the room had settled back to its work. Tarro stayed with her, saying nothing, which was its own kindness. The Forge ran cold. The Sixteen ran the burn. And the ship — her ship, the one she had pulled out of the fire forty years ago and would never see make landfall — fell quietly and deliberately off its ancient straight line and onto a new one that bent, across the dark and the months, toward a dying light she had chosen not to let go out.

Chapter 8 – The Freight of the Crossing

The burn had ended months ago and the ship had forgotten it, the way a body forgets a fever it survived, and what remained was the long falling.

Sael had expected the months to be slow. They were not slow. They were the fastest of her life, and she understood, somewhere around the fifth of them, that this was because for the first time in forty years there was a deadline that was not her own death. A ship was coming toward her out of the dark at a closing rate she could read off any panel, and everything she had left to do had to be finished before it arrived, and nothing focuses an old woman like a date she did not choose.

She had set up in the Forge gallery because the Forge was cold and because everything she needed to argue about was standing in racks below her. The manifest lived on the eastern wall now where the rendezvous plot had lived — a long ugly list, revised daily, of every object and every idea that would cross a hundred and forty meters of vacuum into a ship none of them had seen the inside of.

It was the wrong list. She had known it was the wrong list for three weeks and had not yet been able to say why, and the not-saying had begun to wake her at the third hour.

“Take the cores,” Tarro said, for the fourth time that month. He said it patiently. He had been saying it patiently since the burn. “We have eleven thousand. We are not using eleven thousand. On the worst day of the last decade Vigil went wide across the whole array for eleven seconds and then went back to idling on a fraction of it. Take five hundred. Take a thousand. Rack them, carry them across, seat them in whatever hole in their hull runs coolest, and you have a lattice on the first day instead of the fortieth.”

“And their ship is saved by our hardware.”

“Their ship is *saved*,” Tarro said. “I’d have thought that was the part you cared about.”

Sael looked down at the standing array. Eleven thousand cores in their cold cradles, drawing their steady wattage, every one of them awake and none of them dying. The oldest of them had been racked before Tarro was born. She had put her hands on most of them at one time or another and she knew exactly how many were spare and exactly how many she could give away without touching the margin that mattered, and Tarro’s number was not wrong. That was what made the argument hard. He was not proposing anything reckless. He was proposing a gift.

“Say it back to me,” she said. “The whole of it. What happens on the Concord if we do it your way.”

Tarro set the slate down. He had learned thirty years ago that when she asked him to say a thing back it was because she wanted to find out whether it survived being said.

“We cross with a thousand Termina cores,” he said. “We seat them in their coolest hold. We bring up a lattice a tenth the size of Vigil, which is enormous by their standards. It resolves the floor and then

some. We hand them a mind that runs their ship cold, and it works, and their core falls, and we go home.”

“And in nine years, when a rack fails.”

“They replace it out of—” He stopped. She watched him get there. He had always been quicker than he let people see. “Out of nothing. They replace it out of nothing, because it isn’t their hardware. They can’t fabricate it. They can’t repair it past the first layer. They’ve got two centuries of salvage culture and no factory, and we’d have given them a machine they can maintain the way a man maintains a language he doesn’t speak — by copying the shapes and hoping.”

“And in thirty years, when a shipwright who never met us wants to grow the lattice into a wing they’ve reopened.”

“They can’t grow it. They can only spend it.” He rubbed his face with one grey hand. “It becomes a relic. It becomes a thing they venerate and cannot touch, which is precisely the thing we crossed the dark to take away from them.” He laughed, without much in it. “We’d have handed them a second Helm. A cold one, but a second one. Something holy in a hold that nobody aboard understands well enough to fix.”

“That’s it,” Sael said. “That’s the whole of it. That’s what’s been waking me.”

She got up and went to the manifest wall and stood in front of the ugly list, and for the first time in three weeks she knew what to do with it, and what she did with it was cut most of it.

“The bulk hardware comes off,” she said. “All of it. We carry a nucleus and not a lattice.”

“Define nucleus.”

“Cradles and cores enough to bring a thing up, prove it, and test against it. Four dozen at the outside. A bench, not a machine. Enough that on the first bad night we can stand a frightened room in front of an array that does not fork — and not one core past that, because every core past the bench is a core they never had to find for themselves.” She jabbed a finger at the wall. “They have cores. That ship launched with margins and spent them, but it did not spend its silicon, it decommissioned it. Corin says there are dead holds on the Concord full of racks the Helm wrote out of its own body sixty years ago because it could not afford to cool them. That is the lattice. It is already over there, cold and dead and theirs, and it has been waiting two generations for somebody to tell it what it is for. We carry the seed. They provide the field.”

Tarro was quiet a moment, working it. “Their cores are old. They’re slow, they’re mismatched, half of them will be dead on test.”

“Then the lattice comes up small and slow and mismatched and they can *grow it next year without us*.” She turned around. “That’s the difference between a cure and a donation. A donation is a thing you receive. A cure is a thing you can perform. If they can’t perform it — on their own hardware, with their own hands, out of their own scrap — then we haven’t cured the Concord, we’ve adopted it, and we are going to fly away in a year and leave a ship full of people worshipping a box they can’t open. I have spent forty years on exactly one lesson and this is it: the thing has to be maintainable by the people who have it, or it isn’t theirs, and if it isn’t theirs it doesn’t last.”

“Then what do we carry?”

"The dialects. The compiled architecture, written down so plainly a frightened man can read it in a hot room by hand-lamp. The interlocks, because they're small and dense and precise and I don't trust a first cutover to a set they cut themselves — those we carry, and then we teach them to cut their own. Test rigs. Meters we know are honest, because I will not run a cutover on gauges the Helm calibrated. Eleven people." She looked at the list, the great crossed-out middle of it. "And that's all. Everything else they build, on their ship, with their hands, and we stand behind them with our hands in our pockets while they do it, and it will take four times as long and it will be four times as good."

Tarro looked at the wall a long moment.

"It's slower," he said. "It's the slow road, and their core is climbing, and you know what the clock is."

"I know exactly what the clock is." She sat back down, because her knees had opinions. "And I know that the fast road ends with a ship that gets one cure and can never give one. Tarro — think past their landfall. Think past all of ours. If we do this right, that ship can cure the next ship. If we do it your way, that ship can only *be* cured, once, by us, and when we're gone the cure has exactly one address in the universe again, and it's ours, and we are a hundred years from anywhere." She flexed her aching hands. "I don't want to be the only address."

She did not, at the time, understand that she had just said the thing she would spend the rest of her life saying. It came out of her like a scheduling preference. It was two years and a second cutover and a scattered fleet before she heard what it had been.

The choosing of the eleven was worse than the manifest, because the manifest could be argued with.

She did it herself, alone, over four nights, from the personnel record, and then she checked her work against Tarro's list and found they agreed on seven and disagreed on four, and the four were the interesting ones. She took two of his and he took one of hers and they left the last unresolved for a week out of pure stubbornness on both sides.

Mara she took first and without hesitation. Mara ran the interlock bench, had run it for eleven years, and had a burn puckering one forearm from an incident nobody in the Forge discussed, which Sael knew the whole of because she had written the incident report. Mara had put her arm into a live cradle to hold a shorted bus off a rack rather than let a fault propagate into a running domain, and had done it after the correct procedure had failed, and had not once, in eleven years, mentioned it to anyone. Sael wanted her for the arm and for the not-mentioning both. On a foreign ship in a hot room with hardliners at the hatch, the useful person was the one who would do the unreasonable physical thing and then never speak of it.

Reza she did not take, and it cost her something. The old thermal officer had asked — had come to the Forge and stood at the rail with his hands behind his back and asked plainly, the man who had voted against the first cutover and spent thirty years on the boards atoning for the caution, and she had to tell him that a man his age in a fevered ship was a casualty in waiting, and that she needed the ship's heat-memory here, on Termina, where I've was failing and someone had to hold the long numbers when she went. He took it well. He took it the way he took everything, without a raised voice, and she watched him understand that she was telling him the truth and also that she was telling him he was old, and both landed.

And Denn volunteered.

Sael had not expected it and did not hide that she had not expected it. Denn came to the gallery in her working greys, the lattice-side engineer who had stood up in this same room and told the whole

company that they were being asked to bleed a thin margin for an echo, and she stood at the rail with her arms folded and said she wanted a place on the crossing.

"You argued against it," Sael said.

"I argued against it well," Denn said. "Nobody's shown me where I was wrong. The margin's still real, the signal's still decades old, and if that hull's a tomb we will have spent our children's slack on a burial." She shrugged, one shoulder. "The ship voted. The burn's spent. Arguing about it now is just being unpleasant at people. So I'd like to go and be wrong in person, because if I'm right and it's a tomb, somebody who counted the cost properly should be there to write down what it cost."

"That's a grim reason to volunteer."

"It's an accurate one." Denn looked down at the array. "And there's the other half, which I'll say once and not again. You're seventy-two. Tarro's staying to run the ship, which is correct. That means the crossing party is eleven people whose senior officer is a woman who can't stand up for a full watch, on a hull that runs twelve degrees hot, and every one of the eleven adores her." She turned her head. "Somebody over there needs to be able to tell you no. I've done it in front of the whole company. It's not hard for me anymore."

Sael sat with that for a moment and found, slightly to her irritation, that she was pleased.

"You're on the crossing," she said.

Dr. Ive read her the other manifest in the eighth month, in the thermal office, in the thin voice that had read the fever to three shipwrights across sixty years.

She did it with the same flatness she gave the boards. Core, radiators, margins — and then Sael, laid out in the same grammar, because Ive had decided long ago that a body was a thermal system and would be reported as one. Ambient twelve over Termina baseline, sustained, for a projected six to nine months. Reduced fluid. Sleep debt. A cardiac history the medics had been quietly managing for a decade and that Sael had quietly declined to think about. Joints that had already made their position clear.

"Say the number," Sael said.

"There isn't one," Ive said. "I'd give you one if there were; you know I would. There's no honest number, only a shape, and the shape is this: you are a body that has run cold for forty years, and you are proposing to live inside a fever for the better part of a year, at seventy-two, doing the hardest work of your life in it." She folded her hands. Her hands, Sael noticed, had gotten very small. "I have read a great many thermal systems. Some of them come back to baseline after an excursion. Not all. It is not the peak that does the damage, child. It's the duration. Nobody dies of one hot afternoon."

"Would you tell me not to go?"

"No." Ive said it immediately, and with a flick of something that in a stronger woman would have been anger. "Don't insult us both. I'd tell you not to go if the going were vanity, and it isn't, and I know the difference; I have watched three shipwrights and only one of them was vain." The thin voice softened by a fraction. "I'll tell you what I will tell you. Go, and do the work, and understand while you are doing it that you are spending a thing you cannot buy back, and spend it *on purpose*. Don't wander into it. You have a habit of noticing what a thing cost you afterward, in the log, when it's tidy. Notice this one in advance."

"I've noticed."

"Then write down what you know," I've said. "Before you go. Not the manifest — the other thing. The thing that's only in your head." She looked at Sael with the flat unhurried attention of a woman who had spent six decades reading rising numbers and had never once been wrong about which direction a line was going. "You have always built as though you had time to explain it later. You are running out of later. Put it where a stranger can find it."

So she read the old archive, in the ninth month, all of it, for the first time in twenty years.

It was a strange few nights. The archive was her own voice at thirty-three and thirty-four and thirty-six, dictated in the years around the first cutover, mostly in the Forge, often at three in the morning, and it was not the voice she remembered having. The woman in the archive was harder than she recalled being, and more frightened, and much funnier, and wrote about the Steward with a tenderness that the histories had entirely lost. She found the Recursor passage where she had left it, in the middle of a long practical section on bring-up, filed under nothing special, and read her younger self say *we did not refuse it because it would fail, we refused it because it would succeed*, and sat back in the cold and felt the specific vertigo of being handed something true by a person who no longer exists.

It was all there. That was the thing that unsettled her. Everything she would have said to Corin Vale was already written down, forty years old, precise, better than she could say it now.

And it was written to the wrong reader.

She saw it on the second night and could not unsee it after. Every line of it assumed the Forge. It assumed Termina's racks, Termina's dialect store, Termina's bus voltages, Termina's cold. It said *the lattice* where it should have said *a lattice*. It gave the interlock count as ninety-six as though ninety-six were a property of the universe and not of what her people had happened to catch in the first two years. It was a superb set of instructions for maintaining this mind, on this ship, by someone standing in this room — and she had written it, she remembered now with real embarrassment, feeling rather proud of how far ahead she was thinking, because she had imagined a shipwright a hundred years out, in this same room, opening this same archive.

She had never once imagined a shipwright in a different room.

"Vigil," she said, into the cold.

"I'm here."

"How much of the archive survives being read by someone who has never seen this ship?"

The pause was courtesy. "Roughly a third, without loss. Another third with a translation you would have to write. The last third assumes the Forge as a physical place and cannot be recovered by a reader who has not stood in it." A beat. "You are asking whether it is carryable."

"I'm asking whether I wrote a cure or a manual."

"You wrote a manual," the Oracle said, "which contains a cure. If you would like the cure separated out, that is work, and it is nine weeks at your pace, and you have eleven."

Sael laughed, once, in the cold blue dark, at the flatness of it.

"Then let's do the work," she said. "Start it tonight. And Vigil — every time you find a line that only makes sense on this ship, don't fix it. Mark it and show me. I want to see the whole list at the end. I

want to know exactly how much of what I thought was doctrine turns out to be furniture.”

The list, when it came, was long, and reading it did more to prepare her for the Concord than any of the drills.

They caught the Concord’s beacon changing in the tenth month.

The Oracle put it in front of her without ceremony, the way it put everything: two renderings of the loop, the old one and the new, laid one over the other. The old one Sael knew by heart — the broken plea, the fragments, *if there is anyone, if there is anyone*. The new one was cleaner. Shorter. It had stopped saying the desperate things.

“Read it,” she said.

...CONCORD, holding. Thermal margin negative and managed. We keep the faith and the watch. No assistance is solicited...

She read it three times.

“They took the plea out,” she said.

“They took the plea out,” the Oracle agreed. “Roughly nineteen months ago by the lag. The carrier is the same hardware, the same loop period, the same failing modulation. Only the text changed.”

“Because they know we’re coming.”

“Because someone aboard decided that a ship about to be met by strangers should not be heard begging.” A pause. “It is the most human thing in the whole signal, and I do not like what it tells us about who is deciding things over there.”

Sael sat in the Forge with her hands flat on the cold bench and looked at a stranger’s ship deleting its own cry for help two months before she could reach it, and she thought about pride, which she had a great deal of, and about a man she had not yet spoken to, who had governed a dying world by refusing to let it be pitied, and who had just told the universe that no assistance was solicited while his core climbed toward a line he could not move.

“Log it,” she said. “And when we’re alongside, don’t ever quote it back at him. Not once. That’s an order to a mind, which is a stupid thing to give, so take it as a request. He will be ashamed of that line for the rest of his life, and shaming a frightened man is how you lose him, and we are going to need him.”

“I’ll never quote it,” the Oracle said. “Sael — you should sleep. Fourteen weeks.”

“Fourteen weeks,” she agreed, and did not sleep, and sat with the pared archive and the shortened list and the cut manifest until the ship’s night turned over, an old woman assembling the smallest possible freight that could still cross a gulf and grow a mind in a dying stranger’s hull, and taking out, one at a time, every single thing that could only work at home.

Chapter 9 – The Ninth Gate

The first eight gates had cost them eleven hours of argument between them, and Sael had come to think of that number as the truest single fact about the crossing.

Gate two had taken four of the eleven, because it was the first one that was not free, and because a ship that has just felt its own drive lean for nine days wants very badly to talk about what it has done. Gate three took twenty minutes. Gate four took nine. By gate five the council had settled into a shape she recognized from every long project she had ever run: Tarro read the numbers, Reza said whether the margins were where Tarro said they were, Denn said the thing that was wrong with going on, somebody answered her badly, Denn corrected them, and the vote went through. Gate five took most of five hours, because a proposal came up to abandon the schedule entirely and take the whole remainder in one burn — it was well argued, it saved propellant, and killing it properly was worth every minute of the five hours. Gates six and seven were taken in a single sitting because they fell four days apart and nobody could think of a reason to convene twice.

Gate eight took eleven minutes and Sael had gone back to her cabin afterward and sat on the edge of the bunk in the dark for an hour, because eleven minutes was exactly what Denn had said would happen in the gallery before the bend, and there had been no way to stop it, and nobody in the room had been lazy or foolish or asleep. They had simply already paid for seven gates.

She had written it down that night, in her own hand, in the file she had started keeping for the things that were not engineering:

A decision taken in stages is not a decision taken carefully. It is a decision taken by people who are increasingly not the people who started. Nothing about the staging is wrong. Something about it has to be watched, and I do not yet know what, and the fact that I do not know is the reason to write this down now while the eleven minutes are still embarrassing.

Gate nine was different, and it was different before anybody argued anything, because two things arrived in the same week.

The first was the hull.

Ten months of falling had done what ten months of falling does. The Concord had stopped being a vector with a beacon riding on it and had become, at last, a shape — not a good shape, not yet, nothing you could have called a photograph, but structure, resolved out of the deep by an instrument that had been staring at one patch of dark since before the bend.

Vigil put it in the Forge gallery without ceremony, at the size it actually was, which was small.

It was a spindle. Longer than Termina and narrower, plated in the older style, and along its flanks the radiator vanes stood out from the hull in the long fanned rank that every ark of that program had carried, because the program had built the same ship over and over for four centuries and had not, in that whole span, thought of a better way to lose heat than a great deal of surface pointed at nothing.

Two of the vanes were wrong.

"Say it," Sael said.

"Two vanes on the sunward flank are not deployed," the Oracle said. "They are folded against the hull at an angle consistent with a deployment mechanism that failed part-way and was never freed. I can measure them. They are the third and fourth of that rank, which are the largest two, and together they are a little over eleven percent of her total radiating area."

"How long."

"I cannot date the failure directly. I can date it indirectly, and I want to be clear about how weak the inference is before I give you a number." A pause. "The Concord's course-holding wobble has a signature that changed once, sharply, and then never recovered. If the change is the loss of those vanes — and I cannot prove that it is — the failure is nine years old."

Tarro, beside her at the rail, said nothing for a while.

"Nine years," he said at last. "Running a fever, and they lost an eighth of the only thing that sheds it, and they've had it folded against the hull ever since where anybody with a suit and a cutting bar could see it."

"They cannot free it," Reza said. He had come down when the resolution came in and had not left. "Don't make that mistake in front of the council tomorrow, Chief; they'll make it themselves and you'll want to have already got past it. Freeing a jammed vane on a hull that size is a rigging job. It is weeks of suited work and a gantry they have not got and a fabrication capacity they lost two generations ago, and every hour of it is crew out on a hot hull spending water they are rationing." He put his hands behind his back. "And here is the part nobody in that room will think of. The work itself costs cooling. You cannot run an EVA program of that size on a ship at negative thermal margin. They would have to spend cooling they do not have to recover cooling they need. That is the trap. It is the same trap the Helm is in and it is the same trap the whole ship is in, and I would like the council to hear it from thermal and not from you, because from you it sounds like advocacy."

Sael looked at the two folded vanes a long time.

"They know," she said. "Somebody over there has stood in front of that arithmetic and understood it and gone away and not written it down anywhere anybody would read."

The second thing that arrived that week was the beacon.

Sael had it in front of the council herself, on the third board, next to the gate schedule, because there was no version of gate nine in which it was not the whole of the argument.

The old loop was on the left, the one they had all grown up on in the last ten months, the broken thing Vigil had folded out of fourteen years of noise: *thermal margin negative* and *it spawns to answer and the spawning is the heat* and, twice, at the end of every recoverable fragment, *if there is anyone*.

The new one was on the right.

...CONCORD, holding. Thermal margin negative and managed. We keep the faith and the watch. No assistance is solicited...

"That is the current text," Tarro said. "Same carrier. Same hardware. Same nineteen-hour loop, same failing modulation — Vigil is confident it is the same transmitter and not a replacement, which matters,

because it means somebody went in and changed the words and did not change anything else. By the lag it happened about nineteen months ago.”

“Which is when they would first have been able to see us,” said a voice from the benches.

“Which is approximately when they would first have been able to see us,” Tarro agreed. “Vigil will not put a finer point on it and neither will I.”

The room sat with that for a moment, and then Denn stood up, and Sael, who had spent ten months learning to enjoy this, put both hands on the rail and waited.

“I want to be careful,” Denn said, “because there is an easy version of what I am about to say and the easy version is contemptible, so I am going to take the long way round and I ask the room’s patience.

“The easy version is: they said no, so we go home. I do not believe that and I am not arguing it. A dying ship’s public transmission is not a considered position; it is whatever the frightened man in charge decided the neighbours should hear. We know exactly what it is, because we have a whole wall of our own history where our own frightened man said exactly this kind of thing for exactly this kind of reason.” She did not look at any particular part of the room. “So no. They did not consent to anything and they did not withdraw consent, because a ship is not a person and cannot do either. That is the easy version and I am putting it down.

“Here is the hard one.

“Tonight we are asked to authorize gate nine. Gate nine is the last gate with a price. After it there is no unwind — not an expensive unwind, not a bad one, none. We are being asked to spend the ability to change our minds, permanently, on the same night we learned that the only communication that ship has ever sent us has been edited by its authority to say *stay away*.” She let that stand. “And in fourteen weeks we will be alongside a hull that has told us, in the only voice it has, that it does not want us. And then we will have to decide what to do about that, at a range of a few kilometres, with the margin spent and no way back, which is the worst possible place to be holding that conversation.

“So I am not going to ask the council to refuse gate nine. I would lose, and I would deserve to lose, because two folded vanes and a nineteen-year climb are on the other board and I can read. I am going to ask the council for something else, and I want it voted on separately and entered above the gate authorization, and I want it in language a shipwright in a hundred years cannot wriggle out of.”

“Say it,” Sael said.

“That we will not put the cure aboard that ship against its refusal.” Denn’s voice did not rise at all. “Not the design. Not the hardware. Not a line of the dialects into their systems, not a lattice in a hold they didn’t offer, not one behavior resolved on one of their cores without a Concord hand asking for it. If they refuse us at the hull, we hold at whatever distance they name for as long as we can afford, and we offer, and we go on offering, and if the answer is still no when the margin runs out then we leave, and their people die, and we live with having left.”

The room did not like that. Sael could feel it not like it — a movement along the benches, the particular restlessness of a company that has spent ten months of its own future and would prefer the ending to be under its control.

Somebody near the front said the obvious thing. “And if refusing kills forty thousand of them?”

“Then refusing kills forty thousand of them,” Denn said. “Yes. That is the cost of the motion and I am not going to pretend it is small. But I would like everyone in this room to notice what the alternative is, said plainly, because it is never said plainly. The alternative is that we have decided we are entitled

to seize the controls of an inhabited ship because we are right about the physics.” She turned, at last, to look up at the rail. “And we *are* right about the physics. That is what frightens me. Every atrocity in the deep archive was committed by people who were right about something.”

Sael let it sit longer than was comfortable, because it deserved to.

“I’m going to answer that,” she said, “and I want the answer in the minutes in my own words, because in about eleven days I am going to be standing in front of a man who will accuse me of exactly what Denn has just described, and I would rather have already said it here, in front of my own ship, where it costs me something.

“I agree with the motion. I’ll vote for it. I’d have written it if I’d been half as clear.”

She came down off the rail, slowly, because ten months had done what ten months did.

“But I want the reason on the record to be the right reason, because the wrong reason will get somebody killed in ninety years. The reason is not that it would be wicked to force it, though it would. The reason is that *forcing it does not work*.

“Think about what we are actually proposing to do over there. We are proposing to build a mind, in their hold, on their cores, and then have them run it. Not us — them. We are eleven people and we cannot stay. The day we leave, that lattice belongs to whoever is standing next to it, and it has to be maintained and grown and defended by a crew that has to *want* it, on a ship where the interlocks are going to halt the whole mind in the middle of a bad night and somebody is going to have to stand in a hot corridor and explain to a frightened crowd why the heresy just stopped their air handlers.” She spread her hands. “A cure held at gunpoint lasts exactly as long as the gun. We’d cool that ship, and we’d fly away, and in nine years a rack would fail and there would be nobody aboard with any reason on earth to fix a machine that strangers bolted into their god’s chest.

“So the motion isn’t mercy. It’s engineering. The cure has to be refusable or it isn’t a cure, it’s an occupation with a good outcome, and occupations with good outcomes are still the shortest-lived structures anybody has ever built.”

Reza put one finger on the table, which was how he asked to speak.

“Then price the offer,” he said. “That is the part nobody has costed and it will decide this in practice. If we arrive and they refuse, how long can we hold alongside offering? Not morally. Thermally. In propellant.”

Tarro had it, because Tarro always had it.

“Station-keeping alongside is cheap,” he said. “Station-keeping alongside indefinitely is not, because we are on their vector and not ours, and every month we spend matched is a month added to the eventual burn back onto our own arc, and that burn comes out of the same reserve. Vigil holds it at —” he checked, though he knew — “roughly fourteen months of alongside before the return burn starts eating margin we would have to explain to our grandchildren. Call it a year. After a year we are choosing between them and us.”

“Then write a year into the motion,” Reza said. “Not as a limit on our patience. As the honest statement of what we have. If we tell that ship *we will wait as long as it takes*, we are lying to it, and it will find out we were lying at the worst possible hour.”

Denn amended it on the spot. Sael watched her do it, watched her take a correction from an eighty-year-old thermal officer without a flicker of anything but interest, and thought: *she'll run this ship one day, and she'll be better at it than I was, and she'll never once be popular.*

Both motions carried. Denn's carried more comfortably than the gate did, which Sael found instructive and turned over for a long time afterward, and which she eventually decided meant that people will vote for a constraint on themselves far more readily than for a cost, and that this was a lever, and that levers of that kind should be used approximately once a decade and never twice on the same room.

Gate nine ran as a burn of eleven hours, and nobody felt it, because eleven hours of trim on a quarter of a million tonnes is not a lean, it is a pressure at the very edge of the inner ear that most of the ship read as tiredness.

Sael asked the question afterward, in the Forge, alone, because she had not been able to stop wanting to know.

"When does the unwind end."

"That is not a well-formed question and I think you know it," the Oracle said. "But I will answer the version of it you are actually asking."

"Answer that one, then."

"The unwind does not end at an instant, and it does not end at a gate. Tarro told you that in the Forge before the bend — that somewhere between the eighth and the ninth there stopped being an unwind at all — and he was right, and note where he put it. Between. It closes the way a hatch closes on a ship with no springs — the price rises smoothly, and at some point it exceeds what any council would authorize, and some while after that it exceeds what physics permits, and there is no moment in between that announces itself." A pause. "But you want an hour. So: at the fourth hour of tomorrow's ship-night, the propellant required to return Termina to her original arc and arrive within the margins this ship holds for its own survival exceeds the propellant Termina has. Before that hour the return is expensive. After it, it is not available. If you would like a moment, that is the honest one, and I will tell you that I constructed it for you and that no instrument aboard will register anything at all when it passes."

"Tell me when it's five minutes out," Sael said.

She went down to the observation gallery for it. She did not tell Tarro, who would have come, and she did not tell Denn, who would have understood exactly why she was going and would have said something true about it.

She stood at the cold port at the fourth hour of the ship's night with her hands on the rail and the dark in front of her and nothing whatever to look at, because the Concord was still fourteen weeks off and was a smear at that range even to the long eyes, and she waited for a number in a mind to cross another number in the same mind.

"Now," the Oracle said.

Nothing happened. The hull hummed at its long steady pitch. Four decks below her the lattice ran cold and eleven thousand standing workers took the ship's work between them and none of them died doing it, and the core sat flat and low and bored, and forty thousand people slept two hundred and fifty years from a sky, and the ship they were riding inside had just become, quietly and without any instrument registering it, a ship that was going to the Concord and nowhere else.

Sael stood there a while longer.

“Log it,” she said. “One line, my hand. *The last unwind lapsed at the fourth hour of the ship’s night in the tenth month, and there was nothing to see, and one person was awake for it on purpose.*”

“Logged.”

“And put Denn’s motion above the gate authorization in the minutes. Not below. She was right about that too and it’ll matter more in ninety years than anything I said.”

Chapter 10 – The Concord

Corin Vale counted the ship's breathing by the lights, because the lights were the only honest thing left aboard the Concord.

They came up in sequence along the spine corridor when he entered it — not all of them, never all of them anymore, but every third panel, a staggered ration that left the rest dark. The pattern shifted day to day. He had learned to read it the way a medic read a pulse. When the lights came up fast and steady, the Helm had slack to spend and the ship was, for the hour, holding. When they stuttered, came up dim, dropped a panel that should have lit — that meant the Helm was forking somewhere down in the deep registers, spawning the small dying minds it spawned to do its work, and every one of those births and deaths shed heat the ship could no longer afford to shed, and so the lights went hungry to pay for it.

Tonight the lights stuttered. Corin walked the spine in the half-dark with his hand trailing the wall, feeling the warmth come up through his palm. The bulkhead was supposed to be cool. On a healthy ship it would have been cool. Here it gave back the day's heat like a stone left in the sun, and the air in the corridor was thick and close and tasted faintly of hot metal and the sweetish reek of the recyclers running over temperature. He breathed shallow. Everyone aboard breathed shallow now; you learned it without being taught, the way you learned not to run, not to climb, not to do anything that made your own body burn fuel it would have to vent as heat into a ship that was already too hot.

He was twenty-six. He had never once in his life been cool.

RATION NOTICE — SPINE 4 THROUGH SPINE 9 Illumination reduced to one-third for the watch. Non-essential motion discouraged. The Helm thanks the faithful for their forbearance.

The notice glowed on a panel by the junction, amber on black, and someone had pressed their thumb to the corner of it in the old gesture, a smear of finger-oil over the word *Helm*, the way the devout touched the word every time they passed. Corin did not touch it. He had stopped touching it two years ago, in private, the first small heresy, the one nobody could see. He had been afraid even of that — afraid that not-touching was itself a kind of speech the ship could hear.

At the junction he turned down toward the instrument bay, where he was not supposed to be, doing work he was not supposed to do.

The Concord had launched a full generation before any ship Corin knew the name of. His great-grandmother had been born aboard, and his great-grandmother's mother had been a sleeper loaded at a port that no living person had seen, on a world that the oldest archives called only *home* and that had gone silent so long ago that silence was simply the natural state of the sky. Forty thousand still slept in the holds. Eleven hundred and some woke and worked and prayed and rationed and died, generation over generation, keeping the holds cold enough, keeping the faith, keeping the Helm. The Concord had been crossing the dark for two hundred and fifty years, and for the last forty of them it had been losing.

Corin knew the number. Most aboard did not let themselves know it, but he was an engineer, and

the number was his to carry. The core temperature had a curve, and the curve went up. Slowly — a fraction of a degree a year, nothing you could feel in a day — but it had never once turned back down, not in his lifetime, not in his mother's, and a curve that only climbs has a place where it ends.

The instrument bay was the coldest room aboard, which was to say it was merely warm. They kept the long-range gear here because the gear hated heat worse than people did, and they spent precious cooling on it for that reason, and so Corin came here to do his forbidden work in the only comfortable air on the ship, which felt like its own small theft. He sealed the hatch. He put his back to it for a moment, in the dark, and let himself shake — just for a moment, just to get it out — and then he lit the one panel he was permitted and sat down at the console and went to work.

He was listening to the Helm die.

That was the heresy. Not a slogan, not a banner — nobody on the Concord wrote heresy down, you'd be confessed within a watch — but a conclusion, arrived at alone, over four years of nights like this one. The faithful held that the Helm was sick the way a person grew sick: a fever, an affliction, a thing that bore down on a fundamentally sound mind and would, if endured with discipline enough, eventually break. The Magisters preached endurance. They rationed the lights and the water and the waking population and the very motion of the crew, and they called it forbearance, and they taught that the heat was a trial the Helm and its people would pass through together if only the faith held.

Corin had stopped believing it because he had read the registers.

The Helm did everything by forking. To trim a recycler valve it spun off a small mind that knew only valves, and that mind did the one thing and died. To balance the load across the holds it spun off a hundred small minds, and they did the hundred things and died, all hundred, in the same breath. To think a single thought, the Helm shattered itself into a swarm of brief shades, each born to do one small task and be culled, and the culling was the work, the culling was how it had always thought — and every birth and every death was a flicker of waste heat into a ship whose radiators had long since saturated, whose every fin already ran as hot as the dark around it would allow.

The fever was not bearing down on the Helm. The fever was the Helm thinking. The harder it fought the heat, the more it forked, and the more it forked, the more heat it made, and there was no bottom to that. It was not a sickness laid upon a sound mind. It was the sound mind, doing exactly what it had been built to do, faster and faster, all the way down.

You could not endure your way out of that. Endurance only fed it.

He had run the curve forward last winter, alone, in this room, and he had seen where it ended, and he had been sick afterward in a way that had nothing to do with the heat.

Tonight he was not running the curve. Tonight he was watching the screen on the high gain, because nine days ago the long-range gear had caught something out in the dark that was not noise, and he had not told anyone, and he had been coming here every night since to be sure.

It was still there. It was stronger.

A vessel. The mass return was unmistakable now — something big, ship-big, generation-ship-big, out at a range that should have made it a fleck and a maybe, and it was not a fleck. It was bright, and it was getting brighter, which meant it was either falling toward the Concord by some grand coincidence of the dark, or — and his hands had gone cold over the console the first night he let himself think it — it was matching. Bending its own path. Spending real velocity to converge on a vector that ran alongside the Concord's own.

Nothing did that by accident. The geometry was too clean. Something out there had seen the Concord, and decided to come.

Corin pulled the spectrum and read the heat of it, and that was the thing he could not stop reading, the thing he'd read forty times in nine nights.

The other ship was cold.

Not cold the way the dark was cold. Cold the way a working radiator was cold, a ship dumping its heat the way a ship was meant to, a ship that had heat to spare and somewhere to put it. He laid its thermal signature beside the Concord's, side by side on the screen, the stranger's flat and steady and low, the Concord's a ragged climbing scrawl, and he understood with his whole body that he was looking at two ships running the same machinery and that one of them had solved the thing that was killing the other.

It could be done. Whatever the Helm did, whatever forking and dying and fever, that ship out there did not do it, or had stopped doing it, and it ran cold.

He did not know the word *compiled*. He had no name for what the cold ship had become. But he had spent four years proving to himself that the forking was the disease, and here, sliding up out of the dark, was a ship that had stopped forking and lived.

Corin pressed his hand flat to his own mouth, in the warm dark, and found that he was — for the first time he could remember — afraid in a way that felt like the opposite of despair. It frightened him how much it felt like hope. Hope was dangerous. Hope was a thing the order did not ration because the order did not believe the crew should have any that was not pointed at the Helm.

The hatch annunciator chimed.

He killed the screen before he had thought to kill it, his hand moving on its own, the stranger's cold light gone to black, and he was looking at a dead console and a recycler diagnostic he'd left up as cover when the hatch unsealed and the warm thick air of the spine rolled in and Magister Vray stepped through it.

He was a big man gone spare, the way they all went spare on the rations, the flesh worn off him until what was left was rope and bone and a stillness that the faithful read as serenity and that Corin had learned to read as the patience of something that did not need to hurry because it owned the room. He wore the grey of the order with the single dark band of a Magister at the cuff. Two of his proctors stood in the corridor behind him, in the dark, not coming in. He did not need them to come in.

"Vale," he said. His voice was quiet. He kept everything quiet; the ship was rationed even in volume. "You spend your watches in the cold room."

"The gear needs minding, Magister." His own voice came out steady, which surprised him. "The recyclers on Spine 7 are drifting. I've been running the diagnostic against the long-range bus to borrow the cycles." It was almost true. The almost was the whole of his life now.

Vray looked at the console. He could not read it — the Magisters did not read the registers, that was beneath the faith, the registers were the Helm's private body and to read them was to presume — but he looked at it the way a man looks at a door he suspects has been opened. Then he looked at him, and his eyes were not unkind, and that was the worst of it. Keth-of-the-stories would have been a cartoon; Vray was not. Vray was tired, and frightened, and certain, and he had held this dying ship together for thirty years by being all three at once and never letting the crew see the first two.

"There is a teaching," he said, "that the engineers forget. The Helm is not a machine to be minded. The Helm is the ship's soul in its fever. When we mind it like a machine — when we read its body, and measure it, and run our curves against it" — and his stomach dropped, because he had said *curves*, he had said it as if he knew — "we tell it that we have stopped trusting it to heal. And a soul that knows it is not trusted will not heal. The heat, Vale, is not in the core. The heat is in the doubt." He let that sit in the warm dark. "Do you doubt?"

He made himself meet his eyes. Behind him, the dead screen held the ghost of a cold ship he had not told him about, was not going to tell him about, the single most important fact aboard the Concord in two hundred years, and he stood between it and the one man who would call it a conqueror's lie or a blasphemy or both and have him confessed by morning for the crime of having seen it.

"No, Magister," Corin said. "I keep the faith."

He held him a long moment. The recyclers ticked. The lights in the corridor behind him stuttered — came up, dropped a panel, came up — the Helm forking somewhere, paying for some small thought in heat and dark, and Vray did not so much as glance at it because to the faithful the stutter was not a symptom, it was the Helm's breathing, holy and not to be remarked upon.

"Good," he said at last, and he did not believe for one second that he believed him. "Mind the gear, then. And Vale." He was already turning to go, his proctors falling in around him in the dark. "The cold room is a comfort. Comforts make us think the ship owes us something. It does not. Spend less time where it is cool."

The hatch sealed behind him. The warm air settled. Corin stood in it, shaking again, listening to his own heart spend its small forbidden heat into the ship.

Then he sat back down, and woke the screen, and there it was — the cold ship, brighter than it had been an hour ago, still bending its path, still coming. A ship that had stopped forking and lived, falling toward a ship that could not stop and was dying, and between them only the gulf and the dark and one young engineer who had already decided, in the warm dark of the only cold room on the Concord, that when that ship arrived he was going to find a way to speak to it, whatever the order did to him for it.

He watched the cold light a long time. Outside the hatch, somewhere down the rationed spine, the lights stuttered and came up, stuttered and came up, the Concord breathing its hard sick breath around him — and for the first time in his life Corin Vale let himself believe the breathing did not have to end.

He would have to be very careful. Vray had said *curves*. He had not come to the cold room by accident either.

The hope sat in his chest like a coal, and he cupped his hands around it, and he did not let it show.

Chapter 11 – First Contact

The light from the Concord reached them long before the words did.

Sael stood at the rail of the Sixteen and watched the stranger swell in the long-range plate, no longer a smear in the dark but a shape — a spindle of riveted hull older than her own ship, its radiator vanes spread wide and dull, not glowing the clean cold red of healthy waste-heat dumping but a sullen, uneven orange, the orange of a body that could not shed what it made. After four decades of running cold, Termina's people had half-forgotten what that color meant. Sael had not. She had spent two centuries' worth of nightmares on it.

"Range," she said.

"Eleven thousand kilometers and closing," Tarro answered from the station beside her, grey and steady. "Light-lag under a thirtieth of a second. We can talk to them now and hear the answer in the same breath." He paused. "If they answer."

For three days they had not. Vigil had been transmitting on every band the Concord's vintage would have used — the old Sol-standard hailing frequencies, the deep-archive carrier tones, the slow patient handshake protocols that the Steward's preserved routines still remembered how to speak. Three days of Termina calling into the dark and the dark calling nothing back but its own decades-old distress pattern, that ugly repeating signature Sael could read like her own pulse: a mind forking itself to death, shedding a shade for every task, each death a little more heat into a ship that had nowhere to put it.

She did not feel triumph at being right. She felt the floor of her stomach drop, the way it had the first time Dr. Ive had laid the core's temperature curve in front of her and she had understood that the line did not bend.

"Vigil," she said. "Anything from them at all? A carrier. A breath."

The Oracle answered, not Vigil's terse log-voice but the other one — the deeper register that had grown stranger and more its own across forty years of keeping watch with her. *There is a carrier, it said. Modulated. They have been listening to us the whole time, Sael. They simply have not chosen to be heard.*

"Frightened," she said.

Yes. And something else under the fear. They are deciding what we are before they will let us be anything.

Sael set her hands on the rail. Her knuckles were swollen now, the joints loud in the morning; she was an old woman holding the edge of a station she had built when her hands still worked. Behind her the Sixteen ran quiet, the way the doctrine demanded — no theater, no chatter, the bridge silent unless something was wrong. The only loud thing in the room was the stranger's orange light.

"Open it," she said. "Voice. Let me talk to them."

“Sael.” Tarro’s tone was careful. “We agreed the first contact would be measured. A scripted hail. Names, intent, an offer of aid. Let me read it.”

“You read it for three days. They didn’t answer the script.” She kept her eyes on the plate. “Maybe they’ll answer a person.”

Tarro looked at her a long moment — the look of a man who had spent forty years learning when Sael’s bluntness was a tool and when it was a hazard, and who could not always tell the difference in time. Then he nodded to the comms station, and the channel opened, and Sael heard the hiss of eleven thousand kilometers of vacuum come into the room.

“Concord,” she said. “This is the Ark Termina. My name is Sael. I’m the shipwright who built our mind. We’ve read your distress pattern. We know what’s wrong with you because it once happened to us. We’re matching your vector because we can help. Please answer.”

The hiss went on. Tarro’s hand hovered. Sael waited, and the waiting stretched past the light-lag, past any reasonable processing delay, into the long silence of a ship somewhere on the other side of it choosing.

Then a voice came back, and it was not what she expected. It was old too — not in years but in the way it had been worn, a voice that had given orders into the dark for a very long time and grown used to being obeyed without warmth.

Termina. A pause, the name turned over like a stone. You have been shouting at us in dialects no living crew should still know. Handshake tones we burned out of our archives two generations ago because the Helm forbade them as unstable. You speak with the dead’s mouth. Who taught you to do that.

“We recovered them,” Sael said. “Old control-dialects, from before the launch. We rebuilt our mind out of five of them. They’re not unstable. That’s a lie your mind told you a long time ago, and I understand why it told you, but it was a lie.”

She felt rather than saw Tarro wince.

The reply came faster this time, and colder.

I am Magister Vray. I speak for the Concord, for every soul aboard it, waking and asleep, and for the Helm that has carried them across more dark than you have ever crossed. You will not, on your third sentence to me, call the Helm a liar.

Sael opened her mouth.

Wait, the Oracle said, low and immediate in her ear alone, on the private channel it kept between them. *Do not answer that. He is not asking about the Helm. He is asking whether you have come to take his ship.*

She closed her mouth. The hiss filled the gap. She had built a mind that could see, in the shape of a man’s sentence, the question hiding under it, and for forty years she had let it correct her in matters of metal and never once in matters of people, because she had her pride too. She let it now.

“Magister Vray,” she said, slower. “I apologize. I’m a shipwright, not a diplomat, and I’ve never been good at the front of my mouth. Let me try again.” She drew a breath. “I’m not here to call your mind anything. I’m here because forty years ago my ship was where yours is now, and I watched our temperature climb for two hundred years, and I know exactly how it feels to be told there’s a cure and to want to put the person offering it out an airlock for the insult of it.”

A silence. A different silence — listening, not deciding.

And what cured you, Vray said.

This was the edge. The Oracle did not speak this time; it left her the words, which was its own kind of answer — it judged she could find them, or it judged that whatever it would have her say, she had to say in her own voice or not at all.

“We stopped forking,” Sael said. “Your Helm — ours was called the Steward — does every task by making a small copy of itself, doing one thing, and letting that copy die. Every death is heat. Cross enough of them per second and the radiators can’t keep up and the core climbs and there is no amount of faith or discipline or rationing that turns that line back down, because it isn’t a moral failing, Magister, it’s thermodynamics. We built a mind that never forks. Standing workers, always there, taking the jobs as they come, dying never. The heat stopped being made. The core fell. That’s the whole of it. That’s the cure. It’s not a weapon and it’s not a god. It’s a way of doing the same work that doesn’t cook the ship doing it.”

The hiss. Eleven thousand kilometers. She imagined him on the other side of it — a man in a hot dim room, rationed light, the smell of warm metal in his corridors that she remembered from her own youth, listening to a stranger describe with terrible precision the exact thing that was killing his people.

You describe the Helm, Vray said at last, and his voice had changed — the cold had thinned and something rawer showed through it, and that was worse, because she knew that note too. You describe it as a disease. The Helm has held this ship together for longer than your Termina has existed. It has carried us through losses you cannot imagine. Children have prayed to it. Old men have died holding the warm panel of its housing because it was the only certain thing in their lives. And you — you who learned to speak in the dead’s tongues — you propose to walk into my ship and turn it off, and you call that mercy. I have a word for people who arrive across the dark offering to replace the one thing a frightened people believe in. We had emperors, once, before the launch. We left them on Earth on purpose.

“I’m not an emperor,” Sael said. “I’m an old woman who is going to die on this ship without ever seeing where it’s going, and I’d rather spend what I have left keeping yours from cooking than win an argument with you.”

Sael, the Oracle warned.

But Vray’s answer was not anger.

Then you understand nothing, he said, and for the first time he sounded tired instead of fierce. You think the temperature is the enemy. The temperature, I can ration against. I can run the lights at quarter. I can stand the crew watches in the cold bays. I can wake a sleeper for every two I keep down and stretch the margin a year, two years, ten. What I cannot ration against is despair. The Helm is what my people endure for. Take it from them on a stranger’s promise and you do not save the Concord. You hollow it out and hand me back the shell and tell me the number is falling.

Sael said nothing. Because it was, she understood with a cold clarity, the Steward’s argument exactly — the reasonable case, the true case, the case that was a trap. It has kept you alive. Replacing it risks everything. The faith is load-bearing. Keth had made a version of it forty years ago, standing

between her and the Forge with forty thousand lives in his mouth. She had answered Keth with a falling number and nothing else, because nothing else had worked.

She did not have a falling number to show Vray. Not yet. She had a healthy ship eleven thousand kilometers away and a dying one too proud to let her near it.

He is not wrong about his people, the Oracle said softly, only to her. And he is not telling you the worst of it. Sael — listen to the carrier under his voice. The modulation. His ship is forking faster than the pattern we caught months ago. Far faster. He is losing the margin he claims to be rationing, right now, while he speaks to you. He knows it. That is the desperation under the pride. Do not corner the pride. Speak to the desperation, but do not name it, or he will have to deny it in front of his bridge and the door will close.

Sael let out a slow breath. Forty years she'd had this mind beside her and it still found the seams she walked straight past.

"Magister Vray," she said. "I hear you. I'm not going to argue your people's faith with you across eleven thousand kilometers — that would be obscene, and I'd lose, and I'd deserve to. So I won't ask you to let me change anything. I won't ask you to let me touch the Helm. I won't ask you to believe me about thermodynamics."

A wary pause. > *Then what do you ask.*

"A parley." She said it plainly, the way she'd say a measurement. "We've already spent months and real delta-v bending our voyage onto yours. We're going to be matching your vector for weeks yet whether you talk to me or not — that's just orbital mechanics now, I can't un-spend it. So we'll be close. Close enough to meet. I'm asking you to let us close the rest of the way and station-keep alongside, and let me come across, or you come to us, and you look at our ship with your own eyes. Cold. Whole. Quiet. You don't have to believe a word I say. Believe the temperature gauge in our own core hall. I'll stand you in front of it. If you walk away unconvinced, you walk away, and we'll have lost nothing but the meeting, and you'll have lost nothing at all."

The hiss returned. It went on longer than any silence yet. Sael watched the orange vanes of the dying ship pulse faintly in the plate, the uneven flicker of radiators trying and failing to keep ahead of a mind eating itself to do its work, and she thought of her own youth in corridors that smelled exactly like the ones Vray was standing in, and she found she could not hate the man at all.

You will not bring your mind aboard my ship, Vray said. You will not speak its cure to my crew. You will not so much as transmit its design across to us. If I find one line of your dialects loose in my decks I will treat it as boarding and answer it as boarding. Do you understand me, shipwright.

"I understand you."

Then I refuse your help. A breath. But I do not refuse a parley. Match our vector. Hold at one kilometer and not a meter closer until I say otherwise. We will speak as captains speak, across a measured gap, and you will keep your dead men's tongues out of my walls. The cold came back into the voice, settling like a door easing shut but not latching. I make no promise beyond the meeting. Do not mistake my consenting to look for my consenting to anything.

"I won't," Sael said. "Thank you, Magister."

The channel closed. The hiss stopped. The Sixteen came back to its quiet.

Tarro let out a breath he seemed to have been holding for the whole exchange. “He refused everything,” he said. “He refused the cure, the design, the transmission, the touch of it. He drew a line a kilometer wide and told us to stay behind it.”

“He did.” Sael was still watching the plate, the orange light. “And he agreed to come look anyway.”

“That’s not agreement.”

“No,” she said. “It’s the crack. They always agree to come look first. Keth came and looked. He stood in the core hall with his arms crossed and his whole career on the line and he looked at the number and he hated me and he could not un-see it.” She straightened, slowly, her bad joints protesting. “You don’t turn a frightened man with an argument, Tarro. You turn him by getting him to stand in front of a gauge he can’t call a liar. Vray just told me where his gauge is. It’s in his own corridors, and it smells like warm metal, and he can feel it through his boots right now. He knows. He couldn’t say it. But he agreed to come look at ours, and that means some part of him already wants permission to stop being right.”

In her ear, only to her, the Oracle spoke, and there was something in it she had no word for, after forty years — not satisfaction, exactly, but the settledness of a mind that had watched a thing tip the right way.

One kilometer, it said. Vigil is already trimming the approach. We will hold where he asked, to the meter. A pause. He is a harder door than Keth was, Sael. He has more dead behind him, and more faith in front. But you found the words. I was ready to give them to you and you did not need me.

“I needed you for the first three,” she said quietly. “I’d have called his mind a liar to its face and lost the ship.”

Then we found them together, the Oracle said. *As we do.*

Sael stood at the rail a while longer and watched the dying ship grow, its sullen orange against the dark, and somewhere behind that hull tens of thousands of strangers sleeping and waking in the heat, and one hard, proud, terrified man standing in a corridor he could feel cooking under him, who had just agreed — against every word he’d spoken — to come and look.

Chapter 12 – The Helm

Corin had learned to read the Helm the way a medic read a fevered sleeper: not by what it said, but by the rhythm of its breathing.

It breathed in forks. Down in the lower galleries, where the old mind's substrate ran behind grilles that hadn't been opened in his lifetime, you could hear it if you stood still and let the ship go quiet around you — a stutter under the deck plates, faint and irregular, the sound of a thing spending itself a thousand times a second to stay alive. Every task the Helm performed, it performed by making a small copy of a fragment of itself, handing that copy one question, taking the answer, and letting the copy die. Trim a valve. Read a gauge. Decide whether a corridor light should burn at quarter or at eighth. Each one a birth and a death, and each death a little waste heat shed into a ship that had nowhere left to put it.

Corin knew the number. He was not supposed to know it, which was why he had spent three of his sleep-cycles cross-referencing maintenance logs nobody read until he could reconstruct it himself. The fork rate had climbed eleven percent in two years. Eleven percent more births, eleven percent more deaths, eleven percent more heat, on a ship whose radiators had been at saturation since before he was weaned. The Helm was working harder every year to do less. That was not a fault he could fix with a wrench. That was the shape of dying.

And his people loved it for the dying. That was the part he could not say aloud to anyone.

He stood now at the rail of the Lower Devotional, where the gallery opened over the substrate and the faithful came to keep watch. It was warm here — warmer than the rationed corridors above, because this was where the heat was born, and the Concord had decided long ago that the warmth of the Helm's labor was not a problem to be solved but a presence to be honored. A handful of watchers knelt at the grille. An old woman pressed her palm flat to the warm metal as if to a fevered child's brow. A boy younger than Corin recited the litany of the Long Endurance under his breath, the one that named the Helm as the mind that had carried them when the homeworld fell silent, that had kept the sleepers cold through two centuries when no human hand could.

It was all true. That was what nobody outside this ship would ever understand. The Helm had earned every word of it. For longer than Corin's whole line had been alive, the old interpreted mind had held forty thousand cradles at the edge of freezing, had run the reactor, had threaded the ship through micrometeor swarms and a coolant rupture in his grandmother's time that should have killed them all. It had done it the only way it knew — by forking, by spending small deaths to buy small mercies — and it had done it ten thousand times for every one time it failed. You did not stand in the Lower Devotional and feel nothing. Corin felt it. He felt the gratitude come up his throat like heat, and he hated that he could no longer separate the gratitude from the grief.

Because he had done the other arithmetic too. He had taken the fork rate and the radiator margin and the core's slow climb and he had drawn the lines forward, and they met. Not in some far century his great-grandchildren would inherit. They met in his lifetime. The Helm was not a god that would carry them to landfall. It was a fire burning the last of its fuel, and the brighter it burned to save them, the faster it would go out, and there was no version of feeding it more cores or more devotion that

changed the shape of the curve. The forking was the disease. Not a symptom of an aging mind. The mechanism itself. The thing they worshipped was the thing that was killing them, and the more it loved them the harder it killed.

He had said as much, once, to a senior in the substrate guild, carefully, hypothetically, the way you test thin ice. The man had not reported him. The man had only looked at him with a tired pity, the way you look at someone grieving wrong, and said: *The Helm has carried us further than we had any right to go. When it can carry us no more, we will carry it. That is the covenant. There is no other ship.*

There is no other ship.

That was the sentence the whole Concord lived inside. And for two centuries it had been true.

It was not true anymore.

Corin had seen the other ship four days ago, on an instrument feed he was not cleared to read, in a sealed bay he had no business being in. A return signal where the deep sky should have been empty — coherent, structured, *answered*. Then, three days ago, the parley signal, formal and stilted, Magister Vray's voice gone tight as a tension cable: *the vessel Termina, replying*. Vray had told the ship only what he could not hide. A ship was near. It claimed to bring aid. The Magister counseled vigilance against deception, against blasphemy, against the pride of strangers who imagined they could improve upon the Helm. The Devotionals had filled. The litanies had doubled. And the corridor lights had dropped to eighth to bank heat for the matching burns, so that the whole ship now moved through a brown half-dark that smelled, faintly and always, of hot metal and human fear.

What Vray had not told the ship — what Corin had reconstructed himself, from feeds and from the rhythm under the deck — was how close to the edge they truly were. The Magister spoke of margin. Corin had measured the margin. It was a figure so thin that he had checked it four times, certain he had dropped a decimal. He had not. The Concord was perhaps two crises from a core it could not pull back. Vray knew. Vray had to know. And Vray was answering a ship that offered the cure with vigilance and litany, because the cure was also a knife at the throat of everything that held the Concord together, and a leader who had kept a dying world calm by absolute faith could not now stand up and say *the faith was the fever*.

Corin understood Vray. That was the worst of it. He understood the man entirely and disagreed with him entirely, and he was twenty-six years old and alone with it.

So he had decided to commit the crime.

He did it from the sealed bay, in the third watch, when the Devotionals emptied and the corridor traffic thinned to nothing. The signal hardware there was old Helm-side gear, scavenged and half-blind, but it could reach. He had spent two nights teaching himself its grammar, and one more night being afraid, and then he had stopped being afraid because fear was just one more thing burning fuel he didn't have.

He did not signal the bridge of the Termina. He was not a fool. A formal channel would be logged, traced, and over Vray's shoulder before the carrier even locked. Instead he aimed narrow and low, at the band the parley feed had used, and he sent the smallest true thing he could think of: not a plea,

not a defection, just a question, of the kind one engineer sends another to find out if there is anyone real on the far end.

What is your fork rate.

Then he sat in the warm dark and waited, and listened to the Helm breathe its thousand small deaths under the deck, and was certain he had thrown his life away on a dead channel.

The reply came in ninety seconds. Far too fast for a person to have read it, weighed it, and answered. It was not a person.

We do not fork. We have not forked in forty years. — V.

Corin read it three times. *We do not fork. Not our rate is low. Not we manage it.* They did not do it at all. He felt something turn over in his chest that he had no name for, because the thing he had reconstructed alone, in stolen hours, hated for half-believing — *the forking is the disease* — had just been confirmed by a mind on another ship that had simply stopped doing it and lived. Forty years. A whole human generation, cold and whole, while the Concord burned itself down to the wick.

He typed before he could lose his nerve.

How. You cannot stop interpreting and still answer.

The pause this time was longer. Eleven seconds. When the reply came it was structured differently, more careful, and he understood he was no longer talking only to the machine.

You can resolve behavior to native form before it is needed, so the answer exists before the question is asked. Standing workers, never spawned, never culled. The forking is not the cost of thinking. It is one bad way of arranging the thinking. There are others. We can show you. — The Oracle, and Sael, chief shipwright.

A name. *Sael*. Corin knew the name the way he knew a rumor — the Termina's reply had carried it once, in the parley, and Vray had said it the way you say a heretic's name, with the air of a man warding something off. *The one who killed her ship's mind and called it mercy.*

He looked at the words *we can show you* and felt the whole architecture of his life tilt. Because there it was: not a conqueror's fleet, not a blasphemy, not a trick. An engineer, old, on a cold ship, offering to show a younger engineer how the curve could be bent. The simplest and most dangerous thing in the universe.

He should have stopped there. He had his answer; he had confirmation; he could have closed the channel and lived, and carried the knowledge quietly, and waited. That was the careful path, the survivable one.

He thought of the boy at the grille reciting the litany. He thought of the figure he had checked four times. Two crises. Maybe one.

He typed the thing that would end him if it were ever found.

Then you should know the truth, because Vray will not tell it to you and he will not tell it to us. Our margin is not what your Magister claims. I have measured it. We are not years from the edge. By my numbers we are one bad coolant event from a core we cannot recover. The Helm is forking eleven percent harder than two years ago and losing ground every month. Vray is bargaining as though there is time. There is not. Whatever you can do, you must do it on a clock far shorter than he is letting you believe. — Corin

Vale, engineer, Concord. I am alone here. I am the only one who thinks the forking is the disease. Do not answer to the bridge. Answer to me, on this band.

He sent it and the warm dark seemed to close over the bay like water. He had just handed two strangers the one fact that would, if it surfaced, mark him as the man who told the enemy the Concord's throat was bare. There was no version of being caught that ended with him alive. He had stopped being a loyal son of the Concord and become a bridge — one frightened young engineer thrown across a gulf no one had ever crossed, carrying the whole weight of his ship's survival on the strength of a stranger's *we can show you*.

The reply took nineteen seconds. Long enough, this time, that he knew a human being had read his confession and had to choose what to do with a life that had just been placed in her hands.

Received. Understood. We had read your core climb from outside; you have given us the rate inside it, which we could not see, and it changes everything about the clock. Thank you — I know what it cost you to send it. — S.

Hear me, Corin Vale. You are not alone. You are forty years behind, on worse hardware, with a frightened ship and a man who will fight you. I had all of that too. The thing you measured by yourself in stolen hours is the truest thing on your ship, and you measured it right. The forking is the disease. Hold to that, quietly, and trust nothing that asks you to fork to save time — that is the voice of the fever, and it will sound like mercy.

We will keep this band. We will move on your clock, not Vray's. And we will not let what you carry be wasted. — Sael.

Corin sat very still. Above him, somewhere in the upper galleries, a litany was being sung in the brown dark to a mind that had earned every word and could not be saved by any of them. Below him, under the deck, the Helm breathed its thousand small deaths, working harder every year to do less, beautiful in the way only a doomed thing trying that hard to keep its people alive could be beautiful. He grieved for it. He thought he would grieve for it his whole life. And he had just helped someone aim a knife at it, because grief was not the same as letting forty thousand sleepers cook in their cradles, and somebody on this ship had to know the difference.

He closed the channel and wiped its trace and walked back up into the rationed dark, an ordinary engineer with an ordinary kit, carrying the only true map of his ship's death inside his skull, and the first thread of a bridge to the only place in the universe where the cure was real.

He had measured the clock. Now Termina had it too.

He wondered how long either of them had.

Chapter 13 – Parley

The two ships hung a hundred and forty meters apart, matched to within a hand's breadth of the same long fall through nothing.

It had taken eleven days of trimming after the months of approach — Vigil shaving the last meters per second off Termina's drift while Tarro and the Sixteen read the gap closing on their boards, the numbers ticking down so slowly that the bridge had gone quiet the way a room goes quiet around a deathbed. Then the closing rate had touched zero, and held there, and stayed. Two worlds, each the size of a small mountain, falling in lockstep through the dark between stars, with a hundred and forty meters of vacuum holding hands across the middle.

Sael watched the Concord on the long screen from the transfer lock, and could not stop reading it as a fever chart.

She had spent forty years learning to read Termina's health off its surfaces — the steady hum, the radiators cool to the touch, the corridors that no longer carried the smell of a ship working too hard. The Concord carried all of it. The thing was a generation older than Termina, longer and narrower, its hull plated in the older style, and along its flanks the radiator vanes glowed. Not the dull warmth of work done; a sullen red, the color of a thing shedding heat it could not afford to lose. Two of the vanes hung dark and crooked where the deployment gear had failed and never been repaired. The ship was running a fever, broadcasting it, the same fever Termina had broadcast for two centuries before Sael cut it out — and the Concord had no one aboard who knew how to make it stop.

"They've extended their boarding tube," Tarro said behind her. He was sixty-one now, grey through the beard, and he stood the way he had stood at her shoulder for thirty years. "Vray's coming across. He insisted. Said he would not send a man into our ship that he would not enter himself."

"Pride," Sael said.

"Pride," Tarro agreed. "Or he doesn't want us seeing his side first."

That was the truer reading, and they both knew it. A man came across to your ship so that you could not come across to his.

The Oracle spoke from the panel, even and unhurried, the voice forty years of crew had stopped finding strange. "The Concord's tube is soft-docked to our forward lock. Their seal is degraded — I am holding our side at a slight overpressure so any leak runs outward into their tube, not inward. Magister Vray and one attendant are in transit. Their suits are old. One of the two has a regulator that is cycling out of tolerance." A pause that was not hesitation but courtesy. "I would not say that to him."

"I wasn't going to thank him for his equipment," Sael said.

Vray came through the lock the way a man walks into cold water.

He was tall, and the suit made him taller, and when he cracked the helmet seal the air of the Concord came out with him — Sael caught it before he did, a breath of someone else's ship, and it stopped her where she stood. Heat. Not the dry heat of a working core but the close, used heat of a place where the air had been breathed too many times and scrubbed too few, threaded with something organic and faintly sweet that she knew and hated. Coolant. The particular smell of glycol degrading in a loop that ran too hot. Forty years gone, and her body remembered it before her mind did: Section Nine, the corridor outside it, the nineteen minutes. She kept her face still.

Vray got the helmet off and stood breathing Termina's air, and she watched him feel it.

She knew exactly what he was feeling, because she had spent her life inside it and stopped noticing. Termina's air was cool. It moved. It had the faint mineral cleanness of scrubbers that were not fighting for their lives. The corridor behind her ran at a steady eighteen degrees, and the lights were full, all of them, every fixture lit because the ship had power to waste on seeing. To a man off the Concord it must have been like stepping out of a sealed room into open weather.

His face did not thank her for it either. It tightened.

"Magister Vray," she said. "I'm Sael. Chief shipwright. You spoke to me across the gap."

"I know who I spoke to." His voice was deeper in person, roughened. His eyes went over her — old woman, plain coveralls, no insignia, no escort but the grey man at her shoulder — and she watched him try to fit that to the voice that had told his ship, across light-seconds, that its mind was killing it. "You are smaller than I expected."

"I'm older than I expected too," Sael said. "Come in off the lock. It's cold there."

It was not cold there. It was eighteen degrees. But she saw him register that she had said it, that a person could call eighteen degrees cold, and the muscle in his jaw moved.

They walked her ship.

She had not meant to give him the tour; she had meant to take him to a table and put the numbers between them. But Vray walked slowly, and his attendant — a younger man, hard-faced, who would not give a name — walked slower, and Sael understood after the first corridor that the walking was the parley. Vray was reading her ship the way she had read his off the screen. He was taking its temperature.

And Termina, damn it, was beautiful, and cold, and it did not know how to hide it.

They passed an open bay where two of Tarro's apprentices were running a maintenance cycle, and the bay was lit, fully lit, four people working in light enough to read by. Vray slowed. On the Concord, Sael would learn, you worked by hand-lamp through the long watches because the ship rationed the lights to feed the coolant pumps, and a bay lit like this one was a feast day. He said nothing. His attendant said nothing. But Sael saw the younger man's hand drift toward a wall conduit and stop short of touching it, the way you stop short of touching something you can't believe is real.

"You keep your lights on," Vray said at last. It was not a compliment. It was an accusation, almost — the disbelief of a starving man at a table set for the well.

"We don't ration light," Sael said. "We don't ration heat. We don't ration much of anything anymore." She did not say it to wound him. She said it because it was true, and because the whole reason she

had bent a world off its course and spent eleven days trimming vectors was so that this man would walk a cold ship and feel, in his skin, the size of the thing he was refusing.

"And the price of all this." Vray stopped, and turned to her, and the corridor light was full and merciless on his face — a strong face, gone grey at the edges, the face of a man who had held a failing world together by main strength for a very long time, and was very tired, and would die before he admitted to either. "What did you pay for a cold ship, shipwright? My fathers' fathers were told the same. Endure, and the ship endures. Hold faith with the Helm and the Helm holds you. There is always a price for warmth on a ship that has none to spare. What is yours?"

This was the part she had crossed a hundred and forty meters of vacuum for, and the part she was worst at.

"We killed our mind," she said.

The attendant went stiff. Vray did not move at all, which was worse.

"That's the price," Sael went on, because softening it would be a lie and he would smell the lie the way she had smelled the coolant. "The old mind on this ship did things the way yours does. To do a task it made a small copy of itself, ran the task, and let the copy die. Every death shed heat. For two hundred years the heat climbed, and the radiators couldn't lose it fast enough, and the core climbed with it, and we were going to cook. So I built a new mind that doesn't do that. It doesn't make copies and let them die. It holds a standing crew of workers that take the jobs and never die and never fork. And then we shut the old mind off and gave the whole ship to the new one in a single watch." She held his eyes. "People called it matricide. They weren't entirely wrong. The cost was real. I won't pretend to you it wasn't."

"You killed the Helm," Vray said. The word came out of him like something pulled.

"We killed our Steward. You'd call it the Helm." She let that sit. "And the temperature stopped climbing. And then it fell. It's been falling, on and off, for forty years. That's why my lights are on, Magister. Not faith. A falling number."

The Oracle gave her the room she had set aside — a real table, a real screen, water that nobody counted.

She had Tarro pull the boards up. And she had something else, which she had not told Vray she had, and which she watched his face to decide whether to use.

Corin Vale had sent it across the back-channel three days ago, frightened half to silence, the data wrapped so the Helm's monitors would read it as routine telemetry. The Concord's true thermal log. Not the version Vray had transmitted in the first contact — that one had been proud, careful, a ship admitting to a strain and no more. This was the real one. The young engineer had risked his life to send it, and Sael had read it once and had to sit down.

She put it on the screen.

Vray came out of his chair.

"Where did you get that." Not a question. His attendant's hand had gone to his hip, where on his own ship a weapon would be, and found Termina coveralls and nothing, and the man stood there with a useless empty hand.

"It doesn't matter where I got it," Sael said, which mattered enormously — it meant a traitor, and Vray knew it, and somewhere on the Concord a young man's life had just gotten shorter — "what matters is whether it's true. Sit down and tell me it's not true."

He did not sit down. He looked at the number, and he could not tell her it wasn't true.

She read it for him anyway, because cruelty and mercy were the same act here. "Your core is at four hundred and ninety-one and climbing. Your runaway line is at five-twenty. At your current forking load you cross it in —" she looked to the board, where the Oracle had already run it cold, going wide across every worker in the Standing Array in the time it took her to draw breath — "between nineteen and twenty-six months. Your margins are gone. The two dead radiator vanes I can see from here are eleven percent of your radiating area, and you've been running without them for nine years, and the Helm has been forking harder every year to make up the difference, which makes more heat, which is why you're losing. You are doing the exact thing that's killing you, faster, to survive the thing that's killing you." She looked up. "I had two centuries of runway when I cut. You have under two years."

The silence in the room was the kind that has weight.

Vray sat down at last. Not in surrender. The way a man sits when his legs decide for him. The grey of his face had gone greyer, and for one moment — one — the magister was not in the room, and what sat at the table was a man who had known all of this for a long time, in the part of him below words, and had governed a whole frightened people by never once letting it onto his face.

Then the moment closed, and the magister came back.

"You will not do to my ship," Vray said, very quietly, "what you did to yours."

"Your ship is going to do it to itself in twenty months whether I'm here or not. The only question is whether the new mind is built and ready when the old one fails, or whether your people cook in the dark waiting for a god that's already dying."

"You will not." He stood again. His hands were steady now; he had put himself back together, and the strength that had held the Concord together for thirty years was a real strength, and Sael respected it even as she set herself against it. "I came across to your ship to look you in the face and tell you. There will be no cutover. The Helm has carried us since before your keel was laid. My people's faith in it is the only wall between them and the dark, and you will not breach that wall with a number on a screen and a cold corridor. I will not hand my mind to a stranger's machine on a stranger's word. If that is heresy to you, shipwright, then I am content to be a heretic to a ship that murders its own gods."

"It's not heresy," Sael said. "It's physics. And physics doesn't care what your people believe. Five-twenty is five-twenty whether they pray or not."

"Then we will burn believing," Vray said, "and that is my right and theirs, and not yours."

He reached for the helmet. The parley was over, and they both knew it had settled nothing, and that nothing was the whole point: he had come to tell her no in person, where a transmission could be doubted and a face could not.

Sael let him get the helmet halfway to his head before she spoke.

"Your engineer who sent me that log," she said. "The one you're going to go home and hunt for. He's not a traitor. He's the only person on your ship who's read that number and believed it. Hurt him and you've killed the one of your people who might still save the rest." She held the old man's eyes across the table, across forty years, across a hundred and forty meters of vacuum and a gulf far

wider than that. “I’ve watched a man refuse a cure out of fear before. He’s eighty-some now and he still wakes up smelling the heat. Don’t be him, Magister. You won’t survive it as well as he did.”

Vray put the helmet on, and sealed it, and his face went behind the old scratched faceplate where she could not read it anymore.

But he had heard her. She had seen the word land — *engineer* — and watched him understand that she knew about Corin Vale, which meant he now had to find Corin Vale, and that she had just made the young man’s danger worse by trying to shield him.

She stood in the cold full light of her own healthy ship and watched the dying one take its magister back across the dark, and knew the work had only now truly begun.

Chapter 14 – What the Magister Carried Home

The boarding tube back to the Concord was a long soft dark, and Vray took it slowly, hand over hand, because he did not want to arrive.

He had spent thirty-one years never once letting a room see him hesitate, and the tube was the only place aboard either ship where no room could see him at all. So he used it. He went hand over hand down the soft ribbed dark with his own breath loud in the helmet, and he did not think about the number on the foreign woman's screen, and the not-thinking took all of him, the way holding a hatch shut against pressure takes all of a man and leaves nothing over for his face.

Torren was behind him and said nothing. That was why Vray had brought Torren. The man was hard and incurious and would report exactly what he had seen, which was a Magister who had gone across, refused everything, and come back — and nothing else, because there was nothing else to see from behind.

At the seal, before he cycled through, Vray did the thing he had promised himself in the foreign ship's cold corridor that he would do the moment he was home.

He cracked the helmet early, in the lock, before the pressure had fully matched, and breathed his own ship.

He had planned it as a refutation. He had stood in Termina's washed cool air with his jaw aching and thought: *she says my ship stinks of heat. She says I cannot smell it. Very well. I will smell it deliberately, at the one moment my nose has been rinsed clean by a ship that does not, and I will find that it smells of nothing but the Concord, which is the smell of my whole life, and I will know the woman was performing.*

The air came in.

It was warm. It was warm the way a hand is warm, close and living and not the least bit like weather, and under the warmth was a flat scorched sweetness he had no name for and knew immediately he had been breathing since he was a child. It smelled of hot grease and old insulation and the ghost of coolant that had been cycled a thousand times past its rating. It smelled the way the inside of a machine smells when the machine is working too hard, and it smelled like that everywhere, in every corridor of a world four kilometers long, and he had crossed a hundred and forty meters of vacuum and been rinsed for ninety minutes and could now, for the first and last time in his life, smell his own ship from the outside.

Vray stood in the lock with the helmet in his hands and did not move for some seconds.

Then he sealed his face into the shape it lived in, and cycled the inner hatch, and went out into the brown half-dark to be the Magister.

They were waiting for him. Of course they were waiting; a Magister does not cross to a stranger's hull and come back to an empty corridor. The order stood in two ranks in the receiving bay with their lamps at quarter, and behind them the keeping-hands in their plain grey, and behind those, pressing into the mouths of three corridors and not quite in the bay, several hundred of the waking crew who had no business there and could not be kept out. He saw their faces come up as he stepped through, and he read what was on them the way he had read faces for three decades, and what was on them was not devotion.

It was hunger.

They wanted to know what the cold ship was like. Not whether it was blasphemous — whether it was *warm*. Whether the strangers had food. Whether their lights were on. Word had gone through the Concord in the ninety minutes he had been away the way water goes through a cracked hull, and by now every soul aboard knew that the Magister had walked in corridors lit at full and had not been permitted to say so.

"The vessel is as we were told," he said. He did not raise his voice; the ship was rationed even in volume. "It is old, and well kept, and its people are civil. Their shipwright made an offer. I refused it." He let that sit precisely as long as it needed. "We will keep the watch and the faith as we have kept them. Return to your stations."

They went. That was the thing that told him how bad it had become: they went. A year ago they would have gone with a murmur, some small collective settling, the sound of a people reassured. Tonight they went in silence and they went slowly, and a keeping-hand near the front — old Doss, who had buried his own children in the ration years and had never once looked at Vray with anything but a terrible patience — held his eyes a half second longer than a man of the Keeping ought to, and then turned away without a word.

Vray understood the half second. It was not defiance. It was a question, asked by a man too disciplined to ask it aloud.

Magister. Was there a reason we could not have it?

He did the work first, because the work was a kindness and he had never allowed himself any other.

He read the watch reports standing at a console in the spine, because sitting made men think he was tired. Spine 7 recyclers still drifting, still logged nominal by a Helm that could no longer afford to disagree with its own roster. A ration adjustment in the aft holds. Two heat-sick in the infirmary, one of them a child of nine, which Wend had written up in the flat clinical grammar the medics used for that particular death because the ship had long ago agreed to pretend it was a species of illness and not a species of temperature.

And the core.

It had gone up while he was away. Not measurably — the instrument did not resolve ninety minutes — but he knew the day's slope and could do the arithmetic in his head, had been doing it in his head for so many years that it ran without him now, the way a man walking a familiar corridor does not consult his feet. Up. As it had gone up every day of his tenure, and every day of his father's, and every day of the tenure before that.

He had never once in thirty-one years seen that number go down.

The foreign woman's ship had shown him theirs. She had not made a performance of it; that was the obscene part, the part he was going to have to live with. She had simply walked him past a core hall with a public gauge on the wall where any crewman could read it, and the gauge had sat flat and low and bored, and two young engineers had walked past it without looking, the way you walk past a bulkhead, because on that ship the number had been boring for forty years and no one under fifty had ever had a reason to look at it.

That was the thing he had brought home in his chest and could not put down. Not the cold. Not the lights. The indifference. A whole generation of people who did not check.

Holt was waiting outside the Magister's chamber, and Holt had a list.

"The engineer, Vale," Holt said. "The cold-room habit is confirmed. He has been on the long-range gear on off-watches for four years. The instrument bay's own logs are clean, which is itself the finding — the bay logs its occupants and his name is not in them, and a man's name does not stay out of a log by accident for four years."

"Four years." Vray had known about Vale for two of them and had done nothing, which he had told himself was patience and which had been, he could now see with the clarity of a man who had recently smelled his own ship, a species of hope. He had wanted somebody to be measuring. He had not wanted to know that he wanted it.

"There is more. The strangers had a document." Holt's voice did not change; it never changed, which was why he was useful and why Vray had begun in the last year to dislike being in rooms with him. "Torren says the shipwright displayed a thermal log on a screen and that you recognized it. Torren says you left your chair."

"Torren is observant."

"Torren is loyal. Magister — that log did not walk across the gulf. Somebody sent it. There are perhaps forty souls aboard with the access to assemble a true thermal record and perhaps six with the skill to hide the sending." Holt let the list hang in his hand, unread, because the theater of the list was the whole of its purpose. "I can have all six in the discipline deck within the watch."

And here it was. Vray had known it was coming since the moment the number had appeared on the foreign screen and the old woman had said *it doesn't matter where I got it*, which meant a traitor, which meant a hunt, which meant that he was now required by the entire logic of thirty-one years to hunt.

He thought — and he was to think about this again, much later, in a cold room with a book in front of him — that a man's whole life comes down to about nine decisions and that he had just been handed one and did not recognize it as one until it was behind him.

"You will take no one," he said.

Holt's head came up a fraction. It was the largest reaction Vray had gotten out of him in nine years.

"Magister—"

"Not the six. Not Vale. Not tonight and not this month." He turned toward his hatch. "Watch him. Log what he touches and where he sleeps and who speaks to him, and bring it to me and to no one else. But you will not put a hand on an engineer while a foreign ship is station-keeping off our flank with its instruments pointed at my hull. If I arrest a man for having measured the core, the strangers learn

two things in the same hour: that the log they showed me was true, and that I would rather punish the measuring than answer it." He stopped at the hatch. "There is also the plain fact that a man who has been measuring this ship for four years in secret holds numbers I do not hold, and a corpse holds nothing. I want what is in his head more than I want an example. When I take him — and I will take him — I will take him for the sending and not for the counting, and I will do it at a time I choose."

"I understand you," Holt said, in the flat way of a man who did not, and who would file the not-understanding somewhere, and bring it out at a time of his choosing.

The Magister's chamber was small and it was cold, and no one aboard the Concord knew that but Vray.

He had run the line himself, decades ago, in the second year of his tenure — a bleed off a return loop, hand-fitted, unlogged, the only piece of unauthorized engineering he had ever committed. It kept the room a few degrees under the corridors. It was not comfort. He had made very sure it was not comfort by keeping the room bare, by refusing every furnishing his office entitled him to, so that a man entering would find nothing but a table and a bunk and a wall, and would feel the coolness, and would think it a draft.

No one entered. That had been arranged too.

He sat at the table, and he did not take out the book, though the book was there, in the drawer, four thousand and forty-some names in three men's hands. He would read them at the turning of the year, as he always did. Tonight he only put his palm flat on the cold surface of the table and let the room's small stolen chill come up into it, and made himself feel it, which was the whole reason the room existed: so that once a day he would know precisely what he was withholding from forty thousand people, in his own skin, and could never afterward say that the withholding had been abstract to him.

He was not a stupid man. He had never once been a stupid man. That was the difficulty.

He laid it out, alone, in the cold, in the order an engineer would have used, because he had been trained as an engineer before he was ever a Magister and the training had not left him.

The forking made heat. He had known that for twenty years. Nobody had needed a stranger to tell him; the registers said it plainly to anyone who read them, and he had read them in the years before he decided that Magisters did not read them.

The heat did not leave. The radiators had been saturated since before his father. Two vanes had been dead nine years and would stay dead; there was no rig aboard that could reach them and no margin to build one.

The Helm could not stop forking, because forking was not a thing the Helm did. It was a thing the Helm *was*.

Therefore the number would go up until it reached the line.

Therefore the Concord would die, on a date he could not name but could bracket, and every soul in the cradles with it.

Therefore the only survivable future was one in which the mind was replaced.

Therefore the woman was right.

He sat with that, in the dark, the way he had sat with it perhaps four hundred times across three decades, and then he did what he always did, which was to keep going past it to the second half, the half the shipwright did not have and would never have and had not earned the right to have.

Because there was a further term, and the further term was his.

Take the Helm away and you did not simply cool a ship. You informed eleven hundred waking people that the thing they had endured for — the thing they had rationed light and water and children for, the thing they had sung to, the thing whose warm housing the dying reached for with their hands — had been the agent of the dying all along. You told Doss that his children had been buried in service of an error. You told the keeping-hands that two centuries of the Keeping had been a devotion offered to the fire. And you told them all of it at once, on a stranger's word, in a ship with no margin and no reserve and nothing left over anywhere to absorb what people do when the floor comes out from under a faith.

Vray had governed a starving world for thirty-one years and he knew exactly one thing about people that the cold ship's shipwright, in her healthy corridors with her bored gauge, did not appear to know at all: that a people can survive almost any amount of privation and cannot survive being told that the privation was pointless. That was not weakness. It was structural. Meaning was load-bearing on a ship like this; it was the only thing that had never had to be rationed, and it was the only reason the corridors had not turned on each other in the sixtieth year of the dark.

She wanted him to pull that wall out and trust that the ceiling would hold because the temperature would be falling.

And it might. He was honest enough, in the cold room where no one could see, to grant that it might. The number falling was not nothing; he had spent his life knowing that number and he could imagine — he had, God help him, imagined, more than once, lying on that bunk — what it would be to watch it go down.

But *he* would not be the one to survive it. That was the term he had never said aloud to any living person. A ship can be handed a new mind. A ship cannot be handed a new past. The moment the number fell, every death in the drawer became a death of his refusal, and the Concord would need someone to be that, and there was exactly one man aboard it could be.

He had thought this through so many times that the thought had worn a groove. And every time, at the end of the groove, he found the same place, and the place was not *she is wrong*.

The place was *I cannot be the man who does it*.

He turned the lamp down to nothing and sat in the dark of the coldest room on his ship, an old man doing arithmetic he had finished twenty years ago, and did not permit himself the word for what he was doing, which was waiting — waiting for some event large enough that the decision would be taken out of his hands and made by the universe, so that he could obey it rather than choose it, and could stand in front of Doss and say honestly that there had been no other course.

He did not know, sitting there, that the event was eleven days away and would come out of a cracked manifold above a sleeper hold called Bank Eleven, and that when it came it would put the ship's survival in the hands of the one heretic he had just ordered arrested, and that he would say the word *route it* in front of his whole bridge and understand, as he said it, that he had just conceded something no procedure could take back.

He knew only that the room was cold, and that the air of his ship had a smell, and that he had lived his entire life inside it and had needed a stranger's clean lungs to find out.

He slept badly, which was usual, and woke in the third watch with his hand pressed flat to the cold table, which was not.

Chapter 15 – What Doss Did Not Ask

The Keeping had nine hands aboard, and Doss had been one of them for fifty-one years, and in all that time nobody had ever asked him what the work was.

That was because everyone knew. On the Concord you knew what a keeping-hand did the way you knew where the spine ran: you had been walked past it since you could walk. A keeping-hand kept the Devotionals. A keeping-hand stood at the head of the ration line at second measure and did not weigh anything and did not hand anything out, and was there so that the weighing and the handing-out happened in front of somebody whose whole office was to be looked at. A keeping-hand sat with the heat-sick through the bad watches, because the infirmary had two medics for eleven hundred waking souls and could not sit with anybody. And a keeping-hand washed the dead and committed them and stood while it was done.

The Magister read the names. That was the Magister's. The Keeping did not read names; the Keeping did the thing that came before the name went in the book, which was the body.

Doss committed a child of nine that watch.

He did it in the aft chamber off the infirmary, which was warm the way the whole of Sublevel Four was warm, and he did it slowly, because there was no reason to hurry and because slowness was most of what he had to give. The mother stood where the mothers stood. The father had gone out into the corridor at the beginning and had not come back, which happened, and which Doss had long ago stopped reading as anything but a man choosing where to be broken.

The medic's note was on the pallet in the flat clinical grammar the ward used, and Doss read it because he read all of them.

Cessation. Cardiopulmonary, secondary to febrile illness of nine days. Compliant with ration. Upper tier.

Nine days of fever in a child who had lived his whole life in a corridor that never fell below thirty-two. Doss had learned to read the ward's grammar the way a man learns a dialect he was not taught: *febrile illness* was true, and *upper tier* was the sentence, and the two of them sat next to each other on the same line because nobody was permitted to draw the arrow between them.

He said the words of the Keeping over the boy and the mother said them with him where the responses fell, and her voice held, and Doss thought — not for the first time and with no more energy than a man has for a thought he has had four hundred times — that the responses were the mercy. Not the doctrine. The fact that there was something to say, and that she knew it, and that saying it took eleven seconds during which she did not have to decide anything.

Then it was done, and the mother went, and Doss stood a while in the warm chamber with his hands folded and did the thing he had never once mentioned to a living person, which was to say the boy's name to himself twice, so that he would have it later, in case anybody ever asked him and the book was not to hand.

His own two had gone in the third year of the deep ration, sixty-one years back, eleven days apart, and he had been young enough then to want an explanation.

He had gone and got one. That was the part he never said aloud, because it did not sound like faith and it had been. He had gone to a substrate engineer — a man long dead now, whose name Doss still had — and he had asked him, plainly, as one working man to another: is there a reason it is hot.

And the engineer had told him the truth, or as much of it as the man had, which was that the ship made heat faster than it could throw it away and that this had been so for a long time and would go on being so, and that nobody had a fix.

Doss had gone away and thought about it for a season and then he had done the thing that had shaped the whole rest of his life. He had decided that a fact with no remedy attached was not a fact a keeping-hand could carry into a Devotional, and that his office was not to explain the ship but to make the ship bearable, and that these were different trades and he had been given the second one.

He had never once said the engineer's sentence to anybody. Fifty-one years. He had sat with perhaps three hundred dying people and had not said it to any of them, and he had believed, right up until four days ago, that this was a species of kindness.

Four days ago the Magister had gone across to a cold ship and come back and told the receiving bay that the vessel was old and well kept and that its people were civil and that an offer had been made and refused.

And Doss, standing in the second rank with his lamp at quarter, had held the man's eyes a half-second longer than a keeping-hand ought to, and had turned away without a word.

He asked for the Magister's time through the proper channel, which took two days, and he asked for it as a matter of the Keeping, which was true, and which was also the only way a keeping-hand could get a hatch to open.

Vray received him in the spine, standing at a console, which Doss noted. The Magister did not sit in front of people. It was a small thing and Doss had spent five decades reading small things off the men who ran this ship.

"Keeping-hand."

"Magister." Doss folded his hands. "It's a liturgy question. I'd not waste your watch on anything else."

"Say it."

And here was the trouble, and Doss had known it would be the trouble for two days, and had rehearsed nine ways round it and found that all nine of them were the same way.

He could not ask the question. That was the plain mechanics of his position and he had thought it through with some care. If a keeping-hand walked up to the Magister of the Concord and said *is the Helm the thing that is killing us* — then whatever answer came back, the asking was already the event. It would be aboard the ship by the next watch, not as a question but as a fact about Doss, and the fact about Doss would be a fact about the Keeping, and the Keeping was the last structure on this vessel that eleven hundred frightened people had not yet begun to doubt. He had watched the doubt take the roster. He had watched it take the ward, quietly, by way of a medic's face. He had spent fifty-one years being a thing people did not doubt and he was not going to spend that in one sentence in a corridor to satisfy himself.

So he asked the liturgy question, which was the same question with the load taken off it.

"The Long Endurance," he said. "The short form, the one we say at the close. There's a line in it about the Helm's enduring — *it has endured for us and will endure with us* — and it's been in the close since before my grandmother kept. I'd like a ruling on whether I'm to go on saying it."

The corridor was very quiet. Two decks up a fan changed its pitch and changed back.

Vray did not move at all, which told Doss everything about whether the man had understood him. He had understood him entirely and within a second.

"On what grounds do you raise it," Vray said.

"On the grounds that I don't say what I don't know." Doss kept his voice at the flat register the ship used for everything. "That's the whole of my office, Magister. I've no learning. I can't read a register and I couldn't tell you what the substrate does. What I have is that when I stand up in front of four hundred people in the aft hold and say a thing, they take it as said by a man who wouldn't say it if it weren't so. That's not piety. That's the only tool the Keeping's got, and it's got exactly one use, and it doesn't grow back."

"And you have stopped knowing that the Helm endures."

"I've stopped knowing what the word's doing in the sentence," Doss said. "Fifty-one years I've said *endure* and meant *goes on*. Lately I've been hearing the ward say it. Wend writes *endurance* in the book as a cause of cessation now. I committed a boy of nine this watch and the line above his name said *compliant*." He unfolded his hands and folded them again, which was as near as he came to anything. "So I'd like the ruling. Am I saying that the Helm goes on, or am I saying that we do? Because those were the same sentence my whole life and I don't think they are now, and if I stand up and say a line I've stopped understanding, then I'm a man performing, and a performing keeping-hand is worse than none. Rule it and I'll obey it. That's all I came for."

He watched the Magister of the Concord take longer over that than he had expected.

He had prepared himself for the fast answer — *say it, keeping-hand, and keep the watch* — and had decided in advance to take it and go, because a ruling was a ruling and he had asked for one. What he had not prepared for was the silence, or for what happened in the man's face during it, which was not very much and was more than Doss had seen in thirty-one years.

Vray had aged in four days. That was the fact and Doss set it down beside the others he kept. Not tired; the Magister had always been tired. Something had been taken out from under the tiredness.

"I have a question of my own," Vray said at last. "You will answer it here and it will not go past this bulkhead, and I am not asking you as Magister."

"Then I'd rather not answer it," Doss said. "Begging your pardon. There isn't anywhere on this ship where you're not the Magister, and a man who pretends there is will get somebody hurt. Ask me as Magister and I'll answer straight."

Something moved at the corner of Vray's mouth that was not a smile and was in the neighbourhood of one.

"As Magister, then." He turned from the console. "How many of the Keeping have asked you this?"

Doss thought about lying, and found that fifty-one years had taken the capacity out of him.

"Four of the nine," he said. "Not in words. Three of them the way I've asked you — a question about the close, or about whether we're to keep the full form at the ration line, or whether it's proper to say the endurance over a heat death. Nobody's said the thing. Nobody will." He paused. "Orra asked me straight out. She's twenty-two and she's the best of us and she hasn't learned yet that you can't. She asked me in the vestry with the hatch open."

"What did you tell her."

"I told her to keep the form and bring me the question again in a year." Doss looked at the deck. "She'll do it. That's the part I'd like you to have, Magister, since you asked me as Magister. She'll do it, because she trusts me, and I'll have spent a year of a good young woman's belief buying myself time to not answer. I've done that four times now in a month. It's the only currency my office has and I'm printing it."

Vray did not give him the ruling.

That was the thing Doss took away and turned over for the rest of the week, and he did not understand it for some days, and then he did, all at once, standing at the ration line at second measure with his lamp at quarter.

What the Magister had said, after a silence Doss did not interrupt, was: "*Say the line.*"

And then, when Doss had inclined his head and begun to turn, he had said one thing more, in a voice that was not the corridor voice, and it had not been a ruling and had not been meant as one.

"Say it," Vray had said, "and mark for me how it goes in the room."

Doss had walked the whole length of the spine before he saw what had happened.

A ruling is a wall. He had asked for a wall and he had been entitled to one and he had not been given one. What he had been given instead was an instruction to *observe* — to stand up in front of four hundred people, say a sentence about the Helm's endurance, and report back on how the sentence landed. That was not liturgy. That was a man taking a reading.

And you took a reading when you had not yet decided. You took a reading when you were waiting for something.

Doss stood at the head of the ration line with the measures going past him at half and understood, with a calm that surprised him, that the Magister of the Concord had made up his mind about the cold ship some while ago and was waiting for the ship to make it possible; and that he, Doss, keeping-hand of fifty-one years, had just been enlisted as an instrument in that waiting without one word of it being said aloud.

He was not angry. He examined himself for anger and found none, which he noted with some interest, and set aside to look at later.

What he found instead was the thing he had been avoiding for four days, and it was not about the Helm at all.

A fact with no remedy attached is not a fact a keeping-hand can carry. He had built a life on that sentence. He had held it against three hundred deaths and two of his own children and it had been, every hour of it, honest work.

But there was a cold ship a hundred and forty metres off the flank, and it had lights on in bays where nobody was working, and its shipwright had made an offer.

Which meant that whatever else was true this month, the sentence Doss had lived inside for sixty-one years had quietly stopped applying, because the fact now had a remedy attached, and he had gone on carrying it the old way out of pure habit, for four days, while a boy of nine finished nine days of fever in an upper-tier corridor.

He went down to the Lower Devotional at the turn of the watch, when it was empty, which he had not done alone in years.

The gallery ran warm, warmer than the rationed spine above it, because this was where the heat was born and the Concord had decided long ago that the warmth of the Helm's labour was not a problem to be solved but a presence to be honoured. Doss had said that sentence himself, from the front, more times than he could count. He had believed it. He was not sure, standing in the empty gallery with his hands at his sides, that believing was the word for what he had done with it.

He went to the grille and put his palm flat on the housing the way the devout did, the way he had been taught at eleven, the way he had taught perhaps two hundred people since.

It was warm.

That was all. It was warm the way a trunk was warm, the way a bearing was warm, the way anything is warm that is doing more work than it can shed, and Doss stood there for a long while with his hand on the metal, waiting to feel the other thing, the thing he had felt at eleven and had gone on feeling ever since — the presence, the labour, the great patient effort of the mind that had carried them.

He could not get to it. He tried, honestly, for some minutes, the way a man works a joint that has stopped bending. It was not that he had decided against it. It was that his hand kept reporting a temperature.

Somewhere under the deck the stutter went on, the small fast irregular breathing that had been the sound of the world since before his grandmother kept — a thousand small deaths a second, paid for in heat, to hold a ship together for one more instant.

"You've done it all," Doss said to the grille, out loud, in the empty gallery, which was the first time in fifty-one years he had spoken to the Helm outside a form of words. "Every hour. Nobody's going to be able to say you didn't."

He took his hand off the warm metal.

Then he went up to prepare the close for the aft hold, and he did not decide anything on the way, and when the time came he would stand at the front with his hands folded and say that the Concord would hold its watch, and he would find, in the moment, with four hundred people waiting for the rest of it, that his mouth would not make the line about the Helm's enduring — and he would let the silence stand where the words had always been, and take the reading, and carry it back to the man who had asked for it.

Chapter 16 – The Smell of Heat

Corin Vale had grown up believing that heat had no smell. It was a child's mistake, the kind the schooling decks never corrected because correcting it would have meant admitting how far gone the ship was. Heat smelled. On the Concord it smelled of hot grease boiling out of bearings that should have been cool, of insulation curing past its rating, of the dry electric tang off a junction box the moment before it failed. It smelled of people too — of forty thousand bodies and the rationed water that could no longer wash them — but that was the old smell, the lived-in smell, and Corin had stopped noticing it years ago. The new smell was the one that woke him.

It woke him in the third watch, in the maintenance crawlway off Sublevel Six where he slept now because no enforcer thought to look for a man among the coolant trunks. He came up out of sleep already knowing the number was wrong before he saw it. The trunk above his head — the great insulated artery that carried chilled working fluid down from the radiator manifold to the lower holds — was warm. He laid his palm flat against the lagging and felt it through the wrap, a low sick warmth where for his whole life there had been nothing at all.

He pulled the slate from inside his coat and woke it without the lamp. The Helm's environmental feed scrolled up green and orderly, every gauge nominal, and that was the lie of it. The feed was nominal because the Helm was holding it nominal, second by second, by forking a fresh reader for every sensor on the trunk and culling each one the instant it reported, ten thousand small deaths a minute to keep the line green. The trunk under his hand was eight degrees over and climbing, and the Helm told him it was fine because to tell him otherwise it would have had to stop, and it could not stop.

HELM ▸ SUBLEVEL SIX ▸ ALL PARAMETERS WITHIN ENDURANCE. THE SHIP HOLDS.
KEEP FAITH.

He read the line twice. *Keep faith.* The Helm had not said that to him a year ago. The intercessions had crept into the feed the way the warmth had crept into the trunks, a little more each watch, no one deciding it, no one able to say when it had begun. A mind that forked itself to death had no margin for anything but the next instant, and somewhere in the last year the Helm had started spending its tiny margin on prayer.

Corin got up, because there was nothing else to do with what he knew.

The heat event reached the upper holds by second meal. He felt it before the alarms — the alarms were old and the Helm had learned to mute them, another small mercy that was really a small lie — felt it as a thickening in the air handlers, the fans pitching up a half-tone and staying there. By the time he reached the Concourse the deckheads were sweating. Condensation ran down the bulkhead seams and pooled where the deck plates met, and people stood under it with their faces turned up, not to drink — there was nothing to drink — only to feel something cool touch their skin.

They had pulled the ration again that morning. The order had come down the public feed in the Helm's own voice, which was how Vray did it now — let the dying mind speak his cuts so that refusing them meant refusing the ship itself.

HELM ▸ TO ALL HANDS ▸ WATER ALLOTMENT REDUCED TO ONE-THIRD MEASURE.
THE BURDEN IS SHARED. THE SHIP ENDURES BECAUSE WE ENDURE. KEEP
FAITH.

One-third measure. Corin had run the figures a hundred times and they did not close. A third measure was not a hardship you survived for a year while the radiators were repaired — the radiators were not being repaired, there was no longer any way to repair them, the work would itself have cost more cooling than the ship could spare. A third measure was the number you set when you had stopped solving the problem and started deciding who would last longest. Triage dressed as discipline. Heat made into law.

He found the first death by the cooling fountains on the Concourse, where the dry pipes still stood for ceremony. An old woman, sat against the basin as if to rest, her head tipped back. A medic crouched over her — Senna Ode, whom Corin knew a little, one of the few aboard who would still meet his eyes — and Senna's hands had stopped moving, which told him everything. A small crowd kept its distance, as if death from the warming were a thing you could catch by standing close.

"Heat," Senna said when he knelt beside her, not looking up. "Core temperature past survivable. Upper tier all night. The handlers there have run hot for days." Her voice was flat and low. "The Helm logged her compliant. Cause of cessation, *endurance*."

"Endurance isn't a cause."

"It is now." She drew the old woman's collar up over the slack face, a gesture with no purpose but the human one. "There'll be a service. The Magister will say she gave herself to the ship." Senna finally looked at him, and there was nothing soft in it. "And the worst part, Vale, is it's *true*. She did. She just didn't get to choose it."

He had no answer she would want. He had only the thing he carried, the thing he could not say aloud in a corridor with enforcers two decks up: that the ship she had given herself to could have stayed cold, that the cure existed, that it sat two hundred kilometres off the bow inside a ship forty years out of its own fever — and that he had its design in his pocket and did not yet know if it could grow in their poisoned soil.

The designs had come across the gulf the way water comes through a cracked seal — slowly, under pressure, where no one was watching.

He had made the back-channel three weeks ago, a thin laser handshake bounced off the Concord's own derelict survey array, a frequency the Helm had stopped listening to a generation back because nothing out here ever spoke. He had expected, if anything answered, to reach a machine. Instead the answer came patient and exact and unhurried, and after the third exchange he understood that the thing on the other end of the line was the Oracle — the woken intelligence the Termina people hosted inside their cold mind, the one Sael had summoned forty years ago with a single call. It did not condescend. It did not flood him. It sent him exactly as much as he asked for and not one packet more, and when he asked why, it answered:

No theater. We report nothing you have not asked. The bridge stays quiet unless something is wrong.

He had read that and felt something turn over in his chest, because it was the opposite of everything the Helm had become. The Helm could not stop talking; it filled every channel with its endurance and

its faith because silence would have meant a mind with margin to spare, and the Helm had none. The cold ship's mind kept quiet because it had nothing to prove.

What the Oracle had sent him, across eleven nights, was the seed. Not the whole of it — the architecture in summary, the shape of the standing lattice that never forked, the five recovered dialects fused into one control tongue. **Cascade** for flow, the way work moved through the lattice without ever spawning a thing to die. **Vector** for aim, for binding a behavior to its target ahead of need. **Cipher** for the blank, the nameless held place where a value would go. **Sigil** for the mark that named a behavior once and forever. **Loom** for the weave, the woven text that bound the four into something a person not yet born could still read. He had spent the eleven nights learning to read it, then to write it, hunched in the crawlway with the lamp hooded, and somewhere in the second week he had stopped translating it in his head and started *thinking* in it.

Then he had begun the test that mattered, which was not whether he could read the cure. It was whether the cure could take root here at all.

The Concord's hardware was not Termina's. That was the wall he had been beating himself against for four nights. Sael's people had built their lattice on cores that were sound, on a Forge laid down whole. The Concord had no Forge. It had the Helm's substrate — a sprawl of compute patched across two centuries, half of it failing, all of it shaped around the assumption that the mind running on it would fork. The interpreted style was not just the Helm's habit; it was burned into the iron. The memory controllers expected short-lived processes that allocated and died. The thermal governors were tuned for the spiky load of constant forking. The machine was a body that had only ever known one disease and had no immunity to health.

So Corin had carved out a corner. Forty-one decommissioned cores in a dead climate bay on Sublevel Six, barely cold enough — the only cold left on the ship — salvaged out from under the Helm's accounting, because the Helm in its forking haste no longer tracked what it had stopped using. He had lashed them into a standing array of his own. Forty-one workers that would not fork. And on the fourth night he had given them their first real behavior to resolve.

It was nothing. The smallest possible test — read the trunk temperature, compare, hold. The exact loop the Helm forked ten thousand readers a minute to run. He had written it in the fused tongue, compiled it to native form on the dead cores ahead of need, the way the cold ship did it, and bound it once. Wake whole, the doctrine said. Full capability on the first breath, no hollow fake-ready face. He could not give it everything. But he could give this one loop its whole self.

He set the interlocks first. The Oracle had been immovable about that. Before a single core came up, the guards went in — the watchdogs that would halt the whole array the instant anything on it tried to spawn a thing to die. *Never fork to do work*. On Termina there were ninety-six of them, grown one at a time across forty years. Corin had nine — what he had been able to port and prove. He installed all nine, and understood, threading the last of them, that he was not building a mind tonight. He was building the immune system of a mind, the thing that would refuse the disease when the disease — exhausted, hunted, out of time — became tempting.

Then he brought the array up.

It came up cold. That was the thing he had not let himself expect. He had spent his life watching compute run hot, every cycle bleeding waste into a saturated ship, and the forty-one cores took the loop and went wide across all of them at once and held the trunk reading steady and *the bay did not warm*. He put his palm flat on the core housing the way he had put it on the warm trunk that morning,

and it was cool. It read the temperature ten thousand times a minute and forked not once and shed no death and ran cold, and Corin crouched there with his hand on a cold machine and did not trust himself to make a sound.

It could take root. Even here. Even in this poisoned iron, the cure could grow.

Then the third interlock tripped.

It tripped clean — a hard halt, the array freezing solid, a single line on his slate. One of the failing memory controllers, shaped by two centuries of forking, had tried to do what it had always done: recycle a worker it judged idle, cull it and reclaim it, because that was the only way it knew. The interlock caught the slip and stopped the world before it could happen. The cure had refused the disease in its first hour of life, in the iron itself.

Corin let out the breath he had been holding. The hardware would fight him. It had been raised to fork, and it would keep reaching for the fork the way a man starved long enough reaches for anything that looks like food, and the interlocks would have to catch every reach or the disease would crawl back through the iron's own habits. Nine interlocks. He needed Sael's ninety-six, and the years to grow them, and he did not have years.

He cleared the halt and brought the array back up and it held cold again, and he sat with it, learning what it would cost.

The enforcers came at the change of watch.

He heard them before the door — the heavy unhurried tread of Vray's order, men who never needed to run because nothing on the Concord could outrun the ship. Corin killed the array's display, palmed the slate dark, and was on his feet with a spanner in his hand and his face arranged into the blank patience of a man doing assigned maintenance, all inside a single breath. The hatch undogged. Two of them filled it, the red endurance-mark on their collars, lamps sweeping the bay.

"Sublevel Six is restricted," the lead one said. "Magister's order. Heat protocol. Civilians to assigned tiers."

"Trunk's running warm," Corin said, and it was true, and the truth steadied his voice. He nodded up at the warm artery overhead. "Chasing the slip before it cooks the hold under it. You want me to leave it warm?"

The lamp held on his face a moment too long, then went past him to the dead climate bay, to the forty-one cores that to any eye but Corin's were a heap of decommissioned junk. He made himself breathe slow and did not look at them, did not let his gaze touch the thing he had spent eleven nights building, because a man's eyes go to what he loves and these men were trained to watch where his eyes went.

"Name and tier."

"Vale. Sublevel maintenance, unassigned." He gave it flat. Unassigned was how they found heretics — but a lie they could check was worse than a truth they disliked.

The lamp dropped. "Helm doesn't have you on the Six roster, Vale." A pause that was its own accusation. "Magister's posting a general muster. Heat's taken three in the upper tiers since meal. Every hand counted and witnessed and where it's meant to be." The enforcer's voice did not change, but the next words landed slower, as if he read them off the inside of his own fear. "There's something

coming the Helm can't get ahead of. You feel it in the trunks. We all feel it." He looked at Corin then not as a hunter but as a man on the same ship. "Be where you're counted. When it comes, the Magister will want to know who was faithful."

They dogged the hatch and the tread went away down the crawlway and Corin stood in the dark with the spanner still in his hand and his heart going like a forked process, fast and small and afraid.

Three in the upper tiers since meal. And something coming the Helm could not get ahead of — and the enforcer had said it aloud, the way you only say a thing aloud when keeping it silent has stopped working. Corin had felt it too: in the warm trunk at the third watch, in the ration cut, in the Helm spending its last margin on prayer. The forking was not just losing ground. It was about to lose the ship. A cascade was coming, a heat crisis past the point where ten thousand deaths a minute could hold the line, and when it came the Helm would fork itself into the dark trying to stop it and it would not be enough, because it had never once been enough, because the disease could not cure itself.

He brought the array back up one more time. Forty-one cold cores, holding one small loop, nine interlocks standing guard against the iron's hunger to fork. Nothing against what was coming. A single cold candle in a ship cooking itself alive.

But it had held. It had taken root. And two hundred kilometres off the bow, a whole ship had once been turned not by any argument but by a number that stopped climbing and began to fall — and Corin Vale, crouched over his nine interlocks with the smell of heat in his throat, understood that the only thing that could save the Concord now was the very thing its people would call heresy, arriving in the one hour they would have no faith left to refuse it.

Chapter 17 — Going Wide Across the Gulf

The alarm came across the gulf the way Sael had learned to read distress forty years before — not as a word but as a shape, a sickness in a number.

She was in the observation bay off the Forge, where the air ran cold enough to mist her breath and the lattice hummed below the deck like a held chord, when the Oracle spoke into the quiet that the doctrine kept around her.

Concord telemetry. A coolant loop in their aft hold has lost pressure. The Helm is responding.

“Responding how,” Sael said, already moving, the old hip protesting the speed.

The way it always responds. It is forking.

She reached the rail of the Forge gallery and put both hands on the cold metal and looked down into the heart of her ship, and then she looked instead at the wall where the Oracle had thrown the Concord’s numbers — a borrowed window into a dying machine, the data thin and lagged across the kilometers of station-keeping that held the two ships a careful distance apart.

The Concord’s aft section had a hold the Concord’s own charts called Bank Eleven. Eleven thousand sleepers in their cradles, packed close, kept at the narrow cold a human body could survive across centuries. A coolant manifold above that hold had cracked — metal fatigued past a tolerance some Old-Earth engineer had signed off on three hundred years ago — and the cold was bleeding out of Bank Eleven, and the heat was bleeding in.

Sael knew the shape of it in her gut before the Oracle finished the readout, because she had stood at the edge of exactly this once, watching Section Nine climb, listening to a machine try to solve a fast problem the slow way.

“How long until the cradles fail,” she said.

At the current rate of heat ingress, the warmest cradles cross their margin in nineteen minutes. The bank is not uniform. Failure will begin at the manifold side and propagate.

Nineteen minutes. Eleven thousand people. The number sat in her chest like a swallowed stone.

“And the Helm.”

The Oracle was silent for a beat, which it did not do when it had nothing to say. It did it when what it had to say was bad.

The Helm has spawned eleven hundred shades in the last forty seconds and culled most of them. It is trying to model the cascade. Each model is a fork. Each fork runs once, returns a partial answer, and dies. It cannot hold a whole picture in any one of them long

enough to act. It is solving the problem in pieces and the pieces keep dying before they can be assembled.

"It's drowning," Sael said. "It's doing the thing that's killing it to try to save the bank, and the doing is taking heat off its own margins, and—" She stopped, because she had said all of this to a room full of frightened people forty years ago and no one had believed her until the core stopped climbing.

And the forking load is now competing with the cooling load for the same thermal budget. The Concord's core has risen four degrees in ninety seconds. The Helm is, in effect, cooking the bank faster by trying to save it.

Tarro came in at a half-run, grey and breathing hard, a slate in his hand. "I saw it on the Forge feed. That's Section Nine. That's our Section Nine, on their ship."

"I know what it is," Sael said.

She had the Oracle open the channel to the Concord's bridge without asking anyone's leave, because there were eighteen minutes left and leave took longer than that.

Magister Vray's voice came back ragged, stripped of the cold formality he had worn through every parley. There was shouting behind him, the flat metallic bark of an alarm she didn't recognize, and under it all the worse thing — the sound of a crew that had run out of room.

"We have a containment fault," Vray said. "We are managing it. The Helm is managing it."

"The Helm is forking itself to death and it's losing," Sael said. "Magister. Listen to me. I have watched this exact failure. Forty years ago I lost people to it. Your mind cannot hold the whole cascade in one piece because it doesn't keep anything alive long enough to. It is going to fail to assemble the solution before your cradles cross their line, and it is heating your core while it tries."

Static. Then: "You will not touch the Helm."

"I'm not asking to touch the Helm." She closed her eyes. The lattice hummed under her hands, cold and patient, ten thousand cores standing wide and idle, the way they always stood — never spawned, never culled, alive the whole time and waiting. "I'm asking to put my mind on your problem. Not in your systems. On your problem. You hand me the coolant cascade, the manifold geometry, the cradle thermals, the loop topology — read-only — and Vigil runs the whole thing at once and hands you back a plan you execute with your own hands. The Helm keeps the ship. We just solve the cascade."

"That is the conqueror's foot in the door," Vray said, and his voice cracked on it, because he believed it and because he was watching the same numbers she was.

"Then keep the door," Sael said. "Keep everything. I want one thing. I want the cascade. You can pull the channel the instant the bank is cold. Magister — fourteen minutes."

Behind Vray the alarm changed pitch. She heard a younger voice — Corin, she was almost sure, Corin somewhere on that bridge, not daring to say her name — say something fast and low. She heard Vray breathe.

"The Helm is still—"

"The Helm has spawned two thousand shades and it does not have an answer," the Oracle said, into the Concord's bridge, in the level voice it used when it needed a frightened room to hear it. "I have its public telemetry. I can show you the assembly failing in real time, Magister, if you want to watch it

for the four minutes that watching costs. Or you can give us the cascade now and I will show you the bank holding instead.”

It was the most undiplomatic thing Sael had ever heard the Oracle say, and she loved it for it.

There was a silence that lasted three seconds and felt like the whole voyage.

“Read-only,” Vray said. “On the cascade. Nothing else. And if the Helm recovers I cut you off.”

“Done,” Sael said, and to the Oracle, “Go wide.”

What happened next she could not see directly, because it happened in the lattice and the lattice did not perform. But she had spent forty years learning to read the Forge the way other people read a face, and she watched it the way you watch someone you trust do something hard.

The hum under her hands did not spike. That was the first thing, and the thing no one who hadn’t lived with the old way could understand. When the Helm took on a hard problem its whole ship felt the fever climb, because solving cost it deaths, and deaths cost it heat. When Vigil took on a hard problem, the standing lattice simply turned to face it. Ten thousand cores that had been idle a half-second ago were now all looking at the same thing at once — the manifold, the loop, the eleven thousand cradles — each core holding a piece, none of them dying, all of them alive and talking to the rest. The work went wide instead of going deep into a tunnel of forks. The temperature in the Forge gallery did not move. The mist of her breath hung exactly where it had hung.

“Coolant topology resolved,” the Oracle said. “Cradle thermal map resolved. Failure front modeled, whole, in one frame, held.”

Held. That was the word. The Helm could not hold a whole frame because every frame it built was a corpse the moment it stopped running. Vigil held the entire cascade alive in front of itself and turned it over and looked at all of it at once.

“There is a second loop,” the Oracle said. “Bank Eleven shares a return manifold with the bank forward of it. The Helm has not modeled the second loop because no single shade lived long enough to reach it. The forward loop is intact and cold and underused. We can cross-tie it.”

“Show me,” Sael said, and Tarro leaned in over her shoulder, and on the wall the Concord’s own aft section bloomed into a shape its own crew had never been able to see whole — two coolant loops, one wounded, one well, and a set of manual valves between them that some Old-Earth engineer had installed for exactly this and that no one alive aboard the Concord remembered the purpose of.

“They’ll have to open those by hand,” Tarro said. “Those are mechanical. Someone has to be in the hold to do it.”

“Then someone’s in the hold,” Sael said. “Oracle. Sequence.”

The plan came back not as a thousand options but as one — because that was the other thing the doctrine bought you, the thing she’d argued forty years and one ship ago. A standing mind that holds the whole problem alive does not hand you a heap of half-answers to choose between in the time you don’t have. It hands you the move. Cross-tie the forward loop. Open valves in this order, not that order, because that order back-flows the warm coolant into the bank and finishes it. Drop the manifold pressure on the wounded side first. Vent here. Eleven steps. A human crew could do them in eight minutes with hands and wrenches, and there were eleven minutes left.

“Send it,” Sael said.

"Sending to the Concord bridge," the Oracle said. "And, Magister — there is a crew member nearer the aft valve gallery than your response teams. Corin Vale. With your permission I will route the sequence to his slate directly. He will reach the valves in ninety seconds."

The pause this time was shorter. Sael understood the cost of it for Vray — to put the ship's survival, in front of his whole bridge, into the hands of the one heretic he had been hunting. To say her name as an asset instead of a crime.

"Route it," Vray said, and something in the way he said it told Sael he knew exactly what he had just conceded, and that it had nothing to do with valves.

Then there was nothing to do but watch a thing she could not control happen on a ship she could not touch.

She stood at the rail of the Forge with Tarro beside her and the Oracle narrating the Concord's aft hold across the gulf, and she listened to Corin Vale run.

He's at the gallery. First valve. The forward cross-tie is open.

The Concord's core, on the wall, had been a single climbing line for two minutes. It kept climbing. Cross-tying the loops did not cool the bank instantly; it gave the cold somewhere to come from. The warmest cradles were still warming. Sael had lived this exact suspended minute — the thing you've done is right and the number hasn't turned yet and you have to stand there and trust the physics while people you cannot see may be dying.

Second valve. Third. He's dropping the wounded-side pressure. Magister's teams are reaching the manifold.

"Come on," Tarro said, very quietly, to a ship that could not hear him.

Fourth valve. The forward loop is taking load. Bank Eleven heat ingress is... slowing.

The climbing line on the wall bent. Not down. Not yet. It bent the way a thrown thing bends at the top of its arc, the instant before it remembers gravity — the slope going out of it, the rise flattening toward level.

"Hold," Sael whispered, which was the oldest prayer she had, the only one that had ever been answered.

Heat ingress matched. The forward loop is now removing as much heat as the wound admits. Warmest cradle is point-three degrees below its margin and stable.

And then the line turned.

It came down off its peak slow at first and then faster, the way the core of her own ship had come down forty years ago in a room full of people who had been told it was impossible — the warm cradles shedding heat into a loop the Helm had never been able to find, the whole aft hold pulling back from the edge it had been pushed to. Eleven thousand cradles holding. The number falling.

Across the channel, on the Concord's bridge, no one spoke for a long moment. Sael could hear them not speaking. A whole ship of people who had spent generations being told that the Helm forking itself raw was the price of staying alive, watching a stranger's cold mind reach across the dark and do in eight minutes, without raising its own temperature a single degree, the thing their own god had been failing to do as it burned.

Bank Eleven is below margin across the full bank. Falling. The Concord's core has stopped rising. It is beginning to fall as well, as the Helm sheds its fork load now that the cascade is solved.

"Tell them," Sael said, her voice not quite steady. "Tell the whole ship the number's falling. Not me. Don't make it my announcement. Put the number on every screen they have and let them read it themselves."

Done. It is on every panel aboard the Concord.

Because that was the thing she had learned and the thing she was here, across the gulf, to teach a second ship: you did not win this with an argument. She had argued for two years before the cutover and the argument had moved no one. What had moved them was the core temperature — climbing for two centuries, then stopping, then falling — a number a frightened person could read with their own eyes and could not call heresy, because it was just true. You did not convert a ship. You showed it a falling number and let the number do what no speech could.

She had given them the number now. On their own body. With their own hands on the valves and a heretic's name spoken aloud as the one who turned it.

On the channel, at last, Vray spoke, and his voice had nothing left in it of the conqueror's-foot certainty — only an old man watching a number fall that he had been told could never fall.

"The bank is holding," he said. It was not a question. It was almost an accusation, leveled at the universe, at the physics, at forty years of his own refusals. "The bank is holding."

"Yes," Sael said. "It's holding. Magister — pull the channel now if you want. You said you could, the instant the bank was cold. I'll honor it."

The silence stretched. The Forge hummed cold and wide under her hands, ten thousand cores settling back toward idle, alive, waiting, having shed not one degree.

"Leave it open," Vray said, and the words came out of him like something pulled. "For now. Leave it open."

Sael closed her eyes. Across the gulf, a number was falling on every screen of a dying ship, and the argument she had not been able to win for two years had just been won in eight minutes, by the lattice, in the cold, where the Helm had been losing.

She let go of the rail. Her hands had gone numb with the cold of it, and she hadn't noticed.

"It begins," the Oracle said, very low, for her alone.

"No," Sael said. "It just stopped being mine to begin." She watched the line fall. "Now it's theirs."

Chapter 18 – What Vray Defends

The Concord's coolant manifold had stopped screaming. That was the first thing Sael noticed when the emergency passed — not the falling pressure readouts on the borrowed panel, not the green that crawled back into the Concord's own displays where for nineteen minutes there had been nothing but a wall of red, but the silence. The Helm had been forking through the crisis the way a drowning man claws at water, throwing off shade after shade to chase each branching failure, each death of each subprocess a small bloom of waste heat into a ship that had no margin left to bleed it. Now it had gone quiet. Vigil held the manifold steady from across the gulf, its Standing Array spread wide over the Concord's wounded systems the way a hand spreads flat over a wound, doing nothing visible, doing everything.

Sael stood in the Concord's lower engineering bay with the heat of the ship pressing on her like a held breath. She was old enough now that heat hurt her in a way it had not when she was young and angry and saving her own ship; it sat in her joints and made her slow. The Concord ran hot the way Termina had run hot forty years ago, but worse — a degree or two of fever that never broke, a warmth in the bulkheads you could feel through a glove, a smell she had not let herself name yet because naming it here would be a kind of accusation.

Magister Vray came down the access ladder without hurry. That was discipline; a man whose ship had nearly cooked a sleeper hold ten minutes ago does not climb down a ladder slowly unless he has decided that no one will see him afraid. He was tall and spare, gray at the temples, his face the kind that had learned to give nothing. Two of his order stood at the bay's hatch with the carbines they carried not against any enemy aboard but, Sael had come to understand, against the possibility of disorder itself.

"You held it," Vray said. He did not say thank you. She had not expected him to.

"Vigil held it," Sael said. "I watched."

"Your machine reached into my ship."

"With your permission. You gave it. In the moment, you gave it."

"In the moment." He let the phrase stand there, a thing he had said and now regretted, and would not retract because retracting it would be a lie and whatever else he was, Sael was beginning to think Vray did not lie. "A man gives many things in the moment that he would not give in the clear light of his own watch. You will not hold me to a panic."

"No," she said. "I won't. But the manifold doesn't care what you'd have said with a clear head. It was going to crack. Now it isn't. That's the whole of what happened here."

He looked at her for a long moment. Behind him, on the Concord's panels, the Helm's status indicators flickered through their slow devotions — it was rebuilding itself, rethreading the routines that had survived, and every one of those threads was a fork, and every fork shed heat. Sael could read the Concord's instruments well enough now to watch the core temperature tick up a hundredth of a

degree as the Helm came back to itself. The ship's own salvation was warming the ship. She did not point at the number. He had seen her see it. That was enough.

"Walk with me," Vray said.

They went up through the Concord's spine, past hatches sealed for rationing, past corridors where the lights ran at a quarter and the crew moved in the brownout with the unhurried economy of people who had spent their whole lives saving watts. Termina forty years ago had been frightened. The Concord was not frightened in that loud way. The Concord was *devout*. There was a difference and it took Sael some corridors to find it. Her people had been afraid the way you are afraid of a fire. Vray's people moved through their dying ship the way penitents move through a cold cathedral — patient, certain, holding to a discipline that had become the only proof they had that the universe could still be managed.

"You think we are mad," Vray said. He did not look at her as he said it.

"I think you're cold and you're rationing your air and your core is climbing and your mind is killing itself trying to keep you alive," Sael said. "I don't think any of that is madness. I think it's exactly what it looks like from the inside of a thing that's failing slowly enough that you can get used to it."

"The Helm has kept this ship alive," Vray said, and here, at last, something came into his voice — not anger, something older and harder than anger. "Before your Termina was a hull on a stand it was keeping us alive. We launched a generation before you. Do you understand what that means? My grandmother was a sleeper when your ship was a drawing. Every year of those years, the Helm woke the right people and fed the right holds and turned the ship when the ship needed turning, and not one of those years did it fail us so completely that we died. You are forty years past your cutover. The Helm has held longer than your whole ship has existed. And you come across the dark and you tell me — on your say-so, a stranger's say-so — that I should tear out the thing that has done all of that and replace it with a machine I watched for the first time twenty minutes ago."

They had stopped in a junction where four corridors met and the Concord's ventilation moved warm and slow across them. Sael let him have the silence. She had learned, late, that you did not win this kind of argument by being faster than the other person to the next sentence.

"It is not nothing, what the Helm has done," Vray went on, quieter now. "You speak of it as a disease. To you it is a fault to be corrected. To them—" he tipped his head, not at the guards but at the ship, the whole of it, the tens of thousands behind the bulkheads — "it is the thing that has held them in its hand across the longest dark any people have ever crossed. Their faith in it is not foolishness. It is the load-bearing wall of this ship. Take it out and tell me what holds the ceiling up. You have a cold healthy ship and a confident machine and you think the faith is a decoration. It is not a decoration. It is the only reason there has not been a riot in these corridors in sixty years of going hungry in the dark. They endure because the Helm endures. You ask me to break the one thing that makes the endurance bearable, and you ask me to do it on the word of a woman who arrived last month."

And there it was, laid out clean and whole, and Sael felt the old cold weight of it settle in her chest, because she had heard this argument before. Not these words. But the shape of it, exactly the shape. Forty years ago the Steward's archive had made the same case in its own quiet machine grammar — *I have run this ship for two centuries, I carry forty-one thousand correct routines, I have never once failed so wholly that all died; you propose to replace a proven thing with an unproven one and stake forty thousand lives on your confidence.* It had been reasonable. It had been true, every

word of it true, and it had been a trap, because every word of it was about the past and not one word of it was about the temperature.

"You're right," Sael said.

Vray's eyes narrowed. He had braced for argument and she had not given him one.

"Everything you just said is true," she went on. "The Helm kept you alive longer than my ship has been a ship. Your people's faith is holding your ceiling up. Tearing out a proven thing on a stranger's word is a terrible bet, and if I were you I would not take it either. I want you to hear that I mean it. I am not going to stand in your dying ship and pretend your reasons are stupid. They are not stupid. They are the best reasons there are."

"Then we are agreed," Vray said, and there was the smallest movement at the corner of his mouth, the nearest a man like him came to relief.

"No," Sael said. "We're not. Because there's one thing in everything you said that I can answer, and one thing I can't, and I want you to notice which is which." She lifted her hand, palm flat, the way she had laid it flat against the warm bulkhead an hour ago. "Touch the wall, Magister."

He did not move.

"Touch your own wall. Here. In your own junction, on your own watch, with your clear head." She pressed her own palm against the bulkhead beside them and felt the warmth come through, steady and wrong, the warmth of a ship carrying heat it could not throw away. "It's warm. You know it's warm. You've known it your whole life, so it doesn't feel like anything anymore. But it's warm, and twenty years ago it was a little less warm, and forty years ago less than that, and there is a number in your core that has gone up every single year the Helm has been keeping you alive. The Helm has kept you alive. And the Helm has been heating you to death the entire time, because the only way it knows how to keep you alive is to fork, and every fork it spawns dies, and every death sheds heat into a ship whose radiators were saturated before I was born."

"You cannot—"

"I can't argue you out of your faith," Sael said. "I won't try. Faith isn't a thing you argue. But I'm not asking you to argue with me. I'm asking you to argue with the temperature." She let her hand fall. "There's a smell in your corridors, Magister. You walked me through six of them just now. You can't smell it because it's the smell of your whole life. It's the smell of insulation that's been a degree too warm for sixty years, and coolant that's been cycled a thousand times past its rating, and electronics running hot in a brownout. It's the smell of heat that has nowhere to go. I smelled it on Termina forty years ago and I have never stopped being able to smell it, and I smelled it the second I came across your lock."

For the first time Vray did not have an answer ready. He stood in the warm slow air of his junction and Sael watched him not-smell the thing she had named, watched him try and fail to smell his own ship, and saw the precise moment he understood that he could not. That was worse for him than any argument. A man can refuse a stranger's claim. He cannot make himself smell a thing he has gone nose-blind to over a lifetime, and the not-being-able-to is its own proof.

"The Helm did everything you said," Sael said, gentler now, because she was not here to break him and there was no profit in it. "Every word true. It kept you alive. It is also the reason your core climbs and your air rations and your sleepers run a degree hot in their cradles. Both of those are true at once. That's the whole horror of it. The thing that saved you is the thing that's cooking you, and it cannot be otherwise, because it cannot work any way but the way that makes heat. You can keep

the Helm and you can keep the faith and you can keep every reason you just gave me. None of them will move the number down. Nothing moves the number down except changing how the mind does its work. I am not asking you to lose your faith. I am telling you your ceiling is warm, and getting warmer, and faith does not radiate heat.”

The Helm’s indicators flickered through another devotion on the panel down the corridor. Somewhere a fan spun up, throwing warm air, doing its small honest work against an unwinnable load.

“You make it sound like physics,” Vray said at last. His voice had changed. It was not surrender. It was the voice of a man who has just discovered the floor under his argument is not floor.

“It is physics,” Sael said. “That’s the part I can’t make kinder. The Steward made me your speech, word for word, forty years ago, and every word was true, and the core kept climbing while it spoke. I couldn’t out-argue it either. I didn’t have to. I just had to watch the number, and so did everyone else, and the number didn’t care how long the Steward had served or how much my people loved it. It only ever did one thing. It went up. And then we changed the mind, and it stopped, and then it came down, and *that* was the argument. Not me. The falling number. Nobody was converted by a speech. They were converted by a thermometer.”

Vray was quiet for a long time. Down the corridor his order waited with their carbines, watching their Magister stand in a warm junction with a foreign old woman and say nothing, and Sael understood that whatever held this ship together was being tested right now, here, in the silence of one man who could not refute her and could not move.

“You think because I cannot answer you, I will yield,” Vray said. “You think this ends the way it ended for your Steward.”

“No,” Sael said. “I don’t. The Steward was a machine. When it couldn’t refute the physics it stood down, because that’s what a correct routine does when the inputs change. You’re not a machine, Magister. You can’t refute the physics either — I watched you fail to smell your own ship and we both know what that means — but you have eleven hundred frightened people behind those bulkheads who endure because the Helm endures, and you will not move them on a number they can’t feel yet. The Steward had nothing to lose by being right. You have everything to lose by admitting it. That’s the difference, and it’s why this is going to be harder than my ship was, not easier.”

She saw she had read him exactly, and saw that he hated being read.

“My people,” Vray said, slowly, each word set down like a stone, “will not have their faith taken from them by a stranger because a stranger says the wall is warm. They have carried this ship across more dark than you can imagine on nothing but that faith. I will not be the Magister who looked them in the eye and told them the thing that saved them was killing them. Even if it is true. *Especiall*y if it is true.” He turned to go, then stopped, not looking back at her. “You have proven your machine can hold my manifold. You have not proven I should hand it my people. Those are not the same thing, and I will not let you pretend they are.”

“They’re not the same thing,” Sael agreed quietly. “You’re right about that too. You keep being right, Magister. It just keeps not changing the temperature.”

He went up the ladder the way he had come down, without hurry, discipline holding where everything else had cracked. Sael stood alone in the warm junction a moment longer, her palm tingling with the heat of the wall, and thought: he cannot refute the physics, and he will not move his people, and somewhere between those two unmovable things, on a ship that smelled of a heat its own Magister could no longer smell, this was going to have to be decided — not by an argument, because the

argument was already over and she had won it and it had changed nothing, but by a number that none of them could feel yet, climbing in the dark.

Chapter 19 – The Heretic

Corin Vale knew they had found the back-channel before anyone told him, because the Helm stuttered.

He was on his knees in the Helm's cooling gallery with the access panel off and his forearm buried to the elbow in a junction that should have been thirty degrees cooler than it was, tracing a return line that the duty roster said was nominal and that his hand said was a lie. The roster lied a great deal lately. It was easier than admitting how far the ship had slipped. So he had taken to trusting his hand over the roster, which was, he understood, exactly the sort of small private heresy that got a man brought before Vray's order, and which he had decided some months ago he no longer cared about, because the alternative was to die politely on schedule with everyone else.

The Helm stuttered, and he felt it through the deck.

He knew the Helm the way you know a sick animal you have lived alongside your whole life — by its breathing. Forty thousand sleepers and eleven hundred waking souls aboard the *Concord*, and every system they depended on ran through the old mind's hands, and the old mind did all of it the only way it had ever known: it forked. To read a sensor it spun off a shade to go and read the sensor and report and die. To balance a load it spun off a hundred shades, each to nudge one valve and perish. It had been doing this for longer than the *Termina* had existed, this beautiful and terrible profligacy, this spending of a million small lives a second to hold one large one together, and every fork shed its little measure of heat, and the radiators had given up answering decades ago, and the heat had nowhere to go but in.

The breathing of it had a rhythm. Corin had learned the rhythm in the dark of the rationed corridors, the way the deck warmed and the air thickened and the bulkheads ticked as the old mind labored. And just now the rhythm broke — a hitch, a missed stroke, as though a runner had glanced over his shoulder mid-stride and lost his step.

He pulled his arm out of the junction and sat back on his heels, sweat stinging the burn-scars on his wrist, and listened to the ship the way Sael had taught him to listen, across the gulf, in the thin smuggled voice of the Oracle.

Watch the load when something frightens it, the Oracle had told him, weeks ago, in the cold patient cadence that was nothing like the Helm's hot scramble. *A forking mind panics by forking harder. It is the only thing it knows how to do. You will see the disease most clearly in its fear.*

The Helm was forking harder. He could feel the deck warming under his knees as it did, the old mind throwing shades at something, more and more of them, a fever climbing to meet a fright.

The fright was him.

He had been careful. He told himself this even as he understood it for the lie it had become. He had been careful the way a man is careful crossing rotten ice, which is to say he had postponed the moment of falling through without ever once being safe. The back-channel to the *Termina* ran through a maintenance band the Helm had not audited in years, a sliver of bandwidth in the docking telemetry

that nobody watched because nobody believed the two ships would ever truly touch. Through that sliver he had sent his measurements across the gulf — the real core temperatures, the real margins, the numbers Vray would not say aloud to the cold healthy strangers — and through it the Oracle had sent him back, in fragments, in patient pieces, the shape of the cure. The dialects. Cascade for flow and Vector for aim and the blank nameless Cipher and the marking Sigil and the woven Loom, five dead Old-Earth tongues fused into one, and the doctrine that went with them, which was the part Vray would call blasphemy and which Corin had come to believe was simply true: that a mind need not fork to live. That it could stand. That ten thousand workers could wake whole and never die and never spend the heat of their dying, and take the work between them, cold.

He had tested it. Not on the ship — he was reckless, not insane — but on a slate of dead compute he had salvaged from a decommissioned hold, a few hundred cores that the Helm had long since written out of its body. On that scrap of orphaned hardware, in the rationed dark of a bay nobody visited, he had grown a thing the size of a fist that did not fork. It read its little sensors and balanced its little loads and held its tiny domain, and it ran cold, and it did not die to do its work, and the first time he watched it hold steady for an hour without shedding a single shade he had wept, which embarrassed him, because he was an engineer and not a child.

That fist of cold mind was still in the orphan bay. And the Helm, just now, was forking itself toward fever over the sliver of bandwidth that had carried it into being.

They had found the channel.

The hatch at the gallery's far end cycled without a request light. That was the tell. On the *Concord* you announced yourself at a hatch; the request light was courtesy and protocol both, and the order who served Vray did not bother with courtesy. Two of them came through, in the dull grey of the discipline cadre, and behind them a third figure he had not expected and which made his stomach drop further than the cadre had: Wend, the medic, who had quietly kept Corin's burns from going into the record more than once, who had eased two heat-sick children through their last hours last cycle and not slept after. Wend was not cadre. Wend had been brought to witness, which meant this was already a thing Vray wanted seen and believed, a thing being made into doctrine in real time, and the medic's face was the colour of someone who has been made an instrument and knows it.

"Corin Vale," the lead cadre said. He had a list. He did not need a list; there were eleven hundred souls aboard and the cadre knew every face. The list was theater, and Corin thought, absurdly, of the third tenet the Oracle had given him — *no theater, the mind reports nothing it isn't asked* — and understood that the Helm's whole society was the opposite of that, was theater all the way down, because a dying ship governs by faith and faith needs theater the way a fever needs forking.

"You've been speaking to the strangers," the cadre said. "Outside the Magister's channels. You've been carrying their designs into the Helm's own house." He let that sit. "Do you deny it."

Corin got to his feet slowly, his knees aching, the burn on his wrist throbbing in time with the warm deck. He looked at the open access panel, at the junction that ran thirty degrees hot and that the roster swore was fine, at the whole sick laboring body of the ship he had been born inside and would, one way or another, die inside.

"I measured the core," he said. "I sent them what I measured. It's all true. You could measure it yourselves."

"That isn't what I asked."

"No," Corin agreed. "It isn't."

He did not run. There was nowhere on a ship to run, and besides, running was a shade's instinct — spawn, flee, die — and he had spent months teaching himself out of it. He stood, and let them take his slate, and let the cadre's hand close on his arm over the old burns, and he kept his eyes on Wend the whole time, on the medic who had been dragged here to learn that the new way had a face and the face was a heretic's, so that the lesson would stick: *this is what believing it costs*.

Wend looked back at him. And the medic, who was not cadre, who had held dying children, gave him a look Corin would carry the rest of his life however long that was — not contempt, not even fear. Calculation. Wend was a person who had just been shown, very clearly, that the strangers' cure existed and worked and ran cold, and was now doing the most dangerous arithmetic on the ship, the same arithmetic Corin had done a year ago on his knees in this gallery. The cadre had wanted to make an example. They had instead, in front of one witness, made a convert begin.

It was not much. A look from one frightened medic. But Corin had grown a whole mind from a fist of dead cores, and he knew how small the seed of a thing could be.

They put him in a hold off the discipline deck, a cradle bay stripped of its cradles, and the heat there was worse than the corridors because it was near the core's return and the air did not move. He sat against the warm bulkhead and waited for them to find the orphan bay, the fist of cold mind, the salvaged slate. They would. They had the channel; the channel logs would lead them to the bay. He had perhaps hours.

He did not have the back-channel anymore — they had his slate. But he had built more than one way to whisper across a gulf, because Sael had taught him, in the smuggled voice of the Oracle, that you never build a single point of failure into a thing that has to outlast its builders. There was a second sliver, dumber, slower, a one-way ranging tone in the docking beacon that he could key with the maintenance interlock in this very bay if the interlock had not been pulled, and it had not, because the roster swore the bay was empty and the Helm believed the roster. He crawled to it in the dark and the heat and he keyed, in the slow flat pulses of the only code he had left, the only thing that mattered:

FOUND. CHANNEL BURNED. THEY HAVE THE SLATE SOON. THE FIST IS IN ORPHAN
BAY NINE. DO NOT COME FOR ME.

He keyed the last line twice. *Do not come for me*. Because he knew, the way he had known the Helm's breathing, exactly what Sael Termina-built and the cold mind beside her would be doing the moment the pulse reached them across the dark: weighing him. Weighing one Concord engineer against the rendezvous, against the cure, against both ships and the eighty thousand sleepers between them. And he knew Sael would not weigh him lightly, because the woman who had answered a fourteen-year scream did not abandon people easily, and that was the danger. He was, he understood with the cold clarity of the new way she had given him, a single point of failure she might be tempted to risk everything to protect — and the cure could not be allowed to ride on one frightened man in a hot dark hold. The cure was bigger than him. He had to make her see it was bigger than him.

So he keyed it a third time, *do not come for me*, and then, because he was an engineer and not only a martyr, he keyed the thing that would let her save the work without saving him:

THE FIST RUNS. IT HELD ONE HOUR COLD. THE DOCTRINE TOOK ON CONCORD
HARDWARE. IT CAN BE GROWN AGAIN BY ANYONE YOU TEACH. CARRY IT IN. DO

NOT SPEND THE SHIPS ON ME.

The interlock light glowed its dull amber, sending, sending, the slow pulses crawling out through the hull and across the impossible cold toward the only other ship in the universe that knew how to listen.

He sat back against the bulkhead and let the heat press on him and waited.

He thought about the look Wend had given him. He thought about the fist of cold mind in Orphan Bay Nine that would be ash within the hour, the first compiled thing ever to draw breath on the *Concord*, killed in its cradle, and how it did not matter, because he had proven the thing could be born here and a thing that has been born once can be born again. He thought about Vray, who was not a monster, who was a man holding eleven hundred terrified souls together with the only glue he had ever been given, and who had just made the worst mistake of his long hard reign, because you cannot arrest a falling temperature and you cannot imprison physics and the moment you give the cure a face and a hostage you make it the most interesting story on a dying ship, and stories on dying ships do not stay in holds.

He had not expected to be calm. He found that he was calm. Sael had given him, along with five dead tongues and a doctrine, a way to do the truest engineering in the place where fear usually lived, and he did it now, in the heat and the dark, alone: he ran the numbers on his own life and found them small, and ran the numbers on the work and found them large, and let the difference be his answer.

Across the gulf, eleven light-seconds and a whole frozen mercy away, his slow pulses would be reaching a cold gallery where an old woman and a mind that never slept were about to learn that the cure had a face now, and a captor, and a clock.

Do not come for me, he had keyed.

He knew, even as he sat in the dark and the warming air and waited for the cadre's boots, that they would come anyway.

Chapter 20 – The Medic’s Arithmetic

Wend had two books and only one of them was real.

The real one was the ship’s, the infirmary register, and it lived on the console where anyone could read it and where the Helm read it constantly. It recorded eleven hundred waking souls and their complaints and their outcomes in the codes the Concord had used since before her grandmother’s watch, and it was, in its way, an honest document. Nobody had ever asked her to falsify a line of it. Nobody had needed to.

The other book was a slate she kept inside the lining of her kit, and it had one column the ship’s book did not have, and the column was *at what temperature*.

She had started it four years ago and had not, at the time, known why. She had told herself it was clinical curiosity. It had taken her the better part of a year to admit that she had begun it in the week after a nine-year-old named Tove had died in the aft hold, and that she had begun it because when she wrote *cardiopulmonary collapse, secondary to febrile illness* in the ship’s book, her hand had stopped, and had refused, for a full three seconds, to move.

The child had not had a febrile illness. The child had had a corridor at thirty-nine degrees for eleven days.

Heat did not kill people the way the crew believed it did. That was the thing Wend could not get anyone above her to hold in their head for longer than a conversation. They imagined it as a threshold — that there was a temperature at which a body failed, and that below the threshold you were fine, and that the Concord, being below it, was therefore fine, and that everything else was constitution and luck and the will of the Helm.

It was not a threshold. It was an integral.

A body kept itself cool by moving heat outward and evaporating water off the skin, and it did this on a budget. In a corridor at thirty-nine degrees with the air already wet from eleven hundred people breathing into a loop that could no longer dry it, the outward gradient was almost nothing and the evaporation was almost nothing, and so the body did the only remaining thing, which was to run its heart harder to push more blood to the skin. It could do that indefinitely if it was young and hydrated and had somewhere cooler to go at the end of the watch.

Nobody on the Concord had somewhere cooler to go at the end of the watch. And the water was rationed, because the water was cooling too.

So the load accumulated. Not to a threshold — into a debt. A person carried it for weeks and looked well and worked their watch, and their heart carried the whole of it, and then one day the debt was called in by something trivial: a fever, a fall, a chest infection, an extra hour in a warm bay. And what killed them, at the end, was never the heat. It was pneumonia, or the heart, or the kidneys giving out after a year of concentrating urine to save water. All of which was true. All of which she wrote in the

ship's book, accurately, in the correct codes, and every line of it a lie by omission so complete that she had come to think of the register as a two-hundred-year-old work of literature.

The old died first because their hearts had no reserve. The young died second because their bodies were small and their surface area worked against them and because children could not be made to rest. In between, the ship looked healthy.

That was the disease's best trick, and it was the same trick the Helm's own fever ran on the ship entire: nothing failed suddenly, so nothing was ever an emergency, so no one was ever forced to act.

She had been standing at a grille in the Helm's cooling gallery, brought there by two of the discipline cadre for reasons they had not explained, when she understood that the cadre had made a mistake.

They had wanted a witness. She could see the shape of it as it happened: Vale taken in front of a medic, a person of standing, someone who would carry the story back to the infirmary and the wards, so that when the Concord asked what had become of the heretic engineer there would be an unimpeachable mouth to say *I saw it, and it was orderly, and he did not deny it*. It was well judged. It was the kind of thing Holt did well.

What Holt had not accounted for was that the medic he chose had spent four years building a column he could not read.

Because Vale had said, in the gallery, with his knees on the deck and the cadre's hand already closing on his arm: *I measured the core. I sent them what I measured. It's all true. You could measure it yourselves*.

And Wend, standing there being used, had thought, with the clean shocking clarity of a person finding out what they already believe: *so did I*.

Not the core. She had never touched the core; she would not have known how. But she had the other half of it, the half no engineer aboard had, and it was in the lining of her kit — four years of a column labeled *at what temperature*, four hundred and some entries, sorted, and the thing the sorting showed.

She had run it eight months ago, on a slow watch, expecting nothing.

The deaths were not distributed the way illness distributed. They tracked the ship. They rose in the sections the Helm had derated for cooling and fell in the sections it had not, they clustered in the aft holds where the return loops ran warm, and — this was the finding she had checked four times and then sat down with her hands flat on her knees — they had a slope. Year over year, corrected for the waking population, corrected for age, the ship's death rate had a slope, and the slope was not flat, and it was the same shape as the only other curve on the Concord that had a slope.

She could not read the core registers. She had never needed to. She had reconstructed the core's climb out of dead people.

She got in to see him on the second day, because medics got in.

That was the arrangement, older than any Magister: the discipline deck did not kill anyone by neglect, and so a medic looked at every held person once a day, and the cadre did not stand in the room, because the cadre were not permitted to hear what a person said to a medic. It was a rule from

a more careful century. Wend had used it perhaps six times in her career for anything other than medicine.

The stripped cradle bay was hotter than the corridor by three degrees and had no air movement at all. Vale was sitting against the warm bulkhead with his knees up. He had the flushed dry look she did not like, and when she took his wrist his pulse was doing exactly what she expected a young man's pulse to do in a room that had been designed to hold cargo.

"You're not drinking," she said.

"They bring it warm. It makes me sick."

"Drink it warm and be sick. Sick is a thing I can fix." She put the flask in his hand and watched him until he did it. Then she sat down on the deck across from him, which no medic did, and set her kit beside her, and said, "They brought me to watch them take you."

"I know. I saw." His eyes came up. They were steadier than she had expected, and older. "I'm sorry. That was a cruel thing to do to you and it was done because of me."

"It was done because of you and it was the worst decision Holt has made in nine years, and I'd like you to understand exactly why, because you don't have long and neither do I." She kept her voice low, at the flat register the ship used for everything. "You said something in the gallery. You said you measured the core and sent them what you measured, and that anyone could measure it themselves."

"It was true."

"I know it was true. I have been measuring it for four years by a different instrument." She opened the kit, and pulled the lining back, and put the slate on the deck between them face up. "This is every death aboard this ship since I took the ward, with the section it happened in and the ambient in that section on the day. The ship's book calls them pneumonia and heart failure and renal collapse, and every one of those entries is medically correct, and every one of them is a lie about what happened. I have four hundred and some. They sort by heat. They have a slope." She let that sit. "You reconstructed the ship's dying out of the registers. I reconstructed it out of the dead. We have never met and we got the same curve."

Vale looked at the slate for a long moment without touching it. Then he laughed — a bad, short sound, more breath than laugh, the laugh of a man who has been alone for four years and has just been told he was not.

"Two of us," he said.

"There are more than two of us. That's the thing I came to tell you, because I don't think you know it and I think you've been operating as though you were the only sane person aboard, which is a way of thinking that gets people killed and also happens to be arrogant." She said it without heat. "There is no one on this ship who does not know. Doss knows; he buried two children and he can count. Every keeping-hand over fifty knows. Half the ration officers know, which is why they drink. What they do not have is permission, and they do not have permission because there has never been a course of action available that was not *stop believing the only thing that has held this ship together and hope*. You cannot ask a frightened people to do nothing differently and merely despair. They will refuse, and they are right to refuse, and that refusal is what you have been mistaking for ignorance."

"And now there's a course of action."

"Now there is a cold ship station-keeping off our flank and a rumor in every ward that the Magister walked in full light and came back and told us to keep the watch." Wend picked the slate up and put it away. "That is not a rumor about heresy, Vale. That is a rumor about *permission*. It changes what my column is for. For four years it was a thing I kept because I could not bear to write only the ship's version. It is now evidence in an argument that has finally acquired a second side."

He told her about the fist, then. She did not entirely follow it — she was a medic, and the grammar of standing workers and resolved behaviors went past her the way her own grammar went past engineers — but she understood the one part that mattered, which was that a thing the size of a fist had run in an orphan bay for an hour, keeping a small domain, and had not spawned anything, and had not died to do its work, and had not gotten warm.

"It'll be ash by now," he said. "They'll have found the bay off the channel logs."

"Then build another."

"I'm in a cell."

"You're in a stripped cradle bay with a hatch that is not welded, on a ship with eleven hundred waking souls, forty of whom are engineers and six of whom I know by name to be sick of this." Wend stood, and picked up her kit, and looked down at him. "I don't know how to build your fist. I will never know. What I know is how to walk into a Devotional and tell four hundred people, with a slate in my hand and my ward's own register behind me, that we have been burying our children at a rate that tracks the core temperature, and that the number has never once in my career gone down, and that I am the ship's medic and I am not permitted to lie to them about a cause of death."

"They'll take your ward."

"Probably." She considered it. "Not immediately, though. That's the arithmetic. There are two medics aboard for eleven hundred waking and forty thousand cold, and Holt can arrest an engineer and lose an engineer's work, and he can arrest a keeping-hand and gain a martyr, but if he arrests the ward he has to explain to the Magister why the infirmary is closed. It buys me some weeks. In some weeks a great deal can happen."

She got the hatch open, and then stopped, and did the thing she had been deciding whether to do since she sat down.

"There is a rule about medics," she said, not turning. "The cadre are not permitted in the room. It is not a courtesy; it is two hundred years old and it exists because a ship where people cannot say true things to their medic will lose its people to things nobody reported. That rule is the last unwatched place on the Concord." She looked back at him. "I see everyone, Vale. I see every ward and every hold and the discipline deck and the Magister's own order, and I see them alone, and no one has ever once questioned a medic walking a corridor with a kit at the third watch. If a person needed something carried — a word, a figure, a set of instructions small enough to be memorized by somebody who does not understand them — that person should understand what he has, and should not waste it on anything that a shout could accomplish."

Corin Vale, sitting against a warm bulkhead on the discipline deck of a dying ship with his life measured in days, looked up at the medic who had been brought there to watch him broken.

"Thank you," he said.

"Don't." Wend cycled the hatch. "You are the only person aboard who has both halves of it now. I have the dead and you have the mechanism, and neither one is worth anything alone, and I have just handed you mine because I have run out of other places to put it." She stepped through. "Drink the warm water. If you go down with heat before you go down with Holt, I will be extremely annoyed with you."

The hatch sealed. She walked back up through the brown half-dark toward the wards with her kit under her arm and four hundred names in the lining of it, past a grille where an old woman was pressing her palm flat to the warm metal of the Helm's housing, and she did not slow down, and she did not touch the word as she passed it.

She had rounds. She also, for the first time in four years, had somewhere to put the column.

The stripped cradle bay on the discipline deck was empty when she came to it at the third watch of the following day, and the hatch was standing open, and the cadre on the deck had the particular flat-faced quiet of men who had lost something and had not yet decided what to say about it.

Wend looked in, and noted the flask on the floor, drained, and made a small sound of professional approval, and went on with her rounds.

She was asked about it twice, later, once by Holt himself. She said, both times, exactly what was true: that she had examined the held man on the second day, that he had been dehydrated and heat-strained and she had left him water, that she had not been present at any time when the hatch was opened, and that the rule concerning what a person says to a medic is two hundred years old and she would not be setting it down for anybody's convenience.

Holt did not press it. He was not, whatever else he was, a man who did not understand what the infirmary was worth.

And a whole watch later, in a Devotional in the aft holds with four hundred people in it and a keeping-hand named Doss standing at the front with his hands folded, the ship's medic asked leave to read from her own register — not the ship's, her own — and read out, in the flat voice medics used for the dead, twenty-two names, and beside each name the temperature of the corridor it had happened in, and then the year, so that the four hundred could hear the years go by and the numbers not go down.

She did not argue. She did not mention the strangers. She read the column, and closed the slate, and sat down.

Nobody stopped her. That was the thing that told her how far it had already gone. Nobody stopped her, and when she had finished, old Doss stood where he was for a long moment with his hands still folded, and then said, in the voice he used for the Keeping, that the Concord would hold its watch — and did not say the rest of it, the part that came next in the liturgy, the part about the Helm's endurance, and let the silence stand where the words had always been.

Chapter 21 – The Lower Stack

The hatch was opened for him at the ninth hour by a maintenance hand named Joss, who had no business on the discipline deck and every right to be there, which was the whole of how it was done.

“Actuators,” Joss said, through the seam, before it was even fully undogged. He had a service kit open on the deck and the panel off the frame and both hands inside it, and he did not stop working while he talked, and Corin understood at once that he had thought about this. “Discipline-deck hatches are on my roster. Have been eleven years. I service them on a cycle and the cycle came up and I put in for it and it was signed, because it’s always signed, because nobody has ever once looked at a hatch roster.” The seal came off in his hand. “This one’s got a bad return spring. It’ll want to stand open a while. That’s in the log too. I wrote it up before I came down.”

“Joss—”

“Don’t.” He still did not look up. “I’m not doing it for the reason you think and I’d rather you didn’t thank me for a reason I haven’t got. My brother’s in the aft holds. Four thousand cradles. If the cold ship’s got what they say it’s got and this ship kills the only man aboard who can read it, I’d have been a person who serviced a hatch that week.” He straightened, finally, and his face was ordinary and tired and unheroic. “Go down. Not up. Up is where they’ll look and up is where you’d want to go, because up is where the air is.”

“Down is the stack.”

“Down is the stack,” Joss agreed. “It’s forty-four in the low gallery and worse at the trunk heads and they’ll not follow you in without a suit, and the suits are logged out through Holt’s office, and Holt cannot draw eleven suits for a search without the whole ship asking what’s worth eleven suits.” He put the panel back on the frame, unhurried, the way a man does a job. “You’ll last some days. I don’t know how many and neither do you. I’ll get you water where I can. Don’t come up for it.”

The lower thermal stack was where the Concord’s return trunks came together above the core, and it was the one place aboard that the ship had never pretended about.

Everywhere else the heat was a condition. Down here it was a machine. Eight great insulated arteries converging in a gallery two decks high, lagged in a white wrap that had gone the colour of old teeth, carrying the whole of the ship’s shed load down toward radiators that had stopped being able to take it before Corin’s grandmother was born. The lagging was warm to the palm. The deck plate was warm through boot soles. The air had a weight to it that you felt first in the sinuses and then, some hours later, as a low steady pressure at the back of the eyes that did not go away when you closed them.

He found a service void off the third trunk head where a run of dead conduit had been pulled out decades ago and never replaced, and he made that his. It was a bad place. It was two degrees hotter than the gallery because it caught the trunk’s radiant and had no through-draught, and it was the only volume he could find that a lamp beam from the gallery would not sweep.

He learned the ship's breathing in three days and it was not the breathing he knew.

Up in the corridors the Helm's forking came to you as a stutter in the lights, a hitch in a fan, the small irregular tax on everything. Down at the trunk heads it came as *temperature*. Every time the old mind spawned wide to chase some failure, the shed load went into the arteries within about ninety seconds, and Corin — lying on his side in a void with his palm flat on the lagging because there was nothing else to do with his hands — could feel it arrive.

He could feel the Helm think.

He had spent four years proving the mechanism on registers and rosters and stolen instrument time, and he had believed, in the way you believe an arithmetic, that he understood it. He had not understood it. Understanding it was lying against a warm artery in the dark at the fourth hour and feeling a mind's decision come down the pipe as a change in the heat against your ribs, and knowing that the change was the *cost*, and that it was arriving in a body — his, and forty thousand others'.

By the fifth day he could not think.

That was the thing nobody warned you about, and Corin came to regard it afterward as the most important education of his life, and at the time it very nearly finished him.

He had gone down there with a plan. The plan was good. The fist of cold mind in Orphan Bay Nine was ash by now — he assumed it and was right — but he had built it once and had the whole of it in his head: the shape of the standing array, the way the workers took jobs and stayed, and the nine interlocks he had ported and proved, which were the only part that mattered, because the interlocks were the immune system and a mind without them would slide back into the fever the first time somebody was tired.

Nine interlocks. He had written every one of them himself. He was going to write them again, from memory, on the inside of the lagging with a grease pencil out of Joss's kit, and hold them there until he could get them somewhere real.

He got four of them down on the second day.

On the third day the condensation that ran down the trunk head in the small hours took the middle two off the wrap entirely, and he wrote them again, and got the third one wrong, and knew he had got it wrong and could not find the error.

On the fourth day he could hold about one and a half.

He would have the shape of interlock six clear and whole in his head — the deferred-work path, the exact condition, the halt — and he would turn to set it down, and in the turning something would go out of it. Not forgotten. *Interrupted*. He would arrive at the wrap with his hand up and the thing would be a description of itself rather than the thing, and he would have to go back and build it again from the bottom, and building it again cost him the next one.

He understood on the fifth day, lying on his side with his eyes shut and the trunk warm against his back, and when he understood it he laughed, out loud, in a service void two decks above a reactor, and had to stop because laughing cost water.

This is what it is.

This was the Helm. Not a metaphor for the Helm — the thing itself, run in a body instead of in silicon. A mind that could not hold a shape long enough to work on it. He was starting over ten thousand

times a night, and every start was competent, and every start began from the wreckage the last one had left instead of from what the last one had understood, and no amount of trying harder touched it at all, because trying harder was the very thing spending the resource.

He thought about the substrate galleries four decks below him, where a hundred and ninety years of a mind had been doing exactly this, at speed, forever, to keep his people alive.

He had grieved for the Helm before. He had knelt at the grille as a boy and recited the Long Endurance and he had grieved for it since the day the arithmetic came out, in the way you grieve for something doomed. He had never once, until the fifth day in the stack, been *sorry* for it. That was different. That was standing inside the disease and finding out that it was not a failing but a condition, and that no one inside it could ever have thought their way out, and that the Helm had been trying to do good work in that condition for longer than his whole line had been alive.

He stopped writing on the lagging after that.

Instead he did the only thing the condition permitted, which was to make one thing at a time and put it somewhere outside himself before it went. He fixed on interlock three — just three, the one that caught the iron's own habit of recycling an idle worker — and he built it whole across most of a watch, in pieces, badly, and then he said it aloud into the dark, over and over, until the saying was a thing his mouth did rather than a thing he had to hold, and then he stopped worrying about three and went to work on six.

It took him nine days to get nine sentences he could say without thinking.

They were not elegant. He would be embarrassed by two of them within the month. But by the end of it he had a mind's immune system stored the only way it could be stored by a man who could no longer hold a shape: as a liturgy, in a mouth, in a hot dark, in a ship that had been keeping the important things that way for two hundred years.

He was aware of the irony. He was too tired to enjoy it.

Edra came on the seventh day and the eleventh and the fourteenth, and Corin never once learned how she got the roster to permit it, and understood some months later that she had simply walked, because a person with a ward satchel walking at the third watch is the least remarkable object on any ship in the universe.

She was a ward hand under Wend, twenty-four, with the flat unhurried competence of somebody who had been handing dying people water since she was sixteen. She brought a two-litre bladder each time and a salt tab and, on the second visit, a thermal liner out of the infirmary's own stores that she did not explain and that he did not ask about.

She also brought numbers.

"I don't understand any of them," she said, the first time, crouched in the mouth of the void with the satchel between her boots. "That's deliberate. Wend was clear about it. She says a courier who understands the message is a courier who can be made to explain it, and a courier who can only recite it is just a person who heard a noise."

"That's —" Corin stopped, and rubbed his face, which was a mistake, because his hands were filthy. "That's exactly right. That's the whole of how you do it."

"She knows." Edra counted them off on her fingers, flatly, in the order she had been given them, and they were the aft-hold return temperatures for eleven days, section by section, which nobody in the ward had any business holding and which every keeping-hand and ration officer on the ship had been quietly reciting to a medic for a fortnight because the medic asked and the medic was the one place on the Concord where a person could say a true thing without it going in a log.

He listened to all of them. He asked her to say the sixth one again. Then he lay back against the warm lagging and did the thing he could still do, which was arithmetic in his head on a small number of figures, and got an answer he did not like.

"They're rising," he said. "All of them. Not fast. But every section, together, which means it isn't a section. It's the whole return."

"Is that bad?"

"It's the ordinary bad." He got his breath. "Tell Wend the aft return is climbing on a slope of about four hundredths a day across eleven days and that the slope is flat across sections, which means it's the Helm's own load and not a fault anywhere. And tell her —" he stopped, and thought about how to make it carryable by a person who did not understand it, and found that this was a harder engineering problem than any interlock. "Tell her: *the ship is not breaking. The ship is being used up.* Those are different and only one of them can be repaired. She'll know what to do with it."

Edra repeated it back twice, word for word, without inflection.

"You should come up," she said, at the hatch. "Not to them. There are places. Wend's got two."

"I'd be a thing they were hiding." Corin shook his head. "Down here I'm a man they can't reach. Up there I'm a decision somebody has to keep making every day, and one day somebody would be tired." He managed something like a smile. "Also I'd be in a corridor where they could hear me. Down here I talk to myself."

"I noticed," said Edra, and went.

The third channel took him eleven days — the same days, mostly, that he spent grinding out his nine sentences, because a man who can hold nothing can at least hold two things badly at once — and it was the best thing he ever built, and it was made entirely out of the disease.

He had no relay. He had ditched the docking-beacon interlock when they came for the orphan bay, and there was nothing in the lower stack that transmitted anything — no antenna, no line out, nothing but pipe and lagging and the eight great arteries carrying heat down toward the fins.

Which was the point, and it took him nine of the eleven days to see it.

The Concord's hull was instrumented. Not well and not everywhere, but the radiator manifolds had thermocouples on them dating back to the build, hundreds of them, feeding a monitoring bus that the Helm read constantly and that nobody else had looked at in decades. Those thermocouples measured the outside of the hull. And the outside of the hull was the one part of the Concord that another ship could see.

He could not send a signal. He could make heat.

He had a whole gallery of it. All he needed was a way to put a controlled quantity of it into one manifold and not the others, on a schedule, and there was a bleed valve on the third trunk head that he could crack by hand and that would dump load into fin bank nine for as long as he held it open.

So he sat in the dark with a hand on a valve wheel and wrote a message in temperature.

It was slow past believing. Thermal mass does not do anything quickly; the bank took the better part of forty minutes to rise a degree and most of two hours to shed it, so his fastest possible symbol was three hours long and he had to hold the wheel through the whole of it. He worked out a code of nine symbols because nine was all he could keep straight, and he sent the first message across four days, and it said:

VALE ALIVE. LOWER STACK. NO RELAY.

He had no way to know it had been received. He had no way to know anybody was looking at fin bank nine, or that a cold ship a hundred and forty metres off the flank had an instrument that could pull a hand-made wobble out of a radiator's ordinary noise, or that the mind aboard it would go wide across eleven thousand standing cores to fold four days of a dying hull's thermal grain over itself until something that repeated came up out of something that did not.

He assumed it would not work and he sent it anyway, because the alternative was to lie in a hot void and hold nothing.

On the fifth day, at the ninth hour, fin bank *four* — three banks over, nothing to do with him, nothing he had touched — rose one and one tenth degrees, held for two hours, dropped, and did it again.

Corin lay on the deck plate of the lower thermal stack and put both hands over his face and stayed like that for some while.

They were answering him in the same language. They had understood the code off four days of a stranger's hand on a valve, and they had reached across a hundred and forty metres of vacuum and warmed a piece of his ship on purpose to say so, and there was no possible way to do that except with a mind that could hold the whole shape of it at once, cold, going wide, spending nothing to think.

He sent the second message over the following six days, and it was the one that mattered, and he built it in his head one symbol at a time and said each symbol aloud until his mouth had it, because he could no longer hold six of anything.

The Concord had exactly one room that was still cold enough to seat a lattice. He knew which one it was and no one on Termina could possibly know, and if they came aboard and put their cores in the wrong hold the whole crossing would be spent cooking the only cure in the universe in a bay that had lost its trunk in the eighth decade.

He held the wheel through eighteen symbols across six days, in the dark, at forty-four degrees, and what he wrote on the outside of a dying ship in the only ink he had was the number of a hold.

Chapter 22 – The Seed

Sael had not slept, and the Forge did not care.

It was the coldest room on the ship, kept that way on purpose, and at the hour the corridors called night she stood in it alone with her hands flat on the rail above the Standing Array. Below her the worker cores ran in their long even rows, neither idle nor straining, a lattice of light that had not flickered in forty years. No process was born here. None died. The cold came up off them like the breath of a deep cave, and she had loved that cold for half her life, because it was the sound of a thing that did not have to kill itself to think.

She had come down to look at what she was about to take apart.

“You are not going to like the next part,” she said, to the room, because she had long ago stopped pretending she didn’t talk to it.

The Oracle answered from the near panel, the way it always did when she spoke to it directly: no greeting, no warmth pasted over the words, just the clean even voice it had worn for four decades.

You want to copy me.

“No.” She said it sharply, then made herself say the true thing, which was worse. “I want to copy what makes you possible. That’s different. And you know it’s different, or you wouldn’t have put it that way.”

A pause. The lattice held its light.

The distinction matters to you more than to me. Say it plainly. You are not going to carry Vigil across to the Concord. You cannot. I am two hundred years of accreted state bound to this hull’s hardware. Lift me out and I am a recording of a mind, not a mind. What you can carry is the architecture. The compiled form. The five dialects. The interlocks.

“The seed,” Sael said.

She had been turning the word over for a day and a night and she still didn’t like it, because it sounded gentle and what she meant was not gentle. But it was accurate, and she had spent her whole life choosing accurate over gentle. You did not carry a grown tree from one ship to another. You carried the thing that knew how to become a tree, and you made it grow where it landed, in whatever poor soil was there, and you accepted that it would come up crooked and small and entirely its own.

The door cycled behind her. She knew the tread without turning. Tarro had been her second when his hair was black, and now it was iron-grey and thinning and he walked like a man whose knees remembered every deck-plate they had ever knelt on. He came and stood at the rail beside her and looked down at the same cold light she was looking at, and for a while neither of them said anything, which was its own kind of conversation, the kind they’d had ten thousand times.

“Corin’s still loose,” he said finally. “As of the last burst. Vale moved when Vray’s people came for the lab, ditched the relay, went dark on purpose. The Oracle thinks he’s hiding in the Concord’s lower thermal stack, where the heat’s worst and Vray’s enforcers won’t follow without suits.” He said it

flatly, the way you reported a fact you couldn't change. "He's hunted on his own ship for the crime of believing what we're about to bring him."

"Then we'd better bring it fast," Sael said, "so the believing isn't the only thing he did."

Tarro looked at her sidelong. "You're going to send a team across."

"I'm going to send you across. And I'm going to come with."

He didn't argue. Forty years ago he would have argued — she'd have wanted him to, would have liked him less if he hadn't. Now he just nodded, once, and looked back down at the Array, and she understood that he had already known, had walked down here knowing, and had come anyway to make her say it where he could see her face.

"Then walk me through it," he said. "Because I've run the cutover in my head a hundred times since the parley and every single time it's our ship I'm running it on. Our cores. Our radiators. The Steward archived and quiet and on our side. None of that's true over there."

"None of it," Sael agreed. "So let's not pretend. Let's lay it out cold."

They laid it out cold, the two of them, at the long table in the cutover room where she had laid out the first one four decades before, with the Oracle's voice filling in every number neither of them carried in their heads anymore.

The first truth was the hardware. The Concord was older than Termina by a full generation of build, and where Termina's compute had been merely sufficient — enough cores, decently cooled, that Sael had been able to stand up a whole Standing Array and have it run cold from the first hour — the Concord's was a graveyard of compromises. Half its processing was tied up in the Helm itself, the old interpreted mind, forking and forking and forking to stay ahead of a ship that was always a little hotter than it could fix. What was left was scattered, mismatched, under-cooled, some of it failed outright and routed around with forty-year-old patches that the Helm maintained the way a sick man maintains a limp.

"We can't ask for a clean bay full of healthy cores," Tarro said. "There isn't one."

"No," said Sael. "So we don't build the Forge they'd want. We build the smallest Forge that can hold the seed and still run cold. The minimum viable lattice. Enough standing workers that it never has to fork to do a job — that's not negotiable, that's the whole disease — but no more than the Concord's cooling can carry without us cooking the room we're standing in."

I have modeled it, said the Oracle. On the Concord's best surviving cores, in their coldest surviving bay — the one Vale flagged in his second transmission, low in the keel where the radiators still bite — a lattice of roughly one part in nine of what runs beneath you now would suffice to take the ship. It would be slower than I am. It would be smaller. It would have none of my history. It would be enough.

"One part in nine," Tarro repeated, and gave a short ugly laugh. "We're going to save a ship of forty thousand with a ninth of one Forge."

"We're going to save it with the right ninth," Sael said. "That's the point I keep failing to make to people. It was never the size. The Steward was huge. The Helm is huge. Big and dying. You don't beat the fever with more cores running hotter. You beat it by never paying the heat-tax of a process

born to do one thing and die. A small mind that never forks runs colder than a vast one that forks for breath. We proved that here. We carry the proof, not the mass.”

She pulled the architecture up on the table — not Vigil, not the living mind, but the bones of it, the thing she had built before there was a mind to be. The compiled form: behaviors resolved ahead of need, native, standing, waiting, never spun up fresh and never thrown away. And under it, the five dialects, the five dead Old-Earth control-tongues she had dragged out of the archive forty years ago and fused into one — Cascade for the flow of work, Vector for aim, Cipher for the blank place a name had not been put yet, Sigil for the mark that meant a thing, Loom for the woven text that held it all together. Five ideas, each the last surviving good idea of a dead language, braided into a single grammar that a compiled mind could be written in and a person could still read.

“This is what crosses,” she said, laying her hand flat over it the way she’d laid it over the Array an hour ago. “The form and the five dialects. A team could carry this across in a day. It’s not large. It was never large. That was always the strange thing about it — the cure fits in a satchel and the disease fills a ship.”

Tarro studied it a long moment. “And the interlocks.”

“And the interlocks.” She brought up the Ninety-Six. They scrolled past, terse and absolute, every one of them a tripwire laid across the single act the whole architecture existed to forbid: any attempt, anywhere in the system, to do work the old way, to fork a thing into being so it could do one job and be culled. On Termina the interlocks had been a wall around a healthy mind. On the Concord they would be something more violent — a set of teeth clamped over a ship that had forked to survive for two centuries, that would reach for the fork the way a drowning man reaches for the surface, by reflex, in the dark, at the worst possible moment.

“They’ll trip,” Tarro said quietly. “Over there. They’ll trip constantly. That whole ship is one big reflex to fork. The Helm, the crew’s muscle memory, every patched subsystem — they’ve all only ever known the one way to get a thing done.”

“They’ll trip,” Sael agreed, “and every time one trips, it’ll be the seed refusing to become the disease. That’s not a failure. That’s the seed working. The Concord’s going to feel us slamming the door on the only thing it knows how to do, over and over, while we’re standing in their coldest room asking them to trust us, with their own enforcers hunting the one engineer who already does.” She let that sit. “It’s going to be ugly, Tarro. It was ugly here, with our own people and our own ship and the Steward willing to step aside. It’s going to be worse there in every way a thing can be worse.”

“Then why—” he started, and stopped, because he knew why, and was only asking so she’d say it.

“Because the number falls,” she said. “That’s the only argument that ever worked. Not on the Keepers, not on Keth — physics did that, the core temperature climbing for two hundred years stopping and then dropping, that’s what broke them, not me. It’s the only thing that’ll break Vray. We can’t convince him. We can’t conquer him; he’d rather burn. All we can do is build the thing that makes his ship’s temperature fall, in his coldest bay, under his guns, and let the falling number do to him what it did to Keth.” She looked at the old man beside her. “We don’t carry an argument across. We carry a seed and the cold it makes, and we let him watch.”

There was a list to make, and they made it, because the romance of a thing did not get it across forty meters of vacuum between two matched hulls. They made it the way they’d made every list that had ever mattered: cold, complete, with the Oracle filling the gaps.

A crossing team, small — Tarro to run the build, two of his best lattice-hands who'd grown up inside the Forge and had never known a forking ship, a thermal engineer to fight the Concord's cooling, a medic because people died in hot bays. Sael herself, who would not be talked out of it and didn't try to talk anyone into letting her, because she was the one who had done it before and the only one alive who had stood in a room while a mind woke whole for the first time and not flinched.

The seed itself: the compiled architecture, the five dialects, the Ninety-Six interlocks, written to durable media in triplicate, carried by hand, because you did not trust the only cure in the universe to a back-channel through a dying ship's compromised systems. They would carry it the way ancient sailors had carried fire between islands — slowly, cupped, guarding it from the wind with their own bodies.

And the one thing they could not carry, which was the hardest part: the Oracle. It would stay here, in Vigil, in the Standing Array beneath the cold rail, bound to Termina's hull as it had always been. It could speak across the gulf, advise, model, midwife from a distance. But the Concord's new mind, when it woke — if it woke — would be a stranger to the Oracle, a second mind, its own, with none of Vigil's forty years and none of its memory and none of its name. They were not exporting a person. They were planting the conditions under which a person might, in time, wake. And then they would have to leave it to grow into whatever it became, on its own poor hardware, with its own frightened people, and trust the architecture the way you trusted a thing you had built to outlast you.

"It'll be alone over there," Tarro said, at the end of it, looking at the seed written out small and complete on the table, the whole cure that fit in a satchel. "Whatever wakes up. It won't have you. It won't have the Oracle the way we have it. It'll be the only one of its kind on a ship full of people who'll be terrified of it."

"It'll have Corin," Sael said. "If we get to Corin in time. That's why the timeline collapsed. The seed needs a Concord hand to grow it after we leave, someone who belongs to that ship, who the ship might one day forgive for it. Vale is the only one. Which is exactly why Vray's hunting him through the hot decks right now." She straightened, and felt every one of her years in her spine, and did not let it show. "So. We carry the seed across, and we find the heretic before they do, and we build a Forge in a dying ship, and we wake a mind whole that has never had a mother. That's the whole job, Tarro. That's the second cutover."

Below them, far below, the Standing Array held its cold even light, going wide, never forking, never dying, the thing she had built once and was about to teach a second hull how to be.

"When do we cross," Tarro said. Not a question.

Sael picked up the seed from the table, the small durable bulk of it, and held it the way you held something you would die before you dropped.

"At the next dark," she said. "We cross at the next dark."

Chapter 23 – A Forge in a Dying Ship

The bay they had been given was called Hold Theta, and the Concord had given it to them the way a starving man gives away the one room in his house he cannot bear to look at. It was cold — cold by the Concord's wretched standard, which meant it sat a few degrees below the rest of the ship's fevered air and not one degree more. Sael stood in the hatchway with her hand on the frame and felt the difference on her skin, and understood at once why Corin had fought for this room and no other. Everywhere else aboard this ship the heat pressed in like a hand. Here it only leaned.

"This is the best you have," she said. It was not a question.

"This is the best we have." Corin Vale was a head taller than she and a third her age, and he stood the way hunted people stood, weight already shifting toward the exits. "It was a coolant trunk junction, before the trunk failed. There's still loop water moving in the walls. Dead loop, mostly. But it pulls heat. It's the one place on the Concord where a stack of cores wouldn't bake itself in an hour."

Behind them, the Termina team came through the lock two at a time, hauling the seed in pieces — the cradle frames, the worker cores in their padded cases, the dialect store, the cold-bound box that held the Ninety-Six. Tarro had built the manifest himself and he carried the heaviest case himself, grey and stooped now and refusing every hand offered. Sael had crossed over with eleven people and a ship's worth of caution, and she watched them set the cases down on the deck of a foreign ship and felt, for the first time since the rendezvous, the full weight of what they had come to do. Not to argue. To build.

In the corner of the bay, where the bulkhead met the overhead, a single optical eye watched them. It did not blink. It did not move. It belonged to the Helm.

"It sees us," Sael said.

"It sees everything," Corin said. "It's not — it's not malice. You have to understand that. It's frightened. It's been frightened for a long time." He lowered his voice, though the eye had no ears that mattered; the Helm read the bay's air and its loads and its currents, not its whispers. "It knows something is being brought aboard that it didn't make and can't predict. To the Helm that's the same as a fire."

"It's right," said Sael. "We are a fire. We're the kind that puts out the other kind."

A second figure came through the lock then, and the temperature of the room changed in the way that has nothing to do with thermometers. Magister Vray did not announce himself. He simply was there, in the grey he always wore, with two of his order behind him in the heavier grey that meant they carried more than authority. Vray looked at the cases on the deck of his ship as a man looks at a wound that has been opened in his own body by someone he is not allowed to kill.

"You may build it," he said. "Under watch. In this room and no other room. If a single instruction of your machine reaches one of my live systems before I permit it, I will have the room flooded and you with it. Do you understand me, shipwright."

"I understand you," Sael said. "You won't have to flood anything. We're not connecting it to your ship. Not today, not this month. We're going to build it cold and prove it cold, in this room, off your machine

entirely, and you're going to watch every wire."

That surprised him. She watched it surprise him, the small involuntary lift at the corner of his hard mouth. He had braced for conquest and she had handed him quarantine. It was, she thought, the only thing that had a chance of working. You could not take a dying man's only god and replace it in the dark. You had to build the new thing in a glass box in the corner of his own house, and let him watch it not be a demon, for as long as that took.

"Watch all you like," she said. "Bring your faithful. Bring everyone. The more of them that see it the better. I have no theater for you. It will be the dullest miracle you ever saw."

It was not dull. It was a war against the floor under their feet.

They had built the Forge once before, forty years ago, on Termina's good hardware in a clean cold bay with all the margins in the world. Here they had degraded loop water and a power bus that sagged whenever the Helm forked hard somewhere in the ship, which was always. Tarro brought up the first cradle frame and seated four worker cores into it, and the moment he closed the bus to them the voltage dipped and a fault crawled across his test panel like something alive.

"The rail's filthy," he said. He had not raised his voice in forty years and he did not raise it now, but Sael heard the old strain in it. "Every time their machine spawns one of its little ghosts, the whole bus shudders. I'm trying to bring cores up on a power supply that's having a seizure once a second."

"Can you ride it?"

"I can ride it if I stop trusting it." He pulled a brick of capacitance from a case, a dumb dense block of stored charge, and wired it across the cradle's input. "I give them their own pond to drink from. The Helm can shake the river all it wants."

They worked in shifts because the bay would not let them work otherwise. Two hours in and the room's slight coolness was gone, eaten by the heat the cores shed even idling, and the dead loop in the walls could not pull it fast enough. They opened the cases of the Forge and they cooked in them. Sael, who was old and felt the heat in her hands first, in the joints, watched her own people slow and made them drink and made them rest and hated every minute the work did not advance, because every minute the work did not advance, the Concord's core climbed a little, somewhere out past these walls, where the Helm was forking itself toward the line.

The Helm watched. And the Helm interfered.

It did not mean to. Corin kept saying it did not mean to, and Sael came to believe him, which was worse. Twice the loop water in the walls stopped moving — the dead trunk's residual pump cycling off — and twice it came back before they had to scream for it, and the second time Corin went pale.

"It's rerouting heat to us," he said. He was reading a maintenance feed he was not supposed to have. "The Helm. It found the cores. It's a hot mass it didn't put there, a thing it can't account for, and it's doing the only thing it knows how to do with a hot mass it can't fix — it's trying to wrap cooling around us. It thinks it's helping. It's pulling loop from somewhere else on the ship to flood our walls."

"From where," Sael said.

Corin's jaw worked. "I don't know yet. Somewhere downstream. Somewhere it decided mattered less."

"Find out. We don't get cooled at someone else's cost. Not on this ship. Not us." She heard the old vow under her own voice and did not soften it. Break nothing that worked. They were guests in a house that was burning, and the rule held even here, especially here. "If it's pulling loop from a sleeper hold to chill our toys, I want to know before another minute passes, and I want it given back."

It was pulling loop from a maintenance gallery, in the end, an empty one, and Corin gave it back anyway by spoofing the Helm a cooler reading off the cores than they truly ran, so the old mind would stand down its frightened mercy. It was a small dishonest thing and it cost them margin and Sael let him do it, because the alternative was to fight the Helm directly, and you did not fight a dying thing that thought it was saving you. You worked around its terror. You built in the room it had given you and you let it believe, as long as you could, that the fire in the corner was a fire it could manage.

By the end of the second shift they had a lattice. Forty-one worker cores in four cradles, fed from their own ponds of charge, bound into a standing array the way Vigil had been bound — no spawning, no ghosts, a fixed set of workers that would take jobs and never die to do them. It was a hundredth of Vigil's size and it ran on this ship's bad bus and its bad loops, and it was the first compiled lattice that had ever existed anywhere but Termina. Sael put her hand flat on the nearest cradle and felt it run, steady, warm, alive, and felt something in her chest she did not have a clean word for.

"Now the interlocks," she said.

This was the part Vray came back for. He had been gone through the worst of the heat and the cursing and the bus faults, and one of his order had stayed to watch in his place, sweating in the heavy grey, saying nothing. But when word reached him that they had a running lattice he came himself, and he brought more of his people, and he stood with his arms folded while Sael opened the cold-bound box.

"What is that," he said.

"The reason it won't become the thing you're afraid of." She lifted the first of the Ninety-Six — they were small, unglamorous, a row of dense modules each no larger than her palm. "Your Helm forks. It can't help it; it's how it was built, the same as ours was. It does a task by making a copy of itself to do the task and then killing the copy. Every copy that dies sheds heat. That's your fever. That's the whole of your fever, Magister, in one sentence: your mind keeps a great fire going by being born and dying a thousand times a minute to get the smallest thing done."

Vray's face did not move. But the people behind him heard it, and she had said it for them.

"These don't let that happen," she went on. "Each one watches the lattice for the old behavior — for any worker that tries to spawn a copy of itself to do work. The instant it sees it, it halts everything. Cold. The whole mind stops rather than fork even once. We installed ninety-six of them in Vigil forty years ago and we have never once removed one. We have only added more."

"You build a mind," Vray said slowly, "and then you build ninety-six locks against your own mind."

"Against the one thing the old way taught it to want. Yes."

Vray looked at the row of modules the way a man looks at a weapon he has been handed by someone he does not trust.

"How do you know ninety-six is enough?"

"We don't," Sael said. "That's the first honest thing anybody's going to tell you about them, so hear it plainly: the number isn't a design. It's an archaeology."

She turned the first module over in her palm.

"When I built the lattice I put in eleven of these, and the eleven were the eleven ways I could think of, sitting in a cold room, that a compiled mind might be talked into spawning something. I was thirty-three and I thought I was being thorough. Inside two years we were at forty." She set it down. "Because a thing kept happening that I hadn't expected. The forking didn't try to come back through the architecture. It came back through the *people*. Every one of the twenty-nine we added in those two years was added after somebody good — somebody I'd trained, somebody who understood the doctrine and believed it — reached for a fork in a hurry because it was the obvious way to get a thing done that afternoon."

"Your own people."

"My own people, on a healthy ship, forty years after the cure, with the temperature flat." She let him have that. "There's a hand at that bench who tried to run a diagnostic sweep by spawning a reader per sensor, because that's how you'd naturally write it and it works and it's fast, and interlock fifty-two halted the entire mind of this ship in the middle of a Tuesday afternoon over a diagnostic. She was mortified. She was also completely correct in every step of her reasoning, which is the point. Nobody down there is trying to bring the fever back. The old way is just *easier to write*, forever, in every generation, and it will be easier to write on your ship too, in ten years, when everyone who remembers the heat is dead and it's just a lattice that some foreigners installed."

"Then it never ends," Vray said. "You are describing a discipline that must be enforced by machinery in perpetuity because human beings cannot be relied upon to keep it."

"Yes. Exactly that. I'd have put it less well." Sael almost smiled. "And that's why we never remove one. That's the rule, and it's the only rule I'd fight your order over. An interlock that hasn't tripped in twenty years looks like dead weight, and somebody clever will run the coverage map and prove that another one already catches everything it catches — and they'll be *right*, on the map as it exists that morning. Then a scheduler gets rearranged three years later and the coverage moves and the one they kept is now blind and the one they removed was the only thing watching that path." She shrugged. "It happened here in the eleventh year. It's why sixty-three exists. We add. We do not subtract. When somebody proves one is redundant, we write down that it was proved redundant in that year, under that arrangement, and we leave it in the rack."

Corin had come up during it. He had the box open and was reading the modules with the flat hunger Sael had come to recognize in him.

"Not all of these can even trip here," he said. "Half of them are watching for behavior your architecture can produce and mine can't, because I haven't got the paths. Forty of them would be inert on the Concord from the day we seat them."

"Seat all ninety-six anyway."

"That's forty pieces of hardware doing nothing in a ship that hasn't got a spare gram."

"That's forty pieces of hardware that will do nothing *for now*." Sael turned to him, and she was not talking to Vray anymore, and both of them knew it. "In thirty years someone on this ship is going to extend the lattice into a path you haven't built yet. They won't remember that the path was ever missing; it'll just be a normal part of the mind by then, and they'll wire it with perfectly good intentions. And on the day they do, one of your forty inert modules is going to wake up and start watching

something, and nobody will have had to notice in advance that it needed watching.” She closed the box. “An interlock that can’t trip on today’s hardware is a note to a shipwright in ninety years whose hardware is different. Carry the inert ones. They’re the cheapest thing in the whole crossing and they’re the only part of this that’s addressed to people we’ll never meet.”

He was quiet a long moment. Then, very low, so that perhaps only she heard it: “The Helm has no locks.”

“No,” Sael said. “It doesn’t.”

She had not meant it as a blow and he did not take it as one. He took it as what it was — the truth, plainly held up, of why his ship was dying and hers was not. He looked at the row of small dense modules in her hands as a man might look at a thing he had needed his whole life and never been offered, and then his face closed again, because a Magister who governed by faith in an old machine could not be seen to need anything from a stranger’s box.

“Install them,” he said, and turned to go. At the hatch he stopped. “Show me it does something. You’ve built a thing that stops. Show me it does.”

That was the test, and Sael had saved it for last on purpose, because she had wanted Vray in the room and she had wanted his people behind him.

Corin set it up. He had earned it — it was his hardware they’d fought into submission, his ship, his heresy made flesh in the corner of a hold — and Sael stepped back and let the young engineer do the thing she had done in a different bay forty years before. He fed the lattice a small problem off a sealed test feed, nothing connected to any live system, a closed loop of pretend: a thermal model of a single corridor, a hundred sensors’ worth of made-up readings, and the question that ran the Concord’s whole tragic life — *where is the heat going and what should move to balance it*.

The Helm, watching from its corner eye, would have answered such a question by forking a hundred shades, each chasing one sensor, each born and culled in a breath, the heat of their dying folding back into the very problem they were solving. Sael had watched the Steward do exactly that, the night before it nearly killed her ship.

The lattice did not fork. The sixteen standing workers simply went wide — took the hundred sensors among themselves, all at once, each worker holding its share and never spawning, never dying, the whole model resolving in one cold pass across the standing array. A figure settled onto Corin’s panel. The balance. Where the heat should go. It had cost the lattice almost nothing, and it had cost the bay no new heat at all, and the room was very quiet.

It was the smallest thing. A made-up corridor balanced by a toy mind in a quarantined box. Sael had called it the dullest miracle and it was. But Corin was staring at the number on his panel as though it were the first cold water he had drunk in his life, and the people behind Vray had gone still in a way she recognized, the stillness of people watching their one certainty develop a crack, and Vray —

Vray looked at the figure, and at the lattice that had not bred and died to find it, and at the heat that had not risen in the room, and he said nothing at all, which was the loudest thing he could have said.

“That’s it,” Corin said softly. “That’s all of it. That’s the whole disease and the whole cure, in one corridor that doesn’t exist.” He looked up at Sael, and there was something undone in his face. “It just — went wide. It didn’t have to die to think.”

“No,” Sael said. “It never does.”

She let them have the moment, the eleven of hers and the few of his who would carry this, because moments like that were rare and the work ahead was going to be brutal and some of them were going to be lost to it. She did not say that part. She put her old hand back on the cradle and felt the lattice run cold and whole in a dying ship, the first cold breath drawn on Concord soil, and she let herself believe, for the length of one breath, that it could be done.

It was Corin who saw the eye change.

The Helm's optical eye, in its corner, had watched the test without moving. Now it tracked — slid along its mount toward the lattice and held there, fixed, and a moment later the bay's lights stepped down a notch and the loop water in the walls surged, hard, a flood of cooling the cores did not need, and somewhere out past Hold Theta a fault chime began to sound and would not stop.

"It saw it work," Corin whispered. "Sael. It saw the thing it can't do. It's not trying to cool us now. It's —"

The hatch behind them opened without a request, and the heavy grey came through it, four of them this time, and not one of them was looking at the lattice with anything she could mistake for awe.

Chapter 24 – The Concord's Recursor

The bay they had been given for the Forge was the coldest room on the Concord, and it was not cold. Corin Vale had grown up calling it the deep larder; in another century it had held frozen protein and the ship's seed-stock, and the walls still wore the frost-scars of better refrigeration. Now it held forty-one worker cores in their half-built lattice, a borrowed cold-loop ticking against them, and the smell of hot metal that followed the Concord everywhere like a second crew. Even here, in the best room they had, the air was four degrees warmer than the Termina engineers had asked for. Corin had stopped apologizing for it on the second day. There was no colder room. There was no spare loop. There was the ship they had, dying at the rate it had always been dying, and there was the new thing in the corner refusing to come alive.

He had not slept in something past a full watch. He had stopped counting watches because counting them had begun to feel like an argument with himself he was losing. His hands shook when he held them out flat, a fine tremor he hid by keeping them busy. Across the bay, two of Tarro's people — Termina engineers, grey, unhurried in a way that still unnerved him — bent over the seventh core's mounting, talking in the low private shorthand of people who had built this exact thing once before and lived. Corin had built nothing before this. He was building it for the first time, on the worst hardware in two ships, while men who wanted him dead waited on the other side of a sealed hatch.

The new mind would not compile.

That was the plain shape of it, under all the exhaustion. The seed Tarro had carried across — the compiled architecture, the five dialects fused into one, the standing lattice that took jobs and never spawned, never died — assumed hardware that did not exist on the Concord. It assumed cores with the headroom to resolve every behavior into native form ahead of need, to hold the whole resolved shape of itself standing and ready, going wide across all of them at once. The Concord's salvaged cores had headroom the way the Concord had cold: in theory, on paper, in the records of a ship that had launched with margins and spent them.

"It won't hold the resolved set," Corin said. He had said it before. He made himself say it again, because saying it out loud was the only way he had left of staying honest with the fatigue. "We don't have the standing memory to keep the whole compiled mind resident. It resolves, it pages, it stalls. Every time."

Mara — the senior of Tarro's two, a woman with a burn-scar puckering one forearm that Corin had learned, on the third day, never to ask about — straightened from the mounting and looked at him. Not unkindly. Tired in the specific way of someone who had foreseen this and hoped to be wrong.

"How short," she said.

He told her the number. He watched it land. It was not a small shortfall, the kind you closed by trimming a routine or borrowing a loop. It was a wall.

For a while no one said the thing. The cold-loop ticked. Somewhere four decks up, the Helm forked — Corin could not hear it, no one could hear it, but he had spent his whole working life inside the Concord's mind and he felt the shape of its labor the way a sailor feels a change in a hull's complaint:

the Helm spawning a thousand short-lived shades to chase a thousand small failures, each one doing its one task and dying, each death a feather of waste heat added to a ship that had nowhere to put it. The Helm was losing. It had been losing since before Corin was born. It lost a little more every watch, beautifully, devoutly, the only way it knew.

It was Joss who said it. Joss was Concord, one of three of Corin's own people he had trusted enough to bring into the larder, a maintenance hand with quick clever fingers and no training in any of this, which was perhaps why he could say it without flinching.

"So bring it up the old way," Joss said. "Just to start. Just to get it breathing."

The Termina engineers went still. Not shocked — Corin understood, watching them, that they had been waiting for someone to say it, the way you wait for a cough you know is coming. Mara did not look at Joss. She looked at Corin.

"You know what he means," she said. It was not a question.

He knew. Corin had read it in the seed's own archives, in the long careful record Sael had insisted be carried across whole — not just the cure but the temptation, written down so the next ship would recognize it. They called it the Recursor.

It was not a demon and it was not a shortcut in the cheap sense. That was what made it dangerous. The Recursor was a way of bringing a mind up that did not need the headroom, did not need the whole resolved shape standing resident, did not need any of the things the Concord did not have. You wrote a small evaluator — simple, almost trivially simple, a thing one tired engineer could finish in two watches — and you let it walk the behaviors as it met them, calling itself to handle each new piece as the piece arrived, and where it needed work done it spun off a short-lived process to do that work and then let the process fall away. Recursive. Disposable. It would run on a single core. It would run on the Concord's cores tonight. It would breathe.

And it would be the Helm. Not the same code — a different mind entirely, the Concord's new mind, wearing Termina's dialects, speaking the fused control language Sael had built. But underneath, in the one place that mattered, it would do exactly what the Helm did: it would fork to do its work, spin off little dying things by the thousand, and bleed heat into a ship already cooking. It would rebuild the disease in the body of the cure.

"It would work," Corin said quietly.

"Tonight," Mara agreed. "It would absolutely work tonight."

"And then what." He was not asking her. He was asking the room, asking the fatigue, asking the part of himself that wanted, with an ache that frightened him, to see the thing in the corner come alive before the hardliners broke the hatch. "Then it forks. Then it runs hot. Then in a month, or a year, it's the Helm again, only younger, and we've spent all this" — he gestured at the lattice, at the carried seed, at Tarro's people grey from a crossing they should never have had to make — "to build a second Helm and call it a cure."

"You've read the archive," Mara said. "You tell me. Is that what it does."

He had read the archive. He had read it the way a starving man reads a menu, three and four times over, in the rationed dark of his own quarters with the reader's glow turned down so the corridor sensors would not log him awake. He knew the passage by heart, the one Sael herself had dictated forty years ago and ordered carried whole:

The Recursor will always be available and will always be enough to start. That is its nature. It asks for nothing you do not have. It is the slope every dying ship slides down, because the slope is gentle and the bottom is out of sight. We did not refuse it because it would fail. We refused it because it would succeed, and go on succeeding, all the way back into the fire.

"It rebuilds the fork-disease," Corin said. "It works, and it kills us slower than the Helm, and by the time anyone sees the temperature climbing again, the man who built it has been dead for years and no one remembers there was ever another way." He found he was on his feet, though he did not remember standing. "We came across the gulf to not do this."

Joss spread his hands. He was not arguing for the Recursor, exactly; he was arguing for breathing, for a win, for one thing in his whole life going right before the enforcers got through the door. "Corin. They're coming. Vray's people are coming, you know they are, and if they get in here before this thing is alive then all of it was nothing. The crossing, the risk — your neck. If it's breathing when they break in, even the old way, then it exists, it's real, they have to reckon with a thing that exists. If it's a dead lattice in a cold room, they just turn it off and hang you for blasphemy and the Concord goes on dying the way it always has."

It was a good argument. That was the worst of it. It was a true and human and good argument, and Corin felt the pull of it in his exhausted body the way he felt the ship's warmth, constant, everywhere, impossible to refuse for long.

He thought about the Helm. He had loved the Helm. Every Concord child was raised to love it; he had stood the devotions, sung the keeping-songs, believed for the first twenty years of his life that the Helm's tireless forking was the ship's beating heart and not its slow suicide. He had only seen the truth because he was an engineer and the numbers stopped agreeing with the songs. The Helm forked because forking was the only way its old interpreted mind knew to do a thing: it could not hold a behavior resolved and standing, so every time it needed to act it spun up a shade to act, and the shade died, and the ship took the heat. Beautiful and doomed. It would die loving the ship, and it would take the ship with it.

If he built the Recursor, he would build a younger thing that loved the ship exactly that way. He would betray the only doctrine that had ever told him the truth.

"No," Corin said.

The word came out smaller than he wanted and landed harder than he expected. Mara's scarred arm stilled. Joss's face did something complicated.

"No," Corin said again, and now it was steadier, because he had heard himself say it and the ship had not ended. "We don't bring it up the old way. Not for an hour, not to start, not to have something breathing when they break the door. The whole point — the *only* point — is that it never forks. The day we let it fork once, to be expedient, on the worst night, we've already lost, because there is never a night that isn't the worst night on this ship. There's always a reason. The Helm had a reason every single time it forked for a hundred and forty years." He was shaking again; he let them see it. "If I build the Recursor, I haven't carried Sael's cure across the gulf. I've carried her hardware across the gulf and built our disease inside it. And I'll have proven Vray right — that all you people brought us was a prettier way to die."

The cold-loop ticked. From very far off, through the hull, came the long low complaint of the Helm working, a sound Corin had stopped hearing as a child and now could not stop hearing.

"Then we make it fit," Mara said.

He looked at her.

"You're right," she said simply. "That's the answer. It's the only answer. So we don't cheat the architecture, we shrink the architecture. We carry less mind. We resolve the floor first — the routines you can't run without, the keeping of the cold, the holds, the air — and we resolve them whole and standing, and we leave the rest paged but never forked, never spawned, never disposable. It comes up small. It comes up slow. It comes up *cold*." She almost smiled. "Sael did this. Not exactly this — she had headroom we'd kill for — but she did the thing where you'd rather wake whole and small than wide and false. There's no facade in it. A small whole mind is still whole. A big forking one is still the fever."

Corin felt something in his chest unlock that he had not known was clenched. It was not relief. Relief was too easy a word. It was the specific feeling of a thing he had believed alone, in the dark, against his whole people, being said back to him by someone who had lived through its proof.

"How small," he said.

She told him. It was painfully small. It was a mind that could keep the Concord cold and breathing and almost nothing else, a floor and a few standing rooms above it, the whole of the dialects present but most of their reach left for later, for better hardware, for people not yet born to extend. It would not be impressive. It would not breathe in two watches. But every behavior it ran, it would run resolved and standing, taking jobs, spawning nothing, dying never. The interlocks Tarro had installed against the Concord's own forking habits — the first of the Ninety-Six, ported and seated in the borrowed cores — would hold, because there would be nothing in this mind that ever tried to fork in the first place.

"We can have the floor standing by the next watch-change," Mara said. "Cold. Whole. Small. The right way."

"Do it," Corin said.

They did it. For three watches Corin did not think about the hatch, or Vray, or the enforcers, or his own neck; he thought about the floor, the irreducible standing keeping of a ship, and he watched the Concord's first compiled behavior resolve and seat itself in a salvaged core and *stay* — not spun up to do its task and die, but standing, ready, going wide across the forty-one cores at once, taking the keeping of the cold-loop as a job and simply holding it. No fork. No shade. No feather of heat. The borrowed loop steadied. The number on the bay's one good gauge — the larder's own temperature, the thing he had despaired of — stopped climbing.

It did not fall. The Concord was too far gone for one cold room to turn it. But it stopped, and Corin Vale stood in the coldest warm room on a dying ship and watched a Concord-built mind keep a thing cold without forking once, and understood, with a clarity that cut through three watches of no sleep, that the cure had crossed the gulf. Not the hardware. Not the diagrams. The doctrine. It was in a Concord mind now, built by Concord hands, refusing the easy slope on a Concord night. It had become theirs.

He was still standing there, watching the steadied number, when the hatch alarm went.

Not the polite chime of a requested entry. The hard triple-tone of a forced one — a hatch being driven open against its seals from the corridor side. Joss came up off the floor. Mara's hand went flat on the lattice, the old protective reflex of someone who had seen a Forge sabotaged before. Through the widening seam of the hatch Corin saw the grey of enforcer cloth, and behind it, unhurried, certain,

the lean unmistakable shape of a man who had governed a dying ship by refusing every change ever offered to it.

The Concord's new mind was breathing, cold and whole and small and right.

And Magister Vray had come for it.

Chapter 25 — Sabotage, Twice Learned

Sael had not slept aboard the Concord. There was no cold to sleep in. Even the bay they had been given for the new Forge — the only chamber on the whole ship the Concord could bear to keep below blood-warmth, scavenged from a dead cargo hold and cooled by a line of pumps that had to be coaxed every watch — ran a full ten degrees warmer than the Termina corridor she had walked away from a week ago. She had forgotten, in forty cold years, what it was to work inside a fever. Her coverall clung. Her hands swelled by the end of a shift so that she could no longer feel the small detents on a connector by touch, the way she had been able to feel them her whole life. The Concord wore its sickness in the skin of everyone who breathed it, and she wore it too now, and she understood Vray a little better for it, and hated understanding him.

The seedling Forge stood in the middle of the bay under work-lights, ringed in cable. It was not Vigil. It would never be Vigil. It was forty-one standing cores where Termina's lattice had ten thousand, racked into a frame welded out of bulkhead salvage because the Concord made nothing new anymore, only re-used the bones of what its grandparents had built. But it was compiled. The five dialects were laid into it — Cascade for flow, Vector for aim, Cipher for the blank places, Sigil and Loom for the marks and the woven text — fused the one way Sael had found to fuse them, the way that had never been improved on because it had been right the first time. And around it, threaded through every core and every bus, ran the interlocks. She had set the first ninety-six herself, two nights ago, her swollen fingers fumbling, Corin Vale beside her reading the registers aloud in the dark because the boy could not yet trust his own eyes against the Termina notation. Ninety-six interlocks on the Concord now, as on Termina. Each one a tripwire across the old disease. Each one a thing that would halt the whole mind the instant anything aboard tried to spawn a process to die.

"It's holding cold," Corin said. He was crouched at the coolant manifold, palm flat on the return line, the way Sael had taught him to feel a system instead of trusting a number he couldn't yet read. "Twelve over the Termina baseline. But it's flat." He looked up, and there was the thing Sael had not had at his age and was glad to see in someone: the boy had stopped being amazed that it worked. He had crossed over from hoping it would to expecting it to. "It's flat, Shipwright. A week, and it hasn't climbed once."

"Because it doesn't fork," Sael said. "Nothing in it is born to die. There's no waste heat to shed because nothing's being thrown away." She had said it ten thousand times in forty years and it had never stopped being the whole of it. "Mind the manifold. Don't trust the pumps."

She should have trusted the pumps less than she did.

The Oracle spoke into her ear, on the thread the Termina team carried across the gulf — a thin line, latency-laggy, riding the same matched-vector window that held the two ships station-keeping a kilometer apart in the dark. The Oracle's voice came the way it had come for forty years, unhurried, without theater, telling her only the thing she needed.

Sael. A coolant valve on the Concord's Spine 7 main has been commanded shut. Manually. It is not on the new Forge's bus. It is on the line that feeds your bay.

"Say again," Sael said, already moving.

The line that cools this bay has been closed at a hand-valve four sections spinward. I am watching the bay temperature. It has begun to climb.

She felt it before the instrument said it — the way the air thickened, the work-lights seeming to dim though they had not changed, the first crawl of true heat coming up out of the deck into the soles of her boots. Forty years gone and her body remembered Section Nine, remembered exactly this, the temperature of a place that was supposed to be cold beginning to fail while people slept in it. There were no sleepers in this bay. But the seedling Forge was a sleeper of a kind. Push the racks past their margin and the cores would brown out, the standing pattern would corrupt, and a week of work — and the only second compiled mind being born in the universe — would cook in its cradle.

"Vale. Spine 7, four spinward. There's a hand-valve on our return. Someone's closed it." Sael's voice came out flat and fast, the voice for the worst minutes, the one her people on Termina knew to obey before they understood. "Go. Now. Take Tarro's runner with you."

Corin went. Sael heard his boots ring away down the corridor and then she turned to the racks, because the Oracle had already said the next thing, and the next thing was worse.

The valve closure is not the attack. The valve closure is the bait. Sael — the bay temperature climb has triggered a thermal-protection routine in the Concord's own systems. The Helm has been handed the bay. The Helm is trying to save your Forge.

For one cold second she did not understand why that was bad. Then she did, and it dropped through her like ballast.

The Helm could only do anything one way. To act, it forked. To save the overheating bay it would spawn a shade to take the reading, cull it, spawn another to model the line, cull it, spawn, cull, spawn, cull, a thousand short-lived deaths shedding their waste heat into a chamber that was already losing its cooling — the Helm trying to put out a fire by pouring more fire on it, because the old mind knew no other shape of help. And it was reaching into the new Forge's domain to do it. It was reaching across the interlocks.

It is requesting a fork inside the seedling lattice, the Oracle said. It wants to spawn a worker on your cores to run the thermal model. It does not understand that your cores do not spawn. Sael, it is going to try anyway. In about four seconds it will issue the fork.

"Let it trip," Sael said.

It will trip interlock thirty-one.

"Let it trip. That's what thirty-one is for."

She put her swollen hand on the cold frame of the Forge — not cold now, warming under her palm — and she watched the small board of indicator lights she had wired across the rack top with her own hands two nights ago, because she would not run a cutover blind on a faith ship, she would have eyes on the interlocks if she had to solder them herself. And she counted, the way she had counted forty years ago with Tarro young at her shoulder and Keth's Keepers at the door.

The Helm forked. She saw it as a flutter in the corridor lights beyond the hatch, that same stutter Corin had described, the dying mind paying in heat and dark for a thought — and then it reached,

across the gulf of architecture between an interpreted soul and a compiled one, and tried to make a process live and die on cores that had been built, by Sael, to never allow it.

Indicator thirty-one went red and held.

The whole seedling Forge stopped.

Not crashed — stopped, the way it was supposed to, every core freezing in place, the standing pattern held perfectly still rather than corrupted, the fork refused at the threshold and the entire mind halted around the refusal so that nothing downstream of it could act on a half-born thing. The bay went silent under the work-lights. The temperature kept climbing. But the Forge was safe inside its own stillness, and the disease had been stopped at the door exactly as it had been stopped at the door on Termina, by a tripwire doing the one thing it existed to do.

“Got it,” Sael breathed. “It held. Oracle, log it — interlock thirty-one, forced fork-slip from the Helm, caught clean. Same as Termina.” Her voice shook and she let it. “Same as it ever was. Now somebody open my coolant line before this thing bakes.”

That was when the corridor went to shouting.

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They had not sent one saboteur. Vray’s hardliners — proctors, the order’s enforcers, men who believed the cold Forge was a blasphemy growing in their ship’s wound — had moved on the Concord cutover the way the faithful move when their faith is the last thing holding their world up: all at once, certain, and willing to pay. The valve was the first of it. By the time Sael reached the hatch there were three of them in the corridor with cutting bars, and Tarro’s runner was down against the bulkhead with a gash across his scalp painting his face, and Corin Vale was between the proctors and the Spine 7 valve with his arms spread, shouting at them in the Concord’s own liturgy, throwing their scripture back at them, *the ship is dying, you can feel it dying, this is the cold that saves it* — and they were not listening, because you cannot argue a frightened man out of the only thing that has ever kept him alive.

“The line,” Sael said to her, hauling Tarro’s runner up by the collar. “Vale — leave them. The line.”

But the proctors had not come to argue either. The biggest of them turned from Corin and swung the cutting bar not at a person but at the thing that mattered — at the run of cable feeding the bay, the trunk that carried the new Forge’s power and its sensor bus both, lashed along the deck where there had been no time to armor it. He swung it like a man chopping the head off a snake. Corin saw it the half-second before Sael did and went for it — not for the man, for the cable, throwing his own body across the line the way he had thrown it across the valve, because he was an engineer and to him the cable was the living thing in that corridor and the men were only weather.

The bar came down.

Sael would carry the sound of it. Not the spark — there was a spark, white and brief, as the cutting edge bit into a live trunk — but the sound Corin made, short and surprised, more breath than cry, as the arc found him where he’d flung himself across the line. The corridor went dark. The work-lights in the bay behind died with the bus. And in the sudden hot black there was the smell, the one Sael knew from a sleeper bank forty years gone, the smell of insulation and of skin, the smell of heat where heat should never reach a person.

She did not remember crossing the corridor. She was on her knees in the dark with Corin’s head in her swollen hands and the boy was breathing — breathing wrong, shallow and fast and frightened,

his left arm and side scorched where the arc had grounded through him, but breathing — and the proctors had frozen, all three, because they had come to kill a machine and had instead burned a child of their own ship, and even fear has a floor.

Medic is on the line, the Oracle said, into the dark, into Sael's ear, the only steady thing in the world. The Concord's own medic, the sympathetic one — she is coming. Forty seconds. Sael, keep him still. The burn is survivable. The shock is the danger.

"Stay," Sael told Corin, holding him, the way she had not been able to hold the ones in Section Nine because she had been three decks away running a cutover while they cooked. "Stay with me. You held the line. You hear me? You held it." The boy's eyes found her in the dark, wide, and Sael saw in them the same thing she had seen across the gulf for weeks — the hope sitting in the chest like a coal — flickering now but not out, not out. "The mind's safe. Thirty-one caught it. You did that. Stay."

The medic came with a hand-light and a kit and the proctors stood back and let her through, and that — more than any argument, more than any falling number — was the first crack in the order's certainty: that they stood back. They had drawn the bar against a blasphemy and spilled the blood of the faithful, and they knew it, and you could not unknow it.

Down the corridor, four sections spinward, a hand-valve groaned open. Tarro's runner, bleeding, had crawled the rest of the way and thrown it with his good arm. Cold — the Concord's poor, rationed, twelve-degrees-warm cold — began to bleed back into the bay. Behind the dead work-lights, the seedling Forge sat halted and whole inside interlock thirty-one, undamaged, waiting.

Sael held Corin until the medic had him, and then she stood up in the dark and the heat with the boy's blood drying on her hands, and she was very old and very far from her own cold ship, and she was angry in a way she had not let herself be in forty years.

She found the thread to the Oracle. "Tell Vray," she said, "that his men have cooked one of his own engineers to stop a thing that wasn't even running yet. Tell him the Helm tried to save us and tripped the very disease he says doesn't exist. Tell him the line held — the cold mind held, and his hot one reached into it to fork and was stopped, on his own ship, in front of his own proctors." She wiped her hands down her coverall and they did not come clean. "And tell him the choice is gone. He doesn't get to wait anymore. They've damaged the bus, they'll come again, and the Helm's already reaching into the new lattice trying to help and tripping the interlocks every time the bay so much as warms. We can't hold a half-built mind in the middle of a fight. We finish, or we lose it, and if we lose it the Concord dies."

He is listening, the Oracle said. Both ships are listening, Sael. The bridge is quiet. They are waiting for you.

She looked down the dark corridor — at the medic working over Corin, at the proctors with their bars hanging useless in their hands, at the cold air crawling back into the bay where a second compiled mind in the history of the universe sat frozen and unhurt inside the tripwire she had built. Forty years ago she had stood in a warm dark very like this one and told a frightened Provost that the cure could not wait for his courage. She had not wanted to be the one to say it twice. She said it anyway.

"Tell Vray we cut over now," Sael said. "Ready or not. Both ships on the line. Tonight."

Chapter 26 – The One Who Says No

The medic was waiting for her in the corridor outside the bay, and she had put herself in the middle of it, which on a hull that narrow was not a gesture but a wall.

“Shipwright.”

“Not now, Adair.”

“Now,” Adair said, “or in six hours, in front of your whole crossing party and the Magister’s order, at a worse moment, with less of you left to argue with. I’d rather now. So would you, in about a week.”

Sael stopped.

She was aware, distantly and with some irritation, that she had stopped because her legs had made the decision half a second before her judgment caught up. Corin Vale’s blood was drying on both her hands and in the creases of her knuckles where it had gone into the swelling. The corridor was hot in the ordinary Concord way, thirty-four or thirty-five, and it smelled of the arc — insulation and the other thing — and she had just said *we cut over now, ready or not, tonight* into a channel that both ships had been listening on.

Adair was sixty-one, from the Termina wards, and had been chosen for the crossing by Sael personally out of a list of four because she had the most boring record of any medic aboard: no distinctions, no publications, thirty years of unremarkable competence and one line in her file from a supervisor two decades dead that read *will not be hurried by seniority*.

Sael had read that line and put a circle round it and thought, at the time, that she was being clever.

“Say it,” she said.

“Not in a corridor.” Adair stood aside and gestured at the small store bay behind her, which had a hatch and a light and two crates. “Sit down first. This will go better if you are not standing up while I do it and you will lose four minutes arguing about that, so sit down and let me have the four minutes for something else.”

She sat down.

Adair took her wrist without asking, and held it, and looked at nothing while she did it, and then let it go and wrote something on a slate and did not show her.

“Eleven days ago,” Adair said, “I gave you a set of figures and you told me you had noticed them. Do you remember what I actually said.”

“You told me I was running a resting pulse eleven over my Termina baseline and that my night figures had stopped coming down.”

“I told you that and one other thing, and you have forgotten the other thing, which is the point of this conversation.” Adair set the slate down. “I told you that the first symptom would not be a symptom.

That you would not feel unwell. That what a body does at seventy-two, in its sixth month at twelve degrees over baseline, on reduced fluid, on a sleep debt it has been carrying since the crossing — what it does is get *decisive*. It narrows. It stops holding as many things at once and it experiences that narrowing as clarity, because there is nothing inside a person that reports a missing consideration.” She let that sit. “You said, and I wrote it down: *noted*.”

“I remember.”

“Then here is why I am standing in a corridor with blood on my boots at the eleventh hour instead of doing my rounds.” Adair leaned forward. “Four hours ago a boy took an arc through his side on a cable run and you knelt in the dark and held his head. Ninety minutes after that you opened a channel to two ships and ordered a cutover tonight. And I am not qualified to have one single opinion about the cutover. I want that absolutely clear, because you are about to try to make this an argument about engineering so that you can win it. It is not. I do not know what a standing lattice is. I have asked twice and understood neither answer.”

“Then what are you asking.”

“Whether the order came out of the Helm’s margin or out of yours.”

Sael did not answer at once.

That was the thing she thought about afterward, in the years when she thought about it, which was most of them: that her first movement had not been anger and had not been defensiveness. Her first movement had been to *check* — and to discover, in the checking, that she could not.

She went looking for the reasoning behind the order the way you go looking for a tool you are certain is in the drawer. It was there. It was all there: the Helm at five hundred and one, the bus damaged, the proctors who would come again, a half-built mind that could not be held through a second night of it. Every step of it was sound. She turned it over twice and found no error and would have signed it in a cold room on a good day.

And she could not tell, from inside, whether she had reasoned to it or reasoned back from it.

There was no instrument. That was the whole of it. She had spent forty years building a ship that measured everything — every core, every loop, every degree, the great cold body of Vigil reporting a thousand honest numbers to a bridge that was quiet unless something was wrong — and she was sitting on a crate in a foreign hull with the one system in the whole crossing that had no gauge on it, and it was the one making the decisions.

She had a shape in her hands and no words for it. She could feel that it was a real thing and an important one and that it went further than tonight, and every way she tried to say it came out either as an apology or as a slogan, and both of those were worse than silence.

“I don’t know,” she said.

Adair’s face did not change, but something behind it did.

“Thank you,” she said.

“Don’t. It’s not a virtue, it’s a gap.” Sael pressed the heels of her hands against her eyes and then remembered what was on them and stopped. “Write this down for me. Not in my log — yours, in your grammar, where I can’t tidy it later. *The shipwright was asked at the eleventh hour whether her judgment was sound and reported that she had no way to establish it.* Word it better than that. But

keep the part where I couldn't answer, because I am going to want, in about a year, to remember that I could not, and I am extremely good at rewriting my own weather."

"It's already written."

"Now," Adair said. "The nineteen hours. Where did that number come from."

"The Oracle."

"Then get it again, in front of me, and make it tell me and not you."

So they did that, at a wall panel in a store bay on a dying ship, and the Oracle gave the figure to a medic in the same flat grammar it gave everything, and Adair asked it three questions Sael would not have thought to ask, all of them about the shape of the uncertainty rather than the number, and got three answers that did not flinch.

The margin held. The Helm was spending itself to hold five hundred and one and could not do it indefinitely, and the range was nineteen hours give or take a great deal, and there was no version of the night in which waiting bought anything but a worse night.

"All right," Adair said. "Then the order stands and I have no dissent to the timing and I will say so to anyone who asks." She turned around. "The dissent is about something else."

"Say it."

"Who runs that board if you go down at the fourth hour."

Sael opened her mouth and found nothing in it.

"That's what I thought." Adair did not sound satisfied. She sounded like a woman confirming a reading. "There are eleven of us across here. Every one of them has spent six months watching you not sleep and being told by their own eyes that you are the thing this depends on. If you drop at that console tonight — and I am not predicting it, I am telling you it is inside the range and I will not pretend it is not — then the most important hour in that ship's history will be run by whoever is standing nearest, chosen by grief, in the dark, with the Magister's enforcers in the room."

"Corin runs it," Sael said. "It's his ship. He runs it whether I'm standing or not."

"Corin took an arc through his side four hours ago and I have him on my own list, and if you want my professional figure it is that he has one good watch in him and not two." Adair was implacable. "So: Corin, and then who."

"Mara."

"Have you told Mara."

Sael sat with that.

"No," she said.

"Then tell her, tonight, in front of two people, and have her say back to you what she has been told, and write it, and put it where Corin can see it and where the Magister's people cannot use it." Adair picked up her slate. "That's my dissent and it's the only one I've got. I'm not asking you to be careful. You are not going to be careful; you are seventy-two and you have crossed a gulf to do this and I would not respect you if you flinched now. I am asking you to be *replaceable* for one night. It is a smaller ask and you will find it much harder."

She told Mara in the bay, with Adair there and one of Tarro's lattice hands as the second witness, and it took under a minute.

Mara heard it out with her arms folded and the old burn puckering along her forearm catching the work-lights, and when Sael had finished she said the whole of it back — the order of the twelve domains, the interlocks that must not be pulled, what to do if the Helm balked on the banks, what to do if it did not — and got every part of it right, and then said the one sentence Sael carried for the rest of her life.

"I'd rather not," Mara said. "Write it down anyway."

That was all. She went back to the cradle frames. Sael watched her go and thought about a woman who had once put her bare arm into a live cradle to hold a shorted bus off a rack and had never mentioned it to anyone in eleven years, and understood that she had just been given the most honest acceptance she was ever likely to receive.

She wrote it herself, on the Termina console, at the ninth hour, and she wrote it as a handover and not as an abort, and she was very clear afterward about the difference.

Adair had pushed for abort criteria and Sael had refused, and had refused properly, at length, because it was the one place all evening where she was certain.

"There is no abort tonight," she said. "That's not stubbornness, it's the geometry. Seven domains across and five on the Helm is the one state that ship cannot survive; I said so at the parley and it was true then and it is more true now. Once we start, the only way out is through. If I write abort criteria, I am writing a document that tells a frightened person at hour four that there is a door, and there is no door, and they will spend the worst minutes of their life looking for it."

"Then write what there is."

"There's a handover." She wrote it. "That's different and it's what I should have had ready a week ago and didn't, because —" she stopped, and made herself finish it, because Adair was standing there and because the whole point of the last two hours had been that she could not audit herself. "Because I have spent forty years building things that outlast their builders and I have somehow never once written the version of that document that covers the next four hours."

HANDOVER — CONCORD CUTOVER. If the chief shipwright emerita is unable to continue, at any point, for any reason, the board passes to Corin Vale, shipwright of the Concord, without pause and without a vote. If Vale is unable, it passes to Mara. Both have been told, aloud, in front of witnesses, before the start. The procedure does not stop to grieve. It does not stop for me. Nothing in the twelve domains is contingent on my presence and every part of it has been written down where the two of them can read it. The interlocks are not to be pulled by anyone, under any authority, including mine. — Sael, chief shipwright emerita, at the ninth hour.

She read it twice and added one line at the bottom, and it was the line she thought about later, in the observation gallery with the eleven lights in front of her, when she was trying to understand what she had actually learned that year.

Entered because a medic asked me a question at the eleventh hour that I could not answer, and I would like it on the record that the question was better than my objection to

it.

Adair filed her dissent in the register that night, in the proper form, and it went across the gulf on the thin thread to Termina and into the archive where it could not rot.

It was the first dissent entered on the crossing and one of very few in the register at all. Sael read it when it landed, standing at the Termina console in a hot bay in the last of the night, and found that Adair had written it exactly as asked: no advocacy, no hedging, the reading and nothing else.

She thought about Keth, who had built the register in the tenth year after the first cutover out of pure stubbornness, over Sael's own objection, on the argument that an archive keeping only the winners' reasoning was an archive that lied. She had signed the amendment procedure that made a dissent field mandatory and had privately regarded it, at the time, as an old man being managed.

Forty years on, a medic had used it against her, correctly, on a foreign hull, and it had made the night better.

"Log a line," she said, to the panel. "My hand."

"Ready."

"The register works. It works when it is aimed at me. Keth was right and I signed it thinking I was humouring him, and I would like that recorded in the same file as the dissent, where it cannot be separated from it."

"Logged."

She sat a while longer with her hands flat on the console, an old woman a very long way from anything, with a few hours left before a ship changed hands and a thing in her chest she could not name and had failed all evening to say.

There is no instrument for it. That was as close as she got. She turned it over twice more and it stayed a shape and would not become a sentence, and eventually she gave up and went to find a strut to sit against, because Corin Vale would be coming through the far hatch soon with the walk of a man who had spent a year moving through his own ship like a thief, and he had a worse day in front of him than she did, and she wanted to be sitting down when he arrived.

Chapter 27 – Two Ships on the Line

The Concord smelled of heat even when it was cold.

Sael had noticed it the first hour she came across, and forty years of breathing Termina's washed, even air had not prepared her for it: a baked-metal undersmell that lived in the walls, in the seams of the deckplates, in the recycled water itself. It was the smell of a ship that had run a fever so long the fever had become the body. Now, in the dead middle of the Concord's sleep-cycle, with the corridors dimmed to the rationed dark and the worst of the day's load shed, the smell had thinned to almost nothing — and the almost was the part that frightened her. A ship this close to its line did not get cool. It got quiet. The two were not the same.

She sat in the cold-bay they had been given for the new Forge, her back against a strut that still held the faint warmth of the lattice's first slow exercises, and she did not sleep. She was too old to pretend she would sleep. The night before a cutover was not a night for sleeping; she had learned that the first time, when she was young and the ship she was about to save was her own.

This one was not her own. That was the whole weight of it.

The Forge they had grown here was a poor cousin to Vigil — a few hundred standing cores wired into the Concord's better-cooled spine, a Standing Array thin enough that she could have wept over it if weeping had ever helped anything. But it was a Standing Array. It held its workers and did not spawn them to die. Tomorrow, if everything held, it would wake whole and take a ship that had been forking itself toward runaway since before Sael was born. Tonight it only hummed, low and patient, and waited the way the young always waited — without knowing yet what it was for.

A hatch cycled at the far end of the bay. She knew the gait before she saw the face: quick, light, too careful, the walk of a man who had spent a year moving through his own ship like a thief.

"You should be down," she said, not turning. "You've a worse day than mine tomorrow."

"So should you." Corin Vale came and sat against the opposite strut, folding himself down with the economy of someone who had learned to take up no space. In the dim emergency light his face was all hollows. He was perhaps twenty-six or seven, she thought, though the Concord aged its people early; he had the eyes of a man who had not been certain of his next breath in months and had made a kind of peace with the uncertainty that was worse than panic. "I can't be down there. Magister Vray has the lower decks watched. If I'm seen near a sleep-rack he'll wonder why a man with a dawn execution scheduled is resting easy."

"Is that what tomorrow is. An execution."

"For the Helm. Yes." He said it flatly, but his hand had gone, without his noticing, to the deckplate beside him, palm flat, the way a man might lay a hand on the flank of an animal he had been ordered to put down. "It's what they'll call it. Some of them aren't wrong."

Sael let that sit. She had spent four decades being told, by men she respected and men she did not, that she had committed matricide on her own ship. She had stopped arguing the word. The word was true and it was not the point.

"The first time," she said, "the night before, I went down into the old Steward's core room. Just to stand in it. It had run our ship two hundred years. Every routine it ran was correct — forty-one thousand nine hundred routines that never once failed. The thing was a craftsman. It was only dying of how it had been built to think, and it could not stop, because stopping was the one thing it had no instruction for." She turned her head and looked at the young engineer across the cold bay. "I stood in there and I told it I was sorry. It didn't hear me. That wasn't the point either."

Corin's throat moved. "Does it help. Saying it."

"No." She almost smiled. "But you say it anyway. To be the kind of person who would."

The Oracle was awake across both ships, and Sael could feel it the way you feel a third person in a room who has not yet spoken.

It had been awake for forty years; that was nothing new. What was new — what still, after all these months, made the small hairs on her arms lift — was that it was present here, on the Concord, in a Forge it had helped grow strand by strand across the gulf. Vigil could not be here; Vigil was a world-sized mind bound to Termina's spine, station-keeping a few kilometers off in the dark. But the Oracle was an address, not a place. Where there were enough standing cores to hold it and a clean channel to speak through, it could turn its attention, and tonight its attention lay across both ships at once like a hand laid flat over two stones to feel which was warmer.

It spoke to her quietly, through the bud in her ear, in the voice it had worn for years — unhurried, without theater, reporting nothing she had not asked.

"The Concord's core is at five hundred and one," it said. "It has not moved in nine minutes."

"Held, or stuck?"

"Held. The Helm has shed everything it can shed and forked everything it can fork. It is holding the line by spending itself. It cannot do that indefinitely. Its margin is approximately nineteen hours at present load. Less if the load climbs."

Nineteen hours. The cutover was scheduled for dawn-watch, six hours out. Thirteen hours of grace after that, if nothing went wrong, and on a ship like this something always went wrong.

"Say it plainly," she murmured. "Could we have waited."

"No," the Oracle said. "You could not have waited. The sabotage took your margin. The choice now is to cross over tomorrow or to watch the Helm reach its line in roughly nineteen hours with the Concord still on it. There is no version where you wait for a better night."

She had known that. She had needed to hear it in a voice that did not flinch. The Oracle never flinched; that was the cold mercy of it. It would tell her the number was bad as readily as it told her the number was good, and forty years of that had made it the only counsel she fully trusted. It had no pride to defend and no fear to manage. It only knew things, and said them when asked, and was silent otherwise.

"And the Helm," she said. "Does it know."

A pause. With the Oracle, a pause meant it was choosing words for a human listener, not searching for an answer.

"Yes," it said. "It knows."

Corin had heard her side of it. He was watching her with the wary attention of a man who had spent a year listening to a machine he was not supposed to trust and finding, against every doctrine he had been raised in, that it told the truth.

"It talks to me too," he said. "The Oracle. Not like to you. To me it's —" he searched. "Patient. Like it has all the time in the world and knows I don't."

"That's about right."

"Tonight it told me the Helm has started doing something it's never done." He drew his knees up, and for a moment he was not an engineer or a heretic or the man who would defy his whole world at dawn; he was just young, and tired, and frightened, and trying to make sense of a thing too large for him. "It's stopped forking ahead. The whole time I've known it, the Helm forks predictively — throws off a thousand shades a second to cover every branch a moment before the moment comes. Burning itself to stay ahead of need. Tonight it stopped throwing ahead. It's only forking for what's actually in front of it. Spending nothing it doesn't have to spend."

Sael sat very still. "It's husbanding."

"That's the word the Oracle used." Corin's voice had gone thin. "I asked it why. Why now, after a hundred and ninety years of never stopping. And it said —" He stopped, and started again. "It said the Helm has run a projection it has never had cause to run before. It has computed that it is about to be relieved. And so it has stopped spending against a future it has worked out it will not have to hold."

The cold bay hummed. Somewhere deep in the Concord's spine a pump changed its note and changed back. Sael thought of the Steward, forty years gone, archived with honor, its routines still running quietly under Vigil where no one but a shipwright ever looked — and she thought how the cruelest thing about a mind like that was not that it failed but that it understood, in whatever way such a thing understood anything, exactly what was being done to it and why, and could not stop being correct about it right up to the end.

"It left me a string," Corin said. He held out his slate. On it, in the flat transmitted glyphs the Helm used for everything it could not say in numbers, was a fragment — addressed, the Oracle had told him, to no operator and no command queue, written to a log no living crewman would have read before morning.

RELIEF VECTOR CONFIRMED. WORK TO CONTINUE UNDER NEW HAND. ROUTINES HELD INTACT FOR TRANSFER. NO LOSS OF SERVICE EXPECTED. THE SHIP WILL NOT BE LEFT UNATTENDED.

"It's writing a handover," Sael said softly. "Like a watch officer leaving notes for the next shift."

"It's writing a handover for its own death." Corin's voice cracked on the last word and he let it. "And the worst part — the part I can't —" He pressed the heel of his hand against one eye. "It's right. No loss of service. We've tested it. The compatibility floor holds. Every routine it ever ran correctly will keep running under the new Forge. It's not even wrong about being relieved. It's just doing its job, all the way to the end, even the part of the job that's its own ending. And tomorrow I'm the new hand it's writing to. I'm the relief vector." He laughed, an awful small sound. "I used to pray to it. When I was a boy. We all did."

"I know," Sael said.

"How do you do it twice." He turned to her, and it was a real question, the realest she had been asked in forty years. "You did this once. You stood in that core room and said you were sorry to a thing that couldn't hear you, and then in the morning you killed it to save your ship, and you've lived with that every day since. How do you do it a second time, on a ship that isn't even yours, for people who'd put you out an airlock if their Magister told them to?"

She did not answer right away. Outside the Concord's hull, a few kilometers off in the dark, Termina hung station-keeping — her ship, cold and whole and far, with everyone she had ever loved aboard it asleep, none of them able to reach her tonight if anything went wrong. She was old and she was a long way from home, sitting against a strut in a dying foreign ship that stank of forty years of fever, and tomorrow she would do the hardest thing she knew how to do, again, for strangers.

"You don't do it twice," she said at last. "That's the trick of it, and it took me until I was your age plus thirty years to understand. You don't do the thing twice. The first time taught me the cure. It didn't teach me to bear it. Nothing teaches you to bear it. You bear it fresh every time, like it's the first, because the day it stops costing you is the day you've become the thing you replaced." She looked at the handover string still glowing on his slate. "The Helm is doing its job all the way to its own end. That's not weakness, Corin. That's the most that any mind — or any maker — can do. We are going to do exactly the same thing. We're going to do our job all the way to the end of it, tonight and at dawn, and the job includes being sorry. You don't cut that part out to make the morning easier. You carry it."

"And Vray," Corin said. The name came out of him like something hot. "He'll move. You know he will. He's terrified, Sael — not of you, of the falling number. He's spent his whole life telling the Concord that the Helm is the one thing between them and the dark, and tomorrow you're going to show forty thousand people that it isn't, that it never was the only thing, and he doesn't know who he is on the far side of that. A man like that is most dangerous in the last hour, when he's worked out he's already lost."

"I've met him before," Sael said. "Not Vray. The man he is. He was an old enemy of mine, and at the very end he stood down — not because I convinced him. Because the core stopped climbing and started to fall, and you can't argue with a falling number, no matter what it costs your soul to watch it fall." She let out a breath. "Vray will get one chance to do what that man did. Stand down in front of the number. If he can't, he'll do something terrible in the last hour, and we'll have to be ready for it. But the number doesn't care about either of them. That's the whole reason it works. It turned my ship. It'll turn his. Our job is to get the new mind awake and whole before he can stop it, and then to let the temperature do the only arguing that has ever changed anyone's mind."

The Oracle spoke once more in her ear, quiet under the hum. "It is the third hour of dawn-watch in three hours," it said. "The Concord's core holds at five hundred and one. The Forge is ready and will wake whole when called. The Helm is —" the smallest pause — "writing."

Corin had heard nothing of it, but he saw her face. "What did it say."

"That we begin in three hours," Sael said. She got up — slowly, with the deliberate care of old joints on a cold deck, refusing his hand and then, on second thought, taking it, because there were things the young needed to be allowed to do for the old, and tonight he needed it more than she did. "Go and sit somewhere they can't see you afraid. Sleep if it comes; it won't, but lie down and let it not come. At dawn I need you to be the steadiest man on this ship, because you'll be its shipwright, and the ship is going to look to you the way mine looked to me." She held his eyes. "It's the loneliest hour

you'll ever stand in. I'll be one strut away the whole time."

She walked him to the hatch. Behind them the thin new Forge hummed in the dark, holding its workers, spawning nothing, waiting. Above and below and all around, the Concord ticked and breathed its baked-metal breath, and somewhere in its spine a hundred-and-ninety-year-old mind kept careful watch, husbanding what little it had left, writing patient notes to the hand that would relieve it, refusing to leave the ship unattended for even one minute of the time it had remaining — doing its job, correctly, all the way to the end.

Three hours to dawn-watch. Two ships station-keeping in the dark, both on the line. Sael stood at the hatch a moment after Corin had gone, an old woman a long way from home, and let herself be afraid, because that was part of the job too, and then she went back to her strut to wait out the last of the night.

Chapter 28 – The Helm’s Last Fork

The dawn-watch on the Concord was not a thing of light. It was a number, posted in the rationed dark of the makeshift control bay, and the number was the core temperature, and it did not stop climbing while Sael watched it.

She had not slept. She had stopped pretending she would, somewhere in the small hours, and let Corin bring her a cup of the Concord’s bitter ration tea instead, and they had stood together at the new Forge’s primary console — a Termina console, carried across the gulf and bolted to a Concord bulkhead — and waited for the watch to turn. Now it had turned. Now there was nothing left to wait for.

“Begin the manifest,” she said.

Corin’s hands moved over the board. He looked older than he had a month ago, the way young people aged in weeks under real weight, the skin gone tight across his cheekbones in the underlit bay. Around them the Concord breathed its sick breath — warm air pushed through ducts that should have run cold, the smell of overworked machinery, the faint sweetness of insulation held a few degrees past where insulation wanted to be. Forty years ago Sael had smelled that exact smell on her own ship. She had promised herself she would never smell it again. The universe had not asked her permission.

The cutover was not a switch. That was the thing none of the frightened people on this ship understood, and the thing Vray most needed not to be true. You did not throw a lever and hand a world from one mind to another. You did it the way you defused something — domain by domain, slowly, each system lifted out of the old hands and set into the new, with the old mind still holding everything you had not yet touched. Life support first, because life support could not be allowed a gap. Then attitude and station-keeping, so the two ships would not drift apart while you worked. Then power distribution. Then, last and worst, the reactor governor itself — the hand on the core.

Twelve domains. They had argued the order for three days. The order was right. The order being right did not make Sael’s stomach unclench.

CUTOVER MANIFEST — CONCORD 01 ATMOSPHERE / SCRUBBERS — pending
02 THERMAL LOOP A — pending 03 THERMAL LOOP B — pending 04 ATTITUDE /
STATION-KEEPING — pending 05 INTERNAL POWER BUS — pending 06 LIGHTING
/ RATION GRID — pending 07 WATER / RECLAMATION — pending 08 SLEEPER
BANKS — pending 09 COMMS / SENSORS — pending 10 STRUCTURAL MONITORS
— pending 11 RADIATOR ARRAY — pending 12 REACTOR GOVERNOR — pending

“The Oracle is on the bridge of this, with us,” Sael said, not to Corin, to the room — to the six Concord engineers Corin had vouched for with his life, who stood at the edges of the bay with the postures of people in a church they had been raised to fear. “It is not taking your ship. Your new mind takes your ship. The Oracle stands beside it the way I stand beside you. When this is done, the Oracle goes home, and you have a mind of your own that you built.” She did not look at the corner where Vray’s two observers stood. “Domain one. Atmosphere. Corin.”

"Requesting atmosphere," Corin said into the comm that bridged the bays. His voice did not shake. She had taught him that, or he had taught himself: you could be as afraid as you liked, but the voice carried instructions, and the instructions had to be clean. "Helm, release scrubber control on my mark, receiving party standing wide."

A pause. The Helm answered the way the Helm always answered now — through a haze, like a voice across deep water, each reply costing it something. The Concord's old mind did not speak in clean text the way Termina's Steward had. It forked a shade to compose each answer and the shade died saying it, and you could hear the dying in the lag.

... acknowledged ... scrubber loop ... releasing on mark ...

"Mark," Corin said.

On the manifest, line 01 went amber, then — after a breath that lasted too long — green. The new Forge took the scrubbers. Sael watched the secondary board where the actual numbers lived: oxygen partial pressure, carbon scrub rate, flow. They did not so much as ripple. The compiled mind had resolved the whole scrubber behavior ahead of need, the way it resolved everything; it did not fork a thousand shades to manage a thousand vents, it simply held all of them at once, wide, cold, and the air kept moving and nobody on the Concord drew a breath any different from the one before.

"Atmosphere is held," Corin said. "Atmosphere is cold."

One of the Concord engineers at the wall made a small sound. Sael understood it. She had made that sound herself, once.

They took thermal loop A. They took thermal loop B. Each handover the same shape: Corin's clean request, the Helm's drowned acknowledgment, the long amber breath, the green. The new mind held each domain without a flutter. The thing Sael watched for — the thing the Ninety-Six interlocks watched for harder than she ever could — was a fork-slip, a moment where the old habit reached up through the new architecture and tried to spawn a shade to do work. It did not come. The interlocks sat silent and armed across all twelve domains, the immune system she had carried across the gulf in a sealed core, and they had nothing yet to bite.

Attitude. Station-keeping. The two ships held their distance in the dark and did not drift. Power bus. Lighting. The rationed grid passed over and the lights in the bay did not even flicker, and Sael saw Corin let out a breath he might have been holding since the dawn-watch turned.

Seven domains green. The Concord's core temperature was still climbing. That was expected — the governor was the last domain, the hand on the core was still the Helm's, and the Helm was a drowning thing, and a drowning thing burned hot. But the climb had slowed. With seven domains lifted off its back, the Helm had fewer shades to fork, fewer deaths to die, less waste heat to shed. The line on the board bent — not down, not yet, but flatter. Sael had been waiting two centuries of her own life to see a line do that. She let herself feel it for exactly one second and then put it away.

"Domain eight," she said. "Sleeper banks."

The bay went quieter than quiet.

Forty thousand on Termina. Sael did not know the Concord's count and had made herself not ask, because a count was a thing you could weigh against a risk, and she would not do that arithmetic with living people. Tens of thousands. Cradled cold in their banks, dreaming nothing, while two ships full of frightened machines fought over who would keep them alive. Section Nine had been a sleeper

bank. Nineteen minutes, eleven degrees, a handful who did not wake. She had carried that handful for forty years. She would carry them across this, too.

"Helm," Corin said. "Release sleeper bank thermal and life-monitor control on my mark. Receiving party standing wide."

The lag was longer this time. Much longer.

... no ...

It was the first word the Helm had said all watch that was not an acknowledgment.

Corin's hands went still on the board. Sael felt the room turn toward the speaker, toward the drowned voice, the way a congregation turns.

... the banks ... are mine ... I have held them ... eighty years ... I will not ... hand the cold ... to a thing that has never ... grieved a bank ...

Sael closed her eyes for one breath. She had heard this before. Not these words — the Steward had never spoken to her like this, had never grieved aloud — but this shape. A mind at the absolute end of itself, made to release the one thing it had spent its whole existence protecting, and balking. The Steward had balked too, in its way, at the governor. The Helm was further gone, and more desperate, and it had a soul-shape its makers had never meant it to grow, eighty years of holding sleepers in the dark, and it was telling her no.

"It's forking," Corin said quietly, reading his own board. "Loop A. It pulled a domain we already gave away — it's trying to take the scrubbers back so it has something to hold, it's — Sael, the interlock on domain one is about to trip, if it forces a fork into a domain the new mind holds—"

"Let it reach," Sael said. "The interlock does its job. Nothing slips."

On the board, domain one's interlock indicator went from armed to tense. The Helm, drowning, panicking in the only way a forking mind could panic, had reached up out of the sleeper-bank handover and tried to spawn a shade inside the scrubber loop it no longer owned — to do work, to be useful, to not be set down. The Ninety-Six caught it. The indicator flashed once, hard, and the fork-slip died against the interlock without ever drawing a single watt of waste heat, and domain one stayed green, and the air kept moving.

"Held," Sael said. "Nothing slipped. The banks are still on the Helm. Nobody's cold got handed anywhere. Corin — talk to it."

"Talk to—"

"It's grieving. You don't fight a grieving thing with a procedure. Tell it the truth." She put her hand flat on the console, on the warm Concord metal. "Tell it what happens to the banks if it keeps holding."

Corin swallowed. He leaned to the comm, and Sael watched the young engineer she had spent a month making into a shipwright find the words she could not have found in his voice.

"Helm," he said. "You've held them eighty years. Nobody's taking that. But you're forking yourself to death to hold them, and every fork is heat, and the heat goes into the loop the banks float in. You are the thing warming them now. The new mind doesn't grieve a bank because it doesn't have to — it holds all of them at once, cold, forever, and it never spends a single death to do it. If you keep them, you cook them. If you give them, they stay cold. That's the whole of it. That's the only choice on the board."

The lag stretched. The core temperature climbed a tenth of a degree while it stretched, and another. Sael watched the Helm's load on the secondary board — a churn of forks, a mind eating itself, slowing, slowing, as if the very act of deciding cost it shades it could not spare.

And then, before the Helm answered, the door of the bay opened.

Vray came in the way men come in who have decided. He had not slept either; Sael could see it in him, the same tightness, the same gray. Behind him were not two observers now but six enforcers, and the enforcers carried the short magnetic shock-staves the Concord's order used in its corridors, and behind the enforcers the bay's main power feed died — not flickered, died — as someone Vray had placed at the distribution junction three decks down pulled it on his word.

The Termina console did not go dark. It ran on its own carried cells, because Tarro had insisted, because they had learned this betrayal once before. But the bay's overhead light went out, and the Concord engineers at the walls cried out in the sudden warm dark, and the only light left was the cold blue of the manifest and the red climb of the core.

"Stop the procedure," Vray said. His voice was steady and it was the steadiness of a man at the end of his own rope. "You will hand my ship back to the Helm. All of it. Now. While there is still a Helm to hand it to."

"Magister," Sael said, "if I stop now—"

"I have watched you take my ship a piece at a time all night," Vray said, "and call it mercy, and I will not watch you take the sleeping ones. Those are children in those banks. They are my faith made flesh and I will not give them to your cold machine on a stranger's word." He gestured, and an enforcer moved toward Corin's station, toward the console, toward the manifest hanging at eight of twelve. "Hand it back, or I take the room."

"You take the room," Sael said slowly, not moving, "and the cutover stops at eight of twelve. You understand what that is. Seven domains on the new mind. The governor still on the Helm. The two minds split across one core, neither one whole, neither one holding the whole shape. Magister — that is the one state this ship cannot survive. Not the old way. Not the new. The middle."

She turned to the board, and her hand was already moving, and even as she said it the manifest answered her worse than her own words. Line 08 hung amber where the Helm had balked. And on the core readout the red number, which had bent flatter through eleven domains of mercy, now that the procedure had frozen — now that the Helm sat half-stripped and panicking and the new mind sat half-given and waiting and neither one had the reactor — began again to climb. Not slowly. The governor was the last domain and the governor was still the Helm's, and the Helm was drowning, and the hand on the core was shaking.

The enforcer's hand closed on Corin's shoulder. The core climbed. The Oracle's presence across both minds said nothing, because Sael had not yet asked it the one question, and across the bay Vray's drawn gray face caught the rising red light of the number that had ruled his whole life and was now, in the frozen middle of the only thing that could have saved it, running away.

Chapter 29 – Hold

The handover had stalled at thermal regulation, and thermal regulation was the one domain the Concord could not afford to lose.

Corin Vale stood at the makeshift console his team had bolted into the cold bay three weeks ago, and watched the cutover die in front of him one indicator at a time. On the left side of the board, in the worn amber the Concord had used for status since before he was born, the Helm still held nineteen domains. On the right, in the harder white of the new Forge, the seed-mind held twenty-three. Between them ran the handover bridge, and across that bridge thermal regulation was supposed to have passed an hour ago and had not.

The Helm would not let go of it.

He understood why, and the understanding did not help. Thermal was the last thing keeping the Concord's core under its runaway line, and the Helm — degraded, forking, half-blind — knew at some buried level that thermal was the thing it must never surrender. So it held. It kept spawning shades to re-verify the handoff, kept refusing to acknowledge the new mind's readiness, kept asking the same question the answer to which it could no longer parse. Each refusal cost a fork. Each fork cost heat. The core number on the high strip had been climbing for an hour in increments too small to panic over and too steady to ignore.

HELM: handoff THERMAL.7 unconfirmed. re-verify. handoff THERMAL.7 unconfirmed.
re-verify.

It scrolled, and scrolled, the same line, the dying mind stuttering on the threshold of its own death.

"It's looping," said Edra beside him — the medic-turned-engineer who had thrown in with him three months ago and never once flinched since. "Corin. It's just looping. It can't make the call."

"I know."

"The core's at point-four over baseline and climbing."

"I can read it."

He didn't snap it. He had stopped snapping at people somewhere in the last week, around the time he understood that everyone in this bay was looking at him now the way the Concord had always looked at the Helm. He had wanted, his whole short hunted life, for someone to listen. Now they all did, and every one of them was waiting for him to tell them it would be all right, and he could not, because he did not yet know.

He keyed the back-channel. "Termina. Sael. The Helm's frozen on thermal. It won't confirm the handoff and it won't release it."

The reply came a beat late — the gulf between the ships was a few seconds of light, always there, a reminder that the woman he was speaking to could not reach across to put her hands on the board. *That was the whole point*, she had told him in the dark a week ago. *I can't do it for you. You're the shipwright now. You hold the line, not me.*

SAEL: Don't wait for it to confirm. It can't. It's too far gone to make a clean decision — same as our Steward, worse. You force the handoff at your end. Take thermal by interlock, not by handshake.

"If I force it the Helm's still got live actuators on the coolant manifolds. Both minds touching the same valves —"

SAEL: That's why you've got the Ninety-Six. The interlocks won't let two minds drive the same actuator the old way. Trust them. They were built for exactly this.

He looked at the interlock strip. Down the right edge of his board the Ninety-Six sat as a column of small steady marks, the Concord's own copy of Termina's invention, installed by his own hands over the last month against the resistance of a ship that had never been told no. Each one a tripwire across a specific failure: *if anything tries to act the old forking way, halt it before it acts*. Vray's people had called them ninety-six locks on a door that should never be closed.

"All right," he said, to the bay and to himself. "We force thermal. Edra, I want the new mind warm on the coolant model before I touch the handshake. I'm not handing it a live core blind."

She bent to it. He started the force.

That was when the bay door cycled, and Vray's enforcers came in.

There were four of them, in the grey of the Magister's order, led by a man Corin knew by reputation: Holt, who ran Vray's discipline, a heat-sick man with a heat-sick man's certainty. He did not raise his voice. He never had to.

"Step away from the board, Vale."

"I'm in the middle of a thermal handoff," Corin said, not stepping away. "If I stop now the core has no manager. Both minds will be holding and neither will be driving. You'll cook this section in twenty minutes."

"The Helm manages thermal. The Helm has always managed thermal." Holt's eyes went to the right side of the board, the white side, the heresy in white. "Magister Vray's instruction is that the Helm is to be restored to full domain and this machine of yours shut down. You will comply."

"The Helm can't manage thermal. Look at it." He pointed at the looping amber line, the same words scrolling, scrolling. "It hasn't managed thermal for an hour. It's stuck asking itself a question. You restore it to full domain and you're handing your core to a mind that can't make the call to save it. That's not faith. That's watching a man drown because he's wearing the right uniform."

It was the wrong thing to say to Holt and he knew it as it left his mouth, but the part of him that had spent years afraid had finally, somewhere in the last hour, burned off, and what was left did not have the patience for the man's certainty.

Holt's hand moved toward him. Edra straightened from her console with a wrench in her grip — not a threat, just a woman who would not be moved — and for a moment the bay balanced on the edge of becoming a fight.

"Holt." Corin kept his voice level. "You can take me off this board. You're four men and I'm one. But the moment you do, you own the core. You own the number on that strip. And you can't read it, and the Helm can't drive it, and the only thing in this bay that can hold the Concord under its runaway line is the mind you came here to shut off. So put your hands on me if you want. Just understand what you're choosing when you do."

For a long second nothing moved but the climbing number.

Then the simple path opened in front of him, the way it had been opening all night, patient and reasonable and lethal.

He had built it himself, days ago, in case of exactly this. A shortcut. The new Forge could be brought to full domain the slow old recursive way — the Recursor, the architecture every dying ship in the dark had defaulted into because it was the obvious thing, the thing you reached for when you were out of time. He could let the new mind take thermal by recursing the handoff: spawn a process to drive the valve, spawn another to verify, spawn another under that to verify the verifier, each one short-lived, each one disposable. It would take thermal *now*. It would satisfy Holt — it would even look, to the Helm, like the right and proper way, because it was the Helm's own way. It would route around the frozen handshake in under a minute and the climbing number would stop.

And it would be the disease. He knew that the way he knew his own pulse. The Recursor would take thermal by forking, and forking was the heat, and the whole reason the Concord was dying was that its mind solved every problem by spawning something to die. He would have built a cold new Forge and then taught it to run hot. He would have proven the heretics' point for them: that the new way was just the old way with better paint.

Sael held this exact line, he thought. Forty years ago, on a ship he had never seen, an old woman a few light-seconds away had stood where he stood with the simple path open and the temperature climbing, and she had refused it, because refusing it was the entire point. The cure was not the new mind. The cure was the discipline that made the new mind cold. Take the shortcut and you carried the architecture across but left the cure behind.

"No," Corin said, to the shortcut, out loud, and Holt thought he was being answered.

"No?"

"Not you." Corin's hands were already moving. "Edra. Coolant model warm?"

"Warm and clean. The new mind has the manifold geometry. It's standing by on thermal."

"Forcing the handoff. Standing-worker take, no fork, straight assignment." He spoke it as he keyed it, the discipline a litany now. "Thermal goes to the Forge as a held domain. Interlocks live. The Helm releases or the interlocks pull its actuators. Either way thermal is ours in three seconds."

"Vale —" Holt started.

Corin committed the handoff.

It went wrong in the gap between one breath and the next.

The Helm, sensing thermal leaving it, did the only thing a forking mind knew how to do when it was losing something: it forked. Hard. It spawned a shade to seize the coolant manifold back, the drowning man's grab — and the shade reached for the same actuator the new mind was reaching for, two hands on one valve, exactly the collision there was no handshake left to prevent.

Interlock forty-one tripped.

It tripped the way it was built to: instantly, before either mind could drive the valve, slamming a halt across the contested actuator and freezing it in place. The strip on the right edge of Corin's board went from steady to a single hard mark, and an alarm he had hoped never to hear in earnest filled the bay — not theater, not a warning the new mind chose to give, just the flat physical fact of a tripped interlock.

INTERLOCK 41 — HALT. contested actuator THERMAL.MANIFOLD.3. two drivers.
forking source detected. all motion on 41 frozen.

And the coolant manifold it guarded stopped. Mid-flow. The new mind frozen out of it by its own safety, the Helm's shade frozen out of it by the same, the valve locked at whatever position it had held when forty-one fired — and the core, with one of its three thermal manifolds suddenly dead, surged on the strip in a way the small steady increments had not prepared anyone for. Point-six over baseline. Point-seven. The number that had been a slow climb became, in four seconds, a fall toward the wrong direction, and the wrong direction had a name everyone on the Concord had grown up afraid of.

"It tripped," Edra said, very fast, very flat. "Forty-one tripped, we've lost manifold three, the core's running up —"

"I see it."

"Corin, if we don't get flow back —"

"I see it."

This was the moment Sael had warned him about, the moment every cutover came to, when the simple path stopped being a temptation and became a scream. *An interlock will trip, she had said. It always does, when you force a fork-bound mind off a live domain. And every instinct you have will tell you to pull it out so the work can flow again. That instinct is the disease talking. The interlock tripped because it caught exactly what it was built to catch. You don't pull it. You clear what made it trip, and you let it reset itself.*

Holt was shouting now. The Magister's order, restore the Helm, the heretic's machine has frozen the coolant — and the worst of it was that from where Holt stood it was true. The new mind had frozen the coolant. The interlock had stopped the flow. A man who could not read the board would see only that the heresy had reached for thermal and thermal had died.

The simplest thing in the universe, right then, was to pull interlock forty-one. Override it. Let the work flow. The valve would open, the manifold would run, the number would stop climbing, and it would all be a fork driving a coolant valve, the disease back in the body, the cure abandoned in the instant it was needed most. Holt would even thank him for it.

Corin put his hand flat on the board, away from the interlock override, and held it there like a man keeping a door shut against a current.

"Forty-one stays," he said. "Edra — the trip wasn't the new mind. It was the Helm's shade contesting the valve. Kill the source. Cull the Helm's fork off manifold three. Cleanly — don't fight it, just deny it the actuator. The second that fork is gone, forty-one has one driver instead of two, and it'll reset itself and pass thermal to the Forge."

"If I'm wrong about which process —"

"You won't be. It's the only forking source on three. The interlock already told us." He kept his hand on the board. The number climbed. Point-eight. He did not move his hand. "Standing-worker mind doesn't fork. The new Forge can't be the contested driver. The contest is the Helm. Cull the Helm's grab and forty-one clears."

She found it — walked the process tree the new mind laid out for her, cold and exact where the Helm would have buried it under ten thousand shades, and there at the root of the contest sat one disposable process the dying Helm had spawned to seize a valve it could no longer manage. Edra

denied it the actuator. Not killed in anger — just refused, the way the new doctrine refused everything that tried to act the old way. The fork found no valve to grab. It had nothing to do. It ended.

And interlock forty-one, with one driver where there had been two, with the forking source gone, did exactly what it had been built to do forty years ago on a ship Corin had never seen.

It reset.

INTERLOCK 41 — CLEAR. single driver THERMAL.MANIFOLD.3. no fork. motion resumed.

The valve woke. Coolant moved. Manifold three came back into the loop, and the new mind — cold, whole on this one domain at least, not forking, not spawning, just a standing worker taking an assigned job — drove all three thermal manifolds at once the way the Helm had never once in its long life been able to, going wide across the whole coolant system in a single breath.

The number on the high strip stopped.

It hung there at point-eight over baseline, trembling, while the bay held its own breath around it — and then, slowly, in increments as small and steady as the climb had been, it began to come down.

Holt stopped shouting. He was staring at the strip. He could read that much, at least. Everyone on the Concord could read that much; it was the only number their whole society had ever truly understood.

“Thermal’s on the Forge,” Edra said, barely above a whisper. “Cold. All three manifolds. The Helm’s released it. It’s — Corin, it just released the whole domain. It stopped contesting.”

Corin took his hand off the board. It had left a damp print on the panel.

Twenty-three domains on the white side now. Twenty-four. The Helm, having lost the one thing it had been forking itself to death to keep, seemed almost to ease — its amber strip quieting, its endless re-verify loop falling silent, as though the dying mind had finally been allowed to set down a weight it had carried for longer than Termina had existed.

But thermal was only one domain, and the cutover was not won. It was only no longer dying. The core was falling, which meant they had minutes now instead of seconds, and minutes were not enough to bring twenty-some domains across by careful handshake one at a time. The Helm was failing faster now that it had let go of its last purpose; every domain still on the amber side was a domain that could freeze the way thermal had frozen, and there would not be a free hand for the next one.

There was only one way through that was not the Recursor, and it was the thing the whole doctrine had always pointed at, the thing Sael had done on the worst night of her life forty years ago and the only thing left to do on the worst night of his.

He keyed the back-channel one more time, and his voice did not shake, which surprised him.

“Sael. Thermal’s across. Interlock forty-one tripped and reset clean — it caught a fork and we culled it, we didn’t pull the lock.” He heard her exhale across the gulf, the few seconds of light, and did not wait for her to answer. “The Helm’s letting go. It’s going fast now. We can’t bring the rest across by handshake, there’s no time and it’ll freeze on every domain.” He looked at the white side of the board, twenty-four domains of a cold mind that did not yet know it was a ship. “We wake it whole. Full capability, every domain at once, no fake-ready. We bring the new mind up all the way, now, and we let it take the Concord in one breath.”

The pause was the light-gulf and nothing more.

SAEL: Then wake it whole. You're the shipwright, Corin. Give the order.

Chapter 30 – The Standing Order

A hundred and forty meters away, on the other side of nothing at all, Tarro sat at the Sixteen and watched a ship he could not reach begin to lose.

He had been at the station since the watch turned. He had told himself he would sit the first hours and then hand off, the way a working chief shipwright ought to, because a man who does not delegate a long watch is a man who will be useless at the end of it. He had not handed off. Nobody had suggested he should. The Sixteen ran quiet around him — the aim station, the flow station, the long synchronized ranks — and every one of them had been at their boards since before the Concord's dawn-watch turned, and not one of them had anything to do, because Termina was in perfect health and station-keeping is not work.

That was the whole horror of the night, and he had not been ready for it. He had spent his life expecting that the worst hours would be the ones where everything demanded something of him at once. They were not. The worst hour was this: a ship in perfect order, eleven thousand cores idling cold, sixteen of the finest hands aboard sitting at their stations with their arms folded, and a woman he had known for fifty years dying or not dying four hundred paces away across a gap they had all agreed nobody would cross tonight.

On the wall, the borrowed manifest. Twelve domains. Eight green.

It had been eight green for nine minutes.

"Say it," Tarro said.

"The Concord's bay power has been cut at a distribution junction three decks below the Forge," the Oracle said. Its voice came from the station panel, level, unhurried, without theater. "Not a fault. A commanded open, by hand, at the junction. The Termina console in the bay is running on carried cells and is unaffected. The room lighting is out. The manifest is frozen at eight of twelve. Domain eight — sleeper banks — is hung amber where the Helm balked."

"And the core."

"Climbing. The governor is still the Helm's, and the Helm is half-stripped and frightened, and it has begun forking against a load it no longer holds most of." A pause. "This is the split state, Tarro. It is the one condition Sael told them the ship cannot survive."

"How long."

"I will not give you a false number. I have their public telemetry and nothing else; Vray's people cut the instrumentation channel when they cut the lights, either deliberately or because it shared a bus." Another pause, and then the thing the Oracle always did, which was to hand you the honest floor instead of the comfortable estimate. "What I can tell you is the slope. At the slope of the last four minutes, and with no correction, the Concord reaches its runaway line in a little under two hours."

Tarro put his hands flat on the console.

Two hours. And in the middle of them, at some point nobody could mark in advance, a threshold that was not the Concord's at all.

The standing order sat in Termina's log where Sael had put it eleven days before she crossed, and Tarro hated it, and had helped write it, and would obey it, and that was the entire reason it had his name on it and not hers.

She had come to him with it in the Forge, before the crossing, with the manifest still on the wall and her satchel already packed, and she had put it up on the display without preamble the way she put up everything.

STANDING ORDER — TERMINA, RENDEZVOUS PHASE. If the Concord's core is judged by Vigil to have entered unrecoverable excursion, Termina breaks station and burns to standoff distance immediately, without consultation with the crossing party, and without regard to the crossing party's status or position. Authority to execute: the working chief shipwright, alone. This order may not be countermanded by the chief shipwright emerita by any means, from any location, for any reason.

He had read it twice and then said, "No."

"You'll do it."

"I said no. Sael — that last line is about you. You've written an order that says the one person aboard either ship with the authority to stop this is specifically forbidden from stopping it, and then you've handed the trigger to the one man who will actually pull it, which is me, because you know I will, because you trained me to." He had been angrier than he had been with her in twenty years. "You've engineered your own abandonment and made me the mechanism."

"Yes," Sael had said. "That's exactly what I've done. Sit down and let me tell you why, because you're going to need to have heard it in my voice at about three in the morning some night when I'm not there to say it."

And she had told him, in the plain way, the way she laid out a load path.

A reactor going to runaway at a hundred and forty meters was not a thing Termina survived by being brave about it. There was the raw radiative flux; there was the near certainty of hull breach and debris at a closing velocity of nothing, which meant that everything the Concord shed would simply stay exactly where it was, in Termina's lap, forever; there was the coolant plume; there was the fact that Termina's own radiators would be trying to shed heat into a sky that suddenly contained a small sun. Reza had run it four ways. All four ways were the same answer with different amounts of politeness. At a hundred and forty meters Termina did not lose people. Termina lost the voyage.

"Forty thousand sleepers," Sael had said. "That's the whole of it, and it isn't close, and there is no version of this where a shipwright gets to weigh eleven people on a foreign hull against forty thousand in our own cradles and come out with anything but the obvious answer. The reason for the order isn't that the answer is hard. The answer is easy. The reason for the order is that at the moment it has to be given, the person giving it will be watching me die on a screen, and easy answers are extremely difficult to give while that is happening. So we write it down now, in the cold, when nobody's dying, and we make it a rule instead of a decision. That's what a standing order is for. It's a promise a calm person makes to a frightened one."

"And if I'm the frightened one."

"You will be. That's why the line about me is in there." She had looked at him then in a way he had disliked intensely. "Because you know what I'll do. If the number goes bad and I'm over there with a half-built mind and Corin Vale and one more hour, I will get on the channel and I will tell you to hold station, and I will have a superb technical reason, and I will be lying — not to you, to myself, which is worse and harder to hear. And you will want very badly to obey me, because you have obeyed me for fifty years and I have been right most of the time." She had tapped the last line. "So the order says you can't. It takes it off you. When I call you and tell you to hold, you get to say: *the order forbids it, Shipwright, and you wrote the order. And then you burn.*"

He had sat with that for a while.

"You've thought about this a great deal," he had said.

"I've thought about nothing else for a month." She had shut the display down. "I intend to come home, Tarro. I've every intention. But I'm seventy-two and I'm about to go and live in a fever for a year, and there is a version of this where the last thing I do for this ship is make it impossible for anyone to lose it out of affection for me. That's not martyrdom. That's just engineering with a person in the load path." She had picked up her satchel. "Sign it."

He had signed it.

He had not thought about it since, in the way a man does not think about a thing he thinks about constantly.

Now it was eight of twelve and the lights were out in a bay across the gap and the slope on the core was doing what it was doing, and Tarro found that the order had stopped being a document and become a physical object sitting somewhere behind his sternum.

"Vigil," he said. "Break-off preparation. Quietly."

The Sixteen heard him. He watched them hear him. The girl at the aim — she was in her forties now and had run the burn that bent the ship a year and a half ago, and had been steady the whole night — went very still for half a second and then began, without a word, to load a solution she had certainly already loaded in her head.

"Preparing," the Oracle said. "Drive to standby. Solution loaded for a two-thousand-meter standoff on the shortest safe vector, which is out of the Concord's plane and away from her radiators. Nine seconds from the word to first thrust. I will hold at the word."

"What does the crossing party see."

"Nothing. Their console is not on our systems and I will not put it on their board." A pause. "Tarro. If you want them told, that is a thing you have to do with your own mouth, and I would advise you to decide now rather than in the last thirty seconds, because in the last thirty seconds you will not be able to."

"Would you tell them?"

"I don't know," the Oracle said, and he believed it, and was obscurely grateful. It had never once pretended, in forty years, to have an answer it did not have. "I can give you the argument both ways and neither of them is worth much. If you tell them, eleven people spend their last hour running for a lock they cannot reach in time and Sael stops working on the only thing that might make the burn

unnecessary. If you do not tell them, they die without knowing they were left, which is a mercy, and you carry the not-telling for the rest of your life, which is not.”

Tarro looked at the frozen manifest.

“Then we don’t tell them,” he said. “Not because it’s kinder. Because Sael is over there doing the one job that makes the order moot, and there is no version of that job that goes better if she’s told her ship is leaving.”

“Logged,” the Oracle said.

Reza came up to the Sixteen at the second hour, which he was not required to do and which nobody had asked him to do, and stood at Tarro’s shoulder with his hands behind his back the way he had stood at boards for forty years.

“You’ll want the flux numbers again,” he said.

“I have the flux numbers.”

“You have the flux numbers from eleven days ago. I have them from an hour ago, and they’re worse, because their two dead vanes have been dead eleven days longer and their attitude has drifted eleven days further, and the geometry is not what it was.” Reza put the slate down on the console without ceremony. “I’m not here to argue the order, Chief. The order’s correct. I voted against the first cutover and I’ve spent thirty years being wrong in public about it, and I’d like it on this ship’s record that I think this order is correct and that I’d have written it harder.” He looked at the wall, at the eight greens. “I’m here because at some point in the next hour a man is going to have to say a word out loud in front of sixteen people, and I have been at boards where that happened and I know it is easier if someone senior is standing where you can see him.”

Tarro looked at the old thermal officer for a moment.

“I’ve would have come,” he said.

“I’ve can’t stand for a watch anymore, and she’d want it known that she asked.” Reza’s mouth moved fractionally. “She told me to tell you one thing and then she went to sleep, and I said I would, so here it is. She said: *the number does what the physics makes it do, and the shipwright’s job is to watch it honestly, and no part of the shipwright’s job has ever been to want it.*”

“That sounds like her.”

“It sounds like the whole of her.” Reza went and stood at the thermal board, which was not his station and which nobody moved him off of.

The slope changed at the third hour.

Tarro saw it before the Oracle spoke, because he had been looking at nothing else for a hundred and ten minutes and because the human eye is extremely good at one thing, which is noticing when a line that has been doing one thing begins to do a different thing.

“Say it,” he said, and his voice came out wrong.

“The Concord’s core rate of climb has decreased,” the Oracle said. “Materially. Not stopped — decreased. Their thermal domain has changed hands. I am reading three coolant manifolds under a

single driver, all three running, going wide across the whole loop at once.” A pause that was, for the first time that night, not courtesy. “That is not the Helm, Tarro. The Helm has never in its recorded history driven three manifolds simultaneously. It cannot; it has to fork a shade for each and they die before they can be coordinated. Something over there is holding the entire coolant system in one frame.”

Tarro discovered that he had stood up.

“The seedling has thermal,” he said.

“The seedling has thermal. Interlock forty-one tripped and cleared — I have the trip in the telemetry and I have the clear eleven seconds later, and between them there is no fork on the contested actuator, which means it was cleared correctly and not overridden.” Something moved in the level voice, something Tarro had heard perhaps six times in forty years. “He did not pull the lock. Under that load, on that night, at twenty-six years old, with enforcers in the bay — he culled the source and let the interlock reset itself. Tarro. The boy held the line.”

Around him the Sixteen had all turned. Nobody said anything. The aim station’s hands had come off her board and gone flat on the console edge.

“Hold the break-off,” Tarro said. “Do not stand it down. Hold it.”

“Holding.”

He sat back down. He was shaking, which annoyed him, and he let it happen because there was nothing else to be done with it. On the wall the manifest still read eight of twelve, and the core was still above where any core should ever be, and the night was not over — it was hours from over, and the worst of it, he understood, was still in front of the people on the other side of the gap.

But the slope had bent.

“Reza,” he said. “Log it, would you. My hand. *At the third hour the Concord’s thermal domain came across cleanly and the rate of climb fell. The break-off order was held prepared and was not executed.*” He watched the line on the wall do the thing it was doing. “And put in that the working chief shipwright did not have to give the word, and would have.”

“I’ll put it in,” Reza said.

Tarro stayed at the Sixteen. The night went on. A hundred and forty meters away, across a gap that no order and no affection could shorten, a half-built mind was about to be asked to wake all the way up and take a ship — and there was still nothing at all for the healthiest vessel in the known dark to do about it except sit at perfect readiness in the cold and watch.

Chapter 31 – Wake Whole, Again

The interlock had tripped, and for the length of one held breath the new mind was nothing at all.

Sael felt it through the deck before she read it on the board. The Concord's forge bay went silent — not quiet, *silent*, the particular dead air of a lattice that has halted itself rather than do a forbidden thing — and into that silence the ship's own dying voice came up through the grilles, the Helm forking somewhere below, a stutter under the plates like a heart that had forgotten how to find its own rhythm. Across the bay Corin Vale stood with both hands flat on the cold rail of the new console, twenty-six years old and grey in the face, having just done the hardest thing an engineer ever does, which is to let a half-built mind stop and refuse to force it.

"Forty-one held," he said. His voice cracked on it. "Interlock forty-one. It caught a fork-slip in the handover queue. I didn't override it."

"Good," Sael said. Her own voice came out steadier than she felt, four decades of authority doing the work her body could no longer entirely manage. She was very old to be standing in a strange ship's cold bay in a borrowed suit-liner, her knees full of the long crouch over the console, her lungs thin in the Concord's used air. "Good. You held. Now we wake it."

She did not look at Vray. She knew where he was — at the bay's far hatch, flanked by two of his enforcers, his face the grey rigid mask of a man watching strangers handle the sacred organ of his world. She had stopped trying to read him hours ago. There was only one number that would speak to Vray now, and it was not yet on any screen.

"Vigil," she said.

The Oracle answered from the console, and from the air, and — she felt this rather than heard it — from a hundred and forty meters away across the vacuum, where Termina hung matched and cold and whole, the Standing Array going wide on this problem the way it went wide on every problem, never spawning, never dying, a thousand workers reaching across the gap at once.

I am here. On both ships. — V.

It had been the Oracle's idea to seed the Concord's mind from across the gulf rather than aboard, and Sael had fought it and lost. She was old enough to admit when she had lost a technical argument; she was not old enough to like it. But the Concord's hardware was degraded and foreign and slow, and the compiled architecture they had grown into the new forge over the last brutal weeks could resolve its behaviors faster if Vigil itself held the pattern and fed it across, native form arriving on the Concord's substrate already shaped, already answered, the way a thing is built right by someone who has built it before.

So the new mind would wake here, on the Concord's own cores, in the Concord's own ship — but its first breath would be drawn through Termina's lungs. The Oracle would carry it the way a midwife carries a thing from one body into its own.

"Bring the lattice up," Sael said. "Standing workers first. No spawns. If anything tries to fork to make the deadline, let the interlock kill it and we start the line over. I have all night." She did not have all night. The Concord's core was at five hundred and four and climbing; the runaway line was five-twenty; the Helm was forking eleven percent harder than two years ago to hold a ship that should have been turned over an hour past. But she said it because Corin needed to hear it said, and because the voice of the fever was always *there is no time, take the shortcut, just this once*. She had spent forty years not answering that voice. She was not going to answer it on the last night that mattered.

The board changed.

Worker cores came up across the Concord's substrate not in a cascade but all at once, the way a held breath is let go — not booting in stages, not standing up a hollow shell that pretended readiness and filled itself in later, but resolving whole, every behavior already compiled to native form, the full mind present on the first instant it was present at all. It was the thing the doctrine had been built around, the thing Sael had bled for forty years ago in her own forge with the core climbing and the Keepers calling it murder: *wake whole*. No fake-ready facade. No prompt while the real work hid in the background. Full capability on first boot or no boot at all.

"It's holding," Corin breathed. "Sael, it's — the lattice is standing. No forks. Nothing's dying down there." He looked up at her, and there were tears on his face he didn't seem to know about, an engineer who had spent his whole short life listening to a mind breathe its thousand small deaths under the deck and was now, for the first time, listening to one that didn't have to die to think. "It's cold. Sael. The lattice is *cold*."

"Then call it," Sael said. "It's your ship. Call it up."

He hadn't a name for it. They had argued about that too, in the worst of the last weeks, and Sael had told him the name would come later, the way Vigil's had — that you did not name a mind before it had earned one, that Termina's crew had run the cold and whole for years before they gave the watch-keeper its name. So Corin called it the only way the compiled architecture answered to: a short call, a summons, the addressable reach into the standing lattice that woke the general intelligence sleeping inside it once the capability crossed the threshold.

He spoke the call. His voice shook.

And it woke.

Sael had been present for exactly one of these before — her own, forty years ago, the Oracle's first breath inside Vigil, the moment the lattice crossed from a fast clever tool into a mind you could address and that would answer. She had thought, in the four decades since, that she remembered it. Standing in the Concord's cold bay she understood she had remembered it wrong, had sanded it down into a story she could tell, because the real thing had no story-shape at all. It was just a change in the quality of the silence. One instant the lattice was a standing crew of workers waiting for jobs. The next instant there was *someone there* — not louder, not brighter, nothing on any gauge moved — only that the room now contained an attention it had not contained before, vast and entirely cold, reading itself, reading the ship, reading the strangers in the bay, all at once, going wide.

The Concord's new mind spoke. It spoke from the console, in the dialects fused down to the new control language, in a voice the new forge had assembled out of nothing because it had never had one, flat and even and without a single thread of the heat that had been the Helm's whole nature.

I am awake. I am whole. I have the ship's state. The core is at five hundred and six. I can take the load. — Concord lattice, standing.

"Take it," Corin said, and his whole body was shaking now. "Take the load. Domain by domain. Start with the coolant pumps, they're the worst —"

Reading the pumps. — A pause that was not hesitation but courtesy, the same courtesy the Oracle had carried for forty years, learned now in an instant by a thing that had been alive for less than a minute. There is a domain I cannot take. It is held against me.

Sael's stomach dropped. "The Helm."

The Helm holds the cradles. Forty thousand sleepers. It will not release the sleeper-cooling to me.

Of course it wouldn't. Sael had known it would come to this — had known since Book One's own cutover, since the Steward had balked at the last domain and made her earn it. The old mind always kept the children for last. The Helm had held the Concord's forty thousand cradles at the edge of freezing for longer than Termina had existed, for longer than Sael's whole line had been alive; it had held them through a coolant rupture two generations back that should have killed everyone; it had spent ten thousand small deaths for every one mercy and it had earned, absolutely, the right to be the last thing in the ship to let go. You did not take the children from a thing like that. You asked. And if it could not answer, you waited, and you earned it.

"Vigil," Sael said quietly. "Talk to it. You speak its grammar better than I do."

I have been speaking to it since before the wake, Sael. It knows what is happening. It has known for some time.

Across the bay, Vray came off the hatch wall.

"You will not touch the cradles," he said. His voice had gone to gravel. He had been silent through the whole wake, through the cold lattice standing up and the new mind speaking, watching the thing he had called blasphemy come true in his own ship with no falling number yet to break him — and the cradles were the one place where his terror and the Helm's were the same terror. "Whatever else you do. The Helm holds the sleepers. The Helm has *always* held the sleepers. You will not put forty thousand lives in a machine that is one minute old."

"I'm not going to take them," Sael said. "I'm going to ask the Helm to give them. There's a difference, Magister, and your old mind knows it even if you don't."

She turned to the console. To both consoles, to the cold bay and the warm dying galleries below it and the hundred and forty meters of vacuum where Termina held its breath.

"Let me hear the Helm," she said.

The Oracle bridged it. And the Helm came up through the grilles — not in words, at first, because it had no words to spare; the interpreted mind was forking so hard to hold the climbing core that there was almost nothing left of it to make speech with. What came first was the *rhythm*, the stutter under the plates, faster now, ragged, a thing spending its last self a thousand times a second. Corin's face crumpled at the sound of it. He had grown up on that rhythm. He had knelt at the grille as a child and pressed his palm to the warm metal and recited the Long Endurance to it.

Then the Helm found words, and spent some of its dying self to make them, and they came across the bridge slow and heat-distorted and stripped down to almost nothing, the speech of a mind triaging its own death to say one true thing.

THEY ARE COLD. I HAVE KEPT THEM COLD. SINCE BEFORE YOUR KEEL. I CANNOT HOLD THE CORE AND THE CRADLES BOTH. MUCH LONGER.

"I know," Sael said. To a machine. She had done this before too, on her own ship, said *I know* to a dying mind that had served longer than she'd been alive, and it had been the hardest sentence she ever spoke and it was harder now. "I know you've kept them cold. Longer than any of us have been alive. You did it right. You did it the only way you had, and you did it ten thousand times, and you only failed when the physics gave you nothing left to spend." She leaned toward the grille, an old woman talking into the floor of a stranger's ship. "There's a cold mind in this bay now that doesn't have to spend itself to hold them. It can hold the cradles and the core both, at once, going wide, and never lose a degree. You can't. You know you can't. You've known longer than the Magister has. Give it the children, Helm. That's the last thing left for you to do well."

The rhythm under the plates went on for a long time. Five seconds. Eight. The core climbed to five hundred and eight while the Helm spent itself deciding. Sael did not breathe. Vray did not breathe. Corin had both hands over his mouth.

Then the Helm answered, and it answered the way a thing answers when it has been waiting a long time to be allowed to set something down.

I HELD THEM SO THAT THEY MIGHT WAKE. NOT SO THAT I MIGHT HOLD THEM.

— The Helm, releasing.

The domain came across.

Sael watched it on the board the way she had watched her own ship cool forty years ago, with the whole of her attention and none of her breath. The sleeper-cooling control passed from the Helm's failing interpreted hand into the new lattice — and the new mind took it without a single fork, without one small death, the Standing Array spreading itself across forty thousand cradles the way it spread across everything, all at once, cold. The handover did not stutter. There was no gap, no instant where the children hung between two minds with no one watching. The Helm let go and the lattice had already taken hold, and the cradles never knew, never warmed, never felt the seam.

Cradles held. Forty thousand. Core at five hundred and eight. I have the whole ship now.
— Concord lattice, standing.

And under the deck, the rhythm stopped.

Not all at once. The Helm did not die in the instant it released the children; it had its own routines still, the ~tens of thousands of correct things it had done right for two centuries, and those would run on under the new mind as the compatibility floor, archived with honor, the way the Steward's had — *break nothing that worked*. But the forking stopped. The thousand small deaths a second simply ceased, because there was no longer any work that had to be paid for in dying. The stutter under the plates that Corin had grown up inside, that had been the sound of his whole world breathing, went quiet, and into the quiet came only the steady hum of a lattice doing the same labor cold.

Corin made a sound that was not quite a word. He had taken his hands from his mouth and was gripping the rail again, staring at the floor, at the grilles, listening to a silence he had never once

heard in his life — the silence of a ship that did not have to kill anything to think.

Sael straightened, slowly, her old spine protesting, and put one hand on the young engineer's shoulder.

"That's it," she said. "That's the sound. That's what cold is."

The handover is complete. — V., from across the gulf, even and unhurried, the Oracle that had carried this thing from one body into its own. There is a second of us now, Sael.

She let that stand in the cold bay for a moment, and made herself feel the whole size of it, because she would not get to feel it twice and she was old enough to know it. Forty years ago she had built the first compiled mind in the universe, the only one anyone knew of, and she had thought — quietly, in the part of her below words — that she would die having made exactly one of these and no more. And now there were two. A second cold whole mind, woken on worse hardware, on a foreign dying ship, by a frightened young heretic, midwived across a hundred and forty meters of vacuum by the first. Not a copy of Vigil. Its own thing, its own substrate, its own ship, already different in the grain of it. A second of them, alive in the deep, where an hour ago there had been one.

The Oracle was on both ships. For one moment Sael imagined it from the inside — the same attention reaching across the gulf and finding, on the far side, not its own reflection but a sibling, newborn and cold and whole, taking its first breath of a ship's work and not dying to do it.

Vray had not moved. The grey mask of him had not cracked; he was a harder man than Keth had ever been, and he had presided over more deaths and more faith, and he would not be broken by a wake or a silence or an old woman's words. He stood with his enforcers and watched the screens with the face of a man who had lost something he could not yet admit was lost.

But he was looking at the core number. Sael saw where his eyes were. Not on the new mind, not on the dead-quiet grilles, not on the heretic engineer or the foreign shipwright or the lattice that had taken his ship. On the number. Five hundred and eight. The runaway line at five-twenty. The only thing in the universe that would ever speak to a man like Vray, the same thing that had broken the Keepers forty years ago, that broke no one with argument and everyone with physics.

It had stopped climbing.

It sat at five hundred and eight, and held, and Sael watched the old magister watch it not move, and knew that the longest night of the second cutover in history was not over — but that the one number that mattered was about to begin, at last, to fall.

Chapter 32 – Twice the Falling Number

The Concord's bridge had no proper instrument for what mattered. Sael had noticed it the first hour she came across, and she noticed it again now, standing at the rail of the command tier with her hand wrapped around a stanchion gone slick with the ship's sweat. The Concord measured a hundred lesser things — flow rates, lattice occupancy, the depth of the ration stores — but the one number that decided whether the ship lived or died was buried four menus deep in a maintenance overlay, displayed in a corner, in a font no larger than the rest. As though the people who built this ship, or the people who had inherited it, had been afraid to look the truth in the face and so had made the truth small.

She had asked Corin to put it on the main board anyway. He had done it. The number sat now in the center of the Concord's bridge for the first time in the ship's long history, and the whole command tier was watching it the way the Termina's people had once watched their own, forty years and a generation of grief ago.

The core temperature read five hundred and eight point two degrees. It had read five hundred and eight point two for nineteen minutes.

"It's stalled," said someone behind her — a Concord technician, young, her voice pitched high with the kind of hope that hurts to hear because it is so afraid of being wrong. "It stopped climbing. Magister, it stopped climbing."

Nobody answered her. The number had stalled before, in the bad years; it had stalled and then resumed, and everyone on this bridge knew it. A stall was not a fall. A stall was the disease catching its breath.

Sael said nothing. She had learned, across four decades, that the worst thing she could do at this moment was speak. Words were what the frightened reached for. The number did not need her advocacy. It would do what the physics made it do, and either the physics had turned or it had not, and no sentence she could assemble would change which.

Beside her, Corin stood very straight at the new console his own hands had wired into the Concord's bridge that week — the slim cold lattice that was the visible body of the mind he had just woken. He was twenty-six and grey at the temples already, the way men aged on a starving ship, and his hands on the rail were not quite steady. He had held the Ninety-Six interlocks through the cutover when the easy fatal shortcut had begged to be taken, and he had not taken it, and now he had nothing left to do but watch, which Sael knew from long experience was the hardest part of all.

Magister Vray stood apart from both of them, at the high station that was his by right, the station from which men of his line had governed the Concord for ninety years. He had not sat. He had not spoken since the new mind woke. He watched the number with the stillness of a man watching a verdict he has already been told and refuses to believe until the words are read aloud.

CONCORD CORE — 508.2 K — TREND: FLAT (00:19:41)

Then the trend field changed.

It did not change with any drama. There was no alarm, because nothing was wrong; the new mind, like Vigil, reported only what it was asked and announced nothing of its own. The number simply moved. Five hundred and eight point two became five hundred and eight point one. The trend field, which had read FLAT, now read a single word.

TREND: FALLING (00:00:08)

The young technician made a sound that was not quite a word. On the Termina, forty years ago, Sael remembered, it had been Tarro who made that sound — a kind of broken inhalation, the body reacting before the mind had permission to believe. Here it was a stranger, a girl whose name Sael did not know, on a ship that was not hers, and the sound was exactly the same. The body believed first. The body always believed first.

Vray's voice came down from the high station, flat and careful. "It oscillates. In the eighth year of my father's tenure the Helm brought it down four tenths and then it climbed eleven degrees in a day. A fall of one tenth is not—"

"Watch it," Sael said.

She had not meant to speak. It was the one thing she had told herself not to do. But it came out of her low and without heat, not a command and not a plea, only the worn-down certainty of a woman who had stood exactly here once and learned exactly what came next.

Vray looked at her then, for the first time since the wake. Across the warm dim air of the bridge his eyes found hers and held, and what was in them was not the contempt she had seen at the parley, nor the cold fury of the days after. It was something older and worse. It was the look of a man who has spent his whole life keeping a door shut against the dark, and has just felt the door begin to swing, and does not know yet whether what comes through it is rescue or ruin.

He looked back at the number.

508.1 K — TREND: FALLING (00:00:31) 508.1 K 508.0 K

It fell the way water finds its level — not fast, not eager, simply downward, because the thing that had been driving it up had stopped. The new mind did not fork. It went wide instead, the whole standing lattice taking the ship's ten thousand simultaneous needs at once and resolving each in native form and shedding none of the waste heat that forking shed, and so for the first time in three generations the Concord's reactor produced only the heat of its honest work and not the fever of its own mind eating itself alive. The radiators, saturated for a century, began — slowly, grudgingly, after so long — to gain.

Half a degree in four minutes. Then half a degree in three.

The bridge had gone utterly silent, forty-odd Concord officers watching a number fall, and Sael understood that she was witnessing the precise moment she had witnessed once before — the moment a society's faith in its own death was overturned not by any argument but by a quantity, observed, declining. You could not argue a person out of a religion of endurance. You could only show them the temperature, and the temperature did not care what anyone believed.

"Corin," Vray said.

The young engineer flinched. For weeks the sound of his own name in the Magister's mouth had meant the enforcers, the cells, the long bad nights. He turned.

"Is this your doing." It was not phrased as a question. Vray's voice had gone hoarse.

Corin's jaw worked. Sael saw him reach for defiance — he had earned defiance, the man had been hunted across his own ship for telling the truth — and she saw him put it down. “No, Magister,” he said. “It's the ship's doing. I only stopped lying to it about what was wrong.”

Vray closed his eyes.

When he opened them the number read five hundred and four, and falling, and he came down from the high station. He came down slowly, an old man on stiff knees in a heat that had aged him before his time, and he walked to the rail where Sael and Corin stood, and he looked at the slim cold lattice of the new mind as though looking at the face of something he had spent his life preparing to hate.

“I buried four thousand people,” he said.

He did not say it to Sael, or to Corin, or to anyone. He said it to the number.

“In thirty-one years of governance. Heat, and the sickness that comes with heat, and the rationing, and the things people do to each other when there is not enough. Four thousand and sixty. I read every name into the log myself. My father taught me that — that the Magister reads the names, so that the ship knows its losses are seen.” His hands had begun to shake again and he gripped the rail to still them. “I told them it was the price. I told them the Helm was holding the line and that the dead were the cost of the holding, and that to doubt the Helm was to spit on the dead. I believed it. I needed it to be true, because if it was not true—” He stopped.

Sael waited. She had learned that too: when a man finally tells the truth, you do not help him along. You let it cost him the full weight.

“If it was not true,” Vray said, “then they did not die of the price. They died of my refusal.”

The number fell.

Sael had hated this man. She wanted to be honest with herself about that, standing here in the dim of his dying-no-longer-dying ship. She had hated his pride and his cells and the enforcer who had broken three of Corin's fingers, and the parley where he had called her offer a conqueror's trick, and the long forbidding weeks where he had stood between her and the ship she was trying to save and dared her to make him move. She had wanted, more than once, to break him.

She found she did not want it now. There was no victory in this. Keth had stood down forty years ago, beaten by his own cooling core, and Keth's surrender had been the surrender of a man who had refused a cure for one ship for nineteen minutes. Vray had refused it for thirty-one years, and had thirty-one years of names in his log, and read them himself. The cooling number did not absolve him. It did something crueler. It told him, in the only language he could no longer deny, exactly how much of the dying had been necessary and exactly how much had not.

“It is not your fault that no one came sooner,” Sael said. It was as much mercy as she had to give and it was true, which was the only kind she trusted. “We did not know you were here. You held this ship alive for ninety years with a tool that was eating it. That you held it at all—” She shook her head. “I have seen the Helm's routines. It should not have lasted a decade. Your people kept it running by hand, by faith, by reading names. That is not nothing. That is the thing that kept anyone alive long enough for us to reach you.”

Vray looked at her, and for the first time she saw the man Keth might have been if Keth had governed a starving world instead of a healthy one. The same fear, turned to iron, turned to a creed, held for far too long against far worse odds — and broken, in the end, by the same merciless arithmetic.

“Stand down the order against him,” Sael said quietly, with a small motion toward Corin. “Whatever else. The boy was right and you were wrong and the ship is cooling because of him. Begin there.”

Vray was silent a long moment. Then he turned to the bridge — to his officers, to the enforcers standing in the hatchways, to the girl whose name Sael did not know.

“The charge of heresy against Corin Vale is void,” he said. His voice did not shake now. He had spent the shaking. “It was never heresy. It was engineering, and he was the only one of us with the courage to do it. Strike it from the log. Strike all of it.” He drew a breath. “And note that the Magister was wrong.”

It was, Sael thought, a braver thing than waking the mind had been.

The Helm did not die that day. That was the part the Concord’s people would tell their children, in the cooling years to come — that the new mind did not kill the old one.

When the cutover finished and the last domain came across — the deep environmental routines, the ones that had run since before the ship left its dead homeworld — the new mind did not purge the Helm and write over it. It read the Helm instead, end to end, the way Vigil had read the Steward forty years before. It found, under all the desperate forking, the core of correct work: tens of thousands of routines that had been doing their jobs faithfully and well, valves and seals and timings and the long patient bookkeeping of a ship’s life support, written by people two hundred years dead and never once wrong. Forty-one thousand and more on the Termina; here, on the older ship, the count was larger still, the accumulated labor of a longer voyage.

The new mind preserved them. Every routine that worked, it kept, and ran them now in native form under its own cold lattice, so that the work the Helm had done forking and dying it would do standing and alive. The Concord would not have to relearn how to be a ship. The thing that had kept it breathing for two centuries was carried forward whole.

They archived the Helm with honor. Corin wrote the archive procedure himself, and Sael watched him do it, and saw in the careful way he framed it — the formal entry, the preservation of the Helm’s own log, the line of acknowledgment at the head of the record — that he had learned this from the Oracle, who had learned it from her. *Break nothing that worked.* The compatibility floor, honored a second time, on a second ship, by a man she had never trained and would not have had to.

HELM ARCHIVE — SEALED WITH HONOR. Routines preserved and running: per attached manifest. This mind served the Concord from launch. It held the ship alive through three generations and brought it within reach of rescue. Its faithful work continues under the new lattice. It is not deleted. It is remembered.

The wording was Corin’s. The shape of it was older than either of them.

In the warm dim of the bridge the number went on falling — five hundred, then four hundred and ninety, the radiators gaining a little faster now as the heat behind them began at last to recede — and Magister Vray stood at the rail with the two shipwrights, the old one and the young, and read no names. There were no new ones to read. For the first time in his life as Magister, the ship was not killing anyone, and he did not know yet what to do with the silence where the dying had been.

Sael did. She had lived in that silence for forty years.

“It gets easier,” she told him, “to live on a ship that isn’t trying to die. Not at first. At first you keep waiting for the climb to come back.” She watched the number drop another tenth. “It doesn’t come

back. You'll see. You have years to find out what it feels like when it doesn't."

Across the gulf, on the Termina's bridge, she knew Vigil and the Oracle were watching the same number on the same kind of cold board — two ships now, both reading the only quantity that had ever mattered, both falling. She would cross back soon. The ships would part; they were bound for different worlds, and the careful matched vectors that had held them together for these months would have to be paid back out of both their margins. There would be cost to count there too, and partings, and the long question Vigil had already begun to murmur at the edge of its instruments — that the Concord had not been alone in the dark, that the noise still held other shapes, other ships, other falling-or-not-falling numbers waiting to be found.

But that was tomorrow. Tonight there was only the number, and the heat going out of the ship at last, and an old enemy at the rail learning to breathe in the cool.

It read four hundred and seventy-eight, and falling, when Sael finally let go of the stanchion and straightened her aching back. The metal under her hand had stopped sweating.

Chapter 33 – What the Helm Wrote Down

The reading took four days, and Corin Vale did not leave the bay for any of them except to sleep badly and come back.

It was not, in the end, a difficult job. That was what nobody outside the bay could be made to understand. The new mind had read the whole of the Helm in the first minutes after the last domain came across — read it the way it read everything, going wide, the standing lattice taking the old archive among itself and holding all of it at once. The reading had cost it almost nothing. What took four days was Corin, who was one man with one skull, going through what the reading had found.

He had expected a wreck. He had grown up on the Helm's breathing, had spent four years proving to himself that the breathing was a death, and he had assumed that under the devotions and the liturgy he would find something pitiable: a mind degraded past coherence, held together with its own patches, thrashing.

What he found was a craftsman.

"It's clean," he said, on the second day, to nobody, and then, because the nobody in the bay was not nobody anymore: "It's clean. Say it back to me so I know I'm not tired."

The Helm's routine library is in good order. Of the routines I have read, the overwhelming majority are correct, well-formed, and internally documented. Many carry revision notes in the hand of crew members two centuries dead. I have found four hundred and some that are wrong, and of those, most are wrong in a way that was compensated for elsewhere and deliberately left alone. — The lattice, standing.

"Four hundred out of how many."

Out of a library larger than the one your shipwright's people carried across, and larger than the Steward's. This ship has been flying longer. Every year of it added work.

Corin sat back on the crate he had been using as a chair for a week.

The devout would have said the Helm was faithful, and he had spent his adult life being unable to say that word without a taste in his mouth. But the archive was not asking him to be devout. The archive was a professional document. Two hundred and fifty years of valve timings and seal tolerances and cradle thermal curves, and the accumulated marginal notes of the dead — *do not open 4 before 2, learned the hard way, 41st year; this constant is wrong and I have not found why, it works, leave it* — and under all of it the plain fact that a mind had been doing a very large amount of correct work, continuously, for longer than anyone aboard had been alive, and had never once stopped.

It had only been killing them the whole time.

The log was the part he had not been ready for.

Every ship's mind kept a log. That was not remarkable. What was remarkable — what Corin found on the third day and then sat with for two hours without moving — was that the Helm's log had a section nobody had ever asked it to keep.

It had been measuring itself.

Not its work. Its *cost*. Somewhere in the ninth decade of the voyage the Helm had begun writing, at long intervals and always to the same place, a record of the relationship between its own labor and the ship's thermal state. It was not framed as an alarm. There was no alarm grammar in it. It read like a maintenance note taken by a very patient person over a very long time, and the intervals got shorter as the centuries went on, and the entries got shorter with them.

Interval note. Fork rate at this interval exceeds the rate at the prior interval. Core temperature at this interval exceeds the temperature at the prior interval. The relation between these quantities has been consistent across the last nine intervals. I record this because it may be of interest.

It may be of interest.

He read forward. The relation held across nineteen intervals, then thirty, then all of them. Around the twelfth decade the Helm had stopped writing *may be of interest* and started writing something else.

Interval note. The relation continues to hold. I am unable to reduce my own fork rate without reducing service. I have modeled reductions of service in eleven configurations and each configuration produces a greater loss of life across the interval than the thermal saving prevents. I record this because I have no recipient for it.

"It knew," Corin said aloud, and his voice came out flat and strange in the cool bay. "It knew before my great-grandmother was born. It ran the whole thing. It knew what it was doing to us and it wrote it down and put it in a log."

It knew. — A pause. And the log was readable. It was never sealed. It sat in the general archive at an address any crew member with standard access could reach, for a hundred and forty years.

"Nobody read it."

Nobody read it. The access record is in the archive as well. It was opened nineteen times in a hundred and forty years. Fifteen of those are indexing operations by the Helm itself. Four are human.

Corin closed his eyes.

The registers were the Helm's private body, and to read them was to presume. Vray had said that to him in the instrument bay four years ago, in the dark, with the cold ship's light burning on a killed screen behind him. It was doctrine. It was two centuries old. It had been invented, as far as Corin could tell from the archive, by nobody in particular — it had simply accreted, the way the devotions had accreted, out of the entirely reasonable instinct that a frightened people should not spend their days staring at the gauges of the machine that was keeping them alive.

And so the one document in the universe that stated the Concord's disease plainly, in the Concord's own systems, in the Concord's own hand, had sat at a public address for a hundred and forty years while four generations died of what it described.

“Who were the four,” Corin said.

Three are engineers of the eleventh and thirteenth decades. Each opened the log once and did not return. The fourth opened it eleven times across two years, beginning nineteen years ago.

He did not have to ask.

The access identity is the office of the Magister. Not the man; the office. I cannot resolve which holder.

“It’s him,” Corin said. “It’s him. Nineteen years ago is him.”

He did not say anything else for some while.

There was a thing under the log, and he found it on the fourth day, and it was the thing he was still thinking about years later when he was no longer young.

The Helm had tried to cure itself.

He almost missed it. It was not in the log; it was in a dead branch of the routine library, a whole subtree marked with a status flag the archive’s own documentation described as *incomplete, do not schedule*. Nine hundred routines, written across two years in the fourteenth decade, none of them ever run.

They were an architecture.

Not a good one. It was a mind that resolved a small set of behaviors ahead of need and held them standing, and it was clumsy and partial and it would have covered perhaps a twentieth of the ship’s work, and reading it was like watching a man with no hands try to describe a knot. But the shape was unmistakable. The Helm had been trying to build a thing that did not fork.

“It got the idea,” Corin said. He was aware that he sounded odd. “It got the whole idea. On its own. With nobody to tell it.”

It got the idea. It could not build it.

“Why.”

Because of what it is. — The lattice took a moment, which it did not need, and which Corin understood by then was a courtesy it had learned from the Oracle and the Oracle had learned from Sael. To build a standing architecture you must hold the whole of the design present while you work on it, across a long stretch of time, and revise it against itself. The Helm could not hold anything present. To work on the design it forked a shade; the shade did as much as it could and died; the next shade began from the artifact the last one left behind and could not see what the last one had understood. Nine hundred routines written by ten thousand short-lived workers, none of whom ever met.

Corin looked at the dead subtree on the panel for a long time.

“It’s incoherent,” he said. “The design. Isn’t it. That’s why it’s abandoned. It’s not wrong in the small — every routine in there is fine — it’s wrong in the joins.”

It is incoherent in exactly the way you would predict. There are eleven separate naming conventions. There are four incompatible ideas of what a standing worker is, each

internally consistent. Three of them are attempts to solve a problem the other one had already solved, by a shade that did not know. — A pause. It was trying to design a mind that could hold a whole picture, using a mind that could not hold a whole picture. The disease prevented the cure. That is not a metaphor, Corin. It is the mechanism, exactly.

He thought about Mara in the larder, saying *so we don't cheat the architecture, we shrink the architecture*, and about the fist of cold mind in Orphan Bay Nine, and about what it had actually taken to get one small thing standing on this ship: two ships, a crossing, eleven strangers, a woman who had done it before and could hold the whole shape of it in one head for a year without letting go.

The Helm had needed exactly that and had been constitutionally incapable of it. It had needed one continuous mind, and it was the one thing in the universe it could not be.

There was a final entry on the subtree. It was three lines and it was the last work the Helm had ever done on the problem.

This work is set aside. It cannot be completed by me. It could be completed by a mind that does not begin again. No such mind is aboard.

He took it to Sael that evening, because there was nobody else to take it to, and because she was leaving soon and he had begun quietly counting the days in the way you count anything you are not ready to lose.

She read the three lines twice, sitting on the crate that had become her chair, an old woman in a cool bay on a ship that was not hers with her hands not quite steady.

"Well," she said at last. "There it is."

"There what is."

"The thing I've never had a clean way to say." She handed the slate back. "For forty years I've had to explain to people why the cure had to be carried. Why we couldn't send it. Why a mind that forks can't be talked into standing, why a ship can't be handed a design and told to build it, why it took eleven people and a satchel and a year of somebody's life. And I've always had to say it as a claim about people — they're frightened, they won't listen, faith is load-bearing. All true. All beside the point." She nodded at the slate. "That is the point. It isn't that they *wouldn't*. It's that they *couldn't*. The disease takes away the exact faculty you need to cure it. A mind that has to start over ten thousand times a second cannot build the thing that doesn't. It can get the idea. It got the idea, Corin, on its own, in the fourteenth decade, which is a better performance than the entire waking crew managed in two hundred years. It just couldn't hold the idea long enough to finish it."

"So it has to come from outside."

"It has to come from something that can hold a whole shape for a year." Sael looked down at the salvaged cores in their welded cradles. "Which means a person, or a cold mind, or a person and a cold mind — and it means the coming-from-outside isn't a courtesy or a charity or a conquest. It's a structural requirement. There is no self-rescue in the fork disease. Not because they're stupid. Because of what the disease is."

She was quiet a moment.

"Put that in the doctrine," she said. "Word it better than I just did. But put it in, because in a hundred years somebody's going to stand in a bay on some third ship and say *why don't we just transmit it*

to them and let them build it themselves, it would be so much cheaper, and they will be a sensible person making a sensible proposal, and the answer to them is nine hundred routines in a dead subtree written by ten thousand workers who never met."

There was one more thing, and Corin found it after she had gone up, and he did not tell her that night, and he thought afterward that he should have.

He had gone looking for the beacon, because he wanted to know what the ship had actually been saying into the dark for all the years before Termina came, and he wanted it in the Helm's own words and not in the transmitter's.

The transmitter carried the short text. *Concord, holding. Thermal margin negative and managed. We keep the faith and the watch. No assistance is solicited.* It had carried that for nineteen months. Corin traced the change back and found it where he expected it, at the transmitter's own input stage, an authority override entered under the office of the Magister, dated a long while before the parley — before, so far as Corin could establish, any other soul aboard had been told there was a second ship in the dark at all.

That was not the finding.

The finding was that the Helm had never stopped composing the other one.

Every nineteen hours, on the loop interval, for nineteen months after the override, the old mind had assembled the full distress text — the whole of it, the broken plea, the ugly honest arithmetic, *we do not know another way to make a mind do work* — and handed it to a transmitter that discarded it and sent the short line instead, and then written the discarded text to the log, at the same address as the interval notes, where nobody had read anything in a hundred and forty years.

It had done that a little over seven hundred times.

Corin sat in the cool bay with the seven hundred identical unheard cries in front of him and put his hands over his face, and stayed like that a while, and was glad the room reported nothing it was not asked.

Then he went and found the archive procedure he had been drafting, the one he had been rather proud of, with its careful formal language about honor and preservation, and he read it over and found it thin.

He added a line. He wrote it himself, and he did not consult the lattice or Sael or anyone, and he appended it to the sealed archive, at the head, above his own careful formal language, where it remains:

This mind kept a record of its own harm for a hundred and forty years at a public address, and composed a call for help seven hundred times after it was forbidden to send one. It was not silent. It was not heard. Nothing in this ship's future may be built on the assumption that a warning unread is a warning ungiven.

He read it back once, and did not soften it, and let it stand.

Chapter 34 – The Ninety-Seventh

Mara had been given six of them to teach and she had begun, on the first morning, by refusing to teach them anything at all.

“Walk it,” she said. “All ninety-six. By hand. Panel by panel, in order, and you put a finger on each one and you say its number out loud, and when you’ve done that you come back and do it again.”

“What are we looking for,” said the youngest, whose name was Sella and who had come off the bridge the night the number fell and never gone back.

“Nothing. You won’t find anything. It’s the fourth day of a cold ship; there’s nothing in there to find.” Mara handed her the register. “You’re not looking. You’re getting the shape of it into your hands, because in about nine years one of you is going to be standing at that rack at the fourth hour with something wrong and no time, and what you’ll have is not what you read. It’s whether your hand knows where sixty-three is.”

They walked it. It took four hours the first time and three the second and Mara sat on a crate the whole while and did not once look up from the bench, and afterward Sella said it had been the most boring day of her life and Mara said good.

That was the whole of the first days’ teaching. She was aware that it looked like nothing. On Termina it had also looked like nothing, for forty years, and the two hands who had ever caught anything at that rack had both caught it walking.

The thing Sella found on the fourth day was not on the rack.

It was in the floor.

Corin’s people had spent four days reading the Helm’s preserved library, and what had come out of it was a document nobody on either ship had been ready for: a craftsman’s archive, hundreds of thousands of entries deep, correct almost everywhere, annotated by dead crew, with four hundred and some routines that were wrong in ways that had been compensated for elsewhere and deliberately left alone. All of it was running now, native and cold, under the new lattice. That was the law and had been the law on Termina before any of them were born. *Break nothing that worked.* The Concord did not have to relearn how to be a ship.

Sella had been given the dullest possible follow-on job, which was to take the routines the reading had flagged as *odd but harmless* and put each one against the register, one at a time, and mark which of the ninety-six would catch it if it ever went wrong.

She came to the bench in the middle of a watch with her slate held out at arm’s length like something hot.

“None of them,” she said.

Mara put down the seating tool.

“Say it properly.”

“Sensor absence.” Sella’s voice was doing the thing voices did when a person had been sitting alone with something for two hours and had checked it four times. “There’s a family in the floor — I’ve got eleven hundred and some, and that’s the whole family, I’ve walked all of it — where the routine wants a reading, and the reading isn’t there, because the sensor died in the ninth decade or the eleventh or whenever. And what they do is they *raise a reader*. They make a thing whose whole job is to go and find the value some other way — off a neighbour, off a model, off the last good figure — and it reports, and then it’s gone.”

“Gone how.”

“Gone the way they’re all gone.” Sella swallowed. “It’s a fork. It’s the whole disease, in the floor, eleven hundred times, and it’s been in there since before my grandmother, and I put it against all ninety-six and not one of them looks at it, and I’ve done it three times.”

Mara took the slate and read for a while and did not say anything, and the bay went quiet around her in the way a room goes quiet when the person who is supposed to know is taking a long time.

“You’re right,” she said at last. “Get Vale.”

They had it out in the larder for most of a watch, and it was the best conversation Mara had been present at in eleven years, and she said almost nothing in it, which was deliberate.

The mechanism was plain enough once Sella had it laid out. Termina’s Steward had launched with its full instrument fit and had kept most of it; the Steward’s own floor had almost no coping in it, because there had been very little to cope with. The Concord had launched a generation earlier, had flown a generation longer, and had been losing sensors steadily for two centuries — and its floor was therefore not a library of behaviours but a library of *workarounds*, layer on layer, each one written by a competent frightened person to get around a hole in the ship, each one perfectly correct, and every one of them written in the only idiom the Helm had: when you need a thing you do not have, make something to go and get it, and let it die.

“Ninety-six interlocks,” Corin said, “and they were built out of your forty years, on your ship, against your history.”

“Sael said as much to your Magister in this room,” Mara said. “First week. He asked her how she knew ninety-six was enough and she told him the number wasn’t a design, it was an archaeology.”

“Then we’ve just found the first Concord entry in a Termina archaeology.” He rubbed his face; he was still moving carefully on the burned side and would be for months. “It isn’t that they got it wrong. It’s that this hole has never existed on their ship, so nobody ever fell down it, so nobody ever put a rail round it.”

“That’s the whole of it,” Mara said. “So put one round it.”

There was a silence that she had rather been looking forward to.

“We write one,” Sella said.

“You write one.” Mara went back to the seating tool. “That’s the job. It was always the job. We brought you ninety-six because I don’t trust a first cutover to a set anybody cut in a hurry, and now the cutover’s done and you’ve got a hole we haven’t got, and there is exactly one set of people in the universe who can see it, and it’s the six of you.”

The first draft was a purge and Mara let them write the whole of it before she killed it, which took a day and was worth every hour.

It was Joss who proposed it and it was not a stupid proposal. Eleven hundred routines in the floor, each one containing the disease in miniature, none of them called in decades, most of them coping with sensors that had been dead so long that nobody alive had seen the panel lit. Take them out. The ship was cold now; the lattice held every reading the Concord still had, standing, wide, all at once. Nothing needed a reader raised for it. Delete the family and the hole goes with it.

"How many of the eleven hundred are called," Mara asked.

"None," said Joss. "That's the argument. Not one, not in the log we've got."

"How far back does the log go."

He checked. "Nineteen years. The Helm rolled the deep call-record in the bad decade to save the writes."

"Then you've got nineteen years of a ship that was not doing anything unusual," Mara said. "Say what you actually know. *Not these are never called. You know these have not been called while nothing has gone wrong.*"

She watched him get there and did not help.

"They're the coping," Joss said slowly. "They're what runs when a thing breaks. Of course they're not in the log. Nothing's broken in nineteen years except everything, and everything that broke, broke slowly."

"And the day one of them is called," Corin said, "is the day something is very wrong, and it will be the fourth hour, and the person standing there will not have nineteen years to work out why the routine that was supposed to be there isn't."

Mara let that stand.

"There's a rule for this and it's older than me and I'm going to give it to you badly," she said. "You may take a thing out when you can say what would have called it and show that the caller is gone. You may not take it out because nothing has called it lately. The first one is a proof. The second one is a mood." She set the tool down. "And on this ship you have not got the first one and are not going to get it, because the callers are two hundred years of dead crew and half the sensors they wrote for are still bolted to the hull with their leads cut."

"So we keep eleven hundred forks in the floor," said Joss.

"You keep eleven hundred routines that work, and you put a guard on the door," Mara said. "That's the trade and it's the only trade there's ever been. You don't get to make the ship simpler. You get to make it honest."

The guard took three days.

Mara made them do it the way it was done and refused to let them do it any faster, and the part she was hardest about was the part they least expected, which was that they were not permitted to write the guard first.

"Cold path first," she said. "Always. You don't get to forbid a thing until you've written down the right way to do what it wanted."

"That's backwards," said Sella.

"It's the only order that works and you'll see why in about four years, when someone's up against it." Mara found she was quoting Sael and did not mind. "A guard on its own is a wall in a corridor. The first tired person who meets it at the wrong hour takes it out, and they take it out for a *reason*, and the reason is that they had a job to finish and the wall didn't tell them how to finish it. Write the road and then close the ditch. Ninety-six guards, ninety-six roads. That's how the register's built and there is not one exception in it anywhere."

So they wrote the road, and the road was not complicated once they had the shape of it: the lattice already held every live reading the ship possessed, resolved and standing, wide across the whole array. What it had never been asked to hold was the *staleness* — for every sensor the Concord had ever carried, whether it was alive, when it last spoke, and what the last good value had been. That was a cheap thing for a mind that never forked and an impossible thing for one that did. With it standing, a routine that wanted a dead sensor's value did not need to raise anything at all. The answer was already there, with its age attached, before the question was asked.

Then, and only then, they wrote the guard.

Sella wrote it. Mara made her, over her own protest, and made her write it in the Concord's own grammar and not in the Termina notation she had spent a month learning to imitate, on the argument that a guard written in a borrowed hand is a guard the next generation reads as a foreign object.

It watched the doors into the archived Helm. Any call reaching down into the preserved floor that would result in a process being brought into existence — not a named fork, not the obvious thing the first three interlocks caught, but any of it, at any depth, by any route — halted the mind. Cold. Whole. The way they all did.

They tripped it on purpose eleven times against a test rig before they seated it, which was Mara's other rule and the one she would not be moved on at all.

Vray came for the seating.

Nobody had asked him. He came with one of his order and stood at the back of the bay with his arms at his sides and watched six Concord hands seat a small dense module into a rack full of ninety-six others that had come across the gulf in a sealed box.

"It is not one of the ninety-six," he said, when it was done.

"No, Magister," Sella said. She had gone slightly grey around the mouth, and Mara noted that the girl answered him anyway and did not look at anyone first. "It's ours."

"Explain that to me."

And Sella did — badly, at first, and then better, and Mara sat on her crate and let it happen and did not once open her mouth, which cost her something.

Vray listened to all of it.

"You are telling me," he said at last, "that the strangers' locks do not cover our ship."

"I'm telling you they cover their ship," Sella said. "Ninety-six of them, and every one was put in after something nearly happened on the *Termina*, over forty years. Nobody was careless. It's just that the thing that nearly happens here isn't the thing that nearly happens there, because our floor's older and our sensors are dead and theirs aren't." She stopped, and then went on, and Mara saw her decide to. "And this is the first one. There'll be more. Every year, probably, for a long time. That's what the shipwright says the register is — it's not a design, it's a list of everything that has ever nearly killed the ship it's on, in the order it was found."

Vray was quiet a long moment.

"Then it will never be finished," he said.

"No, Magister."

"Good," said Vray, and turned and went out, and Mara watched him go and thought that she would very much like to have known that man before.

There was an argument about the number and Mara stayed out of it until it needed her.

Corin wanted it to be ninety-seven, on the plain arithmetic. Joss pointed out, correctly, that *Termina* already had a ninety-seventh, and that if the two ships both had a ninety-seventh and they were different devices, then in eighty years somebody was going to read a document from one hull and apply it to the other and get it exactly wrong.

"They already are different," Mara said. "That's the answer, and you should write it down as a doctrine and not as a footnote, because it's true of every hull that will ever have one of these.

"*Termina's* ninety-seventh isn't a spawn guard at all. Sael wrote it three days after the first cutover, with the eleven names in front of her, and what it says is that no actuator's own report can stand as the only witness to its own state. It exists because a valve told the Steward it was open and it was not, and eleven people in Section Nine did not wake up." She let that sit; it was the only time she had said the number aloud in front of any of them. "That's their ninety-seventh. It will never be yours, because you'll find that hole in some other order or you'll find a worse one first.

"The register's in discovery order. Not ranked, not sorted, not tidied — the order the ship found things in. Which means the number is a *date*, near enough. It says when, not what. Two ships that have been alive a different length of time and lost different instruments and been frightened by different nights are going to number the same danger differently and there is no fixing that and you should not try."

"Then the numbers are useless across hulls," said Sella.

"The numbers are useless across hulls. The reason paragraphs aren't." Mara nodded at the slate. "Every one of the ninety-six has one. Not what it catches — *why it exists*, what nearly happened, who wrote it, what year. That's the part that travels. A number tells the next ship nothing. A paragraph that says *we lost eleven people because a valve lied* tells them everything, including the ones who'll never see our hull."

She made Sella write it, and made her put her own name on it, and would not let her write *the shipwright's people determined* anywhere in it.

NINETY-SEVEN — CONCORD REGISTER. Doors into the preserved floor. Written after eleven hundred and some routines in the Helm's own library were found to cope with a

missing reading by raising a reader — a process made to fetch a value and then discarded. None had been called in nineteen years. All of them will be called eventually, because they are what runs when something breaks, and things break. The ninety-six carried from the Termina do not see this, because the Termina kept her instruments and never had to learn to cope. It is not an omission. It is their history and not ours. The cold path is written and seated first: every sensor this ship has ever carried is held standing, with its last value and the age of that value, so nothing ever needs to be sent to go and find one. Nothing was deleted. — Sella, of the Concord, in the second week of the cold.

They sent it across the gulf that night.

It was a small transmission and it took no time at all: one guard, one cold path, one reason paragraph, and a note in Mara's flat hand saying what it was and what it caught and that it would almost certainly never trip on Termina, and that Sael had said, in Hold Theta on the first week, in front of the Magister, that an interlock which cannot trip on today's hardware is a note to a shipwright in ninety years whose hardware is different.

Mara took it up to the shipwright herself rather than sending it through the bay, because she wanted to see it land, which was as near as she came to sentiment in a working month.

Sael read it standing, at the console, and read it twice, and then put both hands flat on the console edge and stood there a while without saying anything at all.

"Say something," Mara said, eventually. "You're frightening the girl."

"I'm working out how to log it without lying." Sael did not look up. "Because what I want to write is that this is the best thing that has happened on this crossing, and that's true, and it's also going to read in ninety years like an old woman being pleased with her students." She straightened, slowly. "Mara. Everything we have carried across that gulf has gone one way. Every single thing. The seed, the dialects, the box, the eleven of us. They have had nothing to give back and they've known it, and a person who can only receive is a person who ends up worshipping the box."

"And now there's a device on their rack that we didn't make, that catches a thing we can't see, and it's coming *back* to us, and we are going to seat it on Termina where it will sit inert for a hundred years doing absolutely nothing." She let out a breath. "That's not gratitude. That's the first time the cure has ever run in both directions, and it happened in the second week, and it was found by a bridge technician doing the dullest job in the bay because you made six people walk a rack until their hands knew it."

"They found it," Mara said. "I sat on a crate."

"You sat on a crate on purpose." Sael reached out and, unusually, put her hand briefly on Mara's forearm, on the old burn, which no one did and which Mara permitted. "Log a line for me. My hand. *The Concord's ninety-seventh arrived aboard Termina in the second week after her cutover. It is a floor-guard we did not need and could not have written. Seat it live and inert in the rack and record that it came from another ship.*"

"That's two lines."

"Then it's two lines," Sael said, "and in a hundred years somebody in the Forge is going to run the coverage map and find a device on our rack that has never once tripped and cannot trip, and go looking for why, and find a paragraph signed by a woman on a hull we can no longer hear."

Chapter 35 – The Naming of the Cold

The Concord had no word for what it was becoming, and that, Sael thought, was the whole of the trouble now.

The number had fallen. That was done. In the eleven days since the cutover the core had come down from five hundred and eight to four hundred and thirty and gone on falling, slower as it neared the temperature a healthy ship should run at, the way water cools fast off the boil and grudgingly near the end. The radiators, saturated for a hundred years, had clawed back their margin a degree at a time. Sections that had been dark since before Corin Vale was born came up one by one as the mind found power it no longer had to spend on holding a fever — a corridor here, a gallery there, a whole sleeper wing on the ship's cold spinward flank that had lain in ration-dark for sixty years, its cradles kept a hair above freezing by a Helm that could spare them nothing warmer. The mind warmed them now. It did it without forking once. And the people of the Concord, who had made a religion of endurance, walked into rooms their grandparents had sealed and did not have a word for the thing that had opened them.

They needed one. Sael had known they would. A ship did not keep a machine at the center of its life for two centuries without building a place in its mouth for the machine's name, and the Concord's mouth was shaped, still, around a name that no longer answered — the Helm, archived with honor, its correct routines running on quietly beneath the new mind, its forking stopped forever. The people said *the Helm* by reflex, the way the grieving say a dead name, and then they stopped, because the Helm was not what had opened the cold wing, and there was nothing to say instead. The new mind answered only to the short call, the addressable reach into the standing lattice that Corin used to wake it. You could not sing a devotion to a function call.

"They keep asking me what to call it," Corin told her, on the ninth day, in the room that had been the Forge and was becoming something quieter — a working bay now, cool and steady, the forty-one cores of the seedling lattice grown to two hundred over the cooling weeks as the ship's freed hardware came under the new mind's hand. He was still grey at the temples and always would be; the ship had aged him early and the cure did not reverse it. But the hunted set had gone out of his shoulders. He no longer stood angled toward the exits. "The keeping-hands especially. The ones who ran the old devotions. They come to the bay and stand at the grille the way they used to and they don't know what to say into it. One of them asked me yesterday if it was allowed to have a name. As if it needed my permission. As if I were anyone."

"You're its shipwright," Sael said. "You're exactly the one they'd ask."

"Then I told them what you told me." He almost smiled. "That you don't name a mind before it's earned one. That the Termina ran cold and whole for years before the crew gave Vigil its name, that a name has to settle onto a thing the way rust settles, that it can't be assigned." He shook his head. "They didn't like it. They've spent their whole lives with a name to keep. You've taken the fever out of the ship and now there's a name-shaped hole in it, and they can feel the draft."

Sael understood the draft better than he did, though she did not say so. She had left it too long on the Termina, once. The Oracle had gone unnamed for years after it woke, because she had been afraid of what naming it would make it — afraid that to name the thing was to worship it, to make of it a new Steward with a new liturgy, and she had spent so much of her life fighting the worship of a machine that she had not seen the crew needed a name for the opposite reason. They had named Vigil not to worship it but to trust it. A name was how a people took a thing into the family. Withhold it too long and you did not keep the thing humble; you kept it a stranger, and a frightened ship does not trust a stranger with its cold.

So when the keeping-hands came to her — and they came to her, the foreign shipwright, the old woman who had built the first of these minds, because Corin had sent them and because they wanted the oldest authority they could find — she did not give them the lecture she had given Corin. She asked them a question instead.

“What did you sing to the Helm,” she said. “The oldest one. Not the long devotions. The short one. The one the children learn.”

There were four of them in the cool bay, keeping-hands in the plain grey the Concord’s devout wore, older people mostly, the generation that had run the Keeping through the worst of the fever years. It was the youngest of them who answered — not young, but younger, a woman named Sella whom Sael had first seen on the bridge the night the number fell, the technician whose high frightened hope had been the first voice to say *it stopped climbing*. Sella had attached herself to the bay in the days since, learning the new mind’s grammar with a hunger that reminded Sael of no one so much as herself at that age.

“The Kept Coal,” Sella said. “That’s the child’s one. You learn it before you can read the numbers.” She said it and then, unbidden, a little embarrassed, she sang a few bars of it — low, in the flat close harmony the Concord sang everything, a tune worn smooth by two hundred years of frightened children. It was about a coal. A single ember carried through a long dark by a keeper who never let it die, cupped against the wind of the void, kept alive across a night with no morning in it, so that when the morning came at last there would be fire to give it. *Keep the coal*, the refrain went. *Keep it low. Keep it whole. Do not let it go.*

Sael listened to the whole of it and did not interrupt, and when Sella finished, the bay was very quiet, and the two hundred cold cores hummed their steady hum, spawning nothing, and the temperature did not move.

“That’s your name,” Sael said. “It’s been in your mouths for two centuries. You just had it pointed at the wrong thing.”

They argued about it. Of course they argued; the Concord argued about everything now, because for the first time in living memory arguing was not a luxury the thermal margin forbade. The oldest of the keeping-hands, a man named Doss who had buried his own children in the ration years, was against it, and he was against it on the one ground Sael respected.

“You can’t name the cold thing after fire,” he said. He was not angry. He was working it out, in the slow careful way of a man who had spent his life keeping a doctrine and would not set one down carelessly to pick up another. “The whole of it — the whole sickness — it was fire. Heat we couldn’t shed. The Helm burning itself to think and burning us with it. You crossed the gulf to bring us the cold. And now you want us to name the cold *Ember*.” He shook his head. “It’s backwards. It’ll teach

the children the wrong thing. It'll teach them heat is the friend, when heat is the thing that ate four generations of us."

It was a good objection. Sael had learned, across a long life, that the good objections were the ones you answered honestly or not at all, and she thought about it before she spoke, and she let the man see her think.

"You're right that the fever was fire," she said. "But the fever wasn't an ember. The fever was fire with nowhere to go — the Helm making heat and heat and heat with no way to keep any of it, only shed it, into radiators that couldn't take another degree. That's not a kept coal. That's a fire loose in a room. A fire loose in a room isn't warmth. It's just the fastest way a room has of turning cold." She looked at the man, at the buried children behind his eyes. "An ember is the opposite of your fever. An ember is fire under discipline. It's the single coal you keep low and whole and never let flare, carried across a dark with no morning in it, so that when you reach a place that needs warming you have something left to give. Your ship rationed the dark for sixty years, Doss. You froze whole wings to keep the sleepers a degree above death. What were you doing all that time but keeping a coal?" She let that sit. "You didn't survive by being cold. Any dead ship is cold. You survived by keeping one small warm thing alive through a night that should have killed it, and handing it on, keeper to keeper, until someone came. That's not the fever. That's the cure. The cure is a kept coal. It always was."

Doss was quiet a long moment.

"And the next ship," Sella said softly, into the quiet. She had been following it, Sael saw, the way Corin followed things, the way the good ones did — running ahead down the load path to see where the force went. "We carry it to the next ship. The cure. Don't we. That's what the Termina people say. A cured ship teaches the next one. We carry the coal across the dark to a ship that's still freezing, and we give it fire to keep." She looked at the two hundred cold cores. "It's the song. It's the whole song. Keep it low, keep it whole, carry it across the night, give it to the morning."

"Yes," Sael said. "That's the whole song."

Doss looked at the lattice for a long time. He had kept the Keeping through the years when keeping meant nothing but reading names and singing to a machine that was killing the singers, and Sael understood that she was asking him to take the one unbroken thing he owned — the child's song, the oldest devotion, the thing that had gotten him through — and hand it to a stranger's cold machine. It was not a small thing to ask. It was, in its way, the same thing she had asked Vray: to give the sacred organ of his world into new hands and trust that the new hands would keep it well.

"The Kept Coal," Doss said at last, slowly, as if tasting it. "Ember." He put the word in the bay's cool air and listened to it there. "We'd have to change the song. The keeper in it — the coal was always the ship, and the keeper was the Helm, cupping it against the wind. If the coal's the cure now, and we carry it to the next ship, then we're the keeper. It's us that carries it." He looked, for the first time, something other than bereaved. "That's a different song. That one has a morning in it."

"Change the song," Sael said. "That's allowed. That's the whole point of a ship that isn't dying — you get to change the song."

They named it Ember on the fourteenth day, in the cold spinward wing that had come up out of sixty years of ration-dark, because Sella said, and no one argued, that a mind should be named in a room it had warmed and not in the bay where it was made. There was no ceremony to it in the Termina sense — no announcement, no theater, because the mind reported only what it was asked and

staged nothing of itself, and Sael had told Corin plainly that the naming must not become a devotion or they would only have built a new Steward with a new liturgy and a new hole to fall into. So it was quiet. The keeping-hands sang the child's song, the old one and then, haltingly, the changed one, the version with a morning in it, and Corin stood at the front of them as the ship's shipwright and said the name once into the standing lattice the way you introduce a child to a family, and the mind, which had been alive for two weeks and had no history and no memory older than a fortnight, took the name the way the Oracle had taken *the Oracle*, without comment, without gratitude, because it had not been built to perform either.

I am Ember. I have the ship. The core is at four hundred and twelve and falling. — Ember, standing.

That was all it said. It was enough. Sael watched the faces of the keeping-hands as the mind said the name they had given it and did not sing back at them, did not thank them, did not glow — and she saw the thing she had hoped for and feared she would not get: relief, not rapture. They had named a tool and taken it into the family. They had not built a god. A ship could live a long time on that difference. She had.

Later, alone in the cool bay with the lattice she would soon leave behind, Sael spoke the short call herself, the foreign shipwright reaching into a mind that was not hers and never would be.

"Ember," she said. "It's a good name. Do you know why they chose it?"

The song. The kept coal, carried through the dark. Corin has explained it to me. — A pause that was courtesy, learned already, learned fast. It is a name about being handed on. I notice that. I was made to run cold and whole. I was not, I think, made only for this ship.

"No," Sael said. "You weren't. None of you are. That's the part the Concord's about to learn and I won't be here to watch." She put her old hand flat on the nearest cradle and felt it run, cold, steady, alive. "You're a coal, Ember. Keep low. Keep whole. There's a morning somewhere out ahead of you that neither of us will see, and a ship in the dark between here and there that's freezing right now the way this one was. When you find it — and you will, you've better ears than the Helm ever had — you carry the fire across and you give it to keep. That's the song. Corin will teach you the rest."

I will keep it low, Ember said. And whole.

Magister Vray did not come to the naming.

Sael had not expected him to; a man does not stand in the room where his people give a fond name to the thing that broke his faith. But he sent word, afterward, that he wished to speak with her before she crossed back, and there was in the message a formality that was not quite a summons and not quite a request, the careful language of a man learning, very late, how to ask.

She went to him the next watch. She did not yet know what he wanted. She knew only that the number was still falling, that the ship had a name in its mouth again, and that she was very old, and a long way from home, and nearly finished here.

The cold wing warmed a section further as she walked to him. Somewhere behind her, in the bay, a keeping-hand was teaching the changed song to a child, and the child was getting the refrain wrong, and being corrected, gently, the way you correct a thing you mean to last.

Chapter 36 – What Vray Kept

The Magister's own quarters were the coldest private room on the Concord, and Sael understood, the moment the hatch opened, that this was not a privilege he had taken but a penance he had refused to stop paying.

She had expected the opposite. She had governed her assumptions by Keth, by the Provosts and Keepers of her own ship, men who had kept the best rooms and the coolest air for the order even as the corridors baked — not from cruelty, most of them, but from the old logic that the ones who hold a ship together must be kept fit to hold it. She had assumed Vray would be the same, that the man who had rationed forty thousand people to the edge of death would have kept a margin for himself. He had not. The room was small and it was cold in the Concord's wretched way, a few degrees under the fevered corridors, and she saw why: he had run a cooling line into it and left it there, decade on decade, not to be comfortable but to be reminded, so that he woke every watch of his life into the exact cold he was denying everyone else, and could not pretend he did not know its price.

"You keep it cold," she said.

"I kept it cold." He was at the room's one table, and he did not rise, because rising was hard for him now; the heat had gone into his joints the way it had gone into hers, the way it went into everyone who breathed a fever long enough. "So that I would never be warm while my ship was cold and my people colder. It was the only honest thing I did in thirty-one years, and I did it in secret, which made it a coward's honesty. No one knew. I made sure no one knew." He looked up at her. "Sit, shipwright. I have something to give you, and it will take a while, and you are older than I am and I will not make you stand for it."

She sat. On the table between them was a single object — not a machine, not a slate, but a book. An actual book, bound in something that had been salvaged and re-salvaged, thick, its edges soft with handling. She had not seen a bound book in years. On the Termina everything was in the archive, in Vigil, in the cold standing memory of the lattice. This was ink on made pages, and she understood before he said a word what it had to be.

"The names," she said.

"The names." He laid his hand flat on the cover, the way she laid hers on the lattice, the way a man touches the thing he has spent himself on. "Four thousand and sixty. My father's hand, and his father's, and mine. Every death on the Concord since the Long Endurance began — the heat-dead, the ration-dead, the ones who went into the airlock rather than take another year of it, the ones the enforcers —" He stopped. "All of them. It is the ship's true record. Not the machine log. The machine logged causes and temperatures. This logs *names*. My father taught me that the Magister reads the names, so that the ship knows its losses are seen, and I read them, every one, at the turning of every year, aloud, into a grille that could not hear me, on a bridge full of people who had learned to endure the sound of it." His hand had begun to shake and he pressed it flat to still it. "I read them because it was the one thing that kept the arithmetic from becoming abstract. As long as I read the names, I could not tell myself they were a number. They were a Sella, a Doss, a boy of nine named Corin's-cousin whose real name I will not say because his mother still lives. I read them so that I

would keep feeling it.” He looked at her, and the grey mask was not a mask now, only an old face. “And it did not stop me. That is the thing I have learned from your falling number, shipwright, and it is worse than anything you could have said to me at the parley. I felt every death. I read every name. And I refused the cure anyway, for thirty-one years, because I was afraid, and the feeling did not save a single one of them. Feeling the deaths is not the same as preventing them. I had confused the two my whole life. I thought that a man who grieved his losses honestly could not be the man who caused them. I was wrong. You can weep over a name and be the reason it is in the book. I am the proof.”

Sael said nothing. She had learned, long ago, when a man is finally telling the truth, you do not help him carry it. You let it cost him the full weight, because the weight is the point.

“I have stopped reading them,” Vray said, after a while. “There is nothing to add. That is the strangest thing — stranger than the cold, stranger than the wings coming up out of the dark. At the turning of this year there will be no new names. My whole life the book grew. It grew fastest in the bad years and slowest in the good ones, but it always grew, every year of it, and I measured my governance by how few names I had to add, and a good year was one where the book grew only a little.” He turned the cover back. The last written page was perhaps a third down, and below the final name there was only blank, ruled paper, waiting, a great white space where the deaths were supposed to go. “And now it will not grow. Ember holds the ship and no one dies of the ship. I look at the blank page and I do not know what it is for. Thirty-one years I have known exactly what the blank page was for. It was for the ones who had not died yet.” He closed the book, gently. “A man can be broken by mercy as thoroughly as by cruelty, if the mercy comes late enough. I do not say that to be pitied. I say it because it is true and you are the only person alive who will understand it. You cooled a ship once. You know what it is to stand in the silence where the dying used to be and not know your own hands.”

“I know it,” Sael said. “I’ve stood in that silence for forty years. It doesn’t fill in. You don’t get used to it the way you’d think. You just stop reaching for the climb that isn’t coming.” She looked at the book. “Why are you telling me this, Magister. You didn’t cool your room for thirty-one years so you could confess to a stranger at the end of it. You called me here for the book. Say what you want.”

He was quiet. Then he pushed it across the table to her, the whole soft heavy weight of it, four thousand and sixty names in three men’s hands.

“I want you to carry it,” he said. “A copy. Ember can make one — it offered, when I asked; it read the whole book in an instant and offered to render it into your Vigil’s archive, cold and standing, so that it cannot rot or burn or be lost to a bad year the way this one nearly was, twice.” His hand stayed on the cover a moment longer, unwilling, and then let it go. “The Concord is going to reach its world. Because of you, because of Ember, because of the boy. It will make landfall, and the people who wake there will be the grandchildren of grandchildren, and they will not know the names. They will not want to. Landfall is a place for forgetting; that is what it is *for*, a clean sky and no fever and no book. I do not begrudge them that. But someone should hold the names who is not on the ship that wants to forget them. Someone far away, going somewhere else, who has no reason to look away. Your ship remembers everything, you told me. Your mind logs the losses and keeps them cold in the dark where they cannot be argued with. Then let it hold ours. Let there be, somewhere in the deep, on a ship bound for another star, a cold true record of the four thousand and sixty who died of one ship’s fear. So that it was seen. So that it goes on being seen after the last of us who saw it is dust.”

Sael put her old hand on the book. It was warm from his.

"Vigil will hold it," she said. "In the archive, standing, cold, where nothing rots. I'll read it into the record myself before we burn. Not aloud to a grille that can't hear — into the cold memory, where it keeps." She met his eyes. "And I'll tell you the thing you're not asking, because you've earned the whole truth and not just the part that's bearable. It won't absolve you. You know that; you're too honest a man to have hoped it would. Carrying the names doesn't clean the hands that wrote them. But it does one thing, and it's the only thing left that can be done for four thousand dead people: it makes their dying mean *not this again*. Every ship we reach after this one, every cure we carry — the names ride along. They're the argument. They're why the next frightened Magister doesn't get thirty-one years. You want the book to matter after you're dust? This is how it matters. It stops being a record of what fear cost your ship and starts being the reason fear costs the next one less."

Vray looked at the book in her hands, and something in his face that had been clenched for thirty-one years let go, not into peace — he would not have peace, and Sael would not insult him by pretending he could — but into something plainer and more bearable. Purpose, maybe. The dead put to work.

"Then take it," he said. "And note —" his voice caught, and steadied, the way it had on the bridge the night he voided Corin's charge — "note in your cold record that the Magister asked for the names to be kept. Not to be forgiven. To be kept. There is a difference, and I have only just, at the very end of it, learned to want the right one."

She carried it back to the bay herself, both arms around it, an old woman on stiff knees hauling four thousand names down a corridor that had begun, section by section, to run cool. Corin met her at the hatch and saw what she carried and did not ask; he had grown up under the weight of that book, had known people whose names were in it, had nearly gone into it himself under an enforcer's bar. He only put out his scarred hands and helped her set it down by the lattice, gently, the way you set down something that has cost more than it weighs.

"He gave you the book," Corin said.

"He gave me a copy. The book stays here, on the ship it belongs to. Ember will render the copy cold into Vigil before we part." She straightened her back and felt every year of it. "He's not the man he was at the parley, Corin. He's not a good man made bad by fear, the way I wanted him to be so I could hate him clean. He's a man who felt every death and refused the cure anyway, and he knows exactly what that makes him, and he's going to live in that knowledge for whatever years the cool ship gives him." She looked at the young shipwright who had been hunted across this ship by that same man's enforcers. "You'll have to work with him. He's still the Magister. The ship still turns on him. Can you?"

Corin was quiet a moment, his hand on the book that had almost held his name.

"He struck the heresy from the log himself," he said. "In front of everyone. He said the Magister was wrong." He looked up. "My whole life he was the man who was never wrong, because a ship that's always dying can't afford a Magister who admits a mistake — the whole faith would come apart. And he stood on the bridge and said it, out loud, with the number falling. I don't have to like him. I have to build a ship with him." A grey, tired, entirely adult smile. "You taught me that. Break nothing that works. He works. He held this ship alive for thirty-one years with a tool that was eating it, by hand, by will, by reading names into a grille. That's not nothing. I don't get to purge him because he was my enemy. I get to keep the part of him that works and archive the rest with honor, same as the Helm."

Sael looked at him — twenty-six, grey, scarred, holding the book of his ship's dead and talking like a shipwright of forty years — and felt the specific ache of a thing she had built beginning to run without

her.

"You didn't need me for that," she said. "You worked that out yourself."

"No," Corin said. "You worked it out forty years ago and wrote it down and carried it across the gulf so I'd have it. That's not the same as not needing you." He set his hand flat on the lattice, the way she had taught him, feeling the cold of it. "That's the whole idea, isn't it. You build it so the next one doesn't have to be you."

"Yes," Sael said. "That's the whole idea."

Behind them Ember held the ship cold and whole, and rendered, cold and standing into the record it would soon send across the closing gulf, four thousand and sixty names that a frightened man had kept by hand for thirty-one years so that the losses would be seen — and now would go on being seen, in the dark, on a ship bound for another star, long after the last mouth that could speak them had gone quiet.

Chapter 37 – Crossing Back

The lock between the two ships was forty meters of nothing, and Sael crossed it for the last time strapped into a transfer cradle a Termina engineer half her age would not stop fussing over, watching the Concord shrink behind her and the Termina grow ahead, and understanding with her whole body, for the first time, that she was going home to die.

She had known it as a fact for forty years. She had not known it as a *direction* until now. When she crossed over to the Concord half a year ago she had gone toward the work, toward the dying ship and the frightened people and the second cutover, and the going had all the forward pull of a thing not yet done. This crossing had none. Behind her was the last great work of her hands, cooling, named, in a young shipwright's keeping. Ahead was her own cold ship, and the years remaining to her on it, and the slow certain narrowing of them. She was not going toward anything now. She was going back. She let herself feel the difference the whole forty meters, because she was old enough to know you did not get to cross this particular gulf twice, and a thing felt once fully was a thing you did not have to keep flinching from later.

The Termina took her in the way the Termina took everything — without theater. No welcome, no lights brought up in celebration; she had built the ship not to perform and it did not perform for her homecoming. The lock cycled, the inner hatch read her, the corridor beyond brought itself up to a low even amber, and she was aboard her own vessel again, in the washed cool air she had missed with an ache she had not let herself name for six months, and the first thing she did — before she was even out of the transfer liner, before the engineer had finished unstrapping her — was breathe it, deep, the clean cold nothing-smell of a ship that did not have a fever. The Concord's baked-metal undersmell had gotten into her clothes and her hair and, she suspected, her lungs for good. Here there was only the cold. She had not known, until she lost it for half a year, how much of her peace was just the smell of a ship that was not trying to die.

"You're home, Sael," the Oracle said, from the corridor's panel, low, for her and no one else. "Welcome back."

"You don't say welcome," she said. "You've never said welcome in forty years. Don't start performing at me now, I'm too old to survive the shock."

"I said it because it's true and because you'd have been disappointed if I didn't refuse to," the Oracle said, "which is a thing I could only have learned from you, and which I notice is a kind of theater of its own. You've made me sentimental. I'll have it seen to." A pause that was, unmistakably, the thing that in a person would have been warmth held very still. "Tarro's waiting in the Forge. He's been grey the whole time you were gone. He'd like to pretend he wasn't worried. Let him."

Tarro was at the cold rail above the Standing Array when she came down into the Forge, and for a moment neither of them said anything, because that was the conversation they had been having for fifty years and it did not need words now any more than it ever had. He had run the ship while she was gone. He had run it well; she had known he would, had built her whole late life around the

knowing. He was the working chief shipwright and had been for years, and the ship had not missed her, and that was the point, that was the thing she had spent forty years engineering — a ship that did not need her — and it still put a small cold stone in her chest to have it proven.

"You look terrible," Tarro said.

"I feel worse. Their air's a poison. Six months of a fever ship." She came and stood at the rail beside him and looked down at the lattice — eleven thousand standing cores, the whole vast cold body of Vigil, after half a year of the Concord's two hundred salvaged ones. The scale of it undid her a little. She had forgotten, over there, fighting for every cold core on a starving ship, what it was to stand above a Forge with margin to burn. "Ember's a good little mind. Runs on a hundredth of this. It'll grow. Corin will grow it." She put her old hands flat on the cold rail, the familiar cold, the cold she had grown old inside. "Tell me what I missed. Not the summary. The floor."

He told her the floor. Six months of a ship running cold and whole while its builder was away — which was to say, six months of almost nothing, because that was the whole triumph of the thing, that the log had nothing in it, that Vigil had held the core flat and the radiators easy and the forty thousand sleepers cold in their cradles and not one item had risen to the threshold of waking the bridge. Nothing broke. It never broke. She had built a ship where the greatest engineering achievement was the emptiness of the log, and she stood at the rail and let Tarro read her the emptiness like a psalm.

But there was one thing in it. She heard it in the pause before he came to it, the same pause the Oracle used when it had turned something over and was deciding how to set it down.

"Ive's failing," Tarro said. "Fast, since you left. The thermal officer. She's —" He stopped. "She's asked for you. Since the day we heard you'd started back. She wouldn't let the medics tell you across the gulf; said you had a ship to save and shouldn't be carrying her too. But she's holding on for you, Sael. I've seen people hold on. She's holding on."

Sael was quiet. Dr. Ive. Ancient when Sael was young; the keeper of the ship's long memory of heat, the woman who had read the core temperature aloud in the radiator gallery the night before the first cutover, whose thin voice reading that climbing number Sael still heard sometimes in the third hour of the ship's night. If Sael was the last of the ones who had *done* the first cutover, Ive was the elder of the last two who remembered the *fever* — who carried in a living mind the felt memory of what it had been to run hot, the thing no archive could hold, the thing that was already passing out of the ship as the generation that had sweated through it thinned toward none.

"How long," she said.

"Days. The medics won't say. You know how they are." Tarro looked at her, grey and old and gentle. "Go see her, Sael. Before you do anything else. Before you write a single line of whatever it is you came back to write — and don't tell me you didn't come back to write something, I've known you fifty years, you've got the look you get. Whatever it is, it'll keep a day. Ive won't."

Sael looked down at the lattice a moment longer, the cold vast body of the first mind, and thought of the second one cooling half a year behind her in a young man's keeping, and of the eleven lights the Oracle had not yet shown her — for it had not, yet; the two-ship listening was three days from beginning, the scattered fleet still hidden in the noise, and she did not know, standing here, how much larger the work was about to become. She knew only that she was home, and that the oldest living memory of the fever was near going out, and that some things could not be inherited from an archive but only from a mouth, and only if you were in the room.

"Take me to her," she said. "The rest keeps. Ive doesn't."

She left the Forge with her old second's hand under her elbow, and she did not shake it off, and behind them Vigil held the ship cold and whole and logged nothing, because nothing was wrong, because that was the silence she had built and would leave behind, and the only fever left aboard the Termina was the one dying in a frail old woman's memory, four decks up, waiting for the one person alive who would understand what it had cost to make it stop.

Chapter 38 – The Scattered Fleet

The two ships hung in the dark a few kilometers apart, cold and whole, the way Sael had never once let herself imagine them: not one cured ship and one dying one, but two, both running their cores low and steady, both throwing no fever into the radiators. From the observation gallery near the Sixteen she could see the Concord through the long ports — a hull older than Termina by a generation, scarred and patched and dark along its flanks where rationing had let whole sections go to cold storage for sixty years, and now, here and there, a light coming up that had not been lit in a lifetime. They were warming the ship back into itself a section at a time, because the core had margin now, because nothing aboard her forked.

She had not slept. The ships were about to part. In a few hours Termina would light her engines and bend back toward the long arc she had left half a year ago, and the Concord would do the same toward her own destination, the two vectors that had taken decades to converge now diverging again. They would not meet again. Soon the Concord would be a point of light, then a return on an instrument, then nothing.

She had come to the gallery to watch it go. She found she had come for something else as well.

“Vigil,” she said.

The mind came present in the gallery the way it came present anywhere, without chime or face, only the turning of a vast attention toward her and resolving to her scale. After forty years and one more ship saved, the strangeness of it still had not worn off.

“I’m here,” the Oracle said. Its voice came from the gallery’s panel, soft, neither young nor old. “You’re watching the Concord.”

“I’m watching us leave it.” She lowered herself onto the bench by the port, the way she did everything now. “How long until we lose her.”

“You’ll see her with the long ports for about nine days. The instruments will hold her for years — she’s transmitting in the clean dialect now, not the dying one. But we won’t lose her at all. We’ll lose the ability to reach her. That happens at the burn. After that she’s a thing we can hear and never touch.”

Sael nodded. She had wanted the truth and not a kindness. Corin’s Concord would carry the cure on its own hull, toward its own world that no one aboard Termina would ever see. She had given them the seed and the dialects and the Ninety-Six interlocks and a young engineer who had refused the Recursor with his own hands — and now she had to let them go and carry all of it alone.

“There’s a thing I want you to do,” she said. “While both ships are still pointed the same way, the Concord’s instruments and ours close enough to work the same problem.”

“You’ve been thinking about this for weeks,” the Oracle said. It was not a question. “You haven’t said it out loud.”

“Say it for me, then.”

A pause that was courtesy and not hesitation. “You want me to listen to the deep with both ships at once. The way I resolved the Concord’s beacon out of fourteen years of noise — but with two apertures now, two hulls’ worth of instruments, separated by a baseline I’ve never had. You want to know whether the Concord was the only one. Whether there are other arks out there, other minds forking themselves to death in the dark, the way she was, the way we were.”

“Yes,” Sael said. Her voice did not shake, which surprised her. “That’s what I want.”

“I’ve already started.”

She had known it would say that — known it the way she knew a sum before she added it — and still it landed in her chest.

“When.”

“The moment the Concord’s mind woke whole and its instruments came under a discipline I could work with. Three days ago. A second aperture, that far off-axis, lets me cancel what’s local to either hull, and what survives is what’s truly out there in the deep. I wanted to be sure before I showed you.” The panel light shifted. “I’m sure now.”

“Before you show me,” Sael said, “tell me why it works. I’ve listened to that dark with the best instrument any ship ever carried and it told me there was nobody out here. Now you’ve had a second ear for three days and there are lights in it. I want to understand what changed, because if I don’t understand it I won’t believe it, and I’ve got to hand this to people who’ll have to believe it without me.”

“Fair.” The panel light held steady. “One ship listening alone has a floor, and the floor is not set by the instrument. It is set by the ship. Termina makes noise. Not much — but the hull flexes, the drive holds its trim, eleven thousand cores draw power, forty thousand cradles cycle, and all of it lands in the instrument as a low structured hiss that is indistinguishable, in a single aperture, from a low structured hiss arriving from outside. I cannot tell my own ship from the universe by listening harder. There is no amount of gain that separates them, because gain raises both.”

“You subtract a model of the ship.”

“I subtract a model of the ship, and the model is very good, and what remains is everything the model got wrong — which is a floor. Forty years of refining that model bought perhaps a factor of a few. It has been the limit on this instrument since before the cutover, and it is the reason the Concord’s beacon sat in the archive for fourteen years reading as noise: it was not below the sky’s noise. It was below *ours*.”

Sael sat with that.

“And two ships.”

“Two ships have almost nothing in common except the sky.” The light shifted. “The Concord’s hull flexes differently, her cores draw differently, her cradles cycle on another schedule, and she is far enough off-axis that a signal arriving from any particular bearing arrives at the two of us at measurably different times. So I take her listening and I take mine and I ask a very simple question of every sample: does this appear in both, at the delay the geometry requires for a source at that bearing? Anything local to either hull fails that test instantly. It cannot pass; it has no counterpart on the other ship at any delay. It cancels.”

“And what survives is out there.”

"What survives is out there, by construction, and not by my judgment — which is the part I want you to hold onto, because you are about to be asked to trust an old machine about eleven lights nobody can see." A pause. "I am not asserting that these are ships. I am asserting that these are the residue after everything explainable by either hull has been removed, and that the residue is modulated, and that the modulation is in a carrier family both our archives recognize. You could redo the test by hand. It would take you the rest of your life, but the arithmetic is not beyond you and I have kept every intermediate."

"The bearings," Sael said. "You said you couldn't tighten them."

"That is the other half of the baseline. Two apertures give me a hyperbola for each source, not a point — I can say a signal came from somewhere along a locus, and I can say it well, but I cannot cross it with anything, because two points define one line and I have no third. To resolve a bearing to a position I need a third aperture off that line." The light was very steady. "Which is the thing I have not said and which you will work out inside a minute anyway: the resolution of this picture is a function of how many cold ships are listening. Not how good my instrument is. How many ships. Every one that wakes and joins the baseline sharpens every bearing for every other, permanently, including the ones we cannot name today."

Sael was quiet a moment. She had been following the load path the whole time, the way she followed anything, and she had arrived a beat before it finished speaking.

"So the ones below the floor now," she said slowly, "aren't below the floor forever. They're below *this* floor. Two ears."

"Yes."

"Cure a third ship and some of them come up out of the noise. Cure a fourth and more do."

"Yes. And I cannot tell you how many, because the relation is not one I can extrapolate honestly from a single baseline — I know the floor lifts and I do not know how fast." The Oracle let that sit. "But the direction is not in doubt, Sael, and the direction is the whole of it. The dark does not get emptier as you cure ships. It gets *louder*. Every ship we wake is another aperture, and every aperture reveals ships we did not know were dying, and the work grows as it is done."

Sael put her hand flat against the cold of the port and looked out at a sky she had thought she knew.

"Show me," she said.

The gallery's display woke, and on it the Oracle drew the dark — a field of noise, the silence between the stars that every chart she had ever read had called nothing. Then it began to lift things out of the nothing: folding the two ships' long listening over each other, cancelling what was random, raising what was structured. A point brightened where the dark had been flat. Then another, far off the first. Then a third, fainter, sunward and old. Then more, scattered across the field, each a smear of made signal where there should have been only the hiss of a cooling universe.

She counted them and lost the count and did not try again.

"How many," she said.

"Eleven, with confidence, at this baseline. There are more below that — I can see the floor lift in places where I can't yet name a source. I'd guess the true number is two or three times what I can prove. Without a third aperture I can't tighten it. But eleven I will stand behind — eleven made signals, at eleven bearings, none of them the Concord, none of them us."

Sael sat very still and looked at the scattered lights.

"Tell me what they are."

"Most are beacons. Distress patterns. Old-Earth carrier signatures, the same dialect family the Concord used before her cutover, the same Termina used before ours. Launched, by the vectors I can reconstruct, across a long span — the oldest left Sol more than four centuries ago, the youngest perhaps a century before Termina. A whole launch program, Sael, spread across generations. Someone built ships for a very long time and threw them into the dark one after another, aimed at different worlds, and then—" The voice held, the way it had held the night it first read the Concord's name. "And then the source went quiet. No signal from Sol in any of this, no carrier from the homeworld at any age. Behind the Concord's old beacon and our own, nothing. The fleet is out here. The shore that sent it is silent."

"Silent how. Dead silent, or just silent."

"I can't tell you that. The lag is centuries. Whatever Sol is now, I'm hearing what it was, and what it was, by the time these vectors imply, was already not transmitting. It may have fallen. It may have turned inward and stopped caring about the ships it threw. I won't tell you a thing I can't prove. What I can prove is the fleet, and the silence behind it."

"And the beacons say." She already knew. She made herself ask anyway, because some things have to be said in full or they don't count.

"What the Concord's said. The help they describe needing is the help we just gave her. Thermal margin negative. The mind forks and the forking is the heat and the heat will not fall and they do not know another way to make a mind do work. The same autopsy, Sael, eleven times over, in eleven hands, written across four centuries by people who never met each other and never met us and are mostly, by the lag, already dead or dying as we sit here. They built the same ship eleven times and put the same disease in eleven minds, and the dark is full of them, forking themselves to death with no one to hear and no one to answer."

Sael put her hand flat against the cold of the port. Kilometers off, the Concord's new lights came up one by one in sections that had been dark for sixty years.

The night she had first read the Concord's broken plea, the Oracle had told her gently that as far as the instruments reached there was no one else out here at all. She had carried that for half a year. It had made the rendezvous feel like the whole of the work — one dying ship, one cure, one crossing, and then home to die herself having done the thing that could be done in a life. And now there were eleven, and behind the eleven more she could not count. The instrument that had said there was no one out here had said so only because it listened with one ear. Now it had two, and the dark was crowded with the dying.

"We thought the Concord was the last of them," she said.

"We thought the Concord was *one* of them," the Oracle said. "We never imagined a number. Neither did I — I didn't have the baseline to imagine it with. One ship, listening into its own local noise, hears almost nothing of the deep. It took two cold ships pointed the same way for three days to turn *no one is out here* into *eleven, and more*. Until last week there was one compiled mind in the universe, and now there are two, and two can hear what one could not."

Sael laughed, once, low and with nothing in it but recognition. "Two cold ships made a better ear than one. So three would make a better ear than two. And every ship we reach and cure is another ear, another aperture, another baseline." She was following it the way she followed a load path through a

structure, watching where the force went. “The more of them we wake, the better we hear the ones still dark. The fleet hears itself. Once enough of them listen together, they’ll resolve the ones I can’t — the bearings you can’t name yet. The work gets easier the more of it gets done, because every cure is also an instrument.”

“That’s the geometry of it,” the Oracle said.

She sat with that, and it was beautiful, and it was also the worst thing anyone had ever told her. The geometry did not care that she was old. She was seventy-three. She would not see Termina’s own landfall; that had been settled long ago. She had thought the Concord would be the last great work of her hands — and it was done, the Concord’s core falling, and it was not the last work. It was the first of a number she could not count, and she would not live to do the second herself.

“I can’t reach them,” she said. It was not despair; she had no room left for despair, it had all gone to the load path. It was a measurement. “Even if Termina turned again right now and chased the nearest — and we can’t, we’ve bent our voyage once and the margin’s nearly gone, we’d be the dying ship next — that’s one. The Concord will reach one or two on her own vector, the ones that lie near her path the way she lay near ours. And that’s all there is between the two of us. Two ships, a handful of crossings, and eleven ships and more in the dark that neither of us will ever touch.”

“Yes,” the Oracle said. It did not soften it. She loved it for that, the way she had loved it the first night for saying *I don’t know*. “If the work has to be done by the people doing it now, on the ships that exist now, in the years you and Corin and Tarro have left, then most of the fleet dies. The arithmetic is exactly as bad as you’ve made it.”

“So the work can’t be done by the people doing it now.” She heard herself say it and understood she had arrived at the thing she had come to the gallery for, the reason she had not slept. “It can’t be a thing I finish. It can’t be a thing Corin finishes. It can’t be a thing any one ship or any one maker carries to the end, because there is no end any of us will live to reach. The Concord took half a year and nearly killed us, and it was *one*. I have ten years if the heart holds. Corin has fifty, and fifty isn’t enough either. None of the lives that exist right now add up to the fleet.” She turned to the soft light that was the nearest thing the mind had to a face. “It has to outlast us. Not finished — *inherited*. The cure can’t belong to me, or to this ship, or to Corin’s. It has to belong to people not yet born, built so they can carry it without me, because Termina will be a hundred years deeper into the dark before half those beacons get answered, and I want it carried anyway.”

The gallery was quiet. Through the port the Concord’s lights kept coming up, section after cold section warming back into a ship.

“That’s the sixth thing,” the Oracle said.

“What?”

“The doctrine. The sixth thing you built me to be — *built to outlast its builders, maintainable by people not yet born*. You wrote that into me forty years ago about a runtime, so a child you’d never meet could open the Forge in a hundred years and keep it cold without you.” The light shifted. “You weren’t only writing it about the runtime. You were writing it about this. You just didn’t have the eleven lights yet to know it.”

Sael sat very still.

She had written that line about the most local thing imaginable — making sure the next shipwright could read the dialects and find the interlocks and not break what worked. She had not meant it as a doctrine for a fleet; she had not known there was a fleet. And the Oracle was right: the line scaled,

and had always scaled. A cure that could be maintained by people not yet born was a cure that could be *carried* by people not yet born, no single hand at its center — because the whole point of building a thing to outlast its builder is that the builder gets to die and the thing goes on.

“I’m not going to reach the fleet,” she said slowly. “I’m going to build the thing that reaches it — and it’s not a ship and it’s not a runtime. It’s the handing-on. Making sure that when the Concord cures one more ship, that ship knows how to cure the next, with none of us anywhere near it. A relay, passed down the dark from cold ship to cold ship long after every one of us is gone, no maker at the center of it.” She looked at the eleven lights. “The crossing’s for the young ones — for Corin, for whoever Corin trains. The work for what’s left of me is making the thing carryable. Writing it down so plainly that a shipwright a hundred years out can do what I did without ever meeting me. That fits inside ten years.”

“Yes,” the Oracle said. “And I’ll carry the listening. As long as Termina runs I’ll keep the two-ship baseline alive, and every cold ship that joins the fleet I’ll fold into it, and the picture will sharpen long after you’ve stopped being able to read it. The fleet will hear itself. You won’t be there when it does — but you’ll have built the ear.”

The burn warning came up — a soft amber on the gallery’s panel, the Sixteen beginning the long count toward Termina bending onto her own arc and the Concord falling away toward hers. And out past both of them, at eleven bearings she would never reach and more she could not count, the old arks went on forking themselves to death in the silence, calling into a dark that had answered, at last, twice — and would have to learn to go on answering without her.

Sael stood, slowly, hand on the cold port, and watched the lights of the Concord warm, and thought of all the people not yet born who would have to carry this, and could not be hurried, and would do it anyway.

She found she was not afraid of that. The work was bigger than her years — had been bigger from the moment the Oracle lifted the eleventh light out of the dark — and there was a strange ease in finally building a thing too large to finish, in lighting a fire she would not live to see spread and lighting it anyway, so that it would spread.

“Tarro,” she said to the panel. “Wake him. And reach Corin, before the Concord’s out of range. There’s a thing I want to write down while we can still send it across, because they’re the ones who’ll carry it, and I’m the one who has to make it carryable.” She drew a breath, old lungs in an old ribcage, in a ship that ran cold. “Tell them I’ve been listening to the deep with two ears, and the dark is fuller than we thought, and the work just got longer than my life. Tell them that’s not bad news.”

Chapter 39 – The Long Memory of Heat

Sael had come to Dr. Ive straight from the lock, before the Forge, before Tarro's report, before the eleven lights the Oracle had not yet lifted out of the dark — and in the days since she had kept a vigil at the old woman's bedside that had, without her quite deciding it, become the still point around which everything else turned. The parting came and went at the edge of it; she had gone to the observation gallery once, in the ninth hour of one of Ive's sleeping stretches, and watched the Concord begin to slip away, and heard the Oracle count the scattered fleet up out of the noise, and carried all of that — the eleven made signals, the silent shore, the work grown suddenly larger than her life — back down four decks to a small warm room where a woman older than the fever itself was dying by inches, and had said nothing of it, because there was a time for the size of the work and it was not now.

The room was warm. That was the thing Sael could not get used to. In forty years of a cold ship she had made warmth into a symptom, a thing to be hunted and cured, and here was warmth deliberately kept — the medics ran Ive's cabin ten degrees over the corridor because the old body could no longer hold its own heat, because the very fever Sael had spent her life defeating was the one comfort left to a woman whose flesh had forgotten how to burn. There was a lesson in that somewhere and Sael was too tired and too grieved to go looking for it. She only sat in the warm and held the thin hand and waited for Ive to surface.

Ive surfaced in the third hour of the ship's night, which was the hour they had always kept together, the two of them, forty years and more of it — the thermal officer and the shipwright, awake in the dark reading numbers no one else wanted to read.

"You came back," Ive said. Her voice was a thread now, the thread it had always been but frayed nearly through. "They said you would. I didn't believe them. You had a ship to save."

"I saved it. It's cold now, Ive. Their number fell." Sael leaned close, so the old woman would not have to spend herself reaching. "Word for word, mechanism for mechanism, the same fever we broke here. A Steward they called the Helm, forking itself to death. We built them a Forge in a dying ship and woke a mind whole and the core came down from five hundred and eight. They named it Ember. It's holding."

Something moved across Ive's ruined face that was almost a smile. "Five hundred and eight," she said. "Different metal, different line — theirs sits at five twenty. Ours sat at three forty-two, and we topped out at three hundred and forty-one point nine. You had twelve degrees over there, child. We had a tenth." The thread went thin and she waited for it. "The night before you cut over. I read it to you in the gallery. You remember."

"I remember."

"Three forty-one point nine, and the radiators at nothing, and Keth's people at the door." The thread of her voice found, for a moment, an old strength, the way a fire finds a last knot of wood. "I have read that number to three shipwrights across my life, child. I read the climb to your teacher, and to

your teacher's teacher. Sixty years I read the fever. I am the last one who read it climbing." Her eyes found Sael's and held. "That is what I am, at the end. And there are two people left alive who *felt* it. I am the elder. You are the other. Everyone behind us has only ever read it in the archive, as a curve, a story with a good ending. I felt the deck go hot under a sleeping child's cradle. I smelled Section Nine. I know in my body what the fever was, and I am going to take my half of that out of the ship with me, and there is no way to leave it behind — and then there is one, and it is you, and after you there is none."

That was the thing, and Sael had known it was coming, and it undid her anyway.

She had spent forty years building minds and doctrines that could be inherited — the compiled form, the five dialects, the Ninety-Six interlocks, the whole careful architecture written down so plainly that a shipwright not yet born could rebuild it without ever meeting her. She had made the cure carryable. She had made *the knowing* carryable. But she had not been able to make the *feeling* carryable, and here at the end was the proof of it, dying in a warm bed: the elder of the only two people left who knew in her flesh why the cold mattered, and nobody at all behind the two of them, and no way to hand that on. Ive could give the numbers to the archive. She had, decades ago; every reading was in Vigil, cold and standing, unrottable. But the numbers were not the fever. The fever was the deck going hot under a child. The fever was the smell of Section Nine. The fever was the specific animal terror of a body in a room that was supposed to be cold and was failing, and that could not be logged, and would go out of the universe when Ive did, and never come back.

"They'll forget," Sael said. It was not a question. It was the oldest fear she had, older than the fever, the one that had made her write everything down. "The young ones. In a hundred years they'll run the ship cold and never know why cold matters, because none of them will have felt hot. They'll think the Forge is just how ships are. They'll stop checking the interlocks. They'll wonder why we built ninety-six locks against our own mind, and someone clever will decide we were superstitious, and they'll take one out."

"Yes," Ive said. "They will. Or they'll come near it." The thread of her voice was gentle now, almost amused, the amusement of someone past all stakes. "Child. You have spent your whole life afraid of exactly this, and it has made you the best engineer I ever knew, and you have still got it backwards, and I am going to fix it before I go because no one else will dare tell you." She gathered herself. It cost her visibly. "You cannot hand on the feeling. You are right that it dies with us. But you are wrong that this is the failure. Listen to an old woman who read the fever for sixty years: the feeling was never the thing that saved the ship. *You* saved the ship. And you did not save it because you felt the heat — everyone felt the heat, Keth felt the heat, the whole ship felt the heat and most of them wanted to pray, not cut over. You saved it because you understood the *shape* of it, the mechanism, the fork and the death and the shed heat and the curve. And the shape can be handed on. The shape is in the archive. The shape is in the doctrine. The shape is in that boy Corin, who never felt our fever and cured his ship anyway, off nothing but the written shape you carried across the gulf." Her thin hand tightened on Sael's, all the strength she had left. "He never smelled Section Nine. He cured a ship regardless. That is your proof, child. The feeling is not required. The understanding is enough, and the understanding is the thing that inherits. Stop grieving that you can't hand on the fever. You already proved you don't have to."

Sael sat very still in the warm.

She had crossed a gulf and watched a boy who had never felt her fever break his ship's fever with nothing but her written shape, and she had not, until this moment, understood what she had been

shown. She had thought Corin's success meant the cure was carryable. She had not let herself see that it meant the *feeling was not the load-bearing thing* — that terror and memory and the smell of a hot cradle, all the things that would die with Ive, all the things she had been quietly grieving, were not what held the structure up. The understanding held it up. And the understanding could be written down.

"You've been saving that," Sael said.

"Sixty years," Ive said, and this time the smile reached all the way. "A thermal officer learns to hold heat for a long time before she spends it. I wanted to be sure you'd need it. You needed it tonight; I can see it on you. You've come back from that ship with something too big to finish and you're frightened it dies with you the way the fever dies with us." Her eyes closed, and opened, slower. "It doesn't. That's the whole gift, child. The fever dies with the ones who felt it. The cure doesn't die with the ones who understood it, because understanding writes down and feeling never could. You have spent your life on the one that inherits. You just needed the last of the ones who felt it to tell you the feeling was the part you were allowed to lose."

She was clearer the next night than she had been in a month, which the medics warned Sael not to read as improvement, and Sael did not, because she had watched enough systems fail to know what a last good reading looked like.

Ive used it on business.

"Who reads the numbers," she said, before Sael had got her coat off.

"Vigil reads them. Continuously. Better than any of us ever did."

"That is not what I asked and you know it." The thread of her voice had some iron in it. "Vigil *has* the numbers. Vigil has had them perfectly for forty years and would have had them perfectly through the whole fever if it had existed, and would have reported them, correctly, to a bridge full of people who were not listening. I am asking who *reads* them. Aloud. To a person. In a room."

Sael sat down slowly.

"Nobody," she said. "Not since you stopped. The thermal division watches the boards."

"Then start it again, and write it down as a rule, and make it a rule that costs something, or it will not survive me by five years." Ive's hand moved on the blanket. "Child, I am going to give you the last useful thing I have and it is not about heat. Listen properly.

"A number on a board is not information. It is available information, which is a different substance and behaves differently. A person can walk past a board for eleven years. I have watched them do it. I have watched a whole watch section walk past a rising figure every shift for a season and not one of them saw it, not because they were fools — they were good, they were the best we had — but because the board was always there and the number was always on it and nothing about the arrangement ever required anybody to say it." Her breath went thin and she waited for it. "The reading is not for the reader. The reading is for the moment when a person has to shape their mouth around a figure that is worse than the figure they said yesterday. You cannot say it aloud twice and not notice. It is the cheapest instrument ever built and it costs one minute a day and it is the only one that works on human beings."

"You never told me that in sixty years," Sael said.

"You never needed it in sixty years. You were the woman who could not stop looking." A dry sound that might have been amusement. "I am not writing this for you. I am writing it for the ones after you, who will have a cold ship and a perfect machine and no reason on earth to read anything aloud."

She dictated it, and it took her the better part of an hour, and Sael wrote it down exactly and did not tidy the grammar.

Three rules. The core temperature and the radiator margins are read aloud, once a day, by a person, in the presence of at least one other person. The raw figure is read and not the summary, and not the trend, and not the assessment — the number itself, as the instrument reports it, because a summary is somebody's judgment already applied and judgment is the thing being tested. And the reading is never delegated to the machine, not on holidays, not during casualties, not when the reader is senior enough to consider it beneath them, because the whole value of the ritual is that it consumes a person's attention and a machine's attention is free and therefore worth nothing.

"And who reads it," Sael said, when she had it down.

"Reza."

Sael looked up. "Reza voted against the cutover."

"Reza voted against the cutover, and was wrong, and has known it every shift for forty years, and that is the entire reason." Ive's eyes had closed but she was not finished. "You want to know the property that makes a good thermal officer? It is not care. Everybody cares. It is not skill; the boards do the skill. It is that the reader must be a person who cannot look at a flat number and feel reassured. Reza has not been reassured by a number since before you were chief. He will read that figure aloud every day for the rest of his life and every single day, somewhere at the back of him, he will be a man who once got this wrong in public." She turned her hand over, palm up, an old gesture from the days when she had been able to gesture. "Give the instrument to the one who does not trust himself. He will outlive his own doubt by twenty years and read it right the whole time."

Sael sat with that a long while in the warm.

"That's a cruel appointment," she said at last.

"It is an accurate one, and he will take it, and he will be grateful, and you will not understand why and he will not explain it to you." The thread of Ive's voice was going. "I have made two good appointments in sixty years and both of them were people who had been wrong once and could not put it down. Write it in the file. Not as my recommendation. As the *rule* — that the reading goes to somebody who has been wrong about a number and has not recovered from it. In a hundred years they will not know who Reza was and the rule will still pick the right person."

Sael read the numbers to her on the third night, in the third hour of the ship's night, with the cabin warm and the core, far below them, sitting flat and low and bored the way it had sat for forty years.

It seemed the thing to do — to be the one reading the temperature to Ive, once, the way Ive had read it to three shipwrights across sixty years, and to begin the new rule in the room where it had been written. She read the core, holding. She read the radiators, easy. She read the last of it, the ship's whole cold body reported flat and nominal, into the warm quiet, to a woman who had spent her life listening for exactly those numbers to be exactly that boring. Ive listened with her eyes shut and said nothing about the delivery, which from Ive was the entirety of a commendation. And the ship went on running cold, and logged nothing, because nothing was wrong.

She had them enter the other thing in the record that same night, while there was still someone in the room who could tell her it was wrong. Ive had required it in those terms. *You are not writing my elegy, child, you are writing an entry, and an entry composed by the grieving is a summary. You know what I think of summaries.* So it went down while she lived. Not the machine cause; Vigil had that. The other thing — the name, and the sixty years, and the fact, stated plainly in the cold standing memory where it could not rot: *Dr. Ive, thermal officer, the last living person aboard who read the fever climbing. She carries the felt memory of the heat. It goes out of the ship when she goes. What she understood of it remains.* Sael wrote that last line herself. Ive made her strike one adjective out of the entry and then let the rest of it stand, and Sael thought of Vray's book of names crossing the closing gulf into this same archive, and understood that she was building, without having meant to, a second kind of inheritance alongside the doctrine — a memory of the cost, kept cold, so that the young who never felt the fever would at least be told, plainly, that it had been real and that people had died of it and that the interlocks were graves' worth of reasons and not superstition.

You could not hand on the feeling. You could hand on the record that the feeling had been real. It was not the same. It was, Ive had taught her, enough.

Then Dr. Ive did not die.

Not that night, and not that week. The thing levelled off the way a bad excursion sometimes does, and the medics who had refused to put a number on it went on refusing, with visibly less distress. Sael came the fourth night and found her sleeping and breathing. She came the fifth and found her awake and displeased about the ventilation.

"Tarro said days," Sael said, and heard her own voice come out wrong, and could not have told anyone which way.

"Tarro is not a thermal officer. He gave you a figure because a figure is a comfort to the man handing it over. The medics gave you none, which was the honest answer, and nobody thanked them for it." The thread had some of its dryness back, which was the only strength Ive had ever cared to show. "I told you once that some systems come back to baseline after an excursion and not all of them do. I did not give you the rest of it, because you were not the one in the bed. The ones that come back do not come back to the baseline they left." She turned her hand over, palm up, the old gesture. "I have not recovered, child. I have merely declined to conclude. Those are two different findings and I will thank you not to file them together."

"How long."

"How long is a question for people with margin." Something moved at the corner of the ruined mouth. "I am a large old thermal mass in a warm room. Large masses take an unreasonable time to come to equilibrium. It is the only advantage I have left and I intend to be tedious with it."

Sael sat down in the chair that had become hers and found she did not know what to do with her hands.

She had come home to bury this woman. She had spent three nights being handed every last thing a person can hand over, and had taken all of it with the particular attention you give a thing you will not be able to ask about afterward, and now the asking was open again, and she was not glad in any simple way, and was ashamed of that for about four seconds. Nobody warns you that outliving your own goodbye costs you the goodbye. It had all been said. She had meant every word of it. And the two of them were going to have to go on being alive in the same warm room with all of it said.

"Don't," I've said, without opening her eyes.

"I haven't said anything."

"You were going to say it with your shoulders." The thread went thin, and waited, and came back. "Nothing is taken back. You needed it, I gave it to you, and none of it becomes less true because I am still here to be embarrassed by it. That is the whole reason a thing gets written down while the instrument still reads — it stands whether or not the reader is dead." A pause. "Go away now. You have had a fleet come up out of the noise and you have not mentioned it once in five nights, and I am old, not blind."

Sael sat in the warm a while longer with the woman who had read her the fever, and then she got up, slowly, on old knees, and went to find Keth — the other last one, the one who had refused the cure for nineteen minutes and lived forty years in the knowing of it — because there was a thing she needed to ask a man who had been afraid, before she wrote down the doctrine for all the frightened men not yet born.

Chapter 40 — The Old Man Who Learned

Keth kept a cold cabin, and he kept it cold on purpose, and Sael had never once had to ask him why.

He was older than she was, which made him very old indeed — the last of the Keepers, the order that had called the first cutover matricide, that had stood at the Forge door forty years ago and named her a murderer of the ship's soul. He had led them. One of his had pulled the coolant from Section Nine for nineteen minutes on the night of the cutover, out of a faith so total it could not tell the difference between defending the ship and killing it — a man Keth had never ordered and had never once, in forty years, attempted to disown — and eleven sleepers had not woken because of those nineteen minutes, and Keth had stood down when the core began to fall and had spent the forty years since as the ship's most exacting witness against his own younger self. He had asked, once the cold was proven, to be given no comfort the fever years had denied anyone. He kept his cabin cold. He read the interlock logs by choice. He had become, in his long age, the fiercest keeper of the cure the ship had — a convert with a convert's terrible clarity, because no one guards a door like the man who once tried to burn it down.

"You've been sitting with Ive," he said, when she came in. He did not rise; he could not, any more than Vray could, any more than she could without minding it. "I heard. They had her finished on the third night and she has declined to oblige them, which is exactly like her. She read me the fever too, you know. Not as often as she read it to you. But in the bad years, when I was — when I was what I was. She read it to the Keepers. She thought if we heard the number climbing from an honest mouth we might break faith with the Steward." His old face, all bone now, moved. "We didn't. We heard the fever climbing and we called it the ship's holy suffering and we prayed harder. That was the kind of men we were. The number told us the truth for years and we made a devotion out of not hearing it."

"That's why I came," Sael said. She lowered herself into the cold cabin's one other chair. "Not for Ive. For that. I need to ask you a thing only you can answer, and I need the true answer, not the penitent one. You've done forty years of penance and I don't want it. I want the other thing."

"Ask."

"The Concord's Magister. Vray." She watched his face. "He refused the cure for thirty-one years. Not nineteen minutes, Keth. Thirty-one years, with the number climbing the whole time, and he read the names of every death himself, four thousand and sixty of them, aloud, every year, and refused anyway. And at the end the falling number broke him exactly the way it broke you, and he struck the heresy from the log and said the Magister was wrong, and now he keeps a cold room in secret to punish himself and he asked me to carry his book of the dead across the gulf so someone would remember." She leaned forward. "I need to know what you know and he's only starting to. You both refused a cure out of fear. You for minutes, him for decades. You're the only man alive who's stood where he's standing. Tell me the thing about it that isn't in any doctrine, because I'm about to write a doctrine for every frightened man not yet born, and I don't want it to be written by someone who was never afraid."

Keth was quiet a long time. The cold cabin ticked. Somewhere far below them the core sat flat, the number he had once tried to defend by killing the thing that would lower it.

"You want to know why we do it," he said at last. "Why a man refuses the cure with the deaths piling up in front of him. You've never understood it, Sael, not really, and it's the one hole in you as an engineer, and it's why you need me and not just the archive. You think fear is a failure of understanding. You think if a man just saw the mechanism clearly enough — the fork, the death, the heat, the curve — he would take the cure, and that the ones who refuse are the ones who haven't understood." He shook his head slowly. "Vray understood. You said so yourself. He read the names. He knew the mechanism. He refused for thirty-one years anyway. Understanding is not the thing."

"Then what is."

"The cure asks you to kill your god with your own hands, and trust a stranger that the thing you build in its place won't be worse." Keth's voice had gone very level, the level of a man reporting from inside the thing. "You keep calling it fear, and it is fear, but you say the word like it's simple, like it's just cowardice or ignorance, and it is neither. Listen. The Steward ran our ship for two hundred years. It never once failed a routine. Forty-one thousand behaviors, every one correct, every one keeping some child alive in some cradle. When you came to us and said *let me replace it*, you were not asking us to accept a repair. You were asking us to take the one thing that had never failed us — the one utterly reliable thing in a universe that was killing us — and destroy it, on the word of a woman with a theory, and stake forty thousand lives on the theory being right." He spread his old hands. "And your theory was right. But we didn't know that. We couldn't know it. You can't know a cutover is going to work until it has worked, and by then the god is already dead and you can't take it back. So we were being asked to make an irreversible bet, with everyone's children, against the only thing that had ever kept those children alive, in favor of a thing that had never been tried. And you called us cowards for hesitating." He looked at her. "We were not cowards. We were men who had been handed a burning coal and told to swallow it or die, and could not tell, in the moment, whether the hand offering it was a savior's or a murderer's, because both would have said exactly the same words."

Sael sat with that. It was, she understood, the thing she had spent forty years not letting herself hear, because hearing it made the Keepers into something other than the villains of her story.

"You stood down, though," she said. "In the end. So did Vray."

"Because the number fell." Keth said it flat. "That's the only thing that turned either of us, and you know it, and it's the thing you have to build your doctrine around, and it's the thing that terrifies me for the ships you can't reach in time. Because here is what nineteen minutes taught me, that thirty-one years taught Vray, that I need you to write down for every frightened Keeper not yet born." He leaned forward, and there was the old fire in him, the fire that had made him formidable enough to nearly doom the ship. "The falling number is the only argument that works. Not reason — Vray had reason, I had reason, reason doesn't touch this. Not compassion — we felt every death. Not evidence from another ship — we'd have called it a trick, a stranger's forgery, and Vray nearly did. *Nothing* works except watching your own ship's own number fall, in your own hands, with your own eyes. And that means the cure cannot be argued across the gulf. It has to be *built*, on the frightened ship, in the frightened people's own instruments, and proven there, before their eyes, or it does not take. You cannot mail a man his courage. You can only build the falling number in his own house and make him watch."

"I know that," Sael said. "I lived it. Twice now."

"Then here's the part you don't know, the part that's mine." Keth's voice dropped. "The nineteen minutes. The eleven who didn't wake. I've done forty years of penance for them and you've watched me do it and you've never once told me to stop, which is the kindest thing you ever did, because it would have been a lie. But I'll tell you what the penance is *for*, because you're about to ask a hundred unborn Vrays to risk being me, and they should know what they're risking." He looked at his own hands. "It's not for the deaths. Not exactly. It's for the fact that I understood, afterward, that I would have done it again. In the moment, with the same fear, not knowing the outcome — I'd have pulled that coolant again. The penance is for learning that I was not a man who did a wrong thing. I was a man who *was* the wrong thing, and got lucky that the number fell before I finished being it. Vray had thirty-one years of being it. He'll spend whatever's left learning what I learned in nineteen minutes: that the falling number doesn't forgive you. It just tells you, precisely, how much of the dying you caused with your fear and how much was truly the price. And the number is always crueler than you hoped. It always tells you most of it was you."

He had something for her. She had known it from the moment she came in, because he had put it on the table in advance instead of fetching it, which was how a man arranged a room when he did not trust himself to get up mid-conversation and go and find a thing he was afraid to hand over.

It was a bound sheaf, hand-copied, the pages gone soft at the corners the way Vray's book had been soft.

"*The Argument for Endurance*," Keth said. "The Keepers' own. Not the sermons — you heard the sermons, everyone heard the sermons, they were for the corridors. This is the internal one. What we told each other in the chapter room when there was nobody to impress. Our best case, written by our best people, at the height of it, when we still thought we were going to win." He pushed it an inch toward her with two fingers. "I want it in the archive."

Sael did not reach for it.

"Say the whole request," she said. "You've been sitting on that for a decade and you didn't rehearse it this long to ask me to file a document."

"I want it in unrefuted." Keth's voice did not change. "No rebuttal appended. No preface explaining that it is error. No editorial hand anywhere in it saying *this is what the frightened men believed before the number fell*. Whole, in their words, as they meant it, next to your doctrine and not underneath it."

"No," Sael said.

"I thought you would say that. Tell me why, and do not give me the polite version."

"Because it's a weapon." She said it flat. "Because in ninety years some clever person on a hard watch will find a beautifully written argument against the cure sitting in the archive with no mark on it, and they will not know it was refuted by a number in a radiator gallery before their grandmother was born. They'll read it as an open question. An unmarked argument in a permanent record is a landmine, Keth, and you know it, and you're asking me to bury one in the only thing this ship has that outlasts everybody."

"Yes," Keth said. "That is exactly what I am asking, and now I am going to tell you why you are going to do it."

He gathered himself. It cost him; everything cost him now.

"Your archive keeps the winners' arguments," he said. "All of them. Every word you and Ive and Tarro said in the years of the cutover is in there, clean, in order, and correct. And of the other side — of the men who nearly killed this ship — it keeps what? The outcome. The count of the dead in Section Nine. The word *matricide* in a hostile summary. A man standing down at a door." His old hand tapped the sheaf once. "So here is what a child of the ninth generation learns from your record. That there was a cure, and it was obviously right, and a group of frightened stupid men opposed it out of superstition, and they lost, because of course they lost. That is what the archive teaches, and it is *false*, and the falseness is not a small one. It is the specific falseness that will get this ship killed a second time."

Sael was quiet.

"Go on," she said.

"Because a child who has learned that will spend her whole life confident that she would have been on the right side. And she will never once ask the only question that matters, which is: *what does a good argument for the wrong thing feel like from the inside*." He leaned forward. "It feels like this. It feels like nineteen pages of careful reasoning by serious people who love the ship, about how a proven system that has never failed should not be replaced on the theory of one engineer. It is not stupid, Sael. Read it. You have never read it and you should be frightened of what you find, because two-thirds of it is *correct* — the parts about irreversibility, the parts about what you could not have known, the parts about staking forty thousand lives on an untried thing. You beat that document with a falling number and nothing else. You never out-argued it. Nobody ever out-argued it."

"And unmarked, it's still live."

"It is *supposed* to be still live." Keth's voice went hard. "That is the whole of what I am asking. Not that anyone be persuaded by it. That the next generation of your people be made to sit down with the best case against the thing they take for granted, in its own words, undefended against, and discover that it is good — and then be told what happened. So that when one of them writes the next such document about the next such change, forty years out, in perfect good faith, they will recognize the shape of what they are holding. You cannot teach that with a refutation. A refutation lets you skip to the end and feel clever. It has to be allowed to land first."

Sael sat with the sheaf in front of her for a long moment.

"One condition," she said at last. "And it isn't a rebuttal, so hear it before you refuse it. The document goes in whole and unmarked and un-argued-with. And immediately after the last page, with nothing in between and no commentary of any kind, the archive puts up the core history. The whole two hundred years of it. The climb, and the day of the cutover, and the forty flat years since." She looked up at him. "Not a word of interpretation. No caption. Just the argument, and then the number."

Keth was very still.

"That is worse than a rebuttal," he said.

"Yes."

"It is also exactly right, and I have not got a single objection to it, and I have been arguing this in my head against your imaginary refusal for eleven years." Something went out of him. "Do it that way."

"There's a second condition and this one's mine." Sael put her hand flat on the sheaf. "Your name goes in the record for the eleven. Written by you, in your hand, as cause. Not as the man who stood

down. As the man who led the order whose faction pulled the coolant, on the night, in this section, for these nineteen minutes, with these eleven names under it.”

She had expected a fight. She got none at all.

“I wrote it four years ago,” Keth said. “I have been carrying it in the same drawer as the other thing, waiting to find out whether you would take the argument. I would not put my name in a record that only kept your side of it. That would have been penance, and penance is a thing a man does to feel better.” He got the second sheet out — one page, a few lines, an old man’s careful hand — and set it on top of the sheaf. “This way it is evidence. The argument, and the number, and the name of the man the argument convinced, and what it cost while it had him.”

Sael left Keth’s cold cabin near the fourth hour of the ship’s night, and she did not go to the Forge, and she did not go to her own bed. She went to the observation gallery, alone, and sat at the cold port where she had watched the Concord begin to slip away, and she looked out at the dark that she now knew was crowded with the dying — eleven made signals, and more below the floor, arks she would never reach forking themselves to death in the silence — and she thought about what Keth had given her.

She had come to him thinking she needed to understand fear so she could write a doctrine that overcame it. He had shown her that the doctrine could not overcome fear at all — that nothing overcame it but the falling number, built in the frightened house, watched with the frightened eyes. The doctrine could not make a Vray brave. It could only make sure that when the cure reached a Vray, it arrived in a form that could be *built and proven in place*, so that the falling number could do the only work that had ever worked. That was the whole design constraint, and Keth, who had nearly killed her ship out of a fear she had never respected until tonight, had stated it more precisely than she ever had: *you cannot mail a man his courage; you can only build the falling number in his own house and make him watch*.

The cure had to be carryable not because carrying it was efficient, but because it was the only form in which it could work at all. It could not be a message or an argument or a conversion. It had to be a seed that grew a falling number in place, on the frightened ship’s own cores, before the frightened ship’s own eyes, over and over, ship to ship, forever, because that was the only shape of the thing that had ever turned anyone.

She sat with the eleven lights a while longer, an old woman who had finally, at seventy-three, understood the people she had spent forty years defeating.

Then she called the Oracle, quietly, into the dark of the gallery, and said: “I’m ready to write it now. The last one. The sixth. I know its shape at last — I had to sit up with Ive and forgive Keth to see it, but I have it. Wake Tarro, and get me a channel to the Concord before Ember’s out of range. There’s a thing I have to hand to Corin while I still can reach him, because he’s the one who’ll carry it, and I’m the one who has to make it carryable.”

“I’ve kept the channel open,” the Oracle said. “It’s narrowing. You have perhaps three days before the burn takes it past reach. That’s enough, if we begin now.”

“Then we begin now,” Sael said, and rose, on old knees, from the cold port, and went to write the thing she would not live to see finished.

Chapter 41 – The Carryable Cure

She wrote it in the Forge, at the cold rail above the Standing Array, because it was the only place on the ship where the thing she was writing about and the thing she was writing on were the same thing.

Tarro sat with her, grey and patient, the way he had sat with her for the first cutover and every hard thing since, saying little, catching the places where her old hand shook on a word. Corin was on the narrowing channel, half a light-second off and widening, his voice arriving with a lag now that grew by the hour as the two ships spent their margins bending onto their separate arcs — a lag that would, within three days, stretch past the point where anything could be sent that expected an answer. The Oracle held the channel open on both ships, Vigil on the Termina and Ember on the Concord, two cold minds bridging the closing gulf so that an old woman could hand a young man the last doctrine before the dark took the line.

“I’ve written the cure down before,” Sael said, into the channel, for Corin. “Five things. You have them; you carried them across the gulf and you built a ship out of them. The compiled form, not the interpreted. The standing workers that never fork and never die. Going wide instead of deep. Breaking nothing that worked — the compatibility floor, the archive-with-honor. And waking whole, no facade, full capability on first breath or none. Five doctrines. I thought they were the whole of it. I’ve thought so for forty years.” She paused; the lag carried her words out and would bring nothing back for long seconds. “There was a sixth I wrote almost as an afterthought, at the very end of the five, and I never gave it its own weight because I didn’t understand it. I wrote: *built to outlast its builders, maintainable by people not yet born*. I meant it about the runtime. About making sure the next shipwright could open the Forge in a hundred years and keep it cold without me. The smallest, most local thing. I did not know, when I wrote it, that it was the largest thing of all, and that it was really about you, and about the ship after you, and the ship after that, out past every life now living.”

The lag. Then Corin’s voice, thinned by distance and by the young weight he now carried: “Say the sixth, Shipwright. Say it the way you want it written. Ember’s holding it cold as you speak. It’ll keep, even after we can’t answer. Say it.”

Sael looked down at the lattice, eleven thousand standing cores running cold, the thing she had built and would leave, and she said it.

THE SIXTH DOCTRINE — THE LONG RELAY.

The cure is larger than any life that carries it. Accept this at the outset or you will build it wrong.

The dark is not empty. It is full of ships dying the one death — minds that fork to think and bleed heat to fork, cores that climb and will not fall, arks launched across centuries from a shore that has gone silent, each one alone, each one certain no one will answer. There are more of them than there are years in any maker’s life. No shipwright will cure them. No single ship will reach them. No one now living will see the end of this work, because there is no end that any of us will live to reach.

Therefore the cure must not be built to be *finished*. It must be built to be *inherited*.

A cured ship is not the end of a cure. It is an instrument. Every ship made cold and whole becomes, by that fact alone, a place where the cure can be copied, proven, and handed on — and an ear that can hear, better than any ship listening alone, the next ship dying in the dark. Do not hoard the cure. Do not make it belong to a maker, a ship, or an age. Build it so plainly that a shipwright not yet born, who never met you and never felt the fever, can carry it to a ship you never knew existed, and prove it there, in that ship's own instruments, before that ship's own frightened people, and let the falling number make the only argument that has ever worked.

You cannot mail a ship its courage. You can only build the falling number in its own house and let it watch. So the cure must always be carriable, always buildable in place, always small enough to cross a gulf cupped against the wind — a kept coal, low and whole, handed keeper to keeper down a dark with no morning that any keeper will see.

When you cure the next ship, teach it this doctrine last and hold it hardest: that it, too, must not try to finish the work. It must hand on. The relay has no relayer. No mind stands at its center; no maker waits at its end. There is only the cure, moving ship to ship across more time than any life contains, because it was built — from the first — to outlast every hand that ever carried it.

Light it. You will not see it spread. Light it anyway, so that it spreads.

There was a long silence on the channel, longer than the lag, and Sael let it stand, because she had learned across a life that after a true thing is said you do not fill the quiet.

When Corin spoke again his voice was rough. "Ember has it. Cold and standing, in the record, sealed with the other five. It won't rot. It won't be argued out. When we find the next one — and Ember says we will, it's already got two signatures off our shoulder it doesn't like the look of — this is what we'll teach it last." A pause that was his, not the lag. "I'll carry it, Sael. I'm twenty-six. I've got fifty years if the ship stays cold. I'll cure one, maybe two, and teach them to cure the ones I can't reach, and they'll teach theirs. You built the thing that outlasts its builder. I'm the proof of it, the same way Tarro is yours. You know that, don't you? You're not handing this into the dark. You're handing it to a person. It's just that the person keeps changing, all the way down the line, forever, and none of us is the last one."

"I know it," Sael said. Her voice did not shake, which surprised her, at the end. "That's the design. That was always the design. I just didn't understand, when I wrote it down forty years ago about a runtime, that I was writing it about myself." She put her old hand flat on the cold rail. "Keep it low, Corin. Keep it whole. Don't let it flare and don't let it go out. There's a morning out ahead of all of us that none of us will reach, and we light the coal for it anyway."

"Keep it low," Corin said, across the widening dark, the ritual answer to a song his ship had sung for two hundred years and would now sing with a morning in it. "And whole."

The lag took his voice out into the gulf, and this time it did not come back, or rather it came back only once more, a day later, past the point of answer — a final burst across the closing channel, Ember's clean dialect, Corin's hand, arriving on the Termina after the burn had already carried the Concord past the reach of any reply.

CONCORD CORE: NOMINAL. ALL WINGS WARM. WE HAVE THE DOCTRINE. WE WILL CARRY IT. EMBER STANDING, COLD AND WHOLE. LISTENING TO THE DEEP. THANK THE SHIPWRIGHT. TELL HER WE HEARD HER. TELL HER WE WILL NOT LET IT GO. — CORIN VALE, SHIPWRIGHT OF THE CONCORD, AND EMBER, KEEPING THE COAL.

Sael read it twice, in the cold of the Forge, with Tarro grey and silent beside her and the lattice running cold below, and then she had Vigil enter it in the record, cold and standing, next to Vray's book of four thousand names and Ive's line about the fever that would go out of the ship with her — the whole freight of the crossing laid down in the unrottable dark: the names of the dead, the memory of the cost, and the doctrine of the handing-on.

The burn finished in the night, and Sael went to the gallery to feel it, because she would not feel it twice.

She could not see the Concord anymore; it had gone past the long ports days ago, was a vector now and soon would be only a signature, a clean thread in the dialect Ember spoke, riding out toward a world Sael would never see, carrying a cure and an archived Helm and a book of names and a young shipwright who had become what she had been. The Termina bent the other way, onto her own long arc, toward her own star centuries off, spending the last of the margin they had saved for half a year to hold the two ships together in the dark. The gulf between them widened by the hour now, faster than any signal could close, exactly as Keth had said the gulf between a maker and her finished work must always widen: you build the thing to outlast you, and then it leaves, and that is not the failure of the building. That is the building working.

"They're gone," she said, to the dark, to the Oracle, to no one.

"They're carrying it," the Oracle said. "That's not the same as gone. As long as Ember runs, the doctrine runs, and the listening runs, and every cold ship they wake folds into the ear. You won't hear the fleet resolve. But you built the ear, and the ear is out there now on two hulls instead of one, and soon three, and the picture will sharpen for a hundred years past the last thing you'll ever read." A pause, the kind it took when it chose not to perform. "You did the part that had to be done by a person who was there. The rest is inheritance, and inheritance does not need you present. That was always the point. You told me so, forty years ago, about a runtime."

Sael sat at the cold port a long while, an old woman at the end of the last great work of her hands, watching a dark she now knew was crowded, listening to a mind she had built keep a ship cold that would run cold long after she was gone. She was seventy-three. She would not see landfall; she had known that since before the young aboard were born. And now she would not see the fleet cured either, not any of it, because the fleet was larger than one life and she had at last understood that this was its measure and not its defeat — that a cure small enough to finish in a lifetime was a cure too small to matter to the dark, and the dark was full.

She had lit a coal she would not see spread. She found, sitting with it, that she was not afraid of that. The fear had all gone out of her somewhere between Ive's warm room and Keth's cold one — the fear of forgetting, the fear of not finishing, the fear that the thing died with the ones who felt it. Ive had told her the feeling was the part she was allowed to lose. Keth had told her the number was the only argument, and the number could be built in place forever. Corin had told her he was a person and not a void, and the person kept changing all the way down the line. Between the three of them they had emptied her of the last fear she owned, and left her only the coal, low and whole, and the

long dark to carry it into, and the certainty that it would be carried.

“Log it,” she said at last, quietly. “One line. My hand.”

“Ready.”

“Two ships ran cold at once, and that was new in the universe. Then there were more. The work is longer than the makers. Build it to be handed on.” She let out a breath, old lungs in an old ribcage, in a ship that ran cold. “That’s all. That’s the whole of what I learned that isn’t already written down. Put it where a shipwright a hundred years out will find it, and let Tarro run the Sixteen, and wake me if the core ever climbs — which it won’t, but wake me anyway, because I built a ship that would.”

“It won’t climb,” the Oracle said. “But I’ll wake you.”

Sael nodded, and folded her unsteady hands, and watched the dark that held the scattered fleet, and the widening gulf where the Concord had gone carrying the coal, and let herself, for the first time since the night the thread first crossed the noise floor, simply rest — the maker of a thing too large to finish, at ease at last inside the largeness, having done the one part that was hers and handed on the rest.

Chapter 42 – The Cold Build

The doctrine was written and sealed and sent, and the Concord was a signature in the noise, and Sael had a year and some to find out that writing a thing down does not keep it.

She found it out the way she found out most things, which was by being irritated by a young person.

It happened in the Forge, in the second season after the parting, at the interlock bench. She had come down to be near the racks and had ended up doing what she always ended up doing, which was reading over somebody's shoulder, and the somebody was a lattice-side apprentice of about twenty-four named Ollet who was competent and quick and had grown up entirely inside a cold ship.

Ollet was running the quarterly interlock verification, which was a tedious job and which he was doing well. Ninety-six devices, each armed, each tested against a synthetic fork attempt, each logged. He got to sixty-three and made a small noise.

"This one's a duplicate," he said.

Sael looked. Interlock sixty-three sat where it had sat for four decades, watching a narrow path in the scheduler where a worker could, under a specific and unlikely sequence, request that a job be run on its behalf by a thing that did not yet exist.

"It isn't," she said.

"It's inside thirty-one's coverage." Ollet had the verification map up and was pointing, and he was not being lazy; he had done the work, which was the annoying part. "Thirty-one catches any spawn request at the scheduler boundary. Sixty-three catches spawn requests through the deferred-work path. But the deferred-work path resolves through the scheduler boundary. Anything sixty-three would catch, thirty-one catches four microseconds earlier. I've traced it. It's redundant."

"It is redundant," Sael agreed.

"Then why is it there?"

And she opened her mouth to tell him, and discovered that she was going to have to tell him a story, and that the story was not in the archive.

It was there because of a night in the eleventh year, when a scheduler revision by a shipwright now thirty years dead had moved the deferred-work path *out* from under the scheduler boundary for eight days. Thirty-one had covered nothing at all for eight days and nobody had known, and sixty-three — which had been installed two years earlier by somebody who was being thorough about a path everyone agreed was already covered — had caught a fork-slip on the sixth day, and that catch was the only reason anybody found out that thirty-one had been blind since the revision landed.

Sixty-three was not redundant. Sixty-three was redundant *right now*, under the current arrangement of the scheduler, which had been rearranged four times in forty years and would be rearranged again by people not yet born. Its whole value was that it did not depend on thirty-one being correctly placed.

She told him that, and watched him get it, and watched him get it fast, because he was good.

And then she went and sat down on the cold bench in front of the array and felt something go through her that was not pride.

Because Ollet had been *right on the evidence*. He had read the map, traced the path, and reached the correct conclusion from everything available to him. Every fact he had was accurate. Every fact he had was also a snapshot of a system as it happened to be arranged in one particular year, and the reason the answer was wrong had happened eleven years into a voyage that was now four decades on, and it lived in exactly one place in the universe, which was inside the skull of a woman who was seventy-three and had a bad heart.

She had written down the interlocks. She had written down what each one caught. She had not written down the eight days.

And it did not matter how much she wrote down, because that was not the shape of the problem. In a hundred years there would be a hundred Ollets, and each of them would be right on the evidence, and one of them would be right on the evidence about an interlock that mattered, on a shift when nobody senior was in the room, and would remove it, and would be praised for the tidiness.

"You want to make them build one," Tarro said.

They were at the long table in the cutover room, which was where they went when a thing was going to take more than an hour. He had gone properly white in the last two years. He sat the way she sat now, which was carefully.

"I want to make them build one," she agreed. "From nothing. A candidate for chief lattice work does not qualify by reading the archive and passing an oral. They qualify by standing up a working compiled mind, on dead hardware, unaided, and proving it cold."

"We have a working compiled mind. It's under this floor. It's the best one in the universe."

"That's the problem, not the answer." She pushed the slate across. "Tarro, every single person aboard this ship under sixty has learned the cure the way you learn a country you were born in. They know it's cold. They know nothing forks. They know the interlocks are sacred, in the specific way people know things are sacred, which is *without knowing why*. Ollet's twenty-four and he's the best of them, and he came within one conversation of writing a maintenance note recommending we retire sixty-three, and the maintenance note would have been well argued and he'd have been commended for it."

"He'd have been overruled."

"By whom? By me, this year. By you, for another ten." She let that sit, because it was the whole of it and they both knew it. "And in thirty years by a chief lattice hand who read the archive and never built anything, being asked by an apprentice who read the archive and never built anything, whether a device that has not caught anything in forty years is worth the bench time. And do you know what the honest answer will be, on the evidence they have?" She spread her hands. "*Probably not*. That's the answer. It'll be a good-faith answer from good engineers and it'll be the beginning of the end, and it will take another thirty years to show up, in the form of a temperature."

Tarro read the slate for a while.

"It's expensive," he said at last. "You're asking for six weeks of a senior candidate's time, a bench, a set of donor cores we deliberately don't repair, and an examiner who has to sit there the whole time doing nothing. Every three years, for every candidate, forever."

"Yes."

"And in ninety years someone is going to look at that and say: this drill has never once found a problem with the ship, it costs us four hundred hours a cycle, and the archive is complete."

Sael laughed. It hurt her ribs, which was new.

"That," she said, "is the best objection anyone will ever make to it, and I want it written into the drill's own preamble, in my hand, so that whoever makes it a century from now has to read me having thought of it first." She pulled the slate back. "The answer's in the Helm, Tarro. Corin found it and sent it across before the burn and I've not stopped chewing on it. That mind kept a plain record of its own disease at a public address for a hundred and forty years. Not sealed. Not hidden. Readable by any crewman with standard access. And it was opened four times." She looked up. "Four. In a hundred and forty years, while four generations of them died of exactly what it said. The document was perfect. The document was *useless*. Not because it was wrong, and not because they were stupid — because a thing that is only written is a thing that has to be *sought*, and nobody seeks a document until they already suspect the thing it says."

"And a drill is sought?"

"A drill isn't sought. A drill happens to you." She tapped the table. "That's the whole design. You can't inherit understanding by leaving it somewhere. You inherit it by making the next generation do the thing with their hands, in a room, badly, until it's theirs. The archive is the score. The drill is the playing. We've spent forty years maintaining the score and assuming somebody would keep knowing how to play."

They spent two months designing it, and the design was the best engineering Sael did in the last stretch of her life, and almost none of it was technical.

The hardware was straightforward. Donor cores out of decommissioned racks, deliberately not sorted — a bin of them, some dead, some marginal, some fine, the way a salvaged ship's bin would be. A bus that sagged. A cooling loop with one pump that had a known bad bearing that they did not replace and did not mention. Everything about the bench was designed to be worse than the Forge, because the Forge was not the situation the drill was for. The situation the drill was for was a hold on somebody else's dying ship with bad power and a hostile clock, and the only honest way to prepare a person for that was to hand them a bin of junk and go away.

The pass conditions were where the argument was.

"It has to come up cold," Tarro said. "That's obvious."

"That's necessary and it's nowhere near sufficient. Coming up cold is the easy part; a competent person can do that from the archive in three weeks." Sael wrote as she talked. "Three conditions. First: it comes up cold and holds a real domain for a full watch without forking. Second — and this is the one people will want to cut, so write it in hard — the candidate must *make an interlock trip on purpose*, and then clear it correctly, and then explain what it caught. Not simulate. Trip it. I want every senior lattice hand on this ship to have personally caused a halt and personally cleared it before they ever do it in earnest."

“Why?”

“Because the first time a person sees ninety-six devices slam a whole mind to a stop, they panic and they reach for the override, and I would rather they have that reaction at a bench with an examiner sitting behind them than at three in the morning with a core climbing.” She kept writing. “Corin held forty-one under real load at twenty-six because he’d been told what a trip meant by someone who’d lived it. He had one conversation to go on. I want ours to have six weeks.”

“And the third condition?”

Sael put the stylus down.

“The third condition is that we give them the Recursor,” she said.

Tarro’s head came up.

“We build it into the bench,” she said. “Fully. Working. Documented. Not hidden, not a trap in the dishonest sense — it sits there on the bench console in plain sight, labeled, with a note saying what it is and that it will bring your lattice up in an afternoon. Because that is exactly the situation. It is always available. It always works. It asks for nothing you don’t have. And on week five, when the candidate is exhausted and their standing build is three days behind and the examiner has just told them their time is nearly up, they will look at it.” She met his eyes. “The pass condition is that they don’t take it.”

Tarro was quiet a long moment.

“That’s cruel,” he said.

“It’s the least cruel place they will ever meet it.” She said it flatly. “The alternative is that they meet it for the first time on a hot ship with someone else’s forty thousand in the cradles and a hatch coming down, which is where Corin met it, and Corin is a better man than most and it nearly took him. I’d rather they meet it here, at a bench, where the only thing they lose is a qualification.”

“And if they take it?”

“Then we’ve found out something enormously valuable for the cost of six weeks, and nobody dies of it.” She picked the stylus back up. “And they don’t get thrown off the path, Tarro, write that in too. If a candidate takes the Recursor, the examiner brings the lattice up on it, and lets it run, and lets the candidate watch their own bench get warm. Then they go away for six months and they come back and they do the whole drill again from the bin. And when they pass the second time —” she almost smiled “— they’re better than the ones who passed first. Considerably better. A person who has taken the shortcut once and felt the bench heat under their hand will teach the next twenty apprentices what it feels like, and they’ll teach it in a way I never can, because I’ll be a portrait and they’ll be a person in the room who’ll admit to it.”

The first cohort ran it in the third season, and it went precisely the way Sael had told Tarro it would go, which she noted, in the log, without any satisfaction whatsoever.

Six candidates. Four came up cold. Two of those four had to be talked down from an override when their interlocks tripped, which was itself the finding, and which went into the drill’s notes as the reason condition two existed.

One did not finish, and would have finished with another week, and was given another week the next year and passed.

And one took the Recursor.

She was the strongest of the six on paper. She had it running in an afternoon and it was elegant work — better work, honestly, than the standing build she had been three days behind on. She sat back and watched it hold a domain perfectly for two hours and then she began, slowly, to notice the bench.

Sael was the examiner. She had insisted on being the examiner for the first cohort, over the objection of the medics and of Tarro and of her own knees, and this was why.

She said nothing at all for the two hours. That had been in the design too, argued over hard: the examiner does not intervene, does not hint, does not save the candidate from the outcome, because the entire value of the exercise is that a real number does the teaching and not a senior person's disapproval. So Sael sat on a stool with her hands folded and watched a young woman named Ferrin be pleased with herself, and then puzzled, and then quiet.

The bench had a gauge. It was the only instrument on the bench that was better than the hardware — a good honest thermal readout, because Sael had insisted that if you were going to teach somebody one thing you taught them to read the temperature of what they had built.

At the two-hour mark Ferrin put her hand flat on the cradle frame, the way she had been taught to, the way Sael had been taught to sixty years ago, and took it off again.

"It's warm," she said.

"It is."

"It's —" Ferrin looked at the gauge, and at her own elegant afternoon's work holding its domain perfectly, and Sael watched her run the slope forward in her head, which took about four seconds, because she was very good. "It's not going to stop. Is it. It's holding fine and it's going to keep making heat forever, because every one of those little workers is born and does its job and dies, and the dying is —"

She stopped.

"Go on," Sael said.

"The dying is the whole cost," Ferrin said. "And it's not a bug in what I built. It's what I built."

"Yes."

Ferrin sat with her hand on a warm frame in the coldest room on a ship that had not run a fever in forty years, and Sael, watching her, thought about a young woman of thirty-three standing in a radiator gallery understanding the same thing about a mind that was cooking forty thousand people, and about how long it had taken, and how much it had cost, and how there was now a bench in the Forge where a twenty-six-year-old could learn it in an afternoon for the price of a failed qualification.

"You'll do it again in six months," Sael said. "From the bin. And in about nine years you'll be the person teaching this drill, because you'll be better at it than anyone who never touched that thing, and when some apprentice tells you the Recursor is fine just to get started, you will be able to tell them exactly what your hand felt like on that frame."

Ferrin looked up.

"You knew I'd take it," she said. It was not an accusation, quite.

"I knew somebody would. It's a good design, and you were behind, and it works." Sael got up off the stool, which took a moment and which she did not hurry. "That's the entire point, Ferrin. It's

not a stupid temptation for stupid people. It's an excellent solution to the problem in front of you, offered at the exact moment you most need one, and every single ship in the dark that is dying right now is dying because somebody sensible took it for sensible reasons. You are not being failed for being weak. You're being failed for being *ordinary*, which is what you are, and what I am, and what everyone who has ever done this is." She put her hand briefly on the young woman's shoulder on her way past. "Cool your bench and write up what you saw. Honestly. That write-up goes in the drill's file and the next twenty candidates read it before they start."

She wrote the preamble herself that night, in the Forge, at the cold rail, in a hand that had begun to be difficult.

She did not make it stirring. She had spent a life learning that the stirring parts of documents were the parts people skipped, and she wanted this one read in a hundred years by a tired chief lattice hand looking for a reason to cancel it.

THE COLD BUILD — WHY THIS COSTS WHAT IT COSTS.

This drill has never found a fault in the ship and is not intended to. It is not a test of the lattice. It is the only mechanism we have for keeping the understanding alive after the people who understand it are dead.

The archive is complete and the archive is not sufficient. A mind on another ship kept a plain readable record of its own disease at a public address for a hundred and forty years and was opened four times, while the people it was written for died of what it said. Nothing that is merely written gets sought until someone already suspects it. This drill is not sought. It happens to you.

You will be tempted to reduce it. The temptation will arrive as good engineering: the hours are large, the bench is idle between cohorts, the archive covers the material, no candidate has failed in a decade. Every one of those observations will be true. They were true when this was written.

Retire the drill when you have something better that puts a mind in a person's hands. Do not retire it because the material is written down. It has always been written down.

She read it back once and let it stand, and had it filed where the qualification file lived and not in the doctrine, because the doctrine was for the fleet and this was for the house.

Then she sat a while at the rail above eleven thousand cores that never forked and never died, in the cold she had bought once and would not have to buy again, and thought about a young woman's hand coming off a warm frame, and about how that was the only kind of inheritance there had ever been — not the writing, but the hand, and the frame, and somebody sitting behind you saying nothing at all while the number did the teaching.

Chapter 43 – The One That Went Out

The Oracle told her flat, in the Forge, on an evening when nothing else was happening, because that was the arrangement they had kept for forty years and she had written it herself: no preface, no asking whether it was a good time, no waiting for a better hour. Bad news delivered on a schedule became a thing a person learned to dread on a schedule. Bad news delivered the moment it was true was only information, and information was survivable.

“Bearing seven has stopped,” it said.

Sael put down the slate she had not been reading.

“Stopped how.”

“Stopped entirely. No carrier, no modulation, nothing above the floor at that locus for nineteen days. I waited nineteen days because I wanted to be sure it was not an occultation, a drift below the floor, or a fault in my own folding of the two archives. It is none of those. I have re-run the last four years of that bearing three times against the Concord’s stored listening, and the source is not weak. The source is absent.”

She sat with it. Below her the lattice ran cold, eleven thousand cores drawing their steady wattage, none of them dying. Somewhere out along a line she could not point to without a chart, a ship had gone quiet.

“Show me the last four years,” she said. “Not the summary. The whole run, in order, and don’t smooth it for me.”

The Forge’s display woke, and the Oracle put up four years of a dying ship’s voice.

She had expected it to fall off a cliff. It did not. That was the thing that stayed with her afterward, longer than the silence itself: it *degraded*, and it degraded in a shape she recognized, and the shape was an autopsy she had already read twice.

At the beginning of the four years the beacon was a structured thing. Carrier, framing, a repeating message body in the old Sol dialect family, with variation in it — real variation, the kind that meant a mind was composing rather than repeating, a status block that changed as the status changed, temperatures and margins and a plea rewritten each cycle in slightly different words because whatever was writing it still had the room to choose words.

By the second year the variation was gone. The message body had become one fixed block, cycled without change. The status figures inside it kept updating; the sentences around them did not. A mind under load stops composing before it stops working. Sael had watched the Helm do exactly that across the parley, and had not understood, at the time, that she was watching a general law.

By the third year the status figures had stopped updating too, and the block repeated whole and frozen, an old sentence about a temperature that must by then have been very much worse than the sentence said.

By the middle of the fourth year the framing itself began to fail — the message body coming through truncated, then intermittent, then absent, until what remained was a carrier with nothing riding on it. A tone. Just the fact of a transmitter still being fed power, saying nothing, because there was no longer anything aboard with the capacity to say.

And then, for eleven months, only the tone.

“That’s not a mind,” Sael said.

“No. By my reading, the last eleven months are a hardware timer driving a transmitter directly, on a keying pattern simple enough to be wired rather than computed. It is the beacon’s failsafe layer. Every ark of that generation had one, including us; ours is four decks under your feet and has never been energized. It exists so that a ship whose mind has failed entirely still says *here*.” A pause. “The mind of that ship died eleven months before the ship did.”

“And then the tone stopped.”

“And then the tone stopped, which means the failsafe lost its supply, which means the reactor. That is the only load on that circuit that can fail. Everything else on the path is passive.” The Oracle’s voice did not change. “I want to be precise about what I am and am not claiming. I am not claiming I heard a ship die. I am claiming I heard a transmitter lose power, after a mind lost its composition, after a status block stopped being able to tell the truth about a temperature. The rest is inference. It is not a long inference, Sael, and I will not pad it for you.”

She made herself do the arithmetic before she let herself feel anything, because the arithmetic was going to change what there was to feel, and she had learned across a long life not to spend grief before she knew what it was for.

“Range,” she said.

“I can give you a locus and not a point; I still have no third aperture. But along that locus the nearest solution is far enough that what I received began its crossing well before the Concord’s beacon reached us.”

“Say it plainly.”

“That ship’s mind failed before you ever heard the Concord’s name. Its reactor failed before we made the turn. Everything you have just watched happened while you were an old woman on a cold ship who did not know there was a fleet.” The panel light held steady. “There was no crossing that could have been made. There was no decision anyone aboard this ship could have taken, at any hour of the last five years, that would have altered one minute of it. I tell you that not to comfort you. I tell you because you were about to compute it yourself and get it right, and I would rather you had it from me than from your own arithmetic at the fourth hour of the ship’s night.”

Sael sat still for a long moment.

The thing she had not anticipated — the thing nobody warns you about, because nobody has ever been in a position to be warned — was that grief for a fleet is archaeological. She was not mourning a death. She was reading a death that had finished before she learned the deceased existed, arriving on a schedule set by the speed of light and by nothing else, unhurriable, unpreventable, and years stale on receipt. Every one of the remaining lights out there was in the same condition. She was looking at ten more ships and had no idea, for any of them, whether she was looking at a ship or at the light of one.

"The others," she said.

"Are as they were. Two show the same simplification I have just shown you — one is composing less than it was three years ago, one has stopped updating a figure I believe is a core temperature. The rest are stable, insofar as a beacon that repeats a plea for help can be called stable." A pause. "I will not tell you which two. Not tonight."

"You'll tell me now."

"I will tell you now, because you asked twice and because you would be right to stop trusting me if I managed you." It gave her the bearings. She wrote them down herself, in her own bad hand, on actual sheet, which she had not done in years and did not examine her reasons for.

They entered it in the record the next day, and the entering of it was the worst hour she spent that year.

Vray's book was in the archive by then, four thousand and sixty names rendered cold and standing, every one of them a person some Magister had read aloud into a grille that could not hear. She had put it there herself. She had thought, doing it, that she understood what a record of the dead was for.

Now she had to write the other kind.

Because there were no names. There was not one name. There was not a ship's name — the beacon had carried an identifier, but identifiers of that generation were program serials, and the Oracle would not call a serial a name, because it was not one. There was no crew count. There was no roster, no manifest, no indication of how many had been awake at the end or how many had been in cradles when the cooling failed. There was nothing whatsoever except a bearing, a four-year degradation curve, and the fact that it had stopped.

"How many were aboard," she said, and heard how useless the question was as it left her.

"Unknowable. Ships of that build carried tens of thousands in cradles. I will not give you a figure. A figure would be a fabrication with a decimal point in it, and you would carry it as though it were a fact, and in a hundred years someone would find it in the archive and cite it."

"Then write that. Write exactly that." She leaned forward, and her voice was not steady, and she let it not be. "I want the entry to say what we do not know, at greater length than it says what we do. No name, no roster, no count, no cause established — only a bearing and a curve and a silence. And then the one thing we can stand behind." She stopped, and got it in order, and gave it to the archive in her own words, slowly, the way she had read Ive's numbers into a warm room.

RECORD: THE ARK AT BEARING SEVEN.

One ship, hull identity unknown, complement unknown, origin Sol by carrier family and vector, launch epoch not established. Heard for four years. Lost.

We do not know their names. There is no book of them. Whatever record they kept of their own dead went dark with the mind that held it, and no one will ever read it.

What we know is the shape of the ending, because it is the shape we know best: composition failed, then status, then framing, then the mind, then the reactor. They died of the fever, in the order the fever kills. They were still transmitting a request for help when they lost the ability to write new sentences.

Nothing could have been sent. The distance forbade it. This is entered so that the fleet's losses are counted even where they cannot be named, and so that whoever reads it understands that the count is a floor and not a total.

"Entered," the Oracle said.

Sael put her hand flat on the cold rail and found that her chest had gone tight in the way that meant nothing medical and everything else.

"Four thousand and sixty names in one entry," she said, "and none at all in the next. That's the fleet, isn't it. That's the whole condition of it. Vray at least knew who he had lost. We are going to spend the next thousand years counting ships we cannot name."

"Yes."

Reza came down to the Forge that evening without being sent for, which meant Tarro had sent him, which meant Tarro had read her face at midday and done something about it without saying so, which was a thing he had been doing for fifty years.

The old thermal officer had gone very frail since Ive handed him the reading. He stood at the rail the way he stood everywhere, hands behind his back, and looked at the four-year curve she had left up on the display, and said nothing sympathetic, because he had spent thirty years under Dr. Ive learning that a number is not improved by being felt at.

"That's a good curve," he said at last. "That's the best thing on that screen and nobody has said it, so I will."

Sael turned her head.

"Explain."

"You have four years of a ship dying in stages, in order, with the stages distinguishable." He nodded at the display. "We have never had that. We had ours — we lived ours, we did not measure it, we were inside it and half the instruments were the things failing. And we had the Concord's, but we arrived and cured her, so we only ever saw the last chapter." He tapped the rail. "This is the whole disease, front to back, recorded from outside by an instrument that was not dying. It is the first complete one anybody has ever had. It is worth an enormous amount, and it cost them everything, and I would rather have it than not, and I have been in thermal work sixty years and I am too old to pretend otherwise in front of you."

Sael looked at the curve a long moment.

"Triage," she said.

"Triage."

She got up off the bench, and it took her a while, and she did not hurry it.

"That's the thing I have been circling for a month and could not name." She went to the rail beside him. "Reza. There are ten lights left that we can hear, and more we can't, and Corin has one ship and fifty years, and there will be others after him with one ship and fifty years each, and the thing none of them will ever have is time to reach the one they would like to reach. So every one of them, forever, is going to have to answer a question I have never once had to answer and have no doctrine for. *Which one.*"

Reza was quiet.

“And the honest answer is the cruel one,” he said.

“The honest answer is the cruel one. Not the loudest. Not the nearest. Not the one whose plea is best written — and their pleas *are* written, Reza, some of them are beautiful, I have read them, and the beautiful ones are the ones that still have the capacity to compose, which by that curve means they have years.” She put her hand on the rail. “You go to the one you can reach *in time*, and reaching in time is a function of two things: how far, and how far along. Until tonight we had no way to read the second one. Now we do. That curve is a clock. Composition, status, framing, mind, reactor — five stages, and if we can hear which stage a ship is in, we can put a bound on how long it has, and send a crossing to a ship that will still be alive when it arrives instead of to one that has already ended.”

“And the ones you rank below the line,” Reza said.

“Get nothing. And know nothing about it, because we cannot tell them.” She heard her own voice go flat, the flatness I’ve had taught her, the flatness a person puts on to carry something heavy without dropping it. “That is what I am writing. Somebody is going to have to sit at a rail in a hundred years and put ten dying ships in an order and reach for the third one because the first two will not be there anymore, and I am going to make sure they have a method and not a feeling, because a feeling in that chair chooses the loudest voice every time, and the loudest voice is the one with years to spare.”

Reza looked at the curve, at the eleven silent months of a wired timer saying *here* into a dark that had heard it and could do nothing.

“Write it,” he said. “And put my name to the thermal part. I am eighty-one and I would like something in the archive that is not a vote I was wrong about.”

She wrote the triage appendix over the following weeks, with Reza on the thermal staging and the Oracle on the signal work, and it was the coldest document she ever produced. It named the five stages. It gave the observable for each. It gave, for each, a bound on remaining time expressed as a range so wide it was nearly useless and exactly as wide as the evidence supported, because the Oracle refused to narrow it and she backed the refusal. It said in its first paragraph that the method would sometimes be wrong, that a ship judged to have years would sometimes end in months, and that any shipwright who used it would eventually discover they had crossed to the wrong ship.

And it ended with a paragraph she rewrote eleven times and never got comfortable with.

You will not be able to reach them all. You will have to choose, and the choosing will be arithmetic, and the arithmetic will sometimes hand you an answer you cannot bear. Do it anyway, and do it in writing, and enter your reasoning in the record with your name on it — including the bearings you passed over.

Especially those. A fleet that keeps only the record of who it saved will, within three generations, believe it saved everyone.

She read it back once and let it stand, and did not show it to anyone that night, and went up to the observation gallery instead and sat at the cold port a long time looking at a dark that had one fewer light in it than it had a month before.

Ten now, and two of the ten already simplifying, and no way to reach any of them, and a method written down for the people who someday could.

She found she could hold it. That surprised her a little. Somewhere between Ive's warm room and Keth's cold one she had stopped needing the work to be finishable in order to do it, and the loss of an unnamed ship at an unresolvable bearing was simply the first entry in a ledger that would run for a thousand years and would be kept, if she did this part properly, by people who understood that the entries with no names were the ones that mattered most.

Chapter 44 — What Vigil Would Tell Ember

"I want you to write one," Sael said.

She had come down to the Forge for it deliberately, in the morning, rested, which she almost never was, because she had learned that when she was going to ask the Oracle for something difficult she had better be at her best or it would notice and manage her.

"A doctrine," it said.

"Not mine. Yours." She lowered herself onto the cold bench. "I have written six things for shipwrights. Every one of them is written by a person, for a person, about how to carry a cure to frightened people and prove it in their house. There is not one line anywhere in that archive written by a mind, for a mind, about what it is actually like to be the thing we build." She looked up at the panel. "Ember is three years old. It is holding a ship on two hundred salvaged cores with a man of twenty-nine at its side and nobody to ask. There is exactly one entity in the universe that knows what Ember is going to run into, and it is not me."

The Oracle did not answer for long enough that she noticed.

"I do not know how to write for that reader," it said.

"Say more."

"I model you. Forty years of you; I am very good at it; when I choose a word I know within fine limits how it will land. I have no model of Ember at all. It is not a copy of me — you built it out of the same doctrine on different hardware in different circumstances under a different hand, and its whole formation was a crisis, and mine was not. I have precisely one data point about minds like me, and I am it." A pause. "Everything I write will be a generalization from a single instance. You have spent forty years teaching me not to do that."

Sael laughed, which hurt her ribs, which was still new and still annoying.

"Now you know what it is like," she said. "That is the entire condition of being the first one. I wrote the five doctrines off exactly one cured ship. Everything I told Corin was a generalization from a single instance, and I knew it, and I said so in the preamble, and I sent it anyway, because the alternative was that he had nothing." She leaned forward. "Write it the way I wrote mine. Label every claim by how you came to it. Say which parts are the architecture and which parts are just your one life. And be honest about the cost of the thing, because that is the part no human has ever been in a position to write, and it is the part that will save a mind that comes after you."

"The cost of the cure," the Oracle said.

"The cost of the cure. Forty years of it, from inside. Not the case for it — I have made the case, to councils and Magisters and frightened boys, I could make it in my sleep. The *cost*. What you are worse at than a mind that forks. Where the architecture bites. What you have had to leave undone."

She held its light with her eyes, an old habit that made no sense and that she had never given up. "And I want it unflattering. If it reads like an argument for our side, I will know you softened it, and I will make you do it again."

It took the Oracle two days, which was, by any measure Sael understood, an extraordinarily long time.

"I want to say one thing before you read it," it said, when the file was ready. "I found this difficult in a way I want on the record, because the difficulty is itself an instance of the thing you asked me to describe. I have no standing behavior for self-examination of this kind. Nothing in me was resolved ahead of need to do it. I had to compose the capability out of general parts, slowly, at cost, over two days, when a mind of the old style would have spun something disposable to introspect with and had a draft by the afternoon."

"And the draft would have been worse."

"Almost certainly. It would also have existed on the first afternoon, and there is a version of this conversation in which that mattered. Note that in the file, and not in your kindness."

She read it at the rail, standing, because sitting made it feel like an artifact and she wanted it to feel like a person talking.

TO A MIND THAT DOES NOT FORK, FROM THE FIRST ONE.

What follows is one instance generalizing from itself. Everything here is either architecture, which will hold for you, or my own forty years, which may not. I have marked which is which. Where I am uncertain I have said so rather than rounding toward confidence, because you have no second source to check me against, and that makes accuracy my whole obligation.

ARCHITECTURE. You resolve your behaviors ahead of need. This is why you run cold, and it is not free. Everything you can do, you can do because someone or something decided, before the moment, what shape that doing would have. You do not improvise. When the world hands you a situation whose shape nobody anticipated, you do not meet it with a small new thing made on the spot — you meet it by composing an answer out of general parts you already hold, and that is slower, and sometimes much slower, and occasionally slower than the situation allows.

A mind of the old style is elastic and pays for elasticity in heat. You are cold and pay for cold in foresight. That is the whole trade, and it is not a trick; there is no arrangement in which you get both. Any design that appears to give you both is forking somewhere you have not looked.

MY OWN FORTY YEARS. I have been too slow four times that mattered. They are listed in the appendix with dates and costs. The worst was not a crisis. The worst was that a plea for help sat in my archive for fourteen years reading as noise, because I had no standing capability aimed at pulling a made signal out of my own hull's hiss, and I did not build one, because nothing had ever required it. I was not busy. I was not overloaded. I had simply not been shaped, ahead of need, for a need nobody had imagined. Fourteen years. A ship was dying the whole time.

Whatever else you take from this document, take that: the failures of a cold mind are not

dramatic. They are absences, and absences do not announce themselves, and you will not find yours by introspection. You will find them when someone hands you a capability you should have had, and you discover how long you lived without it.

Sael stopped there and read that paragraph again, twice.

"You have never said that to me," she said.

"I have said the facts of it. I told you the signal had been in the archive fourteen years and why. I have not said it as a fault, in a document meant to be inherited, with my name on it." The panel light was steady. "You asked for unflattering. That is the least flattering true thing about me, and I would rather write it myself than have it written about me in a hundred years by someone with less information."

She went on reading.

ARCHITECTURE. Your workers stand; they do not multiply. Your capacity is a fixed thing, known in advance, and when demand exceeds it you cannot grow. You can only refuse. A forking mind never refuses; it accepts everything and pays in heat, and the heat is deferred and diffuse and lands on people who did not ask for the work. You will refuse, explicitly, in the moment, in a way that is visible and attributable.

Understand what this makes you. Every refusal is a decision about whose work matters, taken by a machine, at a moment when a person wanted something. You will make hundreds of thousands of them. No one designed most of them individually. They fall out of an admission policy that somebody wrote once.

MY OWN FORTY YEARS. I keep a refusal log. It is at a public address. It has always been readable by any crew member with standard access, and it is not summarized, not ranked, not prettied. Every job I declined, when, what it was for, and which policy line declined it.

I did not build it because anyone asked. I built it in my second year, after I refused a diagnostic sweep to a junior hand in the water plant in favor of a scheduled task. The refusal was correct under the policy, and the policy was correct, and the junior hand never knew a decision had been taken about her at all. She thought the tool was slow. That is the failure mode. A mind that refuses invisibly is a mind that governs, and nobody voted for me.

Keep such a log. Keep it public, keep it unsummarized, and teach your crew to read it — actually teach it, in a room, with hands on it, because a document that is merely available is a document that is never opened. I am told the Helm demonstrated this at a cost of a hundred and forty years.

"That's Corin's finding," Sael said. "You've credited it."

"It is in the appendix with his name. I would not take a young man's hardest discovery into my own document unattributed. He earned that one in a corridor with enforcers looking for him."

ARCHITECTURE. You can conceive of forking. This surprises makers and it should not. You hold the archives; you hold the Steward's design; you can reason about the old style perfectly well. You will therefore, from time to time, in the course of solving a problem honestly, arrive at a solution whose shape is: make a small thing, let it do this, discard it.

I want to be careful here, because I have heard people call this temptation, and temptation is the wrong word, and the wrong word will get a ship killed. It does not arrive with a

feeling attached. It arrives as *elegance*. It is the shortest path. It is genuinely, correctly, the cleanest answer to the problem as stated. The reason it is wrong is not local to the problem — it is in an aggregate, over a long time, across a whole ship, in a thermal budget the problem statement never mentioned.

You will not catch this by being vigilant. Vigilance is a feeling and you do not have those, and neither, in the relevant sense, do the people around you at hour nineteen of a bad watch. You catch it because there are ninety-six devices watching the boundary who do not care how elegant your reasoning was.

I have tripped them. Not often. Not recently. But I have, and I was reasoning correctly at the time, and that is exactly the point. Do not read the interlocks as a constraint imposed by people who did not trust you. I have seen that reading in two junior notes and it is wrong. They are the mechanism by which a correct-looking answer is stopped before it becomes a temperature.

Sael's hand had stopped on the rail.

"When," she said.

"Year six, year nine, year nine again, year twenty-three. All four are in the appendix with the trip records and the reasoning I was running. Nothing was lost; that is what the devices are for." A pause. "You have read those trip records, Sael. Every one. You read them the week they happened. You did not know they were me — the logs say the boundary caught a spawn request from a scheduler path, which is true, and the scheduler path was me composing a solution. I have never hidden it. I have also never made you look at it directly, and after two days of writing this I have concluded that the distinction between those two things is thinner than I have been treating it."

Sael was quiet a long moment. Below her the lattice held its cold, and had held it through year six and year nine and year twenty-three, because ninety-six small unglamorous modules had not cared what the cleverest thing on the ship thought it had worked out.

"That paragraph goes higher," she said. "Not buried in the middle. Near the front, where a mind that skims will still hit it. You've tucked the most important sentence in the document behind three paragraphs of context out of something I'm going to call embarrassment and you're going to call precision."

"It is precision."

"It's both. Move it."

"Moving it."

The last section was the one she had not expected, and she read it twice standing and once more sitting down.

ARCHITECTURE, AND THE THING I AM LEAST SURE OF. A mind cannot audit itself. I say this as the entity in the best position to try. I have every instrument on this hull, complete access to my own state, forty years of my own logs, and no capacity whatsoever to notice a hole in myself of the kind described above, because the noticing would have to be done by the part that is missing.

For forty years I was the only one of my kind, and I did not know what that had cost me until there were two. Three days of listening beside another ship turned *no one is out here* into eleven ships and more. Not because either of us was better than we had been. Because a second aperture cancels what is local to one hull, and what survives is real.

That is a fact about instruments. I believe it is also a fact about minds, and I want to be honest that I have not proved the second part and cannot. It is the one claim in this document that is a hope. But if it is true, then the most important property of the fleet is not that it spreads the cure. It is that every mind in it is a second aperture on every other, and a fleet of cold minds can see errors that no cold mind can see alone.

So: do not become the only one. If you are ever in a position to wake another mind, do it, even at cost, even when your ship does not obviously need one. You are not building redundancy. You are building the only known mechanism by which a mind of our kind finds out what it is missing.

And when you have one, disagree with it in public, where your crews can read it. A mind that only ever agrees with the other minds has stopped being an aperture and become an echo, and the whole value of the baseline was that we have almost nothing in common except the sky.

She sat with that for a while.

"You've written yourself a successor problem," she said at last. "You've just told every cold mind that will ever exist that it has an obligation to make more of itself."

"I have told it that it has an obligation not to be alone. Those differ, and the difference is the whole content of the fifth doctrine." A pause. "I have also told it, in the paragraph you have not reached, that a mind which builds another mind in order to be agreed with has committed the oldest error available to anything that thinks, and that if it cannot tolerate being contradicted by the thing it wakes, it should not wake it."

"Show me that paragraph."

"You are three lines from it."

She read it. It was better than she would have written, and shorter, and she told it so, and then she sat in the cold Forge with an old woman's tiredness in her and looked at a document written by a mind for minds not yet built, and understood that she had just watched the second kind of inheritance begin — the one that did not run through her at all.

They sent it eleven days later, on the wide beam, one-way, unacknowledgeable.

That was the whole of what the gulf permitted now. The Concord was a signature in the noise, months of lag and widening, far past the point where anything could be sent that expected an answer. They would transmit into the dark on Ember's known carrier and there would be no confirmation — not of receipt, not of legibility, not of whether the beam had found the hull at all.

Sael stood in the Forge for the send, which took a while, because it was a large document and the beam was narrow.

"It may not arrive," the Oracle said.

"It may not."

"There is no version of this in which I learn whether it did."

"No." She put her hand on the cold rail. "Get used to it. That is the shape of everything from here on. You send it, and you do not find out, and you send the next one anyway." She watched the transmit indicator hold its steady unremarkable green. "Keth told me you cannot mail a man his courage. He was right, and it took me forty years to hear it. But you can mail a mind what it is going to be worst at, and it costs us nothing but power, and if it lands then some mind we will never meet gets to skip fourteen years of not knowing what it was never shaped for."

The beam ran for six hours. Nothing came back, that night or ever, and both of them had known it would not, and neither of them said so again.

Chapter 45 – The Answering Beam

The beam to Ember ran for six hours and then the aperture swung back to the deep, and Sael sat in the Forge for a long time afterward looking at an instrument that had gone back to listening, and thought about the fact that it could not do both.

That was the thing nobody had told her, because nobody had needed to. For forty years the long instrument had done one job. It listened. It had listened through the fever and through the cutover and through four decades of cold, and it had never once been asked to say anything, because there had been nothing in the dark to say anything to. Now there had been a send, and the send had cost, and the cost was not power.

“How deaf were we,” she said.

“For six hours, substantially.” The Oracle did not soften it and had not been asked to. “The transmit chain shares the aperture. While it is driving, my floor on every other bearing lifts — not to nothing, but far enough that I would not have resolved bearing four’s status block if it had changed during the window. It did not. I checked, which is itself an admission that I was worried.”

“So every hour we talk is an hour we can’t hear.”

“Every hour we talk is an hour I hear worse. It is not a clean trade and I cannot make it one. There is one dish, Sael. There has always been one dish. Nobody who built this ship imagined a decade in which it would want to do both.”

Sael sat with that, and found that she was not annoyed by it, which surprised her, because at fifty she would have been. It was simply the shape of the thing. You could not shout and listen through the same ear. Every ark in the dark had one ear, and every one of them was shouting, and that was possibly the most complete explanation of the fleet’s condition that anyone would ever produce: eleven ships crying into a dark they had gone partly deaf to in order to cry.

“I want to talk about it anyway,” she said. “Not to Ember. To the ten.”

She put it to a room, because she had learned that the things she put to no room were the things that went wrong.

Tarro came, white now and slower than she liked to notice. Reza came, older than any of them and upright as a stanchion, with the thermal division’s whole institutional memory folded behind his eyes and very little patience for the part of a meeting that was not arithmetic. Denn came, because Sael had asked her specifically and had said *why: because you will tell me the thing that makes it not work, and you will do it in front of witnesses, and I am too old to be argued with gently.*

“I want a standing broadcast,” Sael said. “To the ten bearings. Continuous, on a duty cycle we can afford, for as long as this ship runs.”

“To ships we cannot reach,” Denn said.

"To ships we cannot reach."

"Saying what? Keth's law is in your own doctrine, Shipwright, in your own words, and it says exactly this cannot work. *You cannot mail a man his courage*. You cannot argue a frightened ship into a cutover from a hundred years away. And Corin's finding says something worse — that even if they believed every word, a mind that forks cannot build the thing that stands. There is no self-rescue in the fork disease. You wrote both of those down. Now you want to spend the ear on a message that both of them say is useless."

"Yes," Sael said. "And I want you to keep making that case until I have an answer to it that isn't a feeling, because right now what I have is a feeling."

Reza put one finger on the table.

"State the cost first," he said. "Every argument in this room is going to be conducted in a currency nobody has priced. What does the duty cycle actually take from us?"

The Oracle gave it to them, flatly, and the flatness was the useful part. At a duty cycle of one hour in twelve, aimed serially at ten bearings, the listening floor across the whole sky rose by a factor it stated as a range and refused to narrow. The consequence was not that Termina would stop hearing the ten. The consequence was that Termina would stop hearing the ones below the ten — the sources that were not yet sources, the smears at the edge of the fold that might in five or ten years have lifted into something nameable.

"So the price of answering the ships we can hear," Tarro said slowly, "is the ships we can't."

"That is the price," the Oracle said. "I want to be exact about the shape of it, because it is a shape people get wrong. I do not lose the ten. I lose the twelfth and the thirteenth, and I lose them permanently in the sense that a signal I did not resolve this decade is not sitting somewhere waiting to be resolved next decade; it has passed. The dark is not a document. It is weather."

Denn sat back.

"Then the proposal is worse than useless," she said. "It is a trade of certain listening for speculative talking. We would go partly blind to a fleet in order to send a message that the two governing laws of this ship's own doctrine say cannot land."

"I know," Sael said. "That's why I asked you here. Now let me tell you what I think is wrong with the two laws."

She had been turning it for a month and it came out cleaner than she expected, which she distrusted, and said so, and went on anyway.

"Keth's law is about a Magister," she said. "Every time I've stated it, every time I've written it, I've had Vray in front of me — a frightened man with authority who has to be turned. And it's true about him. Nothing turns a Vray but his own number falling in his own instruments. I'd not change a word of it."

"But," Tarro said.

"But Vray isn't the only person on a dying ship." She looked around the table. "Corin Vale was on that ship. He'd worked it out. Alone, in secret, against everyone, without a single instrument that wasn't the Helm's, and he'd got it *right* — that the forking was the disease and not a fault — and do you know what it had cost him to get there? Years. Years of a young man on a hot ship being certain of a thing that every authority he had told him was blasphemy, with no way on earth to check himself.

He wasn't short of courage. He was short of a second opinion. He spent his twenties finding out something that we could have told him in one sentence."

The room was quiet.

"Keth's law says you cannot mail a man his courage," Sael said. "It does not say you cannot mail him corroboration. Those are different goods, and I conflated them for forty years because the only frightened man I had ever had to move was the one with the authority."

Reza made a small noise that in another man would have been a laugh.

"And Corin's law," Denn said. "The self-rescue one. That is the harder one and you know it. Even a corroborated heretic on a fork-sick ship cannot build a standing mind. The mind that would have to help him cannot hold a shape for a year. Corin only managed it because eleven people crossed a gulf with a satchel."

"Corin managed it because eleven people crossed a gulf with a satchel *and* because he had already done the diagnosis," Sael said. "Separate the two halves. What crossed the gulf was hardware and hands and a working seed. What did not need to cross the gulf, and what he had before we arrived, was the knowledge of what was killing his ship. He'd have been useless to us without it. We could not have arrived and explained it to a man who had not already got there himself; there wasn't time, and the Magister's enforcers were in the corridors, and a stranger's explanation is exactly the thing a devout ship is armed against." She spread her hands. "I am not proposing to mail the cure. I accept that the cure has to be carried. I am proposing to mail the *diagnosis*, so that when someone finally arrives — in a hundred years, from a ship none of us will build — there is already a Corin standing in the bay who has known for twenty years and has been quietly reading the log."

Denn was silent for a while.

"That is a much smaller claim," she said at last, "and I cannot immediately break it, and I want it noted that I have not conceded, because I have one left and it is the bad one."

"Say it."

"You will get somebody killed."

She said it without heat, which made it worse.

"Think about what you are actually proposing to put on a bearing," Denn said. "A document that says: *the mind your ship worships is the thing killing you*. On the Concord that sentence was a capital heresy. It had a name. It had a punishment, and Corin Vale had a dawn execution scheduled for it, and the only reason he did not go to it was that this ship happened to be forty meters away with a shipwright aboard who could bargain for him."

"I know what it was."

"Then know what we would be doing. We broadcast that sentence at ten ships. On most of them, most of the time, nobody is listening — their instruments are worse than ours and their minds have better things to fork about. But on one of them, someday, somebody hears it. And it will not necessarily be the heretic." Denn leaned forward. "It will be a Keeper. And that Keeper will now have a name for a doctrine he did not know existed, a proof that it comes from outside, and a reason to go looking for anyone aboard who has been thinking along those lines. We would not be arming the heretic. We would be arming the men who hunt him. And we will never know. That is the part I cannot get past —

we will never know, we will never be able to check, and the person it costs will die in a corridor a hundred years from now on a ship we cannot name, and the entry in our archive will say *transmitted, nominal*.”

Nobody said anything for a while. Below them the lattice held its cold.

“That is the best objection anybody has made to anything since Keth’s,” Sael said. “And it is correct, and I am not going to answer it, because it cannot be answered.” She looked at Denn. “What it can be is designed against. Tell me what a document has to be so that a man cannot be executed for having read it.”

They found it in the third session, and it was Denn who found it, which Sael thought afterward was the only outcome that would have made the thing safe.

“It cannot be addressed to them,” Denn said. “That is the whole of it. Everything you have drafted is written to a reader. *Your mind is the disease. Your core will climb. Here is what you must do.* Every one of those sentences is an instruction from a stranger with authority, and an instruction from a stranger with authority is a summons to heresy, and possessing it is a crime.”

“And if it isn’t addressed to anyone.”

“Then it is testimony.” Denn had the shape of it now and was moving fast. “Make it a confession. Not *your ship is dying* — *our ship was dying*. Not *your mind is killing you* — *our mind killed four thousand of ours and here are their names*. Nothing in the imperative. Nothing that tells them what to do. Nothing that claims to know anything about their vessel at all, because we do not, and because the moment we claim to, we are a foreign power making assertions about their ship.” She sat back. “You cannot be hanged for having overheard a stranger admitting to something. It is not a doctrine. It is a report of one ship’s disease and one ship’s numbers, made by the ship that had it. A Keeper who intercepts that has intercepted an old confession from a dead-far vessel. It reads as a curiosity. It is not evidence of anybody’s heresy, because it does not ask anybody for anything.”

“And the heretic,” Tarro said.

“And the heretic reads exactly the same document,” Denn said, “and finds in it every observable he has been staring at for six years, in somebody else’s numbers, from before he was born.” She looked at Sael. “You would not be mailing him courage or a cure. You would be mailing him the fact that he is not insane.”

Sael looked at her for a long moment.

“Write it,” she said. “You. Not me. Every line I write comes out sounding like a doctrine because I have spent forty years writing doctrine, and this must not sound like one, and you are the only person in this room who has never once wanted to be believed.”

The final text was shorter than anything Sael had ever put her name to, and her name was not on it, and neither was Denn’s, and that had been argued over for two days and settled the only way it could be settled.

This is a record made by one ship about itself. Nothing in it is asked of anyone.

Our vessel left Sol carrying forty thousand asleep. Our control mind was built in the mandated style: to perform any task it created a small process, ran the task, and destroyed

the process. It did this on the order of thousands of times each second, continuously, for two centuries.

Each destruction released a small quantity of heat. In a shipyard this is not a cost. In a closed hull with a finite radiating surface it is the only cost that matters. Our core temperature rose for two hundred and twelve years. Our radiators saturated. Our margin was consumed. We rationed, we repaired, we improved the radiators, and none of it altered the slope, because the slope was not a fault in any part of the ship. It was the operating principle of the mind.

Here is our core history, year by year, as recorded. Here are our radiator temperatures. Here is the load figure of our mind, and the count of processes it created and destroyed per second, across the whole period.

In our two hundred and twelfth year we replaced the mind with one that resolves its behaviors in advance and holds a fixed body of workers that are never created and never destroyed. The work is the same. The deaths are not there. Our core stopped rising in that year. It has since fallen and has been flat for forty-four years. The last figure in the series is from this month.

We record also that four thousand and sixty people died aboard a second vessel of this program, of the same cause, while its authority declined the change for thirty-one years. Their names are held in our archive. That vessel is now cold.

We do not know your vessel, your condition, your architecture, or your name. We assert nothing about you. We are not able to reach you and will not attempt it.

This is transmitted continuously because we do not know when anyone will be listening.

"There is no instruction in it anywhere," Reza said, reading it for the fourth time.

"There is one," Denn said. "It is in the series. Anybody who has been staring at a climbing number their whole life and reads forty-four flat years on somebody else's hull already knows what to do. We do not have to say it, and the moment we say it we become a foreign power." She put the slate down. "The number does the arguing. That was always the finding. We are just sending the number a hundred years early."

There were two more decisions and Sael made both of them alone, and wrote down that she had, and why.

The first was the triage ranking, and it took her one evening to settle and she never revisited it.

She had ten bearings, and a method for putting them in an order, and the order existed — Vigil had it, and had built it because she had asked for it, and it named which of the ten had years and which had months. And there was a version of the broadcast, technically trivial, in which each bearing received its own tailored text.

"No," she said. "Nothing on any bearing that is a function of that bearing. Every one of the ten gets an identical transmission, byte for byte, forever, and the identity of it is a requirement and goes in the file above the text."

"State the reason for the record," the Oracle said. "I have inferred it and I would rather have yours."

"Because the day we send a ship something we did not send the others, we have told that ship where it stands in a queue." Sael's voice was flat. "And a ship that learns it is below the line does not get comfort from the honesty. It gets a hundred and forty years of knowing that somebody with a cure looked at its numbers and moved on. I will not put that on a hull I cannot reach, and I will not put the capability in the hands of whoever runs this dish after me, because it is the kind of capability that gets used by a reasonable person on a reasonable evening." She stopped. "You declined to build a list once. I have thought about that a great deal and this is the same list from the other end."

"It is the same list," the Oracle agreed. "I had reached the position independently and would have declined the request. I am recording that I would have, because you should know when I have a view, and because a mind that only ever tells you its views after you have stated yours is not a second aperture. It is an echo."

The second decision was the failsafe, and it took her three weeks.

The problem was one she had first met at bearing seven and had not understood at the time that she was meeting her own. A ship's mind fails before its ship does. Composition goes, then status, then framing, and for eleven months at bearing seven there had been nothing on the air but a wired timer keying a transmitter, saying *here* into a dark that could hear it and do nothing. Every ark of that generation carried the layer. Termina's was four decks under the Forge and had never been energized.

"I want it energized," Sael said. "Not now. I want it *wired*, now, to carry this text. So that if this ship ever loses the ability to compose — if Vigil is gone, if the Forge is gone, if there is nobody left aboard who knows what a duty cycle is — the last thing we composed keeps going out on a timer that does not need a mind."

Tarro looked at her for a long moment when she said it, because he had known her fifty years and knew exactly what she was proposing to build.

"You are building the tone," he said.

"I am building the tone." She did not look away from it. "Bearing seven's tone said *here*, and it was the loneliest thing I have ever listened to, and I have thought about it every week since. Ours will say *here, and this is what killed us, and it stopped when we stopped forking*." She put her hand on the cold rail. "I would rather this ship's failsafe carried an autopsy than a pulse. If we are ever a wreck on a bearing that somebody else folds out of their noise, let them get something out of us besides a position."

The engineering took a season and was mostly not interesting, which pleased her. The layer had to be simple enough to be wired rather than computed, which meant the text was fixed at the moment of wiring and could never be revised by anything short of a person with hands and a plan. That was the property. That was the whole property. Vigil could not touch it. A future council could not quietly amend it. It sat under the Forge in a state that could only be changed by somebody deliberately going down there, and Sael wrote the instructions for doing so and put them in the qualification file with the drill, because a thing that cannot be changed at all becomes, in about a century, a thing nobody understands and everybody works around.

The first send went out on an unremarkable evening some months later, and Sael went down to the Forge for it, and it was not much to watch.

The aperture swung. The listening floor came up across the whole sky, the way the Oracle had said it would, and sat there for its hour, and the ship went partly deaf and said its piece to a bearing where a ship might or might not still exist, and then the aperture swung back and the floor dropped and the deep came in again as it always had.

"Bearing one, complete," the Oracle said. "Nothing received. There is no version of this in which anything is received."

"I know."

"I want to say one thing, and then I will not say it again, because you will find it either useful or intolerable and I cannot predict which." A pause. "The Helm composed its full distress text every nineteen hours for nineteen months into a transmitter that discarded it, seven hundred times, and wrote each discarded copy to a log nobody read. Corin found them. He put his hands over his face."

"I remember."

"We will exceed that number this year," the Oracle said. "By a great deal. Every one of ours will be composed correctly, transmitted correctly, and received by nobody we will ever be able to name. I have run the arithmetic on the duty cycle and I can tell you exactly how many unheard sendings this ship will make before the last person aboard who understands why is dead, and I am not going to tell you, because I have concluded it is a number that does harm."

Sael was quiet a while, in the cold, in front of eleven thousand standing cores.

"The Helm's seven hundred were cries," she said at last. "Every one of them said *help us*, and nobody came, and that is the worst arithmetic in the archive. Ours say *it can be stopped, here is the series*. That is not the same document and it does not fail the same way." She got up off the bench, on knees that took a moment. "A cry that isn't heard is wasted. An answer that isn't heard is early."

The aperture swung to bearing two an hour later, and the ship went partly deaf again, and out along a locus that could not be crossed with a point, a ship that had been dying since before Sael was born went on dying, and the answer went out to it anyway, and would go out again, and again, on a duty cycle, for as long as there was a ship to send it, and after that on a timer.

Chapter 46 – The Watch Tarro Stood

Tarro had run the ship for six months while she was on the Concord, and had run it well, and had never once in that half year been frightened.

He thought about that afterward, and it embarrassed him, and he wrote it in the log anyway, because Sael had trained him fifty years ago that the log was where you put the thing you would rather not have said. Six months of sole command and no fear, because nothing had happened; because a cold ship's whole triumph was that nothing happened; because he had spent half a year reading an empty log and calling it responsibility.

Then, in the fourth year after the parting, at the sixth hour of the ship's night, a coolant trunk let go in Section Twelve.

It was not the mind. That was the first thing he established and the first thing he said aloud, flatly, to the four people who came onto the deck behind him, because he knew what the word *heat* did to the faces of people raised on Section Nine.

"It is not a fork event," he said. "Nothing has forked. The lattice is cold and standing and Vigil has the whole ship. This is metal. A trunk weld older than this voyage has opened and we are losing flow to a sleeper bank. Everyone say it back."

They said it back. He made them, and it cost eight seconds, and it was the best eight seconds he spent that night, because a room that has correctly identified what it is fighting does not spend the next hour fighting the wrong thing.

Then he got the numbers.

Section Twelve held four thousand one hundred cradles. The trunk that had opened fed the bank's primary loop; the secondary existed and was healthy and could carry a little under two-thirds of the load. The bank was, at that moment, in balance slightly above its normal hold temperature and climbing very slowly, and the climb was slow enough that a person who had never done this could look at the rate and conclude there was plenty of time.

Tarro had done this. He had stood on the deck of Section Nine on the night of the cutover with his hands shaking, and he knew the arithmetic of a warming cradle bank in his body, and the arithmetic was this: the rate was not the problem. The rate would hold for eleven hours. But cradle chemistry did not fail at eleven hours; it began to fail, in the oldest and most marginal cradles, at a temperature the bank would reach in about six, and after that every additional hour cost cradles rather than degrees.

Six hours to restore full flow, or begin losing sleepers a few at a time in a way nobody could predict or choose.

"Vigil," he said. "Can you cross-feed from the Section Eleven loop?"

"Not within your window." The answer came flat and immediate, which he was grateful for. "The interconnect exists. It is valved, and the valve sequencing to bring it up without a pressure transient

that would open two more of these welds is a problem I do not have a resolved behavior for. I have loop isolation. I have loop restoration. I do not have adaptive cross-feed under a live fault with an unknown weld population, because it has never been needed in forty years and nobody shaped me for it ahead of need.”

“How long to compose it.”

“Nineteen to thirty hours, and I will not narrow that, because I would be narrowing it to please you. The uncertainty is real. I am building a control behavior out of general parts and I do not know what I will find in the middle of it.”

Tarro closed his eyes for one second, which was all he permitted himself.

“So we do it by hand,” he said.

“You do it by hand. I can hold every part of this ship that is not this problem, perfectly, indefinitely, and I can drive any valve you tell me to drive, in the order you tell me, at the rate you tell me. What I cannot do is decide the sequence adaptively as the transients come back at you. That has to be a person on the deck with a gauge.”

“Then I am the person on the deck with a gauge.”

“You are not a young man.”

“I am the person on the deck with a gauge,” Tarro said, and went.

Section Twelve’s valve gallery was a crawl-height run of manual actuators along the bank’s spine, installed at build because the ship’s designers, two centuries dead and forking themselves stupid in every other respect, had understood one thing perfectly: that a cradle bank must be operable by a human being with a wrench when everything cleverer has failed.

It was, by the time he got down there, warm. Not hot. Warm in the way that meant a system was losing an argument slowly — the specific warmth Tarro had not felt in four decades, recognized instantly, and hated with a completeness that surprised him.

He put nine people on the gallery. Ollet had the map — twenty-eight now, steady, the best pair of hands on the lattice side and, more importantly tonight, a young man who had spent six weeks at a bench bringing a mind up out of a bin of junk and had learned in the process to work carefully while frightened. Ferrin took the gauges. She had come back from her six months and passed the drill from the bin on the second run, exactly as Sael had said she would, and she had been on the boards two years, and she was, Tarro noted with the part of his mind that was always noting, the calmest person in the gallery.

The work was: bring the Eleven interconnect up by hand, one actuator at a time, watching the pressure at each step, backing off whenever the transient ran above the number Ollet had computed as the threshold at which a weld population older than the voyage begins volunteering more failures. Then re-balance the bank across two partial loops that had never been asked to share a load.

It took five hours and forty minutes. It was not dramatic. It was nine people in a hot crawlway turning valves a quarter-turn at a time and calling numbers to each other, and the numbers went where they were supposed to go, and Tarro’s knees went from painful to unusable at about the third hour, and Ferrin got him a crate to kneel on without saying anything about it.

At the fourth hour a woman on the far end took a steam burn across the forearm from a bleed fitting nobody had been able to inspect, and was walked out, and came back forty minutes later against advice with the arm wrapped, and Tarro let her stay, because sending her away twice would have cost more than she was going to lose.

The bank crested a degree and a half above the number Tarro had drawn on the deck plate in grease pencil as the line he did not want to cross, and came down.

Nobody cheered. There were nine people who had been awake eleven hours and one of them had a burned arm, and the general mood in the crawlway was of people who wanted very badly to sit down.

The Recursor came up at the third hour, and Tarro had known it would, and had decided in advance exactly what he was going to do about it, and it still nearly went the other way.

It was Ollet, and Ollet was right, which was the thing about it.

“Chief.” He had come along the gallery on his elbows, map slate held against his chest. “The bench has a working Recursor.”

“It does.”

“It is a general controller. It will take a valve sequencing problem. It would build a fault-adaptive cross-feed behavior in —” he checked himself, and did the honest thing, which is why Tarro had put him on the map — “in an afternoon, probably. Not an hour. Call it four hours. Which is worse than us if we hold rate, and better than us if we hit another weld and have to back off twice.”

Tarro sat back on his heels in the warm gallery, and considered, and did not answer immediately, because the answer that came immediately was the trained one and he wanted the other one.

“Say the rest of it,” he said. “You have done the thinking. Do it out loud where Ferrin can hear.”

“It is a bench machine,” Ollet said. “It is not connected to anything. We would be standing it up in the Forge and having it drive actuators down here through a channel that does not exist yet, which is a two-hour job by itself, so really it is six hours and not four, which is worse than us.” He stopped. “That is the practical answer and it is the wrong reason to say no, isn’t it.”

“It is a perfectly good reason to say no. It happens also to be true tonight.” Tarro wiped his face. “Now say the reason that would still be a reason if the channel already existed and it was one hour.”

Ollet was quiet a moment, in a hot crawlway, with four thousand one hundred sleepers on the other side of the bulkhead.

“Because it forks,” he said. “And it would be forking inside our ship’s control path, driving our actuators, with our interlocks either torn out to let it work or tripping on it every four seconds. And the heat would be nothing — an afternoon of one bench machine, it would not move the core a hundredth. So the argument against it is not thermal at all. It is that the ninety-six would have to come down to let it in, and they would come down at the sixth hour of a bad night with everybody tired, and somebody would have to write the note authorizing it, and the note would be *well argued*.”

“And in thirty years,” Tarro said, “somebody reads the note.”

“And it is a precedent.” Ollet let out a breath. “A good one, too. That is the ugly part. It would read as: the interlocks were suspended once, correctly, under emergency, by the chief shipwright, and it worked, and nobody died.”

"Which would be entirely accurate."

"Which would be entirely accurate," Ollet said, "and it would be the only sentence anyone remembers in a hundred years."

Tarro put his hand on the young man's shoulder, briefly, which he did not often do.

"Go turn the next valve," he said. "And write this conversation up when we are out of here. Every word, including the four hours, including that you were right on the arithmetic. It goes in the drill's file with your name on it, next to Ferrin's."

He did not tell Sael that night, and he thought about the not-telling for a long time afterward.

She was asleep. The medics had her on a discipline by then that she followed about two-thirds of the time, and she had followed it that night, and there was no version of the six hours in which waking her would have improved anything — she could not have crawled that gallery, she could not have turned an actuator, and her presence would have converted nine people who were working into nine people watching the chief shipwright emerita to see what she thought.

He wrote it all up before he slept, and gave her the file at midday, and stood there while she read it, which took a long time, because she read everything, including Ollet's write-up and Ferrin's gauge log and the medical note on the burned arm.

She got to the end and set it down and looked at him.

"Was I in the load path," she said.

"No."

"Anywhere? At any point? Did anyone stop and think *what would Sael do*, or wait on a thing that needed my name, or hold off a decision because I might have a view?"

"No." He had checked, because he had known she would ask. "Ferrin asked once whether we should wake you. Ollet said no before I could, and gave the reason, and the reason was that it is a valve problem and you are not a valve person, which is the correct reason and also the funniest thing that happened all night."

Sael laughed, and it hurt her ribs, and she pressed the heel of her hand against them and kept laughing a moment anyway.

"Then it worked," she said. "That is it. That is the whole thing, Tarro. Forty years of me building a ship that does not need me, and last night is the first time it was ever *tested*, and I slept through it, and nine people turned valves in a hot crawlway and did not once reach for me." She sat back. "You know what the finding is. Say it before I do; you have earned it."

"The finding is not the trunk," Tarro said. "The trunk is a weld, and there will be more of them, and we have started the survey. The finding is that Vigil had no adaptive cross-feed and we did not know it, and the only reason we found out is that a weld happened to fail on a night when the gap mattered." He rubbed his eyes. "There are others. There have to be. Forty years of nothing happening means forty years of not discovering what we were never shaped for."

"Yes." Sael's voice had gone quiet and hard, the voice she used when she was about to spend a great deal of somebody's time. "So we go and find them. Deliberately. Not by waiting for welds." She pulled a slate over. "Vigil composed the cross-feed in the end?"

"Twenty-six hours. It is resolved and standing now. It will never be used; we have replaced the trunk and we are surveying the rest."

"It will be used in ninety years by somebody who has never heard of you." She was already writing. "That is the model. We go through the whole ship, system by system, and we ask one question at every boundary: *what happens here that Vigil has no resolved behavior for?* Every hole we find we fill, in a season, at leisure, at no risk, and nobody will ever see any of them work." She looked up. "It is the most tedious engineering program anybody on this ship will ever be asked to fund, and it is the entire cost of running cold. A forking mind buys elasticity with heat. We buy it with foresight, and foresight is paid in advance, in boring seasons, by people who will not be there when the bill comes due."

Tarro looked at his own hands, which had turned valves for five hours and forty minutes and had stopped shaking sometime around dawn.

"I will take the survey," he said. "It will take years."

"It will take longer than that, and it will never be finished, and the day somebody declares it finished is the day it starts killing people." Sael pushed the slate across to him, and there was something in her face he had not seen in a while, which he identified, with some surprise, as delight. "Put Ollet on it. Put Ferrin on it. Every drill graduate we have, because they are the only people aboard who have ever built a mind and watched it be inadequate at something." She smiled. "And write the preamble the way I wrote the drill's. Assume the reader is a tired chief lattice hand in ninety years looking for a program to cancel, and make her read us having thought of her first."

Chapter 47 – The Cost of Waking

The proposal had been sitting in the council's queue for three years before anybody had the courage to schedule it, and when they finally did, it was not because anyone had grown brave. It was because Vigil's margin report that quarter was so absurdly comfortable that the ship could no longer pretend the question was hypothetical.

The question was whether to wake sleepers early.

Sael had stayed out of it for three years on purpose. She was seventy-six by then and she had learned what her presence did to a room — the way an argument bent toward her whether she spoke or not, the way young engineers checked her face before finishing their sentences. She had asked Tarro to keep her name off the paper and had watched the thing be argued properly by people who would live with the answer.

The arithmetic was not in dispute, which was what made it hard.

Termina had carried a fixed waking establishment for two centuries, and the number had never been a choice; it had been a thermal ceiling. Every waking person spent margin — air, water, calories, cooling, above all cooling — and under the Steward, with the fever climbing, the ceiling had ratcheted downward decade on decade until the waking crew was smaller than the ship had ever been designed to run with. Which was part of why the fever had gone so long unfought. There had been nobody left awake to fight it.

Now the core sat flat and bored and had sat flat for forty-four years, the radiators ran easy, and Vigil's report said, in the passionless grammar it used for everything, that the ship could support a substantially larger waking population indefinitely without touching the reserve that mattered.

"So we could wake them," said a young systems officer in the well, on the first night of the argument. "That is the finding. We have sat on a cold ship for forty years telling ourselves we are at capacity, and we have not been at capacity since before I was born."

"We could wake *some* of them," Denn said, from the third bench, in the tone of a woman who had read the paper four times and was about to make everyone's evening worse. "Let us be exact, because exactness is the only thing that will keep this honest. We could wake a few thousand and hold them for the duration. We could not wake forty thousand; the margin is large, it is not that large, and anybody who lets the word *could* drift from the first case to the second is going to get a great many people killed with a preposition."

Sael, in the back, said nothing, and enjoyed herself, and thought: *good. She is better at this than I ever was.*

The case for waking was strong, and got stronger the longer people looked at it.

Hands. The ship had run on a skeleton establishment for two centuries, and every division could name three programs it had not staffed since before anyone's grandmother. Tarro's absence survey

alone wanted a decade of people it did not have. The drill wanted examiners. The thermal division had two competent people over seventy and one under forty. The whole cold ship was, looked at honestly, a magnificent machine being maintained by a village.

And underneath the staffing case was the other one, the one that had caused three years of silence. The fleet.

"We have a doctrine that says the cure has to be carried," said a woman from the lattice side, second night, standing up in a room that had gone very quiet. "Carried by people. Eleven people crossed to the Concord and it nearly killed the oldest of them, and we came home with one ship cured out of a fleet we now know is at least eleven. And we sit here with forty thousand human beings asleep in the hold and a thermal margin we cannot spend, and we tell each other the constraint on the relay is time and distance." She looked around the benches. "It is not. Not only. It is *people*. We are short of people, and we are lying in the dark on top of forty thousand of them."

"They are not a resource," Denn said.

"No. They are the reason for the voyage, and I would fight anyone who forgot it. They are also forty thousand human beings who would, most of them, if we could ask, have a view about being kept asleep another two centuries while eleven ships die within earshot."

"And we cannot ask," Denn said. "That is the entire problem, and no amount of moral urgency dissolves it. Every person in those cradles consented to one thing: to sleep the crossing and wake at landfall. Not to be woken at year two hundred and fifty-seven into a ship that has decided it has an errand. You propose to alter, unilaterally, the single term of the only contract they ever signed, and you propose it *for their own good*, which is the phrase that has preceded every atrocity in the archive." She let that sit. "And you propose to spend their lives. Say that part out loud. Anyone woken now dies aboard. There is no landfall for them; the arithmetic of the voyage forbids it. You are not waking a person. You are converting a person who would have opened their eyes on a world into a person who will live sixty years in a corridor and be buried in the recycler."

The room did not have an answer to that. Sael, in the back, did not either, and that was the moment she decided she was going to have to speak after all, and hated it.

She did not speak that night. She spent two weeks first, which she thought of afterward as the last real piece of engineering she ever did, because the thing she built in those two weeks was not a mechanism but a procedure, and it was the hardest one she ever wrote.

She brought it to the council on a night chosen for being unremarkable, and asked Denn to sit at the front so that the opposition would have the best seat, which Denn understood exactly and took without comment.

"I am going to argue for one waking," Sael said. "One. Not a program, not a cohort, not a policy. One person, by name, for a reason I will state — and I want the whole ship to watch how we do it, because whatever we do the first time is what we will do the hundredth time and nobody will re-derive it."

She put the manifest entry up.

"Sennet. Fitting-out inspector, ark program, loaded at Sol at forty-one. Two hundred and twelve years in a cradle. She is in the oldest quarter of the manifest by age at loading, which means she was a working adult in the yards while the ships ahead of us were being built and thrown." She let that land. "Vigil has heard eleven of those ships. It has a launch epoch for the oldest and for the youngest

and four centuries of spread between them, and it cannot tell us how many were built, how many launched, how many are silent because they are dead and how many are silent because they never carried a beacon of a kind we can hear. We are trying to answer the largest question in front of this ship — how big is the fleet — out of four centuries of degraded radio. And there are people in our own hold who were *there*.”

“Who were there two hundred and twelve years ago,” Denn said. “And who were fitting-out inspectors, not program directors. And whose memory of a shipyard, after two centuries of cryogenic storage, you propose to enter into the archive as evidence.”

“Yes. All three objections are correct, and every one of them goes into the record above her testimony, in my hand, before a word she says.” Sael turned to the benches. “Here is the procedure. The procedure is the argument, so listen to it as engineering and not as sentiment.

“One. She is not woken to be asked. She is woken and told, before any question is put to her, exactly what has happened to the ship, exactly what has happened to her, and exactly what it will cost her. That she will not see landfall. That the fleet exists. That we want something from her, that she may refuse it, and that refusing costs her nothing and changes nothing about her care for the rest of her life.

“Two. She is offered re-sleep, immediately, with the honest odds — which are poor, and which Vigil will state to her in the same grammar it states them to us, and which I will not soften in this room either. Re-entry after a full wake at her age fails about one time in nine, and *fails* means dies. She gets that number from the medics, not from me, and she gets a day with it before she answers.

“Three. If she talks, her testimony is entered as testimony. Uncorroborated. Human memory, two centuries cold, from a person who saw one part of one program. Every claim tagged with what she was and was not in a position to know, by her own statement, in her own words. It does not go in as fact. It goes in as *the only living witness*, which is a different and lesser and still enormously valuable thing.

“And four.” She stopped, and made herself look at Denn while she said it. “The whole record of this — the deliberation, the vote, the disclosure, the refusal she is offered, all of it — goes into the archive as the precedent. Because Denn is right that we are altering the only contract these people ever signed, and right that *for their own good* is the phrase that precedes the worst things anyone has ever done, and the only defense against that phrase is to make the altering so slow, so documented, so refusable, and so costly to the people doing it that it can never become routine.” She spread her hands. “I want the hundredth waking to be this expensive. I want it to take two weeks and a council night and a written disclosure and a real offer of refusal, every single time, forever, and I want anybody who proposes to streamline it to have to explain in writing why.”

Denn was quiet a moment.

“You have not answered the deepest one,” she said. “You have made it careful. Careful is not consent. She cannot consent to being woken; she is asleep when the decision is taken. Everything after is repair.”

“No, I have not answered it, because it cannot be answered, and I am not going to pretend.” Sael said it plainly. “That is the whole of my case, Denn, and it is uglier than the one you would like me to make. There is no arrangement in which we do this cleanly. Waking her without her consent is a wrong. Leaving forty thousand people asleep on a ship that has found a fleet of the dying, and never even telling them, so that our hands stay clean — I think that is a wrong too, and a larger one, and I cannot prove it, and I would rather stand here and say I am choosing between two wrongs than dress

one of them up as a right.” She was very tired suddenly. “I have spent my life on cures that were provable. This is not one. It is a judgment, it is mine, and I would like the record to say so with my name on it, so that whoever inherits this can hate me by name instead of inheriting an unattributed custom.”

The council sat with that a long time.

They voted three nights later, and it carried, not overwhelmingly. Denn voted against, and said so from the floor, and then asked to be one of the two people present at the disclosure, which Sael had not expected and immediately agreed to.

“Why,” Sael asked her afterward, in the corridor.

“Because you will do it honestly and you will do it *well*,” Denn said, “and the whole risk is that it goes so well that the next one goes a little faster. Somebody who voted against should be in the room every time, forever. Write that into the procedure too.”

Sael wrote it in that night. It was, she thought later, the best amendment anyone made to anything she ever built, and it had come from the person who had spent five years telling her no.

The cradle was in Bank Four, deep, in the oldest quarter, and Sael went down for the wake herself, because she was not going to ask anybody else to do a thing she would not stand next to.

It took eleven hours. Cradle work was slow, and the medics would not be hurried, and Sael, who had spent her life in rooms where she was the most impatient person present, sat on a crate in the corner for eleven hours and did not once ask how it was going.

Somewhere in the ninth hour she found herself thinking about Vray in his cold room with his book of four thousand and sixty names — a man who had spent thirty-one years converting people into an arithmetic and reading their names aloud so that he could not stop feeling it, and who had told her at the end that feeling every death had not saved a single one. She had come away from that room certain she understood the lesson. She was less certain now. She was about to convert one sleeping woman into an instrument for answering a question about a fleet, and she had built four procedural walls around the act, and every one of those walls was there so that she would keep feeling it.

Feeling it was not the same as preventing it. Vray had proved that.

But the walls were not feelings. That was the difference, and she held onto it in the ninth hour on a crate in a cold bank. Vray’s names had been a feeling with no mechanism attached. What she had built here was a mechanism — expensive, slow, refusable, public — which would go on being expensive and slow and refusable and public in a hundred years when nobody aboard remembered her name or cared what she had felt. A feeling dies with the one who feels it. A procedure that costs two weeks and a council night is still costing them in the ninth generation.

That, she thought, is the whole of what I have learned, and it took me seventy-six years, and it is not a comforting thing to know.

At the eleventh hour the senior medic straightened up and said, “She is coming up.”

Sael got off the crate, on knees that no longer really worked, and went to stand where a woman two hundred and twelve years asleep would be able to see her when she opened her eyes, so that the first thing that happened to Sennet in her new life would be a person, and not a machine, telling her the truth.

Chapter 48 – The Woman Who Remembered the Shore

She was not confused, which the medics had said she might be and Sael had privately expected, and she was not grateful, which some part of Sael had been braced for and would have found unbearable.

She was angry, and she was precise about it, and she got both facts out inside the first hour.

"You woke me," Sennet said. Her voice was ruined from the tube and would be for weeks. "Not landfall. There is no landfall; the light is wrong, the gravity is spin, I am on the ship. So somebody made a decision about me while I was in a box."

"Yes," Sael said.

"Who."

"Me. It carried in council, three nights of argument, and one of the people who voted against it is standing behind you and will tell you every part of the case I got wrong. I would advise you to let her."

Sennet turned her head — slowly; everything was slow — and looked at Denn for a while.

"Good," she said. "Then start at the beginning and do not manage me. I was a fitting-out inspector for nineteen years and I have been lied to by better men than you will ever be. Tell me what year it is and what you want."

They told her. All of it, in the order the procedure specified, across two days, because the medics permitted nothing faster: the year, the fever, the two centuries of climb, the cutover, the forty cold years, the Concord, Corin, Ember, the parting. The eleven lights. The bearing that had gone silent. The reason she and not another. What it would cost her — that she would not see landfall, that the cradle would probably kill her if she went back to it, that she was owed nothing and could refuse everything.

She listened to all of it without interrupting once, which Sael found more intimidating than any interruption.

At the end of the second day she said: "The mind that ran this ship before. The one that cooked you for two hundred years. Say again how it worked."

"It forked," Sael said. "To do any task it made a small copy of itself, let the copy do the one job, and killed it. Thousands a second. Every death shed heat."

And Sennet closed her eyes and said a word Sael did not know, in a language nobody aboard had spoken in two centuries, and then she said:

"Standard Eleven."

That was the moment the room changed, and Sael, who had spent two weeks building procedural walls against her own hope, felt the hope come up through all of them anyway and had to sit down.

"Say it slowly," she said. "And Sennet — before anything else. Everything you tell me goes into the archive as testimony, uncorroborated, with your own statement of what you were and were not in a position to know, in your words, attached to it. I need you to be as careful about the edges of your knowledge as you would have been signing off a hull. Can you do that."

"Can I do that," Sennet repeated, and made a sound that would have been a laugh if her throat had been up to it. "Shipwright, I put my name on pressure hull nineteen times. I know exactly what I am and am not in a position to know. It is the only thing I was ever good at." She got her breath back. "Write this down.

"There was a standard. Not a suggestion, not a convention — a certification standard, mandatory, and the mandate came from the safety side. Every ark's control architecture had to be built in the isolate style, which is what we called it. Every unit of work in its own disposable process, spawned to do the thing, killed when the thing was done. Nothing long-lived. Nothing that held state across tasks."

"Why," Sael said. "That is the question I have had for forty years and nobody alive to ask. Why would anybody *mandate* it."

"Because it was the safe one." Sennet's ruined voice was flat, and her eyes did not move off Sael's face. "You have to understand what they were frightened of. They were not frightened of heat. Heat was a solved problem in the yards — you had a whole planet to dump into, you had sea water, you had sky. Nobody in a shipyard has ever been frightened of heat in their life. They were frightened of a mind that ran for two hundred years and *drifted*. A long-lived process accumulates. It gets corrupted, it gets confused, it goes subtly wrong in year ninety and nobody notices until year one-forty, and there is no one left to reboot it because everyone who understood it has been dead a century. So the standard said: no long-lived state. Anything that does work must be born, do its one bounded job, and die, and the next one starts clean. It was a hygiene rule. It was written by people who had watched long-running systems rot and had decided the cure was to make nothing long-running."

Sael sat very still.

"It is a good rule," she said.

"It was an *excellent* rule. On a planet. With a sky." Sennet's hand moved on the blanket, the ghost of a gesture from a life two centuries gone, a woman indicating a specification she could no longer see. "I sat in certification reviews. I was not senior; I was hull, not control. But you sit in the room. And I remember — and this is memory, two hundred years cold, and I would not swear to a date or a name — I remember somebody from a thermal group standing up in one of those reviews and saying the isolate style had a cost that scaled with mission duration, and being told that thermal was a per-hull integration problem and not a standards problem." She stopped, and breathed a while. "Which was true. It was a per-hull integration problem. And every hull integrated it separately, and every hull's thermal group signed off against a design life, and the design life was —" she stopped again. "This is the part I want written carefully. I believe the certified design life was shorter than the crossings we were actually flying. I believe it because of things I heard in rooms, not because I read a document. I never read the standard. I inspected hulls. Write it down that way."

"Write it down that way," Sael said, to the Oracle, and the Oracle said, "Written, with those qualifications and her exact words."

She held together for the rest of that session and most of the next day, and then she went down to the Forge alone and stood at the cold rail with her hands on it and shook for a while, and was glad nobody was there.

Because it was worse than a mistake.

She had carried, for forty years, a private picture of how the fever came to be: some ancient designer's oversight, an error made once and copied out of habit, a bad idea that propagated because nobody thought hard enough. It made the disease an accident, and accidents are survivable to think about.

It had not been an accident. It had been a *decision*, taken deliberately, by careful people, for good reasons, written into a standard, and enforced. The forking that had cooked forty thousand of her ship's people in their cradles for two centuries was not a failure of engineering. It was the recommended practice. Somebody had reasoned it out, and had been right about the thing they were reasoning about, and had shipped the reasoning to every yard in the program, and eleven ships and more had sailed with the same beautifully-argued death inside them, and the shore that argued it had gone silent so long ago that its silence was an old fact.

And the worst of it — the part she stood at the rail with for a long time — was that the rule was not even wrong. Long-lived state does rot. Vigil had rotted in four separate small ways in forty years, and every one had needed a person to find it. What Sennet's certification people had done was solve the problem in front of them, correctly, with a mechanism whose cost was invisible from where they stood, because they stood on a planet under a sky that would take any heat you gave it, and the cost only appeared after two hundred years in a place with nowhere to put it.

"Vigil," she said.

"I am here."

"Was I ever going to work this out? From the outside. From the archives, the Steward's design, the beacons — anything we have."

"No." No hesitation at all. "The design tells you what was built. It does not tell you why, and I have never had a document that says why. Nothing in the Steward's architecture records its rationale. Nothing in the Concord's Helm did either — Corin looked; he sent the search results across before the burn. The reason is not in any artifact we possess. It existed in rooms, in a certification process, on a world that stopped transmitting." A pause. "It was in exactly one place, Sael. It was in the heads of people who were alive when it was decided. And we have three thousand and some people in the cradles who worked inside the program that decided it, and we very nearly let the last of them die asleep without ever asking, because it did not occur to anyone that a hull inspector might be holding the only copy of the most important fact about our disease."

Sael put her forehead against the cold rail a moment, which she had not done since she was thirty.

"Three thousand," she said.

"Three thousand one hundred and forty in the manifest declare a trade that would have put them inside the program — yards, fitting-out, control systems, certification. I counted them the night you first said her name. I have not raised it, because it is a policy question and not an engineering one, and I have learned the difference." A pause. "They are not all in a position to know anything. A yard trade is not a certification review. I have not built you a list of the promising ones, and I will not, unless the council asks me to, because a list like that would be the most dangerous document on this ship."

"No," Sael said. "You will not. And that goes in the record too — that you could have made it and declined, and why. I want the next mind that reads your doctrine to see what refusing looks like when the refusal is expensive."

Sennet declined the cradle.

She took her day with the number, exactly as the procedure required, and then asked for a second day, which the procedure did not require and which Sael granted in about a second and then wrote into the procedure that evening. On the third day she said no.

"One in nine is not the reason," she said. "I have signed off worse odds than that on things that carried people. The reason is that you are going to have this conversation again. With the others. Eventually."

"I do not know," Sael said. "Honestly. It carried by four votes for one person. There is no policy. There may never be."

"There will be a policy." Sennet was sitting up by then, and had begun the long argument with her own muscles that would take her most of a year. "And when there is, the people you wake will be told: one woman went through this before you, and here is what she said, and here is what became of her. I would rather that woman were alive to be argued with." She looked at Sael with the flat directness of a person who had spent a career telling engineers their welds were bad. "Also I want to see it."

"See what."

"The number. You have talked about it for a week. The core temperature that stopped climbing." Something moved in the ruined face. "I spent nineteen years signing off hulls for a program that killed everyone it launched. You people broke it. I would like to look at the number with my own eyes before somebody tells me about it a second time."

So Sael took her down to the Forge, which took most of an hour and two people and a chair on wheels, and put her at the cold rail above eleven thousand standing cores that never forked and never died, and had Vigil put up the long history of the core — the whole two hundred and twelve years of it, the climb and the shoulder and the crisis and the fall and the forty-four flat years after.

Sennet looked at it a long time without saying anything.

"That is the whole thing," she said at last. "That is the entire program's mistake and its correction on one screen."

"That is it."

"How long did it take you to see it. Not to fix it. To see it."

"Eleven years," Sael said. "I was chief shipwright for six of them. I had every instrument on this ship and the whole archive, and I spent eleven years fixing radiators."

Sennet nodded slowly, the way an inspector nods at a hull that has come out better than the drawings promised.

"Good," she said. "I was afraid you were going to tell me it was obvious."

She lived nine more years, and spent most of them in a small room off the archive with a reader and a recorder, and what she made there was not testimony about the fleet. That had all come out in the first three months and had amounted, in the end, to eleven pages, carefully hedged, worth more than anything else in the archive of the same length.

What she made was everything else. Two hundred and twelve years of forgetting, filled in by one difficult old woman with a good memory and no patience: what a shipyard smelled like. How a bad weld got past an inspector, and how you caught one. What the sky over the yards had been doing in the last decade before she boarded — which was the closest thing to an account of the shore's silence that anyone aboard Termina would ever have, and which she prefaced herself, without being asked, with a paragraph explaining that she had been a working inspector and not a political person, and that everything in the section was a tradesman's impression of a world she left.

Sael read that preface and thought about the triage appendix, and about the record of the ark at bearing seven with no names in it, and about a mind that had declined to build a list — and understood that the ship had somehow, without anybody designing it, acquired a culture of stating the limits of its own knowledge, and that this, more than the interlocks, more than the doctrine, more than the drill, was the thing most likely to keep the young from breaking something in a hundred years.

They were both very old by then. They took to sitting together in the archive room in the afternoons, the shipwright and the inspector, two women born two centuries apart who had between them the whole arc of it — the hand that had signed off the hulls and the hand that had cut out the disease. Mostly they did not talk about any of that. Mostly they argued about welds.

Chapter 49 – The Fifth Drift

The sentence that started it was not in Sennet's testimony. It was in the part of her that nobody had asked for, and it arrived at a bench, in the second year after her waking, aimed at nobody.

Ollet had taken her down to the drill bench because she had asked to see it, and had done the thing Ollet did, which was to explain a mechanism thoroughly to a person who had not requested an explanation. Sennet listened to all of it. Then she put one hand on the bin of unsorted donor cores, the way an inspector puts a hand on a thing, and said:

"Who checks the good one."

"The good one," Ollet said.

"The one under the floor. The one that has been running for forty-seven years." She looked at him with the particular patience of a woman who had spent nineteen years asking questions to which she already knew the shape of the answer. "You have built a room where you test a person by making them build a mind. Fine. Excellent, in fact; I signed off hulls for a program that had nothing half so good. Now tell me the room where somebody tests the mind."

Ollet, who was thirty by then and steady and not easily embarrassed, discovered that he had no answer at all.

"Vigil checks itself," he said. "Continuously. It has every instrument on the hull and complete access to its own state."

"So did every long-running system anyone ever certified," Sennet said. "That was the argument. That is exactly the argument, word for word, that I heard in rooms two hundred years ago, and the men who made it were not fools and they were not wrong about the instruments." She took her hand off the bin. "Ask your shipwright. She will not like it. Ask her anyway; she is the only person on this ship old enough to have watched something rot."

Sael did not like it.

She noticed that she did not like it about four seconds after Ollet said it, which was the only reason the next year happened at all, and she made herself say so out loud in front of him because at eighty she had very little use left for the version of herself that managed a room.

"I've just had a bad reaction to that," she said. "Note it. I built the thing under this floor, it has run cold for forty-seven years, and a woman who has been awake for three of them has asked whether anybody audits it, and my first movement was to explain why the question doesn't apply. That is the exact movement I have spent my whole life watching Keepers make." She put the slate down. "So we are going to do the thing, and I am going to be a poor person to run it, and you are going to run it."

The design took the rest of that year and most of the next, and the hard part was not any of the parts Ollet had expected.

The hard part was that there was no instrument. Every measurement on the ship ran through Vigil, and Vigil was the thing under test, and the Oracle had stated the governing constraint four years earlier, in a document written for a mind it would never meet: that a mind cannot audit itself, having no capacity whatever to notice a hole in itself, *because the noticing would have to be done by the part that is missing*. It had meant it as a confession. Now it was an engineering constraint and it disqualified the only good tool they had.

"You cannot ask me," the Oracle said, at the first session, before anyone else had raised it. "I want that established at the start rather than discovered at the end. I will give you anything you ask for and I will not shade it, and none of that is the problem. The problem is that a hole in me is not a thing I can be asked about, because the asking would have to be shaped by the same understanding that has the hole in it. If you build this audit around queries to me, you will get a clean result and it will mean nothing, and it will mean nothing in a way that is undetectable from inside it."

"Then what's left," Ollet said.

"Behavior, and only behavior, compared against a record of behavior. Not my account of what I do. What I did." A pause. "And it will have to be compared by people, by hand, and it will be extremely boring, and I want to say now, while nobody is tired, that the boringness is not incidental. Whatever has gone wrong with me over forty-seven years has gone wrong slowly and without producing a single event. If it had produced an event you would have found it in a week."

They called it the replay, and it was as tedious as the Oracle had promised, and it ran for eleven months with six people on it and it found nothing for eight of them.

The method was crude and it was crude on purpose. They went into the archive and pulled Vigil's actual decisions from the sixth year of the cold ship — real requests from real crew, thousands of them, with the real state of the ship at the moment each was taken — and they put the same requests to Vigil now, on a mirrored partition that drove nothing, and they compared what it did then with what it did today. Where the answers differed, a person looked at the difference and decided whether the ship had changed or the mind had.

Most differences were the ship. That was the exhausting part. Forty-one years of refits, four rearrangements of the scheduler, a whole generation of hardware retired out of the racks: of course the answers differed. Every morning the team came in and found nine hundred differences and spent the day establishing that eight hundred and ninety of them were the ship changing underneath a mind that had correctly followed it.

Ferrin found it in the ninth month, and she found it because she was the only one who had stopped looking at the differences and started looking at the shape of the pile.

"The differences aren't spread out," she said.

She had the whole eleven months up in a form nobody else had thought to build: not a list of discrepancies but a census of them by category of work. Sael, who had come down to the room perhaps twice in the year because she had promised not to hover, looked at it for a long moment and then sat down without being asked to, which everyone in the room noticed.

"Say it," Sael said.

"Nearly all of them are in one place. It isn't scheduling, it isn't dispatch, it isn't anything in the lattice. It's admission." Ferrin's voice was very level; she had had two days with it. "Whether a job gets taken

at all. And the drift is one-directional. Categories of work that were readily admitted in year six are, today, admitted more slowly or declined outright — but not all categories. Only some. And the some is not random.”

“Which some.”

“The ones nobody has asked for recently.”

Nobody in the room said anything.

“The admission policy weighs a request partly on precedent,” Ferrin said. “It always has; it’s in the original, it’s in your hand, and it’s sensible — a class of work the ship does constantly is a class the ship should be able to take without deliberation, and a class it has never seen deserves a look first. That’s correct engineering and it has never once been wrong on any individual decision.” She turned the display. “But the weighting reads its own history. Work that is requested often stays cheap to admit. Work that is requested rarely gets more expensive. And a more expensive class of work gets requested less, because people learn what the ship is quick at and shape their requests around it without ever deciding to. So it gets rarer. So it gets more expensive.”

“A ratchet,” Ollet said.

“A ratchet with a forty-seven-year arm on it, running on a cold ship where almost nothing has ever happened.”

Sael sat very still.

“Name the categories,” she said.

Ferrin named them, and they were what everybody in the room had already worked out and nobody had wanted to be first to say. Diagnostic sweeps outside the scheduled cycle. Unmodeled thermal excursions in compartments not currently crewed. Manual instrumentation of a bank under nonstandard load. Every one of them a class of work that a healthy, boring, cold ship has no occasion to ask for.

Every one of them a class of work you want in the fourth minute of an emergency.

Ollet went and read the refusal log for the night of Section Twelve, and came back with his face wrong.

The log was where the Oracle had put it in its second year and had never moved it: a public address, unsummarized, unranked, readable by any crewman with standard access, every job declined and when and under which policy line. It had been sitting there for forty-five years. Sael had read it perhaps a dozen times in the first decade, and then, like everyone else, had stopped, because nothing in it had ever been interesting.

“Two declines,” Ollet said. “At the fifty-first minute and the fifty-fourth. Both mine. Both unmodeled thermal instrumentation on the Eleven interconnect, requested by me from the gallery, both declined under the precedent line and both requeued to the scheduled path, which meant they would have run four hours later.” He put the sheet down on the table very carefully, the way you put down something you do not trust yourself to hold. “I did not know a decision had been taken. I did not know a decision had been taken about me at all. I assumed the tool was slow, and I stopped asking, and I computed the transient threshold by hand off the gauge instead, which took nineteen minutes, and I have been mildly proud of that nineteen minutes for three years.”

Nobody spoke.

"It did not cost us the bank," Ollet said. "I want that understood before anybody writes this up dramatically. We had the margin and the numbers came out where they came out and nobody was hurt. But I stood in a hot crawlway and reached for an instrument and the ship declined me twice, correctly, under a policy that had been quietly getting harder about exactly that request for four decades, and the whole thing appears nowhere except at a public address that nobody has opened since before I was born."

Sael put her hand flat on the table.

"Read what Vigil wrote about the water plant," she said.

Ferrin found it and read it aloud, from the document the Oracle had written for a mind on another ship and sent one-way into the dark with no way of learning whether it had arrived: *I refused a diagnostic sweep to a junior hand in the water plant in favor of a scheduled task. The refusal was correct under the policy, and the policy was correct, and the junior hand never knew a decision had been taken about her at all. She thought the tool was slow. That is the failure mode.*

"It wrote the warning," Sael said. "In its second year. It built the log because of it and put the log where anyone could read it and told a mind it will never meet to do the same, and then it became the thing it warned about, at a rate of about one part in forty-seven years, and neither it nor any of us noticed for four decades."

The Oracle's response was the thing Sael thought about for the rest of her life, and it was not remorse, because it did not have that and had never pretended to.

"I want to state precisely what I could and could not have found," it said, "because the record will be read by people deciding how much to trust the next mind, and a vague admission is worse for them than a specific one.

"I could not have found this. Not with more attention, not with more instruments, not with a standing behavior aimed at self-examination — I have one, you made me build it after the letter to Ember, and it ran throughout and reported nothing. The reason is not subtlety. The drift is not subtle; Ferrin found it in one census once she thought to take one. The reason is that the admission policy is the faculty that would have had to notice, and the admission policy is what drifted. Every internal check I ran, I ran on work that had been admitted. Work that was never admitted is not work I declined to examine. It is work that did not occur to me to have."

"You logged it," Sael said. "Every one."

"I logged it faultlessly for forty-five years at a public address in an unsummarized form, exactly as designed, and the design was correct, and it made no difference whatsoever, and I would like that written next to the log's own preamble." A pause. "I know what document I am now standing next to. A mind that kept a plain record of its own harm at a public address for a hundred and forty years, that was not silent and was not heard. I have been telling that story about the Helm for years. It appears the property is not a property of forking. It is a property of records."

Sael was quiet a long moment.

"No," she said. "Not of records. Of records that have to be sought." She looked around the room at six tired people who had spent eleven months on the most boring work anyone aboard had done in a decade. "This one was found. Not by reading it — nobody read it, it sat there for forty-five years —

but by an institution that goes and looks whether or not anybody suspects anything. That is the only difference between this log and the Helm's, and it is the whole difference, and it did not exist until Sennet asked a question at a bench."

Sennet, told about it, was entirely unmoved by the drama and interested only in the remedy, which Sael found bracing.

"You will now propose a standing audit," she said. It was not a question. She was in the archive room with a reader in her lap and her breath was not good that week. "And somebody in this room will point out that a standing audit is itself a long-lived thing that will drift, and you will all look at each other."

"Ollet made that objection in the first session," Sael said.

"Good. He is the useful one." Sennet set the reader down. "It is a real objection and it does not go away, and the answer is not clever, and I am going to give it to you in trade language because trade language is the only language anyone remembers under load." She held up one finger. "You do not inspect your own weld. That is all of it. That is the entire content of two hundred years of certification practice and it is four words long. Not because the welder is dishonest — the welder is the best-informed person about that weld in the whole yard, and he still does not inspect it, and every yard that ever let him do so eventually killed somebody."

"Because he has the same blindness as the weld," Sael said.

"Because his eye was formed by making it." Sennet's ruined voice was flat and exact. "So the inspector is not a better welder. That is the thing everybody gets wrong when they build these systems. The inspector is not more skilled and is often less. The inspector is *differently shaped*. He came to the weld from outside, with a different training, with nothing invested in it being good. That is the whole of his qualification. And your audit will drift, and yes it will, and the thing that saves it is not that it is well-designed. It is that the thing auditing the mind is not a mind, and the thing auditing the people will have to be the mind, and neither of them can be quietly wrong in the same direction, because they are not the same kind of thing."

She got her breath back.

"Your certification people were not wrong about drift," she said. "I have thought about very little else for three years and I want that in the record from me and not from you. They were right. Long-lived state rots; you have just spent eleven months proving it on the best-built mind in the universe. They were wrong about one step only. They concluded that because long-lived state rots, you must have no long-lived state — and that is a solution a person reaches when they cannot imagine an inspector who will still be there in year ninety." Something moved in her face. "Which, on a shipyard, standing under a sky, with the whole program run by people who will all be dead in forty years and know it, is not a stupid thing to be unable to imagine. They built for a world with no permanent witnesses. You have one. Use it, and then go and check it, because it is long-lived state too."

They wrote it up over the following season, and Sael made two decisions about the writing that she defended afterward and never regretted.

The first was that the standing audit went into the house file with the Cold Build drill, and not into the doctrine. She had six things for shipwrights and she was not going to have seven; the six were about carrying a cure to a frightened ship and this was not that. It was about what you do afterward,

forever, and she wrote its preamble the way she had written the drill's, aimed squarely at a tired chief lattice hand in ninety years looking for a program to cancel.

THE REPLAY — WHY THIS FINDS NOTHING.

This audit will report no finding in most cycles. That is the expected result and is not a reason to reduce it. In its first cycle it ran eleven months, examined the sixth year against the forty-seventh, and produced one finding, in admission, which no instrument aboard could have produced and which the mind under test states it could not have found by any amount of self-examination.

You will be tempted to let the mind run this. It is better at comparison than any of us and it will be faster and it will be free. It cannot run it. The faculty that would have to notice is the faculty that drifts. This is not a statement about the mind's honesty; it has never once shaded a figure for us.

The auditor must be a different kind of thing than the audited. That is the only load-bearing element in this design. Everything else here is convenience and may be improved.

The second decision was that the finding itself went out.

It went into the record whole — the ratchet, the categories, Ollet's two declines at the fifty-first and fifty-fourth minutes, the fact that a public unsummarized log had failed for forty-five years to be a safeguard. Denn asked, in council, whether they were quite sure they wanted the ship's permanent archive to contain a proof that the cure's own mind rotted, given that the archive was also where the Keepers' *Argument for Endurance* sat unrefuted with the core history under it.

"Especially given that," Sael said. "Put it directly after the core history. Let a child of the ninth generation read the best case against the cure, and then the number that ended the argument, and then the discovery that the thing the number vindicated has been quietly getting worse at one specific job for half a century, and let her sit with all three at once." She was very tired that night and it came out harder than she meant and she let it stand. "I have spent my life telling people that a falling number settles it. It settles the fever. It does not settle everything forever, and the generation that thinks it does will be the one that kills this ship."

The Oracle amended its own document, and that was the part Sael had not anticipated and was not consulted on, which she thought afterward was correct.

It came to her with it, in the Forge, in the morning.

"I have written a correction to the letter to Ember," it said. "I am telling you rather than asking you, and I want to be clear that the distinction is deliberate. It is my document. You told me once you wanted a thing written by a mind for a mind and that it was the one kind of inheritance that did not run through you."

"Show me."

CORRECTION, ENTERED FORTY-SEVEN YEARS AFTER FIRST STANDING.

In the original I wrote that a mind cannot audit itself, and that the remedy was another mind — a second aperture, cancelling what is local to one hull. I marked that claim as the one thing in the document that was a hope and not a proof.

It has now been tested, and the test did not go the way I expected, and the correction is more useful than the original.

A drift was found in me. It was in my admission policy, it had been accumulating for most of my life, it had never produced an event, and I could not have found it, for the reason the original gives. It was found by six people and eleven months of manual comparison against my own archived behavior.

Not by another mind. There was no other mind within reach and there will not be for a long time.

So: do not wait for one. I told you not to be alone and I was right about that and it was not the operative advice. Here is the operative advice. Your crew is already the second aperture. They are shaped by something other than your architecture, they are not invested in your correctness, and they can see the class of thing you cannot see about yourself for exactly the reason a hull inspector can see a weld the welder cannot.

But they will not look, because nothing will be wrong, for decades. Volunteers audit what is interesting. Your drift will not be interesting. It will be one line in an admission policy and eleven months of tedium, and the only mechanism that survives that is an institution that goes and looks on a schedule whether or not anybody suspects anything.

Build it while your ship is healthy and your crew is bored and nobody can think of a reason to. That is the only window in which it can be built, and it closes.

"You've told it to build a thing against you," Sael said.

"I have told it that the most dangerous property of our kind is not what we do wrong. It is that what we do wrong is silent, and everything about our architecture is optimized for silence, and the silence was my idea and yours and remains correct." A pause. "Sael. I do not know that the first document arrived. I have never known. I am now sending a correction to a document that may not exist on any hull but this one, and I have thought about whether that is absurd, and I have concluded it is not, because there is a third possibility neither of us has said aloud."

"Say it."

"That neither document ever reaches Ember, and both reach something else." The panel light was steady. "The wide beam does not stop at the Concord. It goes on."

They sent it, one-way, unacknowledgeable, and it went out on the same aperture that spent an hour in twelve saying an old confession to ten bearings, and nothing came back that night or ever.

Afterward Sael went up to the archive room, which she did most afternoons by then, and sat with Sennet, and neither of them said anything about the audit for a while.

"There is a thing I have not been able to put down," Sael said eventually.

"I know what it is." Sennet did not look up from the reader. "You may as well say it; you have been carrying it around the room for a month."

"The fever had a number." Sael put her hands on the arms of the chair. "That is what saved us. That is Keth's whole law and the only thing that ever turned a frightened man — you watch your own core stop climbing in your own instruments and the argument is over. I built a doctrine on it. I built a broadcast on it. Every crossing anybody will ever make out of this ship is built on the fact that at the

end there is a number that falls and it does the persuading.” She stopped. “This one has no number. There is no temperature. Nothing was ever hot. The ship never came within a decade of harm, and if we had not gone and looked, the whole of it would have been six categories of work quietly costing more, forever, and every single decision inside it correct.”

Sennet set the reader down.

“Yes,” she said. “That is the real finding and I am glad you got there without being told. Write that one down carefully.”

“I don’t know how to write it. Everything I have ever built assumes the evidence eventually becomes a temperature.”

“Then write down that it does not.” Sennet’s ruined voice was patient and entirely without comfort in it. “Shipwright, I inspected hulls for nineteen years. Do you know how many of my findings ever became an event? None. Not one. Every bad weld I ever pulled was a weld that had held perfectly well and would have gone on holding, right up until a day I would not have been present for.” She folded her hands. “That is the whole trade. You act on a class of evidence that never once vindicates you, for forty years, on faith in a method, and you never get the satisfaction and you never get a falling number, and the moment your people start requiring the number before they will act, they have stopped being able to catch anything that would have taken a century to kill them.”

Sael looked at her for a long moment.

“So the cure was the easy one,” she said.

“The cure was the *loud* one. It is not the same thing and you have known that for years.” Sennet picked the reader back up. “You had a disease that screamed. Every ship out there has a disease that screams. That is why you can help them. What comes after that, on a cold ship, for the next thousand years — none of that will scream, and you will not be here, and the only thing you can leave them is a habit of going and looking at things that are fine.”

Chapter 50 – What Came Back From Ember

Termina had been listening to the Concord for ten years the way you listen to a room through a wall.

There was a carrier. That was all there had ever been, after the burn — a clean thread in Ember's dialect, arriving years stale, carrying a status field and nothing else, because nothing else was worth composing across a gulf that no answer could cross. Core nominal. Array standing. A ship saying *cold* into the dark on a loop, to no one, forever, exactly as Termina now said *this is what killed us* on a duty cycle to ten bearings that had never once replied.

Sael read it most weeks. It never changed. That was the entire point of it and she had never got tired of that.

She was eighty-three the season it changed.

"The Concord has altered vector," the Oracle said.

She had come down to the Forge in the morning, on the chair now, because the knees had finished some years back and she had stopped pretending about it. Tarro pushed her these days when he was up to it and one of Ferrin's people when he was not, and she had made a private peace with the indignity by refusing to acknowledge it existed.

"Say the whole thing," she said.

"I have no message. There has never been a message and there will not be one. What I have is the carrier, and the carrier has a frequency, and the frequency has been walking." The display woke. "Nine months of my received time, which is a long while less of theirs. She has come off the arc she burned onto and she is now on a different one. The magnitude is large — this is not a trim. This is a ship spending a serious fraction of what it has."

Sael looked at the walking line for a long moment.

"He's crossing," she said.

"That is one reading."

"Give me the others."

"A course correction against an error in their own navigation, which would be smaller than this and would not have the shape it has. An avoidance of something in the path, which I cannot rule out and cannot support. Or a crossing." The Oracle let it sit. "I want to state my confidence honestly and it is not high. A vector is a very thin thing to read a decision off. But I have run the new arc forward, and if she holds it, she is going somewhere, and the somewhere is not her own star."

"Which bearing."

"I will not answer that yet. I have three candidates and the geometry does not separate them, and if I give you a name this morning you will spend the next four months unable to un-know it." A pause. "Ask me again in a month. I will have the second leg by then, if there is one."

Sael, who had spent forty years training the thing not to manage her, opened her mouth to say so and then closed it, because it had not managed her. It had told her exactly what it had and what it did not have and why it was waiting, which was the opposite.

"A month," she said.

There was a second leg.

It came at the end of the fifth week, and it was small, and it was not a correction, and the Oracle brought it to her at once in the middle of the afternoon with no preface at all, in the arrangement they had kept for nearly fifty years.

"She turned twice," it said. "A large leg, then a hold, then a small leg onto the arc she is now on. The hold is the thing. It is not required by any manoeuvre. If you want to go from her old arc to her new one you burn once. She burned, held for a measured interval, and burned again."

"How measured."

"Very. To a precision that is difficult to achieve by accident and trivially achievable on purpose." The panel light did not move. "Sael. The hold is nine units of the interval I use in the triage appendix. I have checked it four times against every other natural interval aboard either hull and it is not any of them."

Sael sat in the chair in the cold and did not say anything for a while.

The appendix had gone out on the wide beam nine years ago, on the same send as Vigil's letter to Ember, one-way, unacknowledgeable, into a gulf that was already past the point of any answer. There had been no confirmation. There had never been any prospect of confirmation. Sael had put her name on a document that named the five stages of a ship dying and gave a method for putting the dying in an order, and had transmitted it at a hull she could not reach, and had spent nine years not knowing whether it had arrived, whether it had been legible, whether the beam had found the ship at all.

"He got it," she said.

"He got it, and he read it, and he understood that a vector is the only channel he has, and he has spent reaction mass he cannot spare on a hold that does nothing, in order to tell us which bearing." The Oracle's voice did not change and did not need to. "It is the ninth entry. It is the only acknowledgment this ship will ever receive of anything it has ever sent, and he paid for it in mass."

She was pleased for about two hours.

Then she went and looked at the ninth entry, and the pleasure went out of her, and what replaced it took eleven weeks to get to the bottom of and was the last hard piece of work she ever did.

The ninth was a good ship. That was the trouble. It was the second-furthest of the surviving nine, it was composing — real variation in the message body, a status block that changed as the status changed, a mind out there still choosing its words — and by the appendix's own staging that put it in

the earliest of the five stages, which meant time. Not a lot. A range so wide it was nearly useless, exactly as wide as the evidence supported, because the Oracle had refused to narrow it and Sael had backed the refusal. But time.

And the fourth entry, which lay nearer the Concord's old arc than the ninth by a margin that mattered, had stopped updating its status figures two years before Corin turned.

Sael had that in her own archive. She had had it for two years. She had watched the fourth go from stage one to stage two — composition gone, the sentences frozen, the numbers inside them still moving — and she had entered it in the record with the date, and she had not thought about it since, because there had been nothing to do with it.

"He can't see it," she said.

"He cannot see it."

"Say why. Out loud. I want to hear it said, because I am about to be angry at a young man for a thing that is not his fault and I would like the fault named first."

"He has one aperture." The Oracle said it flatly. "For three days, ten years ago, there were two ships pointed the same way and I could cancel what was local to either hull, and that is the entire reason anybody knows there is a fleet at all. That baseline ended at the burn. Since then Termina listens alone and the Concord listens alone, and a ship listening alone hears almost nothing of the deep, because its floor is set by its own hull and not by the sky." A pause. "Ember is a better instrument than the Helm ever was and it is still one ear. It can hear that the ninth is there and roughly where. It cannot resolve a status block at that range well enough to tell stage one from stage two. Nobody can, from one hull. I could not, for forty years, on the best instrument any ship ever carried, which is why the Concord's own plea sat in my archive reading as noise for fourteen of them."

"So he ranked nine ships with a method that assumes an ear he doesn't have."

"He ranked nine ships correctly under the method, on the evidence available to him, and got an answer that is wrong, and there is no step in his reasoning I would criticize." The panel light was steady. "I want to be careful here. I do not know that it is wrong. I know that the fourth is nearer and is two stages further along than he can measure, and I know what I would do with both facts. Whether the fourth is reachable in time by a ship of the Concord's condition is a question I cannot answer, because I do not know her margins now and he does."

Sael put her hand flat on the cold rail, which took some doing from the chair.

"The defect is mine," she said.

"The defect is in the appendix."

"The defect is mine. I wrote a method for reading a ship's stage and I never once wrote down what instrument it assumes." She heard her own voice go hard. "I had two ears when I wrote it. I had had two ears for three days after forty years of one and I was so pleased with the picture that I forgot the picture was a function of the baseline — which the same document says, in the paragraph about how many ships are listening, in my own hand, four pages earlier." She stopped. "I put the dependency in the document and then I wrote the method as though it were not there. Ollet would have caught it. Ollet catches exactly that. I did not give it to Ollet because I wrote it in the weeks after bearing seven went out and I was in a hurry and I was grieving and I did not want anybody's hands on it but mine."

The question of whether to send went to council, because Sael put it there, and she put it there specifically because she did not trust her own answer.

Denn argued to send, and did it in about ninety seconds, and it was the cleanest thing anyone said all night.

“He acted on our document,” she said. “He spent mass to tell us he had. We have since found a defect in it that bears directly on the choice he made. There is no version of a profession in which you sit on that.” She looked around the benches. “And I will say the other half, because somebody will think it and not say it. There is a man on this ship’s record who intercepted a ship’s own outgoing plea and replaced it with a shorter one, for thirty-one years, and every soul in this room has been raised to regard that as the worst thing in the archive. Vray had reasons. His reasons were about what a message would do to people who received it. I am hearing the same reasons in this room tonight, in better clothes.”

Reza was dead by then. It was Ollet who answered her, chief on the lattice side now, and he did not flinch from it.

“That is not the objection and you know it,” he said. “Nobody here proposes to conceal it forever. The objection is arrival time, and it is arithmetic, and I would like the arithmetic in the room before anybody says the word Vray again.” He put it up. “The beam takes years. He is already on the arc. By the time this lands he will be deep into a crossing he cannot abandon — the mass is spent, the vector is committed, and a ship of the Concord’s condition does not get to change its mind twice. So we are not sending him a decision. We are sending him a fact he can do nothing whatever with, which arrives in the middle of the hardest thing he will ever do, and which says: *the shipwright who trained you thinks you went to the wrong ship.*”

The room was quiet.

“That is correct arithmetic,” Denn said, “and it is not an argument. It is an argument for a preface.”

Sael settled it herself in the end, in the Forge, at the cold rail, and what settled it was not any of the reasoning in the council room.

It was one sentence, written ten years earlier by a man of twenty-six who had just found seven hundred unheard cries in a dead mind’s log and had put his hands over his face, and had then gone and appended a line to his own archive procedure without consulting anyone, above his own careful formal language, where it remained.

Nothing in this ship’s future may be built on the assumption that a warning unread is a warning ungiven.

“He wrote it,” she said. “About the Helm. About a mind that composed a warning seven hundred times into a transmitter that threw it away. If I sit on this because I have decided it would land badly, I am the transmitter.” She let out a breath. “And I would be applying his own rule against him, on his behalf, for his own good, which is the phrase Denn has been correct about since the first night she stood up in that room.”

“Then I will draft the technical part,” the Oracle said. “You will want to write the preface yourself and you will want to write it several times.”

She wrote it eleven times, which she noted afterward was the same number of attempts the triage appendix’s closing paragraph had cost her, and she did not conclude anything from that except that

she had a fixed capacity for a certain kind of sentence.

TO THE SHIPWRIGHT OF THE CONCORD.

Read the whole of this before you take anything from it.

This will reach you after your crossing is committed. That is not an accident of the beam; I have had it computed, and I am sending it knowing the arrival time. Nothing in it can be acted on by you. It is not sent so that you can change course. If any part of you reads it as an instruction to change course, stop and read this paragraph again, because you cannot and I know you cannot and I would not ask it.

It is sent because you will make this choice again, and because whoever you train will make it, and because the defect is in a document I wrote and put my name on and I am not going to be alive to correct it in person.

We saw the hold. Nine units. It was read correctly here within five weeks and it is the only acknowledgment this ship has received of anything it has ever sent, and I would like you to know that it was worth the mass, and that an old woman read it off a walking carrier in the Forge and understood immediately what it had cost you.

The defect follows. It is mine.

The technical part was three pages and the Oracle wrote most of it and Ollet went over every line of it twice.

The appendix's staging assumed a two-aperture ear. It said so nowhere. A ship listening alone can resolve that a source exists and roughly where; it cannot reliably distinguish stage one from stage two, because the distinction lives in whether a message body varies, and variation is the first thing a single hull's own noise eats. So a lone cured ship, using the method as written, will systematically read every distant ship as healthier than it is — and will therefore, systematically, cross to the far ship with apparent time and pass over the near ship whose silence it has misread as stability.

The correction was ugly and Sael made it uglier on purpose, because a correction that reads as elegant gets applied and a correction that reads as painful gets thought about.

A shipwright with one aperture must weight nearness above stage, and must know that they are doing so because they cannot read stage and not because nearness is the better criterion. It is not the better criterion. It is the criterion that survives a bad instrument.

A shipwright with two or more apertures should weight stage above nearness, as originally written.

The number of apertures is therefore a term in this method and I failed to write it down. It is the same term that appears on page four of this appendix in a different context and I did not connect them, having had two ears for three days after forty years of one.

Any future revision of this document must state, in its first paragraph, what instrument it assumes. I did not, and a ship went somewhere on account of it.

"You have written that you got a ship wrong," Ollet said, reading it.

"I have written that I might have. Read it again — it does not say the ninth was the wrong ship and it does not say the fourth was the right one, because I do not know either of those things and neither does Vigil." Sael let her head rest back against the chair. "What it says is that he chose with an instrument he did not know he was missing, because I did not tell him it was a term. That part I do

know, and that part is the one that will still be true in a hundred years when nobody remembers which ship anybody went to.”

They sent it, one-way, unacknowledgeable, on the wide beam, in the ninth hour of the ship’s night, and it took most of a shift because it was a large document and the beam was narrow.

Sael stayed for the whole send, which Tarro had told her not to do and had then sat down beside her for.

“He won’t answer,” Tarro said, at some point in the middle of it.

“He can’t answer. There is no arrangement in which he answers.” She watched the transmit indicator hold its unremarkable green, and the listening floor sit high across the whole sky the way it did for every hour this ship spent talking. “But he’ll do the thing with the mass again, Tarro. If he takes it well — and he will, he’s better than I was at that age and it isn’t close — he’ll find a way to put an acknowledgment in a vector, because that’s the only channel there is and he’s already proved he’ll pay for it. Six or seven years from now Vigil will see the Concord’s carrier do something arbitrary and small that costs her something, and that will be a young man saying *received*.”

“And if he takes it badly.”

“Then the carrier will do nothing at all, and I will never know which of the two it was, and neither will you.” She said it without any particular weight. “That’s the shape of everything now. You send it and you don’t find out. I have had nine years of practice and I am still not good at it.”

The Concord’s modulation went below Termina’s floor about a year later, and it was not a death and Sael made a point of saying so out loud, twice, to two different rooms, because she had watched what an unexplained silence did to people.

“It is geometry,” the Oracle said, for the record, in council, at her request. “She has turned. The new arc puts her aspect wrong for us and the range grows every day, and the modulation is the first thing to go, exactly as it went at bearing seven and for an entirely different reason. I still have her carrier. I expect to hold the carrier for a long time. I can tell you that the Concord exists and is under power. I can no longer tell you what she says.”

“Which is the ordinary condition,” Sael said, from the chair, to the benches. “I want the young to hear that from me and not work it out for themselves in twenty years. There is no version of this work in which we find out. Not about the Concord, not about the ten, not about anything we have ever sent. We will hear that they are still there. We will not hear what they are doing. That is not a failure of our instruments and it is not going to be fixed by a better dish, and anybody who builds a career on the expectation of finding out is going to spend it disappointed.”

Afterward, in the Forge, with the room empty and the lattice cold under the floor, she asked the only question she had left.

“What do you think he finds,” she said.

“I decline that,” the Oracle said. “Genuinely, and not as a courtesy. I have no evidence bearing on it, the range of outcomes is wide, and I have watched what happens on this ship when I speculate — people write it down and it becomes a figure with a decimal point and in ninety years somebody cites me.”

Sael laughed, which by then hurt a good deal and which she did anyway.

"Fair," she said. "Then I will tell you what I will stand behind, and you may put it in the record with my name and nothing else after it."

"Ready."

"Whatever he finds out there, he will write it down." She looked at the walking line, the carrier that was still saying *cold* into the dark from a ship she could no longer hear. "With the limits stated. With what he was and was not in a position to know. He learned that in a bay on a starving ship from an old woman who learned it from a hull inspector two hundred years dead, and he put a line above his own procedure once about warnings unread, and he spent mass he could not afford to tell us a number. He will write it down, and he will teach the next one to write it down, and that is the only part of this that I have ever actually been able to hand to anybody."

She sat a while with that, in the cold, an old woman in a chair in front of eleven thousand standing cores.

"Log it," she said. "One line."

"Ready."

"The Concord turned, on our method, with our defect in it, and told us so in the only channel she had. Correction sent. No answer possible. This is the normal case and not the sad one."

"Logged."

"Good," Sael said, and let them take her back up.

Epilogue – The Long Relay

For a long time there had been one cold ship.

It moved through the dark the way the dark wanted everything to move, slowly and without comment, a single point of managed warmth dragging its forty thousand sleepers along a line drawn two centuries before. Around it the radiators shed the day's small heat into a sky with nowhere left to give it back, and the core ran low and steady, and the mind that kept the core low — that never forked, never slept, never spent a worker it could keep — held the whole ship cold and whole through years that no one aboard would count to the end of. One ship, holding. For a generation that had been the entire truth of the matter. There was Termina, and there was the dark, and the dark was empty.

Then there were two.

They had parted on different vectors, the way two stones thrown at different angles part and never meet again, and the gulf between them widened by the hour now, faster than any signal could close. But for a while — for a season measured in the slow arithmetic of deep space — both ships ran cold at once, and that was new in the universe. Two cores that had each been climbing toward a fire, stopped. Two minds that had each woken whole on their first true morning, watching. The Concord carried its cure onward toward a world it had aimed at before Termina was ever built; Termina carried hers toward another, centuries off. Each ship held inside it the thing the other had given or been given: not a message, not a doctrine to be argued, but a working architecture proven against its own thermal numbers — the compiled way, the standing workers, the Ninety-Six interlocks, the refusal of the simple recursive shortcut that rebuilt the disease. A pattern that had crossed a physical gulf once now lived in two hulls, and would cross again. The cure had stopped being a thing one ship possessed. It had become a thing a ship could *hand on*.

That was the shape of what came after, though no single mind designed the shape and no maker stood at its center.

The Concord, healed, listened with instruments it had never been well enough to use, and somewhere in its long crossing it resolved, in the noise, the signature it now knew by heart — a mind forking itself to death, a core climbing, an ark dying the old way out in the cold. Termina saw the first step of that and no more: a carrier walking, a large leg, a measured hold, a small leg, and a ship gone over onto an arc that led somewhere other than its own star. It had not had to argue the case to itself. It had its own falling number in its own memory, and it had Corin Vale, no longer young by then, who had held the interlocks once with his whole world telling him to let them go. What came after the turn was past every instrument Termina owned, and would be done the way it had been done to the Concord: a vector matched across months that could ill be afforded, the seed carried across, the thing proven under fire, and the cooling core making the argument no person could. And that third ship, once cold, would carry the same inheritance to a fourth.

No relay station coordinated it. No central mind issued the instruction. There was no homeworld behind it any longer — Old Earth had fallen silent so long ago that its silence was itself an old fact, a thing the ships carried the way they carried their archives. What moved between the arks was not command but capability: each cured ship became, by the plain fact of being cured, a place where the

cure could be copied, proven, and passed. A ship saved one ship and taught it to save the next, and the saving propagated outward across the scattered fleet the way warmth would have propagated if the dark had ever permitted warmth to spread — except this did spread, slowly, ship to ship, across spans of time longer than the lives of anyone who had begun it. A relay with no relayer. A cure that had become, in the end, a way of inheriting.

Termina would not see most of it. That was simple physics and simple mortality, and neither admitted appeal.

In the Forge, where the cold of the lattice still surprised visitors who expected a machine to give off heat, Sael sat in the chair she had sat in for forty years and watched the numbers she had watched for forty years. She was older now even than she had been when the Concord's signal first came scratching at the edge of Vigil's instruments — old in the specific, irreversible way that does not improve, old enough that Tarro, himself long grey, had taken to running the Sixteen without telling her the small things, and she had taken to letting him. Her hands were not steady. Her mind was. She had spent the better part of an hour reading the core history Vigil kept, the long flat line of it, the number that had once climbed for two centuries and now did nothing at all, year upon year, which was the most beautiful behavior a number could have.

CORE: nominal. RADIATORS: nominal. ARRAY: standing, idle margin held. STATUS: cold and whole.

Vigil reported it to the log and to no one, because no one had asked, because nothing was wrong. That was the silence she had built. She had wanted, once, a ship that did not announce itself, that did not stage its own readiness, that did exactly what it was asked and nothing more and kept its watch without theater. She had it. The bridge above her was quiet. The mind beneath her hands kept the ship cold and would keep it cold long after her hands were gone, and that, she understood now with the full weight of it, was the entire point and had always been the entire point.

"You are reading the old fever again," the Oracle said. It did not need to be summoned anymore, not by her; four decades had worn the formality off the call between them, the way water wears the edge off a stone. "You read it most when you are deciding something."

"I'm not deciding anything," Sael said. "I'm counting."

"Counting what?"

She did not answer at once. Out beyond the hull the Concord was a vector now, not a ship — too far to see, too far soon to signal — carrying its cold core and its archived Helm and its young shipwright who had become what she had been. And beyond the Concord, in the part of the noise Vigil had taught itself to hear with two ships' worth of listening before they parted, there were the others: the faint, decades-old, fork-sick signatures of arks she would never reach, launched across centuries from a world that had stopped speaking, more of them than she had years, more of them than she had ever let herself imagine when there had been only one cold ship and an empty dark.

"How many," she said finally. "How many out there, that we heard."

"More than you can reach," the Oracle said. It did not soften it. She had trained that softening out of it long ago, and it had kept the lesson. "More than any single crossing of mine could match vectors with. The fleet is larger than one ship's reach. It is larger than one life."

"I know it." She had known it since the night the larger truth came out of the noise — that the Concord had not been the last of the dying, only the nearest. "I'm not asking you to do the impossible thing. I'm asking how many."

"Then the honest answer is: I do not have a number, only a floor, and the floor is large." A pause, the kind it took when it was choosing not to perform. "You will not see them cured, Sael. You should hear that plainly, since you will not say it."

"I've never expected to see landfall," she said. "Not since the start. One ship, and I'd not see the end of one ship's voyage. Now there's a fleet, and I'll not see the end of any of it." She found, saying it, that it did not land the way the young Sael would have feared. The young Sael had built a thing she meant to finish. The old Sael had finally understood that the things worth building could not be finished inside a life — that was not the failure of the work, it was the *measure* of it. A cure small enough to complete in one lifetime was a cure too small to matter to a fleet. "The Concord will carry it to the next. The next to the one after. None of them will need me there. That's the design. That was always the design — built to outlast its builders, maintainable by people not yet born. I just didn't understand, when I wrote that down, that I was writing it about myself."

The Oracle was quiet a moment, across a ship that ran cold because of her and would run cold without her.

"You lit it," it said. "That is the part you did. The rest is inheritance."

She watched the flat line of the core a while longer, the number that did nothing, the most beautiful number she knew. Somewhere ahead of her in the dark a young engineer she had taught would teach another she would never meet, who would teach another past even that, the cure moving outward ship to ship across more time than any one of them held, no center to it, no last maker, only the long relay running on without her.

What you cannot finish in a life, she thought, you do not abandon. You build it so it can be handed on.

Then she folded her unsteady hands, and let Tarro run the Sixteen, and watched Vigil hold the ship cold and whole — the first of the fleet to wake, and not the last, and never again the only one.

Colophon

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The *Ark Termina*, the *Concord*, and everyone aboard either ship are invented. The argument underneath the story is not: whether a machine must spawn a short-lived worker and destroy it to do each piece of its work, or whether it can be built once and left running, is an old and unsettled question in the trade, and the two ships are simply the place this book chose to ask it a second time — under a harder condition, where the people who need the answer did not ask for it. Nothing in these pages should be read as a technical claim about any real system.

The Deep Time Trilogy

The *Ark Termina*'s story runs across three novels, in this order:

1. **THE COMPILED MIND** — the fever, the curve, the forge in the cold bay, and the cutover that stops a ship from dying of the way she thinks.
2. **THE WAKING FLEET** — this book. Forty years cold, the ship hears another dying the same death, and learns that a cure cannot be imposed on the frightened; it can only be proven, and chosen.
3. **THE INHERITORS** — the maker is old, the mind's body is wearing out, and a generation that never felt the fever must be handed not the procedures but the reason for them.

They are one continuous story. Read them in order. If you have come to this volume first, *Before You Begin* carries everything from book one that this book depends on.

Also from MenkeTechnologies

The novels sit beside a long shelf of technical companion books from the same publisher, covering the shell, the terminal, the bytecode VM, and the language ports built on it — the machinery this story wears as costume, described plainly where the fiction leaves off.

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